

CURRICULUM-2023 (C-23)

DIPLOMA IN AUTOMOBILE ENGINEERING



STATE BOARD OF TECHNICAL EDUCATION &
TRAINING ANDHRA PRADESH

**DIPLOMA IN AUTOMOBILE ENGINEERING
CURRICULUM- 2023 (C-23)**

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PREAMBLE

Technical Education is a key driver of economic development and plays a crucial role in providing individuals with the skills and knowledge necessary to thrive in the workplace. As technological advancements continue to reshape industries and create new opportunities, it is critical that technical education curricula remain relevant and up-to-date.

The curriculum has been designed with this in mind, with a focus on practical skills, critical thinking, and problem-solving. We believe that these skills are essential for success in both academic and professional spheres. The revamping of the technical education curriculum is made with collaborative effort from educators, industry experts, policymakers, and students.

At the heart of the curriculum, is the belief that the technical education should be **student-centered**, empowering learners to take ownership of their learning and pursue their passions. We aim to create a learning environment that is safe, supportive, and nurturing, where every student has the opportunity to reach their fullest potential. We acknowledge that learning is a lifelong journey, and our curriculum is designed to provide a solid foundation for continued growth and development. We hope that our students will not only leave with a diploma but with employability and passion for learning.

The State Board of Technical Education and Training, (SBTET) AP, has been offering Diploma programmes to meet the above said aspirations of the stake holders: industries, students, academia, parents and the society at large. **The Curriculum should be flexible, adaptable, and responsive to the changing needs of the industry and society.** As such, it has been the practice of SBTET, A.P., to keep the curriculum abreast with the advances in technology through systematic and scientific analysis of current curriculum and bring out an updated revised version at regular intervals.

The design of Curriculum C-23 was started in the month of January - 2023. Feedback was collected from all stake holders: Students, Lecturers, Senior Lecturers, Head of Sections and Principals for all programmes for this purpose. Accordingly, a workshop was convened on 15th February 2023 by Smt. C. Naga Rani, I.A.S, Director of Technical Education & Chairperson, SBTET, AP to discuss on revamping of C-20 curriculum to meet the needs of industries and for improvement of placements.

The meeting was attended by Sri. Saurab Gaur, I.A.S, Principal Secretary, Skill Development & Training, Smt. Lavanya Veni, I.A.S, Director, Employment & Training. Thirteen Representatives from Industries and Fourteen Academicians from Higher Level Institutions and officials of ITI, Skill Development, CTE & SBTET attended the workshop.

Smt. C Naga Rani, I.A.S., Commissioner of Technical Education while addressing in the workshop, emphasized the necessity of industrial training and on-hand experience, that the students need to undergo to support the industries and the Gaps in the Curriculum need to be fixed to make the students passionate to work in the industry in order to support economy of the country.

The committees of each branch consisting of experts from Industries, Higher Level Institutions and Faculty of Polytechnics are informed to study the possibility of incorporating the following aspects while preparation of the curriculum so as to improve employability.

- **To bring out industry-oriented Diploma Engineers.**
- **Internet of Things (IoT) for all branches**
- **Theoretical & Practical subjects 50: 50 Ratio**
- **Industry 4.0 concepts.**
- **5G Technology.**
- **Critical Thinking (Quantitative Aptitude, Data Interpretation, Quantitative reasoning etc) to face the written tests conducted by the industries during placements.**
- **Dynamic, Student centric to suit the needs of the industry.**

In continuation, series of workshops with subject experts followed in the subsequent weeks for thorough perusal for preparation of draft curriculum. Also, the suggestions received from representatives from various industries, academic experts from higher level institutions, subject experts from Polytechnics, have been recorded, validated for incorporation into the **Curriculum C-23**. Finally, the draft curriculum was sent to academicians of higher-level institutions, industrial experts for Vetting.

The design of new Curricula C-23 for different diploma programmes has thus been finalised with the active participation of the members of the faculty teaching in the Polytechnics of Andhra Pradesh, and duly reviewed by Expert Committee constituted of academicians and representatives from industries. Thus, the primary objective of the curriculum change is to produce employable diploma holders in the country by correlating the growing needs of the industries with relevant academic input.

The outcome-based approach as given by NBA guidelines has been followed throughout the design of this curriculum and designed to meet the requirements of NBA Accreditation, too.

The Revised Curriculum i.e., Curriculum-2023 (C-23) is approved by 45th Academic Committee of SBTET, A.P for its implementation with effect from Academic Year 2023-24. Also, the SBTET, A.P under the aegis of the Department of Technical Education, Andhra Pradesh in it's 62nd Board Meeting held on 13-07-2023 (vide item no: 17) Approved to update the Polytechnic Curriculum C-23 with effect from the academic year 2023-2024 onwards after revamping the present C-20 curriculum, to meet the latest industrial technological developments including Industry 4.0 concepts.

2. HIGHLIGHTS OF CURRICULUM C-23

The following Courses/ Topics are incorporated in this curriculum C-23 as per the suggestions received from Industrial Experts, Faculty of Higher Level Institutions and Polytechnics to improve the Employability Skills of the Polytechnic Students.

1. Duration of course for regular Diploma and for sandwich Diploma is 3 years.
2. The Curriculum is prepared in Semester Pattern. However, First Year is maintained as Year-wise pattern.
3. 6 Months Industrial training has been introduced for 3 years Diploma Courses.
4. Updated and new courses relevant to the industry are introduced in all the Diploma courses.
5. The policy decisions taken at the State and Central level with regard to environmental science are implemented by including relevant topics in Chemistry. This is also in accordance with the Supreme Court guidelines issued in Sri Mehta's case.
6. Keeping in view of the increased need of communication skills which is playing a major role in the success of Diploma Level students in the Industries, emphasis is given for learning and acquiring listening, speaking, reading and writing skills in English. Further as emphasized in the meetings, Communication Skills lab and Life Skills lab are continuing for all the branches.
7. CAD specific to the branch has been given emphasis in the curriculum by introducing additional CAD-II course with existing CAD-I. Preparing drawings using CAD software has been given more importance.
8. The C-20 curriculum had more focus on theory than practical skills, but the new C-23 curriculum will have an equal emphasis on both theory and practical's. This will help students learn practical skills better by giving them more opportunities to work in laboratories and workshops.
9. With increased emphasis for the student to acquire Practical skills, the course content in all the subjects is thoroughly reviewed and structured as outcome based than the conventional procedure based.
10. Curricula of Laboratory and Workshops have been thoroughly revised based on the suggestions received from the industry and faculty, for better utilization of the equipment available in the Polytechnics. The experiments /exercises that are chosen for the practical sessions are identified to confirm to the field requirements of industry.
11. Aim to generate interest and provide an understanding of the program based on industry and academia suggestions in C-23 Curriculum a core theory course, "Elements of Automobile Engineering," and a lab course, "Automobile Engineering Lab-I," are introduced in the first year of the program.
12. In response to the developments in the field of automobile engineering and based on industry-academia suggestions, two new theory courses have been added to the curriculum. These courses include "Autotronics" and "Two & Three Wheeler Technology (with E-bikes & E-Rickshaws)".
13. With the increasing demand for fuel and emissions regulations, the C-23 curriculum has replaced the chapter on this topic in the C-20 curriculum with a new course named Alternative fuels & Emission Controls.
14. To complement the new theory and existing courses in the C-23 curriculum, two new lab courses, Autotronics lab and Electric & Hybrid Electric vehicles lab have been

added. These lab courses aim to provide practical experience and support the theoretical concepts learned in the classroom.

15. Production Technology of C-20 is revamped with addition of latest manufacturing techniques like additive manufacturing process (3-D Printing), advanced machining process” etc. and the subject is renamed as Advanced Manufacturing Process” in C-23.

3. ACKNOWLEDGEMENTS

The Members of the working group are grateful to Smt C. Naga Rani I.A.S., Commissioner of Technical Education & Chairman of SBTET, for continuous guidance and valuable inputs during process of revising, modifying and updating the Curriculum C-20 to Curriculum C-23.

We are grateful to Sri. S. Suresh Kumar, I.A.S, Principal Secretary, Skills Development & Training for his valuable suggestions to bring the revamped curriculum C-23 in to a final form to meet latest Industry 4.0 concepts.

We are grateful to Sri. Saurab Gaur, I.A.S, former Principal Secretary, Skills Development & Training who actively participated in the Industry-Academia workshop conducted on 15th February, 2023 and offered valuable suggestions and insights into the learning needs and preferences so that the curriculum is engaging, inclusive, and effective.

It is pertinent to acknowledge the support of the following in the making of Curriculum C-23. A series of workshops in different phases were conducted by SBTET, AP, Guntur involving faculty from Polytechnics, Premier Engineering Colleges & representatives from various Industries and Dr. C. R. Nagendra Rao, Professor & Head, NITTTR-ECV to analyse the Previous C-20 Curriculum and in designing of C-23 Curriculum, is highly appreciated and gratefully acknowledged.

We also extend our sincere thanks to Sri. V. Padma Rao, Joint Director of Technical Education, Sri K.V. Ramana Babu, Secretary, SBTE&T, Andhra Pradesh, Sri K. Vijaya Bhaskar, Deputy Director (Academic), Andhra Pradesh, officials of Directorate of Technical Education and the State Board of Technical Education, Andhra Pradesh and all teaching fraternity from the Polytechnics who are directly or indirectly involved in preparation of the curricula.

4. RULES AND REGULATIONS OF C-23 CURRICULUM

4.1 Duration and pattern of the courses

All the Diploma programs run at various institutions are of AICTE approved 3 years or 3½ years duration of academic instruction. All the Diploma courses are run on year wise pattern in the first year, and the remaining two or two & half years are run in the semester pattern. In respect of few courses like Diploma in Bio-Medical course, the training will be in the seventh semester. **Run-through system is adopted for all the Diploma Courses, subject to eligibility conditions.**

4.2 Procedure for Admission into the Diploma Courses:

Selection of candidates is governed by the Rules and Regulations laid down in this regard from time to time.

- a. Candidates who wish to seek admission in any of the Diploma courses will have to appear for the Common Entrance Test for admissions into Polytechnics (POLYCET) conducted by the State Board of Technical Education and Training, Andhra Pradesh, Vijayawada. Only the candidates satisfying the following requirements will be eligible to appear for the Common Entrance Test for admissions into Polytechnics (POLYCET).
- b. The candidates seeking admission should have appeared for S.S.C examination, conducted by the Board of Secondary Education, Andhra Pradesh or equivalent examination thereto, at the time of applying for the Common Entrance Test for admissions into Polytechnics (POLYCET). In case of candidates whose results of their Qualifying Examinations is pending, their selection shall be subject to production of proof of their passing the qualifying examination in one attempt or compartmentally at the time of admission.
- c. Admissions are made based on the merit obtained in the Common Entrance Test (POLYCET) and the reservation rules stipulated by the Government of Andhra Pradesh from time to time.
- d. For admission into the following Diploma Courses for which entry qualification is 10+2, candidates need not appear for POLYCET. A separate notification will be issued for admission into these courses.
 - i). D.HMCT ii). D. Pharmacy

4.3 Medium of Instruction

The medium of instruction and examination shall be English.

4.4 Permanent Identification Number (PIN)

A cumulative / academic record is to be maintained of the Marks secured in sessional work and end examination of each year for determining the eligibility for

promotion etc., A Permanent Identification Number (PIN) will be allotted to each admitted candidate to maintain academic records.

4.5 Number of Working Days Per Semester / Year:

- a) The Academic year for all the Courses shall be in accordance with the Academic Calendar.
- b) The Working days in a week shall be from Monday to Saturday
- c) There shall be 7 periods of 50 minutes duration each on all working days.
- d) The minimum number of working days for each semester / year shall be 90 / 180 days excluding examination days. If this prescribed minimum is not achieved due to any reason, special arrangements shall be made to conduct classes to complete the syllabus.

4.6 Eligibility (Attendance to Appear for the End Examination)

- a) A candidate shall be permitted to appear for the end examination in all subjects, if he or she has attended a minimum of 75% of working days during the year/Semester.
- b) Condonation of shortage of attendance in aggregate up to 10% (65% and above and below 75%) in each semester or 1st year may be granted on medical grounds.
- c) A stipulated fee shall be payable towards condonation for shortage of attendance.
- d) Candidates having less than 65% attendance shall be detained.
- e) Students whose shortage of attendance is not condoned in any semester / 1st year and not paid the condonation fee in time are not eligible to take their end examination of that class and their admissions shall stand cancelled. They may seek re-admission for that semester / 1st year when offered in the next subsequent academic semester/year.

For INDUSTRIAL TRAINING:

- i) During Industrial Training the candidate shall put in a minimum of 90% attendance.
- ii) If the student fails to secure 90% attendance during industrial training, the student shall reappear for 6 months industrial training at his own expenses.

4.7 Readmission

Readmission shall be granted to eligible candidates by the respective Principal/ Regional Joint Director.

- a) (i) Within 15 days after commencement of class work in any semester (Except Industrial Training).
- (ii) For Industrial Training: before commencement of the Industrial training.
- b) Within 30 days after commencement of class work in any year (including D. Pharmacy course or first year course in Engineering and Non-Engineering Diploma streams). Otherwise, such cases shall not be considered for readmission

for that semester / year and are advised to seek readmission in the next subsequent eligible academic year.

- c) The percentage of attendance of the readmitted candidates shall be calculated from the first day of beginning of the regular class work for that year / Semester, as officially announced by CTE/SBTET but not from the day on which he/she has actually reported to the class work.

4.8 Scheme of Evaluation

a) First Year

Theory Courses: Each Course carries Maximum marks of 80 with an end examination of 3 hours duration, along with internal assessment for Maximum of 20 marks. (Sessional marks). However, there are no minimum marks prescribed for sessionals.

Laboratory Courses: There shall be 40/20 Marks for internal assessment i.e. sessional marks for each practical Course with an end examination of 3 hours duration carrying 60/30 marks. However, there are no minimum marks prescribed for sessional.

b) III, IV, V, VI and VII Semesters:

Theory Courses: End semester evaluation shall be of 3 hours duration and for a maximum of 80 marks.

Laboratory Courses: Each Course carry 60/30 marks of 3 hours duration 40/20 sessional marks.

4.9 Internal Assessment Scheme

- a) **Theory Courses:** Internal assessment shall be conducted for awarding Sessional marks on the dates specified. **Three-unit tests shall be conducted for I year students and two Unit Tests for semesters. The details are presented below.**

S. No.	Type of Assessment	Weightage Assigned
(i)	Testing of knowledge through mid-examination for year/ sem as (Mid-1+Mid-2+Mid3) or (Mid-1 + Mid-2)	40
(ii)	Assignments	5
(iii)	<i>Dynamic Learning activities : Project Work/ Seminar/Tech-fest/Group Discussion, Quizzes etc./Extra-curricular activities/NSS/NCC/ IPSGM/Cleaning & Greening of Campus etc.</i>	5
	TOTAL	50

Internal Assessment shall be of 90 minutes duration and for a maximum of 40 marks for each test.

At least one assignment should be completed for each unit which carries 10 marks. The total assignment marks should be reduced to 5.

The dynamic learning activity is to be conducted which carries 10 marks. The total marks should be reduced to 5.

The total 50 marks assigned to internal assignment is to be scaled down to 20 marks.

b) Practical Courses:

(i) Drawing Courses:

The award of Sessional marks for internal Assessment shall be as given in the following table:

Distribution of Marks for the Internal Assessment Marks			
First Year (Total:40 Marks)		Semesters (Total:40 Marks)	
Max:20 Marks	Max:20 Marks	Max:20 Marks	Max:20 Marks
From the Average of THREE Unit Tests.	From the Average of Assessment of Regular Class work Exercises.	From the Average of TWO Unit Tests.	From the Average of Assessment of Regular Class work Exercises.

- For first year engineering drawing each unit test will be conducted for a duration of 2 hours with maximum marks of 40.
- (Part - A: 4 questions x 5 marks = 20 Marks; Part -B: 2 questions x 10 marks = 20 marks).
- For the semester drawing examinations, Two Unit tests shall be conducted as per the Board End Examination Question Paper Pattern.
- All Drawing exercises are to be filed in serial order and secured for further scrutiny by a competent authority

(ii) Laboratory Courses:

- (a) Student's performance in Laboratories / Workshop shall be assessed during the year/ semester of study for 40 marks in each practical Course.
- (b) Evaluation for Laboratory Courses, other than Drawing courses:
 - i. Instruction (teaching) in laboratory courses (except for the course on Drawing) here after shall be task/competency based as delineated in the Laboratory sheets, prepared by SBTET, AP & NITTTR- ECV and posted in SBTET website.
 - ii. Internal assessment for Laboratory shall be done on the basis of task/s performed by the student as delineated in the laboratory sheets, prepared by SBTET, AP & NITTTR- ECV and posted in AP, SBTET website.
 - iii. Question paper for End semester Evaluation shall also be task/s based and shall be prepared and distributed by SBTET as done in case of theory courses be prepared as per SBTET rules in vogue.
- c) Internal assessment in Labs / workshops / Survey field work etc., during the course of study shall be done and sessional marks shall be awarded by the concerned Teacher.
- d) For practical examinations, except in drawing, there shall be two examiners. External examiner shall be appointed by the Principal in consultation with respective Head of Section preferably choosing a qualified person from in the order of preference.
 - i) Nearby Industry
 - ii) Govt / Semi Govt organization like R & B, PWD, PR, Railways, BSNL, APSRTC, APSEB etc.

- iii) Govt / University Engg College.
 - iv) HoD/Senior Lecture (Selection Grade-II) from the Govt.Polytechnic
- Internal examiner shall be the person concerned with internal assessment as in (c) above. The end examination shall be held along with all theory papers in respect of drawing.
- e) Question Paper for Practicals: Question paper should cover (the experiments / exercise prescribed to test various) skills like handling, manipulating, testing, trouble shooting, repair, assembling and dismantling etc., from more than one experiment / exercise
 - f) Records pertaining to internal assessment marks of both theory and practical Courses are to be maintained for official inspection.
 - g) In case of Diploma programs having Industrial Training, Internal Assessment and Summative Evaluation, shall be done as illustrated in the following table:

Assessment no	Upon completion of	By	Based on	Max Marks
1	12 weeks	1.The faculty concerned (Guide) and 2. Training in charge (Mentor) of the industry	Learning outcomes as given in the scheme of assessment ,for Industrial Training	120
2	22 weeks			120
3.Final summative Evaluation	24 week	1.The faculty member concerned, 2.HoD concerned and 3.An external examiner	1.Demonstration of any one of the skills listed in learning outcomes	30
			2.Training Report	20
			3.Viva Voce	10
TOTAL				300

- h) Each staff member including Head of Section shall be assigned a batch of students 10 to 15 for making assessment during industrial training.

4.10 Minimum Pass Marks

a) Theory Examination:

For passing a theory Course, a candidate has to secure a minimum of 35% in end examination and a combined minimum of 35% of both Sessional and end examination marks put together.

b) Practical Examination:

For passing a practical Course, a candidate has to secure a minimum of 50% in end examination and a combined minimum of 50% of both sessional and practical end examination marks put together. In case of D.C.C.P., the pass mark for typewriting and short hand is 45% in the end examination. There are no sessional marks for typewriting and Shorthand Courses of D.C.C.P course.

C) Industrial Training:

- I. Monitoring: Similar to project work each teacher may be assigned a batch of 10-15 students irrespective of the placement of the students to facilitate effective monitoring of students learning during industrial training.
- II. Assessment: The Industrial training shall carry 300 marks and pass marks is 50% in assessments at industry (first and second assessment) and final summative assessment at institution level put together i.e. 150 marks out of 300 marks. And also student has to secure 50% marks in final summative assessment at institution level.
- III. **In-Plant Industrial Training for 3-Year Diploma (C-23) Courses is scheduled as per the Academic Calendar of the SBTET every year.**

4.11. Provision for Improvement

Improvement is allowed only after he / she has completed all the Courses from First Year to Final semester of the Diploma.

- a) Improvement is allowed in any 4 (Four) Courses of the Diploma.
- b) The student can avail of this improvement chance **ONLY ONCE**, that too within the succeeding two examinations after the completion of Diploma. However, the duration including Improvement examination shall not exceed **FIVE** years from the year of first admission.
- c) No improvement is allowed in Practical / Lab Courses or Project work or Industrial Training assessment. However, improvement in drawing Course(s) is allowed.
- d) If improvement is not achieved, the marks obtained in previous Examinations hold good.
- e) Improvement is not allowed in respect of the candidates who are punished under Mal-practice in any Examination.
- f) Examination fee for improvement shall be paid as per the notification issued by State Board of Technical Education and Training from time to time.
- g) All the candidates who wish to appear for improvement of performance shall deposit the original Marks Memos of all the years / Semesters and also original Diploma Certificate to the Board. If there is improvement in performance of the current examination, the revised Memorandum of marks and Original Diploma Certificate will be issued, else the submitted originals will be returned.

4.12. Rules of Promotion From 1ST YEAR TO 3rd, 4th, 5th, 6th and 7th Semesters:

A) For Diploma Courses of 3 Years duration

- i). A candidate shall be permitted to appear for first year examination provided he / she puts in 75% attendance (which can be condoned on Medical grounds up to 10%) and pay the examination fee.
- ii) A candidate shall be promoted to 3rd semester if he/she puts the required percentage of attendance in the first year and pays the examination fee. A

candidate who could not pay the first year examination fee has to pay the promotion fee as prescribed by State Board of Technical Education and Training, AP from time to time before commencement of 3rd semester.

A candidate is eligible to appear for the 3rd semester examination if he/she puts the required percentage of attendance in the 3rd semester and pays the examination fee.

iii) A candidate shall be promoted to 4th semester provided he/she puts the required percentage of attendance in the 3rd semester and pay the examination fee. A candidate, who could not pay the 3rd semester exam fee, has to pay the promotion fee as prescribed by State Board of Technical Education and Training AP from time to time before commencement of 4th semester. A candidate is eligible to appear for the 4th semester examination if he/she puts the required percentage of attendance in the 4th semester and pays the examination fee.

iv) A candidate shall be promoted to 5th semester provided he / she puts the required percentage of attendance in the 4th semester and pays the examination fee. A candidate, who could not pay the 4th semester examination fee, has to pay the promotion fee as prescribed by State Board of Technical Education and Training from time to time before commencement of 5th semester.

A candidate is eligible to appear for the 5th semester examination if he/she puts the required percentage of attendance in the 5th semester and pays the examination fee.

v) A candidate shall be sent to Industrial training / VI semester provided he/she puts in the required percentage of attendance in the 5th semester and pay the examination fee/ promotion fee as prescribed by SBTET.

A candidate is eligible to appear for Industrial Training assessment (Seminar/Viva-voce) puts the required percentage of attendance, i.e., 90% in 6th semester Industrial Training.

For IVC & ITI Lateral Entry students:

i.) A candidate shall be permitted to appear for Third Semester examination provided he / she puts in 75% attendance (which can be condoned on Medical grounds up to 10%) and pay the examination fee for Third semester.

ii) A candidate shall be promoted to 4th semester provided he/she puts the required percentage of attendance in the 3rd semester and pay the examination fee. A candidate, who could not pay the 3rd semester exam fee, has to pay the promotion fee as prescribed by State Board of Technical Education and Training AP from time to time before commencement of 4th semester.

A candidate is eligible to appear for the 4th semester examination if he/she puts the required percentage of attendance in the 4th semester and pays the examination fee.

- ii) A candidate shall be promoted to 5th semester provided he / she puts the required percentage of attendance in the 4th semester and pays the examination fee. A candidate, who could not pay the 4th semester examination fee, has to pay the promotion fee as prescribed by State Board of Technical Education and Training from time to time before commencement of 5th semester.

A candidate is eligible to appear for the 5th semester examination if he/she puts the required percentage of attendance in the 5th semester and pays the examination fee.

- iii) A candidate shall be sent to Industrial training / VI semester provided he/she puts in the required percentage of attendance in the 5th semester and pay the examination fee/ promotion fee as prescribed by SBTET.

A candidate is eligible to appear for Industrial Training assessment (Seminar/Viva-voce) puts the required percentage of attendance, i.e., 90% in 6th semester Industrial Training and pays the examination fee.

B) For Diploma Courses of 3 ½ Years duration (MET/ CH/ CHPP/ CHPC/ CHOT/ TT):

- i. A candidate shall be permitted to appear for 1st year examination provided he / she puts in 75% attendance (which can be condoned on Medical grounds upto 10%) i.e. attendance after condonation on Medical grounds should not be less than 65% and pay the examination fee.
- ii. A candidate shall be promoted to 3rd semester if he/she puts the required percentage of attendance in the 1st year and pays the examination fee. A candidate who could not pay the 1st year examination fee has to pay the promotion fee as prescribed by State Board of Technical Education and Training from time to time before commencement of 3rd semester.
- iii. A candidate shall be promoted to 4th semester provided he/she puts the required percentage of attendance in the 3rd semester and pay the examination fee. A candidate, who could not pay the 3rd semester exam fee, has to pay the promotion fee as prescribed by State Board of Technical Education and Training from time to time before commencement of 4th semester.
A candidate is eligible to appear for the 4th semester exam if he/she puts the required percentage of attendance in the 4th semester

For IVC & ITI Lateral Entry students:

- a) Puts the required percentage of attendance in the 4th semester
- iv. A candidate shall be promoted to 5th semester industrial training provided he / she puts the required percentage of attendance in the 4th semester and pays the examination fee. A candidate, who could not pay the 4th semester examination fee, has to pay the promotion fee as prescribed by State Board of Technical Education and Training from time to time before commencement of 5th semester.
- v. Promotion from 5th to 6th semester is automatic (i.e., from 1st spell of Industrial Training to 2nd spell) provided he/she puts the required percentage of attendance, which in this case ie.,90 % of attendance and attends for the VIVA-VOCE examination at the end of training.
- vi. A candidate shall be promoted to 7th semester provided he / she puts the required percentage of attendance in the 6th semester and pays the examination fee. A candidate, who could not pay the 6th semester examination fee, has to

- pay the promotion fee as prescribed by State Board of Technical Education and Training from time to time before commencement of 7th semester.
- vii. A candidate shall be promoted to 7th semester of the course provided he/she has successfully completed both the spells of Industrial Training.
A candidate is eligible to appear for 7th semester examination if he/she
- a) Puts in the required percentage of attendance in the 7th semester

For IVC & ITI Lateral Entry students:

- a) Puts in the required percentage of attendance in the 7th semester .

C) For Diploma Courses of 3 ½ Years duration (BM):

The same rules which are applicable for conventional courses also apply for this course. The industrial training in respect of this course is restricted to one semester (6 months) after the 6th semester (3 years) of the course.

- i. A candidate shall be permitted to appear for first year examination provided he / she puts in 75% attendance (which can be condoned on Medical grounds upto 10%) i.e. attendance after condonation on Medical grounds should not be less than 65% and pay the examination fee.
- ii. A candidate shall be promoted to 3rd semester if he/she puts the required percentage of attendance in the first year and pays the examination fee. A candidate who could not pay the first year examination fee has to pay the promotion fee as prescribed by State Board of Technical Education and Training from time to time before commencement of 3rd semester.
- iii. A candidate shall be promoted to 4th semester provided he/she puts the required percentage of attendance in the 3rd semester and pay the examination fee. A candidate who could not pay the 3rd semester examination fee, has to pay the promotion fee as prescribed by State Board of Technical Education and Training from time to time before commencement of 4th semester.

A candidate is eligible to appear for the 4th semester examination if he/she

- a) Puts in the required percentage of attendance in the 4th semester

For IVC & ITI Lateral Entry Students:

A candidate is eligible to appear for the 4th semester examination if he/she puts the required percentage of attendance in the 4th semester

- iv. A candidate shall be promoted to 5th semester provided he / she puts the required percentage of attendance in the 4th semester and pays the examination fee. A candidate, who could not pay the 4th semester examination fee, has to pay the promotion fee as prescribed by State Board of Technical Education and Training from time to time before commencement of 5th semester.

A candidate is eligible to appear for the 5th semester exam if he/she

- a) Puts in the required percentage of attendance in the 5th semester.

For IVC & ITI Lateral Entry students:

- a) Puts in the required percentage of attendance in the 5th semester.

- v. A candidate shall be promoted to 6th semester provided he/she puts in the required percentage of attendance in the 5th semester and pays the examination fee.

A candidate who could not pay the 5th semester examination fee, has to pay the promotion fee as prescribed by State Board of Technical Education and Training from time to time before commencement of 6th semester.

A candidate is eligible to appear for 6th semester examination

- a) Puts in the required percentage of attendance in 6th semester

For IVC & ITI Lateral Entry students:

- a) Puts in the required percentage of attendance in 6th semester.

- vi. A candidate shall be promoted to 7th semester provided he/she puts in the required percentage of attendance in 6th semester and pay the examination fee. A candidate, who could not pay the 6th semester examination fee, has to pay the promotion fee prescribed by SBTET from time to time before commencement of the 7th semester (Industrial Training).

A candidate is eligible to appear for 7th semester Industrial Training assessment (Seminar/Viva-voce) if he/she

- a) Puts in the required percentage of attendance, i.e., 90% in 7th semester Industrial Training.

For IVC & ITI Lateral Entry students:

- a) Puts in the required percentage of attendance, i.e., 90% in 7th semester Industrial Training.

4.13. Students Performance Evaluation

Successful candidates shall be awarded the Diploma under the following divisions of pass.

- a) First Class with Distinction shall be awarded to the candidates who secure an overall aggregate of 75% marks and above.
- b) First Class shall be awarded to candidates who secure overall aggregate of 60% marks and above and below 75% marks.
- c) Second Class shall be awarded to candidates who secure a pass with an overall aggregate of below 60%.
 - i. The Weightage of marks for various year/Semesters which are taken for computing overall aggregate shall be 25% of I year marks + 100% of 3rd and subsequent Semesters.
 - ii. In respect IVC & ITI Lateral Entry candidates who are admitted directly into diploma course at the 3rd semester (i.e., second year) level the aggregate of (100%) marks secured at the 3rd and subsequent semesters of study shall be taken into consideration for determining the overall percentage of marks secured by the candidates for award of class/division.
- d) Second Class shall be awarded to all students, who fail to complete the Diploma in the regular 3 years/ 3 ½ years and four subsequent examinations from the year of first admission.

4.14. EXAMINATION FEE SCHEDULE:

The examination fee should be as per the notification issued by State Board of Technical Education and Training, AP from time to time.

4.15. Structure of Examination Question Paper:

I. Formative assessment (Internal examination)

a) For theory Courses:

Three-unit tests for first year and two-unit tests for semesters shall be conducted with a duration of 90 minutes for each test for maximum marks of 40. It consists of part A and Part B.

Part A contains five questions and carries 16 marks. Among these five questions first question consists of four objective items like one word or phrase answer/filling-in the blanks/true or false etc with one mark for each question. The other four questions are short answer questions and carry three marks each.

Part B carries 24 marks and consists of three questions with internal choice ie., Either/Or type , and each question carries 8 marks.

The sum of marks of 3 tests for I year and 2 tests for semesters including assignments and Dynamic learning activities (50 marks) shall be reduced to 20 marks in each Course for arriving at final sessional marks.

b) For drawing Courses:

For I year:

Three-unit tests with duration of 90 minutes and for maximum marks of 40 marks shall be conducted for first year. It consists of part A and Part B.

Part A consists four questions for maximum marks of 16 and each question carries four marks (4×4 marks=16 marks).

Part B carries maximum marks of 24 and consists of five questions while the student shall answer any three questions out of these five questions. Each question in this part carries a maximum mark of 8, (3×8 marks=24 marks).

The sum of marks obtained in 3-unit test marks shall be reduced to 20 marks for arriving at final sessional marks. Remaining 20 marks are awarded by the Course teacher based on the student's performance during regular class exercise.

For semester:Two-unit tests with duration of 90 minutes and for maximum marks of 40 marks shall be conducted. The sum of marks obtained in 2-unit test marks shall be reduced to 20 marks for arriving at final sessional marks. Remaining 20 marks are awarded by the Course teacher based on the student's performance during regular class exercise.

c) For Laboratory /workshop: 50% of total marks for the Course shall be awarded based on continuous assessment of the student in laboratory/workshop classes and the remaining 50% shall be based on the sum of the marks obtained by the students in two tests.

II. Summative assessment (End examination)

The question paper for theory examination is patterned in such a manner that the Weightage of periods/marks allotted for each of the topics for a particular Course be considered. End Examination paper is of 3 hours duration.

a) **Each theory paper consists of Section 'A' and 'B'**

Section 'A' with Max marks of 30, contains 10 short answer questions. All questions are to be answered and each carry 3 marks, i.e., $10 \times 3 = 30$.

Section 'B' with Max marks of 50 contains 8 essay type questions. Only 5 questions are to be answered and each carry 10 marks, i.e., Max. Marks: $5 \times 10 = 50$.

Thus, the total marks for theory examination shall be: 80.

b) **For Engineering Drawing Course (107) consist of section 'A' and section 'B'.**

Section 'A' with max marks of 20, contains four (4) questions. All questions in section 'A' are to be answered to the scale and each carries 5 marks, ie. $4 \times 5 = 20$.

Section 'B' with max marks of 40, contains six (6) questions. The student shall answer any four (4) questions out of the above six questions and each question carries 10 Marks, i.e., $4 \times 10 = 40$.

c) **Practical Examinations**

For Workshop practice and Laboratory Examinations, Each student has to pick up a question paper distributed by Lottery System.

Max. Marks for an experiment / exercise	:	50
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Max. Marks for VIVA-VOCE	:	10
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Total Max. Marks	:	60
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In case of practical examinations with 50 marks, the marks shall be distributed as

Max. Marks for an experiment / exercise	:	25
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Max. Marks for VIVA-VOCE	:	05
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Total Max. Marks	:	30
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In case of any change in the pattern of question paper, the same shall be informed sufficiently in advance to the candidates.

d) **Note: Evaluation for Laboratory Courses, other than Drawing courses:**

- I. Instruction (teaching) in laboratory courses (except for the course on Drawing) hereafter shall be task/competency based as delineated in the Laboratory sheets, prepared by SBTET, AP and posted in its website.
- II. Internal assessment for Laboratory shall be done on basis of task/s performed by the student as delineated in the laboratory sheets, prepared by SBTET, AP and posted in its website.
- III. Question paper for End semester Evaluation shall be prepared as per SBTET rules in vogue.

4.16. ISSUE OF MEMORANDUM OF MARKS

All candidates who appear for the end examination will be issued memorandum of marks without any payment of fee. However candidates who lose the original

memorandum of marks have to pay the prescribed fee to the Secretary, State Board of Technical Education and Training, A.P. for each duplicate memo from time to time.

4.17. MAXIMUM PERIOD FOR COMPLETION OF DIPLOMA PROGRAMMES:

Maximum period for completion of the diploma courses is twice the duration of the course from the date of First admission (includes the period of detention and discontinuation of studies by student etc) failing which they will have to forfeit the claim for qualifying for the award of Diploma (They will not be permitted to appear for examinations after that date). This rule applies for all Diploma courses of 3 years and 3 ½ years of engineering and non-engineering courses.

4.18. ELIGIBILITY FOR AWARD OF DIPLOMA

A candidate is eligible for award of Diploma Certificate if he / she fulfil the following academic regulations.

- i. He / She pursued a course of study for not less than 3 / 3 ½ academic years & not more than 6 / 7 academic years.
- ii. He / she have completed all the Courses.

Students who fail to fulfil all the academic requirements for the award of the Diploma within 6 / 7 academic years from the year of admission shall forfeit their seat in the course & their seat shall stand cancelled.

For IVC & ITI Lateral Entry students:

- i. He / She pursued a course of study for not less than 2 / 2 ½ academic years & not more than 4 / 5 academic years.
- ii. He / she has completed all the Courses.

Students who fail to fulfil all the academic requirements for the award of the Diploma within 4 / 5 academic years from the year of admission shall forfeit their seat in the course & their seat shall stand cancelled.

4.19. ISSUE OF PHOTO COPY OF VALUED ANSWER SCRIPT, RECOUNTING& REVERIFICATION:

A) FOR ISSUE OF PHOTO COPIES OF VALUED ANSWER SCRIPTS

- I. A candidate desirous of applying for Photo copy of valued answer script/s should apply within prescribed date from the date of the declaration of the result.
- II. Photo copies of valued answer scripts will be issued to all theory Courses and Drawing Course(s).
- III. The Photo copy of valued answer script will be dispatched to the concerned candidate's address as mentioned in the application form by post.
- IV. No application can be entertained from third parties.

B) FOR RE-COUNTING(RC) and RE-VERIFICATION(RV) OF THE VALUED ANSWER SCRIPT

- i. A candidate desirous of applying for Re-verification of valued answer script should apply within prescribed date from the date of the declaration of the result.

- ii. Re-verification of valued answer script shall be done for all theory Courses' and Drawing Course(s).
- iii. The Re-verification committee constituted by the Secretary, SBTETAP with Course experts shall re-verify the answer scripts.

I. RE-COUNTING

The Officer of SBTET will verify the marks posted and recount them in the already valued answer script. The variations if any will be recorded separately, without making any changes on the already valued answer script. The marks awarded in the original answer script are maintained (hidden).

II. RE-VERIFICATION

- (i) The Committee has to verify the intactness and genuineness of the answer script(s) placed for Re-verification.
- (ii) Initially single member shall carry out the re-verification.
- (iii) On re-verification by single member, if the variation is less than 12% of maximum marks, and if there is no change in the STATUS in the result of the candidate, such cases will not be referred to the next level ie., for 2-Tier evaluation.
- (iv) On re-verification by a single member, if the variation is more than 12% of maximum marks, it will be referred to 2-Tier evaluation.
- (v) If the 2-Tier evaluation confirms variation in marks as more than 12% of maximum marks, the variation is considered as follows:
 - a) If the candidate has already passed and obtains more than 12% of the maximum marks on Re-verification, then the variation is considered.
 - b) If the candidate is failed and obtains more than 12% of the maximum marks on Re-verification and secured pass marks on re-verification, then the status of the candidate changes to PASS.
 - c) If a candidate is failed and obtains more than 12% of the maximum marks on Re-verification and if the marks secured on re-verification are still less than the minimum pass marks, the status of the candidate remain FAIL only.
- (vii) After Re-verification of valued answer script the same or change if any therein on Re-verification, will be communicated to the candidate.
- (viii) On Re-verification of Valued Answer Script if the candidate's marks are revised, the fee paid by the candidate will be refunded or else the candidate has to forfeit the fee amount.

Note: No request for Photo copies/ Recounting /Re-verification of valued answer script would be entertained from a candidate who is reported to have resorted to Malpractice in that examination.

4.20. Mal Practice Cases:

If any candidate resorts to Mal Practice during examinations, he / she shall be booked and the Punishment shall be awarded as per SBTETAP rules and regulations in vogue.

4.21. Discrepancies/ Pleas:

Any Discrepancy /Pleas regarding results etc., shall be represented to the SBTETAP within one month from the date of issue of results. Thereafter, no such cases shall be entertained in any manner.

4.22. Issue of Duplicate Diploma

If a candidate loses his/her original Diploma Certificate and desires a duplicate to be issued he/she should produce written evidence to this effect. He / she may obtain a duplicate from the Secretary, State Board of Technical Education and Training, A.P., on payment of prescribed fee and on production of an affidavit signed before a First Class Magistrate (Judicial) and non-traceable certificate from the Department of Police. In case of damage of original Diploma Certificate, he / she may obtain a duplicate certificate by surrendering the original damaged certificate on payment of prescribed fee to the State Board of Technical Education and Training, A.P.

In case the candidate cannot collect the original Diploma within 1 year from the date of issue of the certificate, the candidate has to pay the penalty prescribed by the SBTET AP from time to time.

4.23. Issue of Migration Certificate and Transcripts:

The Board on payment of prescribed fee will issue these certificates for the candidates who intend to prosecute Higher Studies in India or Abroad.

4.24. General

- i. The Board may change or amend the academic rules and regulations or syllabi at any time and the changes or amendments made shall be applicable to all the students, for whom it is intended, with effect from the dates notified by the competent authority.
- ii. All legal matters pertaining to the State Board of Technical Education and Training, AP are within the jurisdiction of Mangalagiri.
- iii. In case of any ambiguity in the interpretation of the above rules, the decision of the Secretary, SBTET, A.P is final.

VISION

Develop Automobile Engineering professionals competent to face global challenges in a progressive environment conducive to learning technical knowledge, and skills blended with ethics and values, to serve society and to better it for a happy and comfortable living.

MISSION

M1	To provide a competitive learning environment, through a need-based curriculum designed in collaboration with industry, conducive to high-quality education emphasizing on transfer of knowledge and skill development essential for the profession and the society as well.
M2	To nurture higher-order leadership qualities and ethics and values in students to enable them to be leaders in their chosen professions while maintaining the highest level of ethics.
M3	To encourage the spirit of inquisition to promote innovation and entrepreneurship strengthened with life skills to sustain the stress.
M4	To foster effective interactions and networking with all the stakeholders so as to work towards the growth and sustainability of the society and environment.

Programme Educational Objectives (PEOs)

On completion of the Diploma in Automobile Engineering programme, the students should have acquired the following characteristics

PEO1	To become an expert in specialized areas of automobile engineering such as design and manufacturing.
PEO2	To become a successful entrepreneur with profound knowledge in core engineering to serve society with ethical values and social responsibilities.
PEO3	To obtain competencies required to become an authorized professional in Automobile engineering like service engineer, supervisor etc.,
PEO4	To be able to pursue advanced education and shall engage in the process of lifelong learning.

PROGRAMME OUTCOMES (POs)

PO 1	Basic and Discipline-specific knowledge: Apply knowledge of basic mathematics, science and engineering fundamentals and an engineering specialization to solve engineering problems.
PO 2	Problem analysis: Identify and analyse well-defined engineering problems using codified standard methods.
PO 3	Design/ development of solutions: Design solutions for well-defined technical problems and assist with the design of systems components or processes to meet specified needs.
PO 4	Engineering Tools, Experimentation and Testing: Apply modern engineering tools and appropriate techniques to conduct standard tests and measurements.
PO 5	Engineering practices for society, sustainability and environment: Apply appropriate technology in the context of society, sustainability, environment and ethical practices.
PO 6	Project Management: Use engineering management principles individually, as a team member or as a leader to manage projects and effectively communicate about well-defined engineering activities
PO 7	Life-long learning: Ability to analyse individual needs and engage in updating in the context of technological changes

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO 1	An ability to understand the concepts of Automobile Engineering and to apply them to various areas like design and manufacturing
PSO 2	An ability to become a successful entrepreneur with profound knowledge in core engineering to serve society with ethical values and social responsibilities
PSO 3	An ability to engage in lifelong learning to remain updated in their profession and be leaders in the technological society

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DIPLOMA IN AUTOMOBILE ENGINEERING
SCHEME OF INSTRUCTIONS AND EXAMINATION
I YEAR

Subject Code	Name of the Subject	Instruction period/week		Total Periods / year	Scheme of Examination			
		Theory	Practical/ Tutorial		Duration (hours)	Sessional Marks	End Exam Marks	Total Marks
THEORY:								
A-101	English	3	-	90	3	20	80	100
A-102	Engineering Mathematics-I	5	-	150	3	20	80	100
A-103	Engineering Physics	3	-	90	3	20	80	100
A-104	Engineering Chemistry &Environmental Studies	3	-	90	3	20	80	100
A-105	Basic Mechanical Engineering	4	-	120	3	20	80	100
A-106	Elements of Automobile Engineering	3	-	90	3	20	80	100
PRACTICAL:								
A-107	Engineering Drawing	-	6	180	3	40	60	100
A-108	Workshop Practice-I	-	3	90	3	40	60	100
A-109	Physics Lab	-	1.5	45	3	20	30	50
A-110	Chemistry Lab		1.5	45	3	20	30	50
A-111	Computer Fundamentals Lab	-	3	90	3	40	60	100
A-112	Automobile Engineering Lab – I	-	3	90	3	40	60	100
Student Centric Activity (Library, Mini project, Physical Education, Preparation for competitive exams/ placements, Seminars, Reporting etc)-			3	90	-	-	-	-
TOTAL		21	21	630+630=1260		320	780	1100

A-101,102,103,104,109,110,111 Common with all branches

A-107 Common with DME.

DIPLOMA IN AUTOMOBILE ENGINEERING
SCHEME OF INSTRUCTIONS AND EXAMINATION
III SEMESTER

Subject Code	Name of the Subject	Instruction period/week		Total Periods/Semester	Scheme of Examination			
		Theory	Practical /Tutorial		Duration (hours)	Sessional Marks	End Exam Marks	Total Marks
THEORY								
A-301	Engineering Mathematics–II	4	-	60	3	20	80	100
A-302	Automobile Power Plant	5	-	75	3	20	80	100
A-303	Automobile Transmission System	4	-	60	3	20	80	100
A-304	Autotronics	4	-	60	3	20	80	100
A-305	Engineering Materials & Manufacturing Processes	4	-	60	3	20	80	100
PRACTICAL								
A-306	Machine Drawing	-	6	90	3	40	60	100
A-307	CAD Lab –I	-	3	45	3	40	60	100
A-308	Automobile Engineering Lab –II	-	3	45	3	40	60	100
A-309	Workshop Practice – II	-	3	45	3	40	60	100
A-310	Autotronics Lab	-	3	45	3	40	60	100
Student Centric Activity (Library, Mini project, Physical Education, Preparation for competitive exams / placements, Seminars, Reporting etc)			3	45				
TOTAL		21	21	315+315=630		300	700	1000

A-301 is common to Group-I branches A/AA/CER/M/MET/MNG/TT

DIPLOMA IN AUTOMOBILE ENGINEERING
SCHEME OF INSTRUCTIONS AND EXAMINATIONS
IV SEMESTER

Subject Code	Name of the Subject	Instruction period/week		Total Periods /Semester	Scheme of Examination			
		Theor y	Practical /Tutori al		Durat ion (hour s)	Sessional Marks	End Exam Mark s	Total Mark s
THEORY								
A-401	Two and Three Wheeler Technology	4		60	3	20	80	100
A-402	Strength of Materials And Fluid Mechanics	5		75	3	20	80	100
A-403	Automobile Chassis and Body Engineering	4		60	3	20	80	100
A-404	Vehicle Service and Maintenance Technology	4		60	3	20	80	100
A-405	Farm Tractors and Earth Movers	4		60	3	20	80	100
PRACTICAL								
A-406	Automobile Engineering Drawing	-	6	90	3	40	60	100
A-407	Vehicle Service and Maintenance Technology Lab	-	3	45	3	40	60	100
A-408	Communication Skills	-	3	45	3	40	60	100
A-409	Hydraulic and Pneumatic Controls Lab	-	3	45	3	40	60	100
A-410	CAD Lab-II	-	3	45	3	40	60	100
Student Centric Activity (Library, Mini project, Physical Education, Preparation for competitive exams/placements, Seminars, Reporting etc)			3	45				
TOTAL		21	21	315+315	-	300	700	1000

A-408 Common with all branches

DIPLOMA IN AUTOMOBILE ENGINEERING
SCHEME OF INSTRUCTIONS AND EXAMINATION
V SEMESTER

Subject Code	Name of the Subject	Instruction period/week		Total Periods /Semester	Scheme of Examination			
		Theory	Practical/ Tutorial		Duration (hours)	Sessional Marks	End Exam Marks	Total Marks
THEORY								
A-501	Industrial and Transport Management	4	-	60	3	20	80	100
A-502	Design of Machine Elements	5	-	75	3	20	80	100
A-503	Advanced Manufacturing Processes	4	-	60	3	20	80	100
A-504	Alternative Fuels & Emission Controls	4	-	60	3	20	80	100
A-505	Electric and Hybrid Electric Vehicles	4	-	60	3	20	80	100
PRACTICAL								
A-506	IC Engines Testing & Vehicle Diagnostics Lab	-	6	90	3	40	60	100
A-507	Electric & Hybrid Electric Vehicles Lab	-	3	45	3	40	60	100
A-508	Life Skills	-	3	45	3	40	60	100
A-509	Project Work	-	6	90	3	40	60	100
Student Centric Activity (Library, Mini project, Physical Education, Preparation for competitive exams/placements Seminars, Reporting etc.)			3	45				
TOTAL		21	21	315+315	-	260	640	900

A-508 Common with all branches

INDUSTRIAL TRAINING

Subject Title	Subject Code	Duration
Industrial Training	A-601	6 Months

Time schedule

S.NO	Code	TOPICS	Duration
1	A-601	1 Practical training in Industry 2 Training Report Preparation 3 Report Preparation: Title Page, Certificate, Acknowledgements, Abstract, Contents (introduction of Industry, Plant Layout, Organization Chart, List of Major Equipment's) List of Processes: Skills Acquired, Conclusions and References	Six Months

Scheme of evaluation

Sl.No.	Subject	Duration	Scheme of evaluation		
			Item	Nature	Max. Marks
1	Industrial Training	6 months	1.First Assessment at Industry(After12 Weeks)	Assessment of Learning outcomes by both the faculty and training Mentor of the industry	120
			2.Second Assessment at the Industry(After22 weeks)	Assessment of Learning outcomes by both the faculty and training Mentor of the industry	120
			Final Summative assessment at institution level	Training Report	20
				Demonstration of any one of the skills listed in learning outcomes	30
				Viva Voce	10
TOTALMARKS					300

FIRST YEAR

DIPLOMA IN AUTOMOBILE ENGINEERING
SCHEME OF INSTRUCTIONS AND EXAMINATION
I YEAR

Subject Code	Name of the Subject	Instruction period/week		Total Periods / year	Scheme of Examination			
		Theory	Practical/ Tutorial		Duration (hours)	Sessional Marks	End Exam Marks	Total Marks
THEORY:								
A-101	English	3	-	90	3	20	80	100
A-102	Engineering Mathematics-I	5	-	150	3	20	80	100
A-103	Engineering Physics	3	-	90	3	20	80	100
A-104	Engineering Chemistry &Environmental Studies	3	-	90	3	20	80	100
A-105	Basic Mechanical Engineering	4	-	120	3	20	80	100
A-106	Elements of Automobile Engineering	3	-	90	3	20	80	100
PRACTICAL:								
A-107	Engineering Drawing	-	6	180	3	40	60	100
A-108	Workshop Practice-I	-	3	90	3	40	60	100
A-109	Physics Lab	-	1.5	45	3	20	30	50
A-110	Chemistry Lab		1.5	45	3	20	30	50
A-111	Computer Fundamentals Lab	-	3	90	3	40	60	100
A-112	Automobile Engineering Lab – I	-	3	90	3	40	60	100
Student Centric Activity (Library, Mini project, Physical Education, Preparation for competitive exams/ placements, Seminars, Reporting etc)-			3	90	-	-	-	-
TOTAL		21	21	630+630=1260		320	780	1100

A-101,102,103,104,109,110,111 Common with all branches

A-107 Common with DME.

ENGLISH

Course Code	Course Title	No. of Periods per Week	Total No. of Periods	Marks for FA	Marks for SA
A-101	English	3	90	20	80

S. No.	Title of the Unit	No of Periods	COs Mapped
1	English for Employability	8	CO1, CO2, CO3, CO4,CO5
2	Living in Harmony	8	CO1, CO2, CO3, CO4,CO5
3	Connect with Care	8	CO1, CO2, CO3, CO4,CO5
4	Humour for Happiness	8	CO1, CO2, CO3, CO4, CO5
5	Never Ever Give Up!	8	CO1, CO2, CO3, CO4, CO5
6	Preserve or Perish	9	CO1, CO2, CO3, CO4, CO5
7	The Rainbow of Diversity	8	CO1, CO2, CO3, CO4, CO5
8	New Challenges- Newer Ideas	8	CO1, CO2, CO3, CO4, CO5
9	The End Point First	8	CO1, CO2, CO3, CO4, CO5
10	The Equal Halves	8	CO1, CO2, CO3, CO4, CO5
11	Dealing with Disaster	9	CO1, CO2, CO3, CO4, CO5
Total Periods		90	

Course Objectives	- To improve grammatical knowledge and enrich vocabulary.
	- To develop effective reading, writing and speaking skills.
	- To comprehend themes related to Personality, Society, Environment to exhibit Universal Human Values.

CO No.	Course Outcomes
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CO1	Learn and apply various grammatical concepts to communicate in academic, professional and everyday situations
CO2	Use appropriate vocabulary in various contexts
CO3	Read and comprehend different forms of academic, professional and general reading material
CO4	Communicate effectively in speaking and writing in academic, professional and everyday situations.
CO5	Display human values by applying the knowledge of themes related to Self, Society, Environment, Science and Technology for holistic development and harmonious living through communication.

CO-PO Matrix

Course Code Common-101	Course Title: English Number of Course Outcomes: 5				No. of Periods: 90
POs	Mapped CO No.	CO Periods Addressing PO in Column 1		Level of Mapping (1,2,3)	Remarks
		Number	Percentage		
PO1		Not directly Applicable for English course, however, the language activities make use of the content from Science and Technology relevant to the programme to enhance English communication skills.			
PO2					
PO3					
PO4					
PO5	CO5	16	18%	Level 1	Up to 20%: Level 1 21%-50%: Level 2 >50%: Level 3
PO6	CO1, CO2, CO3, CO4,	52	58%	Level 3	
PO7	CO1, CO2, CO3, CO4,CO5	22	24%	Level 2	

Level 3 – Strongly Mapped, Level 2- Moderately Mapped; Level 1- Slightly Mapped

Learning Outcomes

1. English for Employability

- 1.1. Perceive the need for improving communication in English for employability
- 1.2. Use adjectives and articles effectively while speaking and in writing
- 1.3. Write simple sentences

2. Living in Harmony

- 2.1. Develop positive self-esteem for harmonious relationships
- 2.2. Use affixation to form new words

2.3. Use prepositions and use a few phrasal verbs contextually

3. Connect with Care

- 3.1. Use social media with discretion
- 3.2. Speak about abilities and possibilities
- 3.3. Make requests and express obligations
- 3.4. Use modal verbs and main verbs in appropriate form
- 3.5. Write short dialogues about everyday situations

4. Humour for Happiness

- 4.1. Realize the importance of humour for a healthy living
- 4.2. Improve vocabulary related to the theme
- 4.3. Inculcate reading and speaking skills
- 4.4. Frame sentences with proper Subject – Verb agreement
- 4.5. Understand the features of a good paragraph and learn how to gather ideas as a preliminary step for writing a good paragraph.

5. Never Ever Give Up!

- 5.1. Learn to deal with failures in life
- 5.2. Use the present tense form for various every day communicative functions such as speaking and writing about routines, professions, scientific descriptions and sports commentary
- 5.3. Write paragraphs with coherence and other necessary skills

6. Preserve or Perish

- 6.1. Understand the ecological challenges that we face today and act to save the environment.
- 6.2. Narrate / Report past events and talk about future actions
- 6.3. Develop vocabulary related to environment
- 6.4. Write e-mails

7. The Rainbow of Diversity

- 7.1. Appraise and value other cultures for a happy living in multi-cultural workspace
- 7.2. Understand the usage of different types of sentences
- 7.3. Ask for or give directions, information, instructions
- 7.4. Use language to express emotions in various situations
- 7.5. Write letters in various real life situations

8. New Challenges – Newer Ideas

- 8.1. Understand the functional difference between Active Voice and Passive Voice
- 8.2. Use Passive Voice to speak and write in various contexts
- 8.3. Understand the major parts and salient features of an essay
- 8.4. Learn about latest innovations and get motivated

9. The End Point First!

- 9.1. Understand the importance of setting goals in life
- 9.2. Report about what others have said both in speaking and writing
- 9.3. Write an essay following the structure in a cohesive and comprehensive manner
- 9.4. Apply the words related to Goal Setting in conversations and in life

10. The Equal Halves

- 10.1. Value the other genders and develop a gender-balanced view towards life
- 10.2. Identify the use of different conjunctions in synthesising sentences
- 10.3. Write various types of sentences to compare and contrast the ideas
- 10.4. Apply the knowledge of sentence synthesis in revising and rewriting short essays
- 10.5. Develop discourses in speech and writing

11. Dealing with Disasters

- 11.1. be aware of different kinds of disasters and the concept of disaster management
- 11.2. Generate vocabulary relevant to disaster management and use it in sentences
- 11.3. Analyze an error in a sentence and correct it
- 11.4. Learn and write different kinds of reports

Textbook: '**INTERACT**' (A Text book of English for I Year Engineering Diploma Courses) - by SBTET, AP

Reference Books:

Martin Hewings: *Advanced Grammar in Use*, Cambridge University Press

Murphy, Raymond :*English Grammar in Use*, Cambridge University Press

Sidney Greenbaum :*Oxford English Grammar*, Oxford University Press

Wren and Martin (Revised by N.D.V. Prasad Rao) : *English Grammar and Composition*, Blackie ELT Books, S. Chand and Co.

Sarah Freeman: *Strengthen Your Writing*, Macmillan

	End Exam (80 Marks)	1,2,3 Unit Tests (20 Marks each)
Part A	10 Question @ 3 Marks	5 Questions @ (1Q X4M) + (4Q X3M =12)
	Total = 30 Marks	Total = 16 Marks
Part B	5 Questions (+ 3 Choice) @10 Marks	3 Questions (with internal choice) @ 8 Marks
	Total = 50 Marks	Total = 24 marks
Grand Total	80 Marks	40 Marks

Time Schedule : C23-A-101 : ENGLISH

S.no.	Title of the Unit	Periods allotted	Weightage of Marks	No. of Short answer questions	No. of Long Answer questions	Mapping of COs
1	English for Employability	8	16	2	1	CO1, CO2, CO3, CO4, CO5
2	Living in Harmony	8				CO1, CO2, CO3, CO4, CO5

3	Connect with Care	8	26	2	2	CO1, CO2, CO3, CO4, CO5
4	Humour for Happiness	8				CO1, CO2, CO3, CO4, CO5
5	Never Ever Give Up!	8	10	1	1	CO1, CO2, CO3, CO4, CO5
6	Preserve or Perish	9	23		2	CO1, CO2, CO3, CO4, CO5
7	The Rainbow of Diversity	8		2		CO1, CO2, CO3, CO4, CO5
8	New Challenges -Newer Ideas	8	19	1	1	CO1, CO2, CO3, CO4, CO5
9	The End Point First	8				CO1, CO2, CO3, CO4, CO5
10	The Equal Halves	8	16	1	1	CO1, CO2, CO3, CO4, CO5
11	Dealing with Disaster	9		1		CO1, CO2, CO3, CO4, CO5
	Total	90	110	30	80	

C23-A-101 :English : Bifurcation of Syllabus for UNIT TESTS 1,2,3			
Unit Test	Lessons / Chapters	Grammar / Language aspects (Topics or Short Answer questions)	Writing Skills (Topics for Long answer/ Essay Questions)
U.T 1	Chapters 1,2,3	a) articles & prepositions, b)Vocabulary: Affixes, synonyms, Antonyms, matching meanings, words & phrases, one word substitutes) c)Adjectives (degrees of comparison) d) Main& Auxiliary Verbs e) phrasal verbs/ word order	a) Theme based Paragraph (focus on LSRW skills, importance of English, Self-esteem, SWOC analysis, Social media) b) Dialogue on themes of lessons 2&3 / Dialogue on General topic / a situation c) Reading comprehension
U.T 2	Chapters 4,5,6,7	a) concord b) Tenses c) Types of sentences d) Framing questions e) words &phrases, linkers	a) Theme based paragraph (Humour for happy living, learning from failures, Environmental protection, multi- culture /global culture) b) Letter writing (formal& informal), c) instructions/ directions, E-mail

			writing
U.T 3	Chapters 8,9,10,11	a) Voice (active &passive) b) Speech(direct& indirect) c) Synthesis of sentences (simple, complex, compound sentences) d) Error analysis e) words &phrases, linkers	a) Theme based paragraph/ Essay writing (Technical innovations, Goal setting, gender sensitivity, dealing with disaster) b) Essay writing, Report writing c) Reading Comprehension
Unit Test Question Paper pattern	Total=40 Marks (Part A=16 Part B =24)	Short Answer questions (Part-A) Q. 1 = 4 marks Q. 2 to 5 = 3 Marks each Total=16 Marks	Long Answer Questions: (Part-B) Q. 6,7,8 @ 8 marks each ; Each question with Internal choice Total: 8X3 = 24 Marks

ENGINEERING MATHEMATICS-I

Course Code	Course Title	No. of Periods/week	Total No. of periods	Marks for FA	Marks for SA
A-102	Engineering Mathematics-I	5	150	20	80

S.No.	Unit Title	No. of periods	COs mapped
1	Algebra	31	CO1
2	Trigonometry	44	CO2
3	Co-ordinate Geometry	23	CO3
4	Differential Calculus	34	CO4
5	Applications of Derivatives	18	CO5
Total Periods		150	

Course Objectives	(i) To apply the principles of Algebra, Trigonometry and Co-ordinate Geometry to real-time problems in engineering. (ii) To comprehend and apply the concept of Differential Calculus in engineering applications.
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Course Outcomes	CO1	Identify functions as special relations, resolve partial fractions and solve problems on matrices and determinants.
	CO2	Solve problems using the concept of trigonometric functions, their inverses and complex numbers.
	CO3	Find the equations and properties of straight lines, circles and conic sections in coordinate system.
	CO4	Evaluate the limits and derivatives of various functions.
	CO5	Find solutions for engineering problems using differentiation.

Learning Outcomes:

UNIT - I

C.O. 1 Identify functions, resolve partial fractions and solve problems on matrices and determinants.

- L.O.**
- 1.1 Define Set, Ordered pair and Cartesian product of two sets - examples.
 - 1.2 Explain Relations and Functions – examples
 - 1.3 Find Domain & Range of functions – simple examples.
 - 1.4 Define one-one and onto functions.
 - 1.5 Find the inverse of a function – simple examples.
 - 1.6 Define rational, proper and improper fractions of polynomials.
 - 1.7 Explain the procedure of resolving proper fractions of the types mentioned below into partial fractions

$$i) \frac{f(x)}{(ax+b)(cx+d)} \quad ii) \frac{f(x)}{(ax+b)^2(cx+d)}$$
 - 1.8 Define a matrix and order of a matrix.
 - 1.9 State various types of matrices with examples (emphasis on 3rd order square matrices).

- 1.10 Compute sum, difference, scalar multiplication and product of matrices. Illustrate the properties of these operations such as commutative, associative and distributive properties with examples and counter examples.
- 1.11 Define the transpose of a matrix and state its properties – examples.
- 1.12 Define symmetric and skew-symmetric matrices with examples. Resolve a square matrix into a sum of symmetric and skew-symmetric matrices and provide examples.
- 1.13 Define determinant of a square matrix; minor, co-factor of an element of a 3x3 square matrix with examples. Expand the determinant of a 3 x 3 matrix using Laplace expansion formula. State and apply the properties of determinants to solve problems.
- 1.14 Distinguish singular and non-singular matrices. Define multiplicative inverse of a matrix and list properties of adjoint and inverse. Compute adjoint and multiplicative inverse of a square matrix.
- 1.15 Solve a system of 3 linear equations in 3 unknowns using Cramer's rule and matrix inversion method.

UNIT - II

C.O. 2 Solve problems using the concept of trigonometric functions, their inverses and complex numbers.

- L.O.**
- 2.1 Define trigonometric ratios of any angle - List the values of trigonometric ratios at specified values.
 - 2.2 Draw graphs of trigonometric functions - Explain periodicity of trigonometric functions.
 - 2.3 Define compound angles and state the formulae of $\sin(A \pm B)$, $\cos(A \pm B)$, $\tan(A \pm B)$ and $\cot(A \pm B)$.
 - 2.4 Give simple examples on compound angles to derive the values of $\sin 15^\circ$, $\cos 15^\circ$, $\sin 75^\circ$, $\cos 75^\circ$, $\tan 15^\circ$, $\tan 75^\circ$ etc.
 - 2.5 Derive identities like $\sin(A+B) \sin(A-B) = \sin^2 A - \sin^2 B$ etc.
 - 2.6 Solve simple problems on compound angles.
 - 2.7 Derive the formulae of multiple angles $2A$, $3A$ etc and sub multiple angle $A/2$ in terms of angle A of trigonometric functions.
 - 2.8 Derive useful allied formulae like $\sin^2 A = (1 - \cos 2A)/2$ etc.
 - 2.9 Solve simple problems using the multiple and submultiple formulae.
- Syllabus for Unit test-I completed
- 2.10 Derive the formulae on transforming sum or difference of two trigonometric ratios into a product and vice versa - examples on these formulae.
 - 2.11 Solve problems by applying these formulae to sum or difference or product of two terms.
 - 2.12 Explain the concept of the inverse of a trigonometric function by selecting an appropriate domain and range.
 - 2.13 Define inverses of six trigonometric functions along with their domains and ranges.
 - 2.14 Derive relations between inverse trigonometric functions so that the given inverse trigonometric function can be expressed in terms of other inverse trigonometric functions with examples.
 - 2.15 State various properties of inverse trigonometric functions and identities like $\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$, etc.
 - 2.16 Apply formulae like $\tan^{-1}x + \tan^{-1}y = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$, where $x \geq 0, y \geq 0, xy < 1$ etc., to solve Simple problems.
 - 2.17 Explain what is meant by solution of trigonometric equations and find the general solutions of $\sin x = k$, $\cos x = k$ and $\tan x = k$ with appropriate examples.
 - 2.18 Solve models of the type $a \sin^2 x + b \sin x + c = 0$ and $a \sin x + b \cos x = c$.
 - 2.19 State sine rule, cosine rule, tangent rule and projection rule and solve a triangle using these formulae.
 - 2.20 List various formulae for the area of a triangle with examples.
 - 2.21 Define a complex number, its modulus, conjugate, amplitude and list their properties.

- 2.22 Define arithmetic operations on complex numbers with examples.
- 2.23 Represent the complex number in various forms like modulus-amplitude (polar) form, Exponential (Euler) form with examples.

UNIT - III

Coordinate Geometry

C.O. 3 Find the equations and properties of straight lines, circles and conic sections in coordinate system.

- L.O.** 3.1 Write different forms of a straight line – general form, point-slope form, slope-intercept form, two-point form, intercept form and normal form (or perpendicular form).
- 3.2 Find distance of a point from a line, acute angle between two lines, intersection of two non-parallel lines and distance between two parallel lines.
- 3.3 Define locus of a point and circle.
- 3.4 Write the general equation of a circle and find its centre and radius.
- 3.5 Find the equation of a circle, given (i) centre and radius, (ii) two ends of the diameter (iii) three non collinear points of type (0,0) (a,0), (0, b).
- 3.6 Define a conic section - Explain the terms focus, directrix, eccentricity, axes and latus-rectum of a conic with illustrations.
- 3.7 Find the equation of a conic when focus, directrix and eccentricity are given.
- 3.8 Describe the properties of Parabola, Ellipse and Hyperbola in standard forms whose axes are along the co-ordinate axes and solve simple examples on these conics.

Syllabus for Unit test-II completed

C.O.4 Evaluate the limits and derivatives of various functions.

- L.O.** 4.1 Explain the concept of limit and meaning of $\lim_{x \rightarrow a} f(x) = l$ and state the properties of limits.
- 4.2 Evaluate the limits of the type $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ and $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)}$
- 4.3 State the Standard limits $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a}$, $\lim_{x \rightarrow 0} \frac{\sin x}{x}$, $\lim_{x \rightarrow 0} \frac{\tan x}{x}$, $\lim_{x \rightarrow 0} \frac{a^x - 1}{x}$, $\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$, $\lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}}$, $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x$ (without proof) and solve simple problems using these standard limits.
- 4.4 Explain the concept of continuity of a function at a point and on an interval
- 4.5 State the concept of derivative of a function $y = f(x)$ – definition, first principle as $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ and also provide standard notations to denote the derivative of a function.
- 4.6 Explain the significance of derivative in scientific and engineering applications.
- 4.7 Find the derivative of standard algebraic, logarithmic, exponential and trigonometric functions using the first principle.
- 4.8 Find the derivatives of inverse trigonometric, hyperbolic and inverse hyperbolic functions.
- 4.9 State the rules of differentiation of sum, difference, scalar multiplication, product and quotient of functions with simple illustrative examples.
- 4.10 Explain the method of differentiation of a function of a function (Chain rule) with illustrative examples.
- 4.11 Explain the method of differentiation of parametric functions with examples.
- 4.12 Explain the procedure for finding the derivatives of implicit functions with examples.

- 4.13 Explain the need of taking logarithms for differentiating some functions of $[f(x)]^{g(x)}$ type – examples on logarithmic differentiation.
- 4.14 Explain the concept of finding the second order derivatives with examples.
- 4.15 Explain the concept of functions of several variables, finding partial derivatives and difference between the ordinary and partial derivatives with simple examples.
- 4.16 Explain the concept of finding second order partial derivatives with simple problems.

C.O. 5 Evaluate solutions for engineering problems using differentiation

- L.O.** 5.1 State the geometrical meaning of the derivative - Explain the concept of derivative to find the slopes of tangent and normal to a given curve at any point on it with examples.
- 5.2 Find the equations of tangent and normal to a given curve at any point on it – simple problems.
- 5.3 Explain the derivative as a rate of change in distance-time relations to find the velocity and acceleration of a moving particle with examples.
- 5.4 Explain the derivative as a rate measurer in the problems where the quantities like areas, volumes vary with respect to time- illustrative examples.
- 5.5 Define the concept of increasing and decreasing functions - Explain the conditions to find points where the given function is increasing or decreasing with illustrative examples.
- 5.6 Explain the procedure to find the extreme values (maxima or minima) of a function of single variable- simple problems for quadratic and cubic polynomials.
- 5.7 Apply the concept of derivatives to find the errors and approximations - simple problems.

Syllabus for Unit test-III completed

CO/PO – Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
CO1	3	2	2	3				3	2	2
CO2	3	3	2	2				3	2	2
CO3	3	3	2	2				3	2	2
CO4	3	3	3	3				3	3	3
CO5	3	3	3	3				3	3	3
Avg.	3	2.8	2.4	2.6				3	2.4	2.4

3 =Strongly mapped (High), **2** =moderately mapped (Medium), **1** =slightly mapped (Low)

Note: The gaps in CO/PO mapping can be met with appropriate activities as follows:

For PO5: Appropriate quiz programmes may be conducted at intervals and duration as decided by concerned faculty.

For PO6: Seminars on applications of mathematics in various engineering disciplines are to be planned and conducted.

For PO7: Plan activities in such a way that students can visit the Library to refer standard books on Mathematics and access the latest updates in reputed national and international journals. Additionally, encourage them to attend seminars and learn mathematical software tools.

PO- CO – Mapping strength

PO No	Mapped with CO no	CO periods addressing PO in column I		Level (1,2 or 3)	Remarks
		No	%		

1	CO1, CO2, CO3, CO4, CO5	150 (31+44+23+34+18)	100%	3	>40% Level 3 Highly addressed 25% to 40% Level 2 Moderately addressed 5% to 25% Level 1 Low addressed <5% Not addressed
2	CO1, CO2, CO3, CO4, CO5	80 (8+23+12+22+15)	53.3%	3	
3	CO1, CO2, CO3, CO4, CO5	61 (9+14+9+14+15)	40.6%	3	
4	CO1, CO2, CO3, CO4, CO5	61 (14+9+9+14+15)	40.6%	3	
PSO 1	CO1, CO2, CO3, CO4, CO5	150 (31+44+23+34+18)	100%	3	
PSO 2	CO1, CO2, CO3, CO4, CO5	62 (10+14+9+14+15)	41.3%	3	
PSO 3	CO1, CO2, CO3, CO4, CO5	62 (10+14+9+14+15)	41.3%	3	

COURSE CONTENT

Unit-I Algebra

1. Functions:

Definitions of Set, Ordered pair, Cartesian product of two sets, Relations, Functions, Domain & Range of functions – One-one and onto functions, inverse of a function.

2. Partial Fractions:

Definitions of rational, proper and improper fractions of polynomials. Resolve rational fractions (proper fractions) into partial fractions covering the types mentioned below.

$$i) \frac{f(x)}{(ax+b)(cx+d)}$$

$$ii) \frac{f(x)}{(ax+b)^2(cx+d)}$$

3. Matrices:

Definition of a matrix, types of matrices - Algebra of matrices, equality of two matrices, sum, difference, scalar multiplication and product of matrices. Transpose of a matrix, Symmetric, skew-symmetric matrices- Determinant of a square matrix, minor and cofactor of an element, Laplace's expansion, properties of determinants - Singular and non-singular matrices, Adjoint and multiplicative inverse of a square matrix-System of linear equations in 3 variables-Solutions by Cramer's rule and Matrix inversion method.

Unit-II Trigonometry

4. Trigonometric ratios:

Definition of trigonometric ratios of any angle, values of trigonometric ratios at specified values, draw graphs of trigonometric functions, periodicity of trigonometric functions.

5. Compound angles:

Formulas of $\sin(A \pm B)$, $\cos(A \pm B)$, $\tan(A \pm B)$, $\cot(A \pm B)$, and related identities.

6. Multiple and sub multiple angles:

Formulae for trigonometric ratios of multiple angles $2A$, $3A$ and sub multiple angle $A/2$.

7. Transformations:

Transformations of products into sums or differences and vice versa.

8. Inverse trigonometric functions:

Definition, domains and ranges-basic properties.

9. Trigonometric equations:

Concept of a solution, principal value and general solution of trigonometric equations:

$\sin x = k$, $\cos x = k$, $\tan x = k$, where k is a constant. Solutions of simple quadratic equations and equations of type $a \sin x + b \cos x = c$.

10. Properties of triangles:

Relations between sides and angles of a triangle- sine rule, cosine rule, tangent rule and projection rule-area of a triangle.

11. Complex Numbers:

Definition of a complex number, modulus, conjugate and amplitude of a complex number - Arithmetic operations on complex numbers - Modulus-Amplitude (polar) form, Exponential form (Euler form) of a complex number.

UNIT-III**Coordinate geometry**

12. Straight lines: Various forms of a straight line - Angle between two lines, perpendicular distance from a point, intersection of non-parallel lines and distance between parallel lines.

13. Circle: Locus of a point, Circle definition-Circle equation given (i) centre and radius, (ii) two ends of a diameter (iii) three non-collinear points of type $(0,0)$, $(a,0)$, $(0,b)$ - General equation of a circle -its centre and radius.

14. Definition of a conic section - Equation of a conic when focus, directrix and eccentricity are given - Properties of parabola, ellipse and hyperbola in standard forms.

UNIT-IV**Differential Calculus**

15. Concept of Limit- Definition and Properties of Limits and Standard Limits -Continuity of a function at a point.

16. Concept of derivative- Definition (first principle)- different notations- Derivatives of standard algebraic, logarithmic, exponential, trigonometric, inverse trigonometric, hyperbolic and inverse hyperbolic functions - Derivatives of sum, difference, scalar multiplication, product, quotient of functions - Chain rule, derivatives of parametric functions, derivatives of implicit functions, logarithmic differentiation - Second order derivatives - Functions of several variables, first and second order partial derivatives.

UNIT-V**Applications of Derivatives**

17. Geometrical meaning of the derivative, equations of tangent and normal to a curve at any point.

18. Physical applications of derivatives - Velocity, acceleration, derivative as a rate measurer.

19. Applications of the derivative to find the extreme values - Increasing and decreasing functions, maxima and minima for quadratic and cubic polynomials.

20. Absolute, relative and percentage errors - Approximate values due to errors in measurements.

Textbook:

Engineering Mathematics-I, a textbook for first year diploma courses, prepared & prescribed by SBTET, AP.

Reference Books:

1. Shanti Narayan, A Textbook of matrices, S.Chand & Co.
2. Robert E. Moyer & Frank Ayers Jr., Schaum's Outline of Trigonometry, 4th Edition, Schaum's Series.
3. G.B.Thomas, R.L.Finney, Calculus and Analytic Geometry, Addison Wesley, 9th Edition, 1995.
4. Frank Ayers & Elliott Mendelson, Schaum's Outline of Calculus, Schaum's Series.
5. M.Vygodsky, Mathematical Handbook, Mir Publishers, Moscow.

TIME SCHEDULE

S.No.	Chapter	No. of Periods	Marks Allotted	Short type	Essay type	COs mapped
Unit - I: Algebra						
1	Functions	6	3	1	0	CO1
2	Partial Fractions	5	3	1	0	CO1
3	Matrices and Determinants	20	16	2	1	CO1
Unit - II: Trigonometry						

4	Trigonometric Ratios	2	0	0	0	CO2
5	Compound Angles	5	3	1	0	CO2
6	Multiple and Submultiple angles	8	3	1	0	CO2
7	Transformations	6	5	0	1/2	CO2
8	Inverse Trigonometric Functions	6	5	0	1/2	CO2
9	Trigonometric Equations	6	5	0	1/2	CO2
10	Properties of triangles	5	5	0	1/2	CO2
11	Complex Numbers	6	3	1	0	CO2
Unit III: Co-ordinate Geometry						
12	Straight Lines	5	3	1	0	CO3
13	Circles	6	5	0	1/2	CO3
14	Conic Sections	12	5	0	1/2	CO3
Unit – IV: Differential Calculus						
15	Limits and Continuity	6	3	1	0	CO4
16	Differentiation	28	23	1	2	CO4
Unit – V: Applications of Derivatives						
17	Geometrical Applications	4	5	0	1/2	CO5
18	Physical Applications	6	5	0	1/2	CO5
19	Maxima and Minima	4	5	0	1/2	CO5
20	Errors and Approximations	4	5	0	1/2	CO5
	Total	150	110	10	8	
Marks				30	80	

Unit Test Syllabus

Unit Test	Syllabus
Unit Test-I	From L.O. 1.1 to L.O. 2.9
Unit Test-II	From L.O. 2.10 to L.O. 3.8
Unit Test-III	From L.O.4.1 to L.O. 5.7

ENGINEERING PHYSICS

Course code	Course title	No. of periods per week	Total no. of periods	Marks for FA	Marks for SA
A-103	Engineering Physics	03	90	20	80

TIME SCHEDULE

S.No	Major topics	No. of Periods	Weightage of Marks	Short Answer type (3 marks)	Essay type (10 marks)	COs mapped
1.	Units and measurements	09	03	1		CO1
2.	Statics	11	13	1	1	
3.	Gravitation	12	20	1	2	CO2
4.	Concepts of energy	10	13	1	1	
5.	Thermal physics	10	13	1	1	CO3
6.	Sound	12	16	2	1	
7.	Electricity & Magnetism	13	16	2	1	CO4
8.	Modern physics	13	16	2	1	
	Total:	90	110	10	8	

Course title : Engineering Physics

Course objectives	(1) To understand the basic concepts of physics for various Engineering applications as required for industries. (2) To equip the students with the scientific advances in technology and make the student suitable for any industrial or scientific organization.
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COURSE OUTCOMES	CO1	Familiarize with various physical quantities, their SI units and errors in measurements; understand the concepts of vectors and various forces in statics.
	CO2	Understand the concepts of gravitation with reference to applications in satellites, provide the knowledge of various forms of energy and their working principles.
	CO3	Familiarize with the knowledge of transmission of heat and gas laws; provide the knowledge on musical sound and noise as pollution and also the concepts of echo and reverberation.
	CO4	Provide basic knowledge of electricity and concepts of magnetism and magnetic materials; familiarize with the advances in Physics such as photoelectric cell, optical fibers, semiconductors, superconductors and nanotechnology.

CO-PO Mapping Strength

Course code A-103	Engineering Physics No of Course Objectives : 4				No of periods 90
POs	Mapped with CO No	CO periods addressing PO in Col 1NO %		Level 1,2,3	remarks
PO1	CO1,CO2,CO3,CO4	44	48.9 %	3	>40% level 3 (highly addressed) 25% to 40% level2 (moderately addressed) 5% to 25% level1 (Low addressed) < 5% (not addressed)
PO2	CO1,CO2, CO4	11	12.2%	1	
PO3	CO1, CO2,CO3, CO4	10	11.1%	1	
PO4	CO1, CO2,CO4	8	8.9%	1	
PO5	CO2,CO3, CO4	8	8.9%	1	
PO6					
PO7	CO1, CO2, CO3, CO4	9	10.0%	1	

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
CO1	3	2	2	1			2	2		2
CO2	3	2	2	2	2		2	1		2
CO3	2		1		2		1		1	1
CO4	3	2	3	2	2		3	2		2

3 = strongly mapped, 2 = moderately mapped, 1 = slightly mapped

Note: The gaps in CO and PO mapping will be achieved by one or more appropriate activities from the following.

- (i) Seminars (ii) Tutorials (iii) Guest Lecturers (iv) Assignments
 (v) Quiz competitions (vi) Industrial visits (vii) Tech fest (viii) Mini project
 (ix) Group discussions (x) Virtual classes (xi) Library visit for e-books

Learning outcomes

Upon completion of the course the student shall be able to

1.0 Understand the concept of units and measurements

- 1.1 Explain the concept of units
- 1.2 Define the terms
 - a) Physical quantity b) Fundamental physical quantities and
 - c) Derived physical quantities
- 1.3 Define unit
- 1.4 Define fundamental units and derived units
- 1.5 State SI units with symbols for fundamental and some derived quantities
- 1.6 State Multiples and Submultiples in SI system
- 1.7 State rules of writing S.I units
- 1.8 State advantages of SI units
- 1.9 What are direct and indirect measurements
- 1.10 Define accuracy and least count
- 1.11 Define error in measurement

- 1.12 Define absolute, relative and percentage errors with their formulae
- 1.13 Solve simple problems on absolute, relative and percentage errors

2.0 Understand the concepts of statics

- 2.1 Explain the concept of Vectors
- 2.2 Define scalar and vector quantities with examples
- 2.3 Represent vectors geometrically
- 2.4 Define the types of vectors (equal, negative, unit, co-initial, co-planar, position vector)
- 2.5 Resolve the vector into rectangular components
- 2.6 State and explain triangle law of addition of vectors
- 2.7 Define concurrent forces, co-planar forces and equilibrant.
- 2.8 State and explain Lami's theorem
- 2.9 State the parallelogram law of addition of forces with diagram.
- 2.10 Write the expressions for magnitude and direction of resultant (no derivation)
- 2.11 Illustrate parallelogram law with examples (i) flying of bird and (ii) working of sling.
- 2.12 Define moment of force and couple.
- 2.13 Write the formulae and S.I units of moment of force and couple.
- 2.14 Solve simple problems on (i) Resolution of force and (ii) Parallelogram law of forces (finding R, α and θ).

3.0 Understand the concepts of Gravitation

- 3.1 State and explain Newton's universal law of gravitation.
- 3.2 Define G and mention its value.
- 3.3 Explain the acceleration due to gravity (g)
- 3.4 Explain the factors affecting the value of g
- 3.5 Derive the relationship between g and G .
- 3.6. State and explain the Kepler's laws of planetary motion
- 3.7 Define a satellite.
- 3.8 What are natural and artificial satellites, Give examples.
- 3.9 Define orbital velocity and write its formula.
- 3.10 Define escape velocity and write its formula.
- 3.11 Write a brief note on Polar satellites.
- 3.12 Write a brief note on Geo-stationary satellites.
- 3.13 Mention the applications of artificial satellites.
- 3.14 Solve simple problems on (i) Newton's law of gravitation and (ii) calculation of orbital and escape velocities.

4.0 Understand the concepts of Energy.

- 4.1 Define work done and energy. Mention their SI units.
- 4.2 List various types of energy.
- 4.3 Define P.E with examples. Write its equation.
- 4.4 Define K.E with examples. Write its equation.
- 4.5 Derive relationship between K.E and momentum.
- 4.6 State the law of conservation of energy. Give various examples.
- 4.7 Write a brief note on solar energy.
- 4.8 Explain the principle of solar thermal conversion.
- 4.9 Explain the principle of photo voltaic effect
- 4.10 Solve simple problems on (i) work done (ii) P.E & K.E and (iii) Relation between K.E& momentum.

5.0 Understand the concepts of thermal physics

- 5.1 Define the concepts of heat and temperature
- 5.2 State different modes of transmission of heat
- 5.3 Explain conduction, convection and radiation with two examples each.
- 5.4 State and explain Boyle's law
- 5.5 Define absolute zero temperature

- 5.6 Explain absolute scale of temperature
- 5.7 State the relationship between degree Celsius, Kelvin and Fahrenheit temperatures
- 5.8 State Charle's law and write its equation
- 5.9 State Gay-Lussac's law and write its equation
- 5.10 Define ideal gas
- 5.11 Derive ideal gas equation
- 5.12 Explain why universal gas constant (R) is same for all gases in nature
- 5.13 Calculate the value of R for 1 gram mole of gas.
- 5.14 Solve simple problems on (i) Interconversion of temperatures between °C, K and F
(ii) Gas laws and (iii) Ideal gas equation.

6.0 Understand the concepts of Sound

- 6.1 Define the term sound
- 6.2 Define longitudinal and transverse waves with one example each
- 6.3 Explain the factors which affect the velocity of sound in air
- 6.4 Distinguish between musical sound and noise
- 6.5 Explain noise pollution and state SI unit for intensity of sound
- 6.6 Explain sources of noise pollution
- 6.7 Explain effects of noise pollution
- 6.8 Explain methods of minimizing noise pollution
- 6.9 Define Doppler effect.
- 6.10 List the Applications of Doppler effect
- 6.11 Define reverberation and reverberation time
- 6.12 Write Sabine's formula and name the physical quantities in it.
- 6.13 Define echoes and explain the condition to hear an echo.
- 6.14 Mention the methods of reducing an echo
- 6.15 Mention the applications of an echo
- 6.16 What aneutronic
- 6.17 Mention the applications of ultrasonic, SONAR
- 6.18 Solve simple problems on echo

7.0 Understand the concepts of Electricity and Magnetism

- 7.1 Explain the concept of P.D and EMF
- 7.2 State Ohm's law and write the formula
- 7.3 Explain Ohm's law
- 7.4 Define resistance and specific resistance. Write their S.I units.
- 7.5 State and explain Kichoff's first law.
- 7.6 State and explain Kirchoff's second law.
- 7.7 Describe Wheatstonebridge with legible sketch.
- 7.8 Derive an expression for balancing condition of Wheatstone bridge.
- 7.9 Describe Meter Bridge experiment with necessary circuit diagram.
- 7.10 Write the formulae to find resistance and specific resistance in meter bridge
- 7.11 Explain the concept of magnetism
- 7.12 What are natural and artificial magnets (mention some types)
- 7.13 Define magnetic field and magnetic lines of force.
- 7.14 Write the properties of magnetic lines of force
- 7.15 State and explain the Coulomb's inverse square law of magnetism
- 7.16 Define magnetic permeability
- 7.17 Define para, dia, ferro magnetic materials with examples
- 7.18 Solve simple problems on (i) Ohm's law (ii) Kirchoff's first law (iii) Wheatstone bridge (iv) meter bridge and (v) Coulomb's inverse square law

8.0 Understand the concepts of Modern physics

- 8.1 State and explain Photo-electric effect.
- 8.2 Write Einstein's Photo electric equation and name the physical quantities in it.

- 8.3 State laws of photo electric effect
- 8.4 Explain the Working of photo electric cell
- 8.5 List the Applications of photoelectric effect
- 8.6 Recapitulate refraction of light and its laws
- 8.7 Define critical angle
- 8.8 Explain the Total Internal Reflection
- 8.9 Explain the principle and working of Optical Fiber
- 8.10 List the applications of Optical Fiber
- 8.11 Explain the energy gap based on band structure
- 8.12 Distinguish between conductors, semiconductors and insulators based on energy gap
- 8.13 Define doping
- 8.14 Explain the concept of hole
- 8.15 Explain the types of semiconductors , Intrinsic and extrinsic
- 8.16 Explain n-type and p-type semiconductors
- 8.17 Mention the applications of semiconductors
- 8.18 Define superconductor and superconductivity
- 8.19 List the applications of superconductors
- 8.20 Nanotechnology definition, nanomaterials and applications

COURSECONTENT

1. Units and measurements

Introduction – Physical quantity – Fundamental and Derived quantities – Fundamental and derived units - SI units – Multiples and Sub multiples – Rules for writing S.I. units-Advantages of SI units – Direct and indirect measurements – Accuracy and least count – Errors : Absolute, relative and percentage errors –Problems.

2. Statics

Scalars and Vectors– Representation of a vector - Types of vectors - Resolution of vector into rectangular components – Triangle law of vectors – Concurrent forces - Lami's theorem - Parallelogram law of forces : Statement, equations for magnitude and direction of resultant, examples – Moment of force and couple – Problems.

3. Gravitation

Newton's law of gravitation and G – Concept of acceleration due to gravity (g) – Factors affecting the value of g – Relation between g and G - Kepler's laws – Satellites : Natural and artificial – Orbital velocity and escape velocity – Polar and geostationary satellites – Applications of artificial satellites – Problems.

4. Concepts of energy

Workdone & Energy-Definition and types of energy - potential energy - kinetic energy-- K.E and momentum relation – Law of Conservation of energy, examples - Solar energy, principles of thermal and photo conversion – Problems.

5. Thermal physics

Modes of transmission of heat – Expansion of Gases - Boyle's law – Absolute scale of temperature - Thermometric scales and their inter conversion - Charle's law - Gay-Lussac's law - Ideal gas equation - Universal gas constant (R) - Problems.

6. Sound

Sound - Nature of sound - Types of wave motion, Longitudinal and transverse – Factors affecting the velocity of sound in air - musical sound and noise - Noise pollution – Causes & effects - Methods of reducing noise pollution - Doppler effect - Echo- Reverberation -Reverberation time-Sabine 's formula –Ultrasonics& applications – SONAR - Problems.

7. Electricity & Magnetism

Concept of P.D and EMF - Ohm's law and explanation-Specific resistance - Kirchoff's laws – Wheatstone's bridge - Meter bridge.

Natural and artificial magnets – magnetic field and magnetic lines of force – Coulomb's inverse square law – Permeability – Magnetic materials – Para, dia, ferro – Examples – Problems.

8. Modern Physics

Photoelectric effect – laws of photoelectric effect – photoelectric cell – Applications of photoelectric cell - Total internal reflection - Fiber optics - Principle and working of an optical fiber - Applications of optical fibers– Semiconductors : Based on Energy gap – Doping – Hole - Intrinsic and extrinsic semiconductors (n-type & p-type)

– Applications of semiconductors – Superconductivity – applications – Nanotechnology definition, nano materials, applications.

REFERENCES

- | | |
|--|----------------------------------|
| 1. Intermediate physics - Volume - I & 2 | Telugu Academy (English version) |
| 2. Unified physics Volume 1, 2, 3 and 4 | Dr. S.L Guptha and SanjeevGuptha |
| 3. Concepts of Physics, Vol 1 & 2 | H.C. Verma |
| 4. Text book of physics Volume I & 2 | Resnick&Halliday |
| 5. Fundamentals of physics | Brijlal&Subramanyam |
| 6. Text book of applied physics | Dhanpath Roy |
| 7. NCERT Text Books of physics | Class XI & XII Standard |
| 8. e-books/e-tools/websites/Learning Physics software/eLMS | |

Table showing the scope of syllabus to be covered for unit tests

Unit test	Learning outcomes to be covered
Unit test - 1	From 1.1 to 3.14
Unit test - 2	From 4.1 to 6.18
Unit test - 3	From 7.1 to 8.20

ENGINEERING CHEMISTRY AND ENVIRONMENTAL STUDIES

Course code	Course Title	No. of Periods per week	Total No. of Periods	Marks for FA	Marks for SA
A-104	Engineering Chemistry and Environmental Studies	3	90	20	80

S.No	Unit Title/Chapter	No. of Periods	COs Mapped
1	Fundamentals of Chemistry	14	CO1
2	Solutions, Acids and Bases	16	CO1
3	Electrochemistry	12	CO2
4	Corrosion	8	CO2
5	Water Treatment	8	CO3
6	Polymers & Engineering Materials	12	CO4
7	Fuels	6	CO4
8	Environmental Studies	14	CO5
	Total	90	

Course Objectives

Course Title: Engineering Chemistry & Environmental Studies	
Course Objectives	1. To familiarize with the concepts of chemistry involved in the process of various Engineering Industrial Applications. 2. To know the various natural and man-made environmental issues and concerns with an interdisciplinary approach that include physical, chemical, biological and socio cultural aspects of environment. 3. to reinforce theoretical concepts by conducting relevant experiments/exercises

Course outcomes

Course Outcomes	CO1	Explain Bohr's atomic model, chemical bonding, mole concept, acids and bases, pH and Buffer solutions.
	CO2	Explain electrolysis, Galvanic cell, batteries and corrosion
	CO3	Explain the chemistry involved in the treatment of hardness in water.
	CO4	Explain the methods of preparation and applications of Polymers and Elastomers, chemical composition and applications of Alloys, Composite Materials, Liquid Crystals, Nano Materials and Fuels.
	CO5	Explain Global impacts due to air pollution, causes, effects and controlling methods of water pollution and understand the environment, forest resources, e-Pollution and Green Chemistry Principles.

A-104	Engineering. Chemistry and Environmental studies No of Course Outcomes:5				No of periods 90
POs	Mapped with CO No	CO periods addressing PO	%	Level 1,2,3	remarks

		in Col NO. 1			
PO1	CO1,CO2,CO3	42	46.7 %	3	>40% level 3 (highly addressed) 25% to 40% level 2(moderately addressed) 5% to 25% level 1 (Low addressed) < 5%(not addressed)
PO2	CO2,CO3	16	17.8%	1	
PO3	CO4	12	13.3%	1	
PO4	CO4	6	6.7%	1	
PO5	CO5	14	15.5%	1	
PO6	-	-	-	-	
PO7	-	-	-	-	

COs-POs mapping strength (as per given table)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
CO1	3	-	-	-	-	-	-	-	-	-
CO2	3	1	-	-	-	-	-	-	-	-
CO3	3	1	-	-	-	-	-	-	-	-
CO4	-	-	1	1	-	-	-	-	-	-
CO5	-	-	-	-	1	-	-	-	-	-
Average	3	1	1	1	1	-	-	-	-	-

3=strongly mapped 2= moderately mapped 1= slightly mapped

Note: The gaps in CO and PO mapping will be achieved by one or more appropriate activities from the following:

i) Seminars ii) Tutorials iii) Guest Lectures iv) Assignments v) Quiz competitions vi) Industrial visit vii) Tech Fest viii) Mini project ix) Group discussions x) Virtual classes xi) Library visit for e-books

Time Schedule:

S.No	Unit Title/Chapter	No of Periods	Weight age of marks	Question wise distribution		Mapped with CO
				Essay	Short	
1	Fundamentals of Chemistry	14	21	1½*	2	CO1
2	Solutions, Acids and Bases	16	21	1½*	2	CO1
3	Electrochemistry	12	13	1	1	CO2
4	Corrosion	8	13	1	1	CO2
5	Water Treatment	8	13	1	1	CO3
6	Polymers & Engineering materials.	12	13	1	1	CO4
7	Fuels	6	3	0	1	CO4
8	Environmental Studies	14	13	1	1	CO5
Total		90	110	8	10	

*One question of 10 marks should be given with 50% weightage from unit title 1 and 2

Upon completion of the course, the student shall be able to learn out

ENGINEERING CHEMISTRY AND ENVIRONMENTAL STUDIES

1.0 Atomic structure

- 1.1 Explain the charge, mass of fundamental particles of an atom (electron, proton and neutron) and the concept of atomic number and mass number.
- 1.2 State the Postulates of Bohr's atomic theory and its limitations.
- 1.3 Explain the significance of four Quantum numbers and draw the atomic structures of Silicon and Germanium.
- 1.4 Define Orbital of an atom and draw the shapes of s,p and d-orbitals.

- 1.5 Explain 1. Aufbau principle, 2. Pauli's exclusion principle 3. Hund's principle.
- 1.6 Write the electronic configuration of elements up to atomic number 30.
- 1.7 Explain the significance of chemical bonding.
- 1.8 Explain the Postulates of Electronic theory of valency.
- 1.9 Define and explain Ionic and Covalent bonds with examples of NaCl, H_2 , O_2 and N_2 . (* Lewis dot method).
- 1.10 List out the Properties of Ionic compounds and covalent compounds and distinguish between their properties.

2.0 Solutions, Acids and Bases

- 2.1 Define the terms 1. Solution, 2. Solute and 3. Solvent.
- 2.2 Classify solutions based on solubility.
- 2.3 Define the terms 1. Atomic weight, 2. Molecular weight and 3. Equivalent weight.
Calculate Molecular weight and Equivalent weight of the given acids (HCl, H_2SO_4 , H_3PO_4) Bases (NaOH, Ca(OH)_2 , Al(OH)_3 and Salts (NaCl, Na_2CO_3 , CaCO_3).
- 2.4 Define mole and solve numerical problems on mole concept.
- 2.5 Define molarity, normality and solve numerical problems on molarity and normality.
 - a) Calculate the Molarity or Normality, if weight of solute and volume of solution are given.
 - b) Calculate the weight of solute, if Molarity or Normality with volume of solution are given.
 - c) Problems on dilution to convert high concentrated solutions to low concentrated solutions.
- 2.6 Explain Arrhenius theory of Acids and Bases and give its limitations.
- 2.7 Define ionic product of water, pH and solve numerical problems on pH (Strong Acids and Bases).
- 2.8 Define buffer solution and classify buffer solutions with examples. Give its applications.

3.0 Electrochemistry

- 3.1 Define the terms 1. Conductor 2. Semiconductor 3. Insulator, 4. Electrolyte 5. Non-electrolyte. Give two examples each.
- 3.2 Distinguish between Metallic conduction and Electrolytic conduction.
- 3.3 Explain electrolysis by taking an example of used NaCl and list out the applications of electrolysis.
- 3.4 Define Galvanic cell. Explain the construction and working of Galvanic cell.
- 3.5 Distinguish between electrolytic cell and galvanic cell.
- 3.6 Define battery and list the types of batteries with examples.
- 3.7 Explain the construction, working and applications of i) Dry cell (Leclanche cell), ii) Lead storage battery, iii) Lithium-Ion battery and iv) Hydrogen-Oxygen fuel cell.

4.0 Corrosion

- 4.1 Define the term corrosion.
- 4.2 State the Factors influencing the rate of corrosion.
- 4.3 Describe the formation of (a) composition cell (b) stress cell (c) concentration cell during corrosion.
- 4.4 Define rusting of iron and explain the mechanism of rusting of iron.
- 4.5 Explain the methods of prevention of corrosion by
 - (a) Protective coatings (anodic and cathodic coatings).
 - (b) Cathodic protection (Sacrificial anode process and Impressed-voltage process).

5.0 Water Treatment

- 5.1 Define soft water and hard water with respect to soap action.
- 5.2 Define and classify the hardness of water.
- 5.3 List out the salts that causing hardness of water (with Formulae).
- 5.4 State the disadvantages of using hard water in industries.
- 5.5 Define Degree of hardness and units of hardness (mg/L and ppm).
- 5.6 Solve numerical problems on hardness.
- 5.7 Explain the methods of softening of hard water by (i) Ion-exchange process and (ii) Reverse Osmosis process.

6.0 Polymers & Engineering materials.

A) Polymers

- 6.1 Explain the concept of polymerization.
- 6.2 Describe the methods of polymerization (a) addition polymerization of ethylene (b) condensation polymerization of Bakelite (Only flowchart).
- 6.3 Define plastic. Explain a method of preparation and uses of the following plastics:
 1. PVC 2. Teflon 3. Polystyrene 4. Nylon 6,6.

6.4 Define elastomers. Explain a method of preparation and applications of the following:

1. Buna- S
2. Neoprene.

B) Engineering Materials

6.5 Define an alloy. Write the composition and applications of the following:

1. Nichrome
2. Duralumin
3. Stainless Steel.

6.6 Define Composite Materials and give any two examples. State their Properties and applications.

6.7 Define Liquid Crystals and give any two examples. State their Properties and applications.

6.8 Define Nano Materials and give any two examples. State their Properties and applications.

7.0 Fuels

7.1 Define the term fuel.

7.2 Classify the fuels based on occurrence.

7.3 Write the composition and uses of the following:

1. LPG
2. CNG
3. Biogas
4. Power alcohol

7.4 Write the commercial production of Hydrogen as future fuel. Give its advantages and disadvantages.

8.0 ENVIRONMENTAL STUDIES

8.1 Explain the scope and importance of environmental studies.

8.2 Define environment. Explain the different segments of environment.

1. Lithosphere
2. Hydrosphere
3. Atmosphere
4. Biosphere

8.3 Define the following terms:

1. Pollutant
2. Pollution
3. Contaminant
4. Receptor
5. Sink
6. Particulates
7. Dissolved oxygen (DO)
8. Threshold Limit Value (TLV)
9. BOD
10. COD
11. Eco system
12. Producers
13. Consumers
14. Decomposers with examples.

8.4 State the renewable and non-renewable energy sources with examples.

8.5 State the uses of forest resources.

8.6 Explain the causes and effects of deforestation.

8.7 Define air pollution and explain its Global impacts 1. Greenhouse effect, 2. Ozone layer depletion and 3. Acid rain.

8.8 Define Water pollution. Explain the causes, effects and controlling methods of Water pollution.

8.9 Define e-Pollution, State the sources of e-waste. Explain its health effects and control methods.

8.10 Define Green Chemistry. Write the Principles and benefits of Green Chemistry.

COURSE CONTENT

ENGINEERING CHEMISTRY AND ENVIRONMENTAL STUDIES

1. Fundamentals of Chemistry

Atomic Structure: Introduction - Fundamental particles – Bohr's theory – Quantum numbers – Atomic structure of Silicon and Germanium - Orbitals, shapes of s, p and d orbitals -Aufbau's principle - Hund's rule - Pauli's exclusion Principle -Electronic configuration of elements.

Chemical Bonding: significance–Electronic theory of valency- Types of chemical bonds – Ionic and covalent bond with examples–Properties of Ionic and Covalent compounds.

2. Solutions, Acids and Bases

Solutions: Types of solutions - Mole concept – Numerical problems on mole concept -Methods of expressing concentration of a solution – Molarity and Normality – Numerical problems on molarity and normality.

Acids and Bases: Arrhenius theory of acids and bases – Ionic product of water- pH–Numerical problems on pH– Buffer solutions – Classification- applications.

3. Electrochemistry

Conductors, semiconductors, insulators, electrolytes and non-electrolytes – Electrolysis of fused NaCl– Applications of electrolysis - Galvanic cell – Battery-Types- Dry Cell (Leclanche Cell), Lead- Storage battery- Lithium-Ion battery -Hydrogen-Oxygen Fuel cell.

4. Corrosion

Introduction - Factors influencing corrosion - Composition, Stress and Concentration Cells– Rusting of iron and its mechanism – Prevention of corrosion by Protective Coating methods, Cathodic Protection methods.

5. Water treatment

Introduction– Soft and Hard water– Causes of hardness– Types of hardness– Disadvantages of hard water – Degree of hardness (ppm and mg/lit) – Numerical problems on hardness - Softening methods – Ion-Exchange

process– Reverse Osmosis process.

6. Polymers & Engineering materials

Polymers:

Concept of polymerization – Types of polymerization – Addition, condensation with examples – Plastics - Preparation and uses of i).PVC ii) Teflon iii) Polystyrene and iv) Nylon 6,6.

Elastomers: Preparation and application of i)Buna-s and ii) Neoprene.

Engineering Materials:

Alloys- Composition and applications of i) Nichrome, ii) Duralumin and iii) Stainless Steel.

Composite Materials- Properties and applications.

Liquid Crystals- Properties and applications.

Nano Materials- Properties and applications.

7. Fuels

Definition and classification of fuels – Composition and uses of i) LPG ii) CNG iii) Biogas

and iv) Power alcohol – Hydrogen as a future fuel-production- advantages and disadvantages.

8. ENVIRONMENTAL STUDIES

Scope and importance of environmental studies – Environment - Important terms related to environment– Renewable and non-renewable energy sources–Forest resources – Deforestation –Air pollution–Global impacts on environment –Water pollution – causes – effects – control measures- e-Pollution –Sources of e-waste - Health effects - Control methods - Green Chemistry- Principles -Benefits.

Table specifying the scope of syllabus to be covered for Unit Test- 1, Unit Test- 2 and Unit Test -3

Unit Test	Learning outcomes to be covered
Unit Test – 1	From 1.1 to 2.8
Unit Test – 2	From 3.1 to 5.7
Unit Test – 3	From 6.1 to 8.10

REFERENCE BOOKS

- | | |
|---------------------------|---------------------------------|
| 1. Telugu Academy | Intermediate chemistry Vol. 1&2 |
| 2. Jain & Jain | Engineering Chemistry |
| 3. O.P. Agarwal, Hi-Tech. | Engineering Chemistry |
| 4. D.K.Sharma | Engineering Chemistry |
| 5. A.K. De | Engineering Chemistry |

BASIC MECHANICAL ENGINEERING

Course Code	Title	No. of Periods/Week	Total Periods/Year	Marks for FA	Marks for SA
A-105	Basic Mechanical Engineering	04	120	20	80

TIME SCHEDULE

Chapter No	Title	Periods	Marks	Part – A Short Type (3 Marks)	Part – B Essay Type (10 Marks)	CO's Mapping
1	Statics	20	16	2	1	CO1
2	Simple Machines	24	26	2	2	CO2
3	Basic Link Mechanisms	12	13	1	1	CO3
4	Transmission of Power	26	26	2	2	CO4
5	Fundamentals of Thermodynamics	18	13	1	1	CO5
6	Thermodynamic Processes	20	16	2	1	CO5
	Total	120	110	10	8	

CourseCode:A-105	Title: BASIC MECHANICAL ENGINEERING	Total No. of Periods:120
Course Objectives:	Upon completion of the course the student shall be able to	
	1. Understand the resolution of forces, laws of forces and equilibrium. 2. Apprehend the working principles of Simple Machines, Link Mechanisms and power transmission drives. 3. Comprehend the basics of thermodynamics and different Thermodynamic processes.	

CO No.	Course Outcome
C01	A-105.1 Apply laws of equilibrium to solve simple problems in mechanics
C02	A-105.2 Describe simple machines and solve related problems.
C03	A-105.3 Describe various basic link mechanisms
C04	A-105.4 Explain various methods of power transmission and solve numerical problems
C05	A-105.5 Comprehend the basics of thermodynamics and different thermodynamic processes.

Learning Outcomes: Upon completion of the course the student shall be able to

1.0 Understand the concept of Statics

- 1.1 Explain the meaning of mechanics in Engineering and state its importance.
- 1.2 Explain the concept of force.
- 1.3 List the types of forces.
- 1.4 Explain the system of forces, (a) Co-planar and Non-Coplanar (b) Parallel and Non-Parallel
- 1.5 State resolution, composition of forces and solve simple problems.
- 1.6 Explain the concept of equilibrium and equilibrant.
- 1.7 State (a) Parallelogram law of forces (b) Lami's theorem – solve simple problems.

2.0 Comprehend the Principles involved in Simple Machines

- 2.1 Define the important terms in simple machines a) Machine b) Mechanical Advantage c) Velocity Ratio and d) Efficiency.

- 2.2 Illustrate the use of three classes of simple lever.
- 2.3 State the expression for Velocity Ratio of: Wheel & Axle, Differential wheel and axle, Differential pulley Blocks, Pulleys, Worm & Worm wheel, Winch crabs, Screw jack and Rack & Pinion.
- 2.4 Solve simple problems on above machines.
- 2.5 Interpret the law of machine.
- 2.6 State the conditions for self-locking and reversibility.
- 2.7 Calculate effort lost in friction and load equivalent of friction.
- 2.8 Evaluate the conditions for maximum Mechanical Advantage & Maximum efficiency.
- 3.0 Understand the concept of Basic Link mechanism**
- 3.1 Define important terms of basic link mechanism a) Link b) Kinematic pair c) Kinematic chain and d) Mechanism & Machine.
- 3.2 Explain kinematic pair and kinematic chain with the help of legible sketch.
- 3.3 List examples for Lower and Higher pairs.
- 3.4 Explain four bar chain and its inversions.
- 3.5 Explain slider crank chain and its inversions.
- 4.0 Understand the concept of transmission of power.**
- 4.1 Identify various power transmitting modes like belt, chain, rope and gears drives.
- 4.2 Select suitable material for belt and explain the belt fasteners.
- 4.3 State the expression for velocity ratio of a belt drive by considering slip & thickness of belt (No derivations) and solve Simple problems
- 4.4 Explain the use of jockey pulley.
- 4.5 Distinguish between open belt drive and cross belt drive
- 4.6 State the expression for length of the belt for open belt and cross belt drives (No derivations) and solve simple Problems.
- 4.7 Explain the tensions in tight and slack sides of a belt and Calculate velocity ratio, angle of contact, Length of the belt and power transmitted by the belt drive.
- 4.8 Explain the rope drive, chain drive & gear drive and state their advantages and limitations.
- 4.9 Classify the gears and explain nomenclature of gear.
- 4.10 Discuss the simple & compound gear train and solve simple problems.
- 4.11 List the advantages and limitations of gear drive.
- 5.0 Fundamentals of Thermodynamics**
- 5.1 Define the State and System.
- 5.2 Name the types of thermodynamic Systems.
- 5.3 Explain the closed, open & isolated systems with Examples.
- 5.4 List the thermodynamic Properties of System.
- 5.5 Define the various thermodynamic properties.
- 5.6 State quasi-static work, flow of work, Zero the law of thermodynamics, first law of thermodynamics and second law of thermodynamics.
- 5.7 Solve problems dealing with conversion of heat into work and vice-versa.
- 5.8 Write non-flow energy equation and steady flow energy equation.
- 5.9 Solve simple problems on non-flow energy equation.
- 5.10 Define the term Perfect Gas.
- 5.11 State Boyle's Law, Charles's Law, Avagadro's Law, Regnault's Law and Joule's Law.
- 5.12 Write Characteristic gas equation and Universal Gas Equation.
- 5.13 Define Specific heat, Specific heat at constant pressure (C_p) and Specific heat at constant volume (C_v) and their relation.
- 6.0 Thermodynamic Processes**
- 6.1 List different types of thermodynamic processes.
- 6.2 Derive expression for work done in Constant Volume (Isochoric process), Constant Pressure (Isobaric Process), Constant Temperature (Isothermal), Polytrophic process (Isentropic process) and Adiabatic Process.
- 6.3 Compute the change in internal energy of gas during a process.
- 6.4 Explain the relationship between heat supplied, internal energy and work done.
- 6.5 Solve simple problems on above processes.

COs–POs Mapping

CourseCode:A-105		Course Title: BASIC MECHANICAL ENGINEERING		No.of COs:5	No.of Periods:120
POs	Mapped with CO No.	CO Periods Addressing POinCol.1		Level(1,2,3)	Remarks
		Number	Percentage		
PO1	CO1,CO2,CO3,CO4,CO5	31	25	2	>40%Level3(Highly Addressed) 25%to40%level2 (Moderately Addressed) 5%to25%level1(Low Addressed) <5%NotAddressed
PO2	CO1,CO2,CO3, CO5	20	17	1	
PO3	CO1,CO2,CO3,CO4,CO5	23	19	1	
PO4	CO1,CO2,CO3,CO4,CO5	25	21	1	
PO5	CO1,CO2,CO3,CO4,CO5	21	18	1	
PO6					
PO7					

CO-PO/PSOMATRIX(1.Slight(Low)2.Moderate(Medium)3.Substantial(High))

CO No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
A-105.1	1	3	2	2	2			2	3	2
A-105.2	1	3	2	2	2			2	3	2
A-105.3	1	3	3	2	2			2	3	2
A-105.4	1		2	3	2			2	2	
A-105.5	1	3	2	3	2			2	3	2
Average	1	2.4	2.2	2.4	2			2	2.8	1.6

Note: The gaps in CO and PO mapping will be achieved by one or more appropriate activities from the following:

- i) Seminars ii) Tutorials iii) Guest Lectures iv) Assignments v) Quiz competitions vi) Industrial visit vii) Tech Fest viii) Mini project ix) Group discussions x) Virtual classes xi) Library visit for e-books

COURSE CONTENT

1.0 Statics

Definition and Application of Mechanics- Definition and specification of force – types of forces - System of forces - Resolution of forces and composition of forces - Equilibrium and Equilibrant - Parallelogram law of forces - Lami's theorem - Related numerical simple problems.

2.0 Simple Machines

Definition, types and uses of Simple machine - Types of Levers - Mechanical advantage, Velocity ratio and Efficiency - Velocity Ratio of Simple pulley, Differential pulley, system of pulleys, wheel & axle, differential wheel & axle, Worm & Worm wheel, Rack & pinion, Winch crabs, & Screw jack (No derivation) - Conditions for reversibility and self-locking - Laws of Simple Machine - Effort lost in friction, Load equivalent of friction, Maximum Mechanical Advantage and Maximum efficiency- Numerical Problems.

3.0 Basic Link Mechanism

Definition of terms: Link, Kinematic pair, Kinematic chain, Machine and Mechanism - Differences between structure and machine, machine and mechanism – examples of lower & higher pairs – four bar chain and its inversions - Slider Crank chain and its inversions.

4.0 Transmission of power

Belt drive - Materials for the belt drive and belt fasteners - Explanation of the terms: Velocity ratio - Slip - Percentage of slip - Thickness of belt - Tight side and Slack side of the belt - Jockey pulley - Open belt and Cross belt drive - Effective tension ($T_1 - T_2$) - Velocity of the belt and Power transmitted by belt drive - Length of open belt and cross belt drives – Problems related to Velocity ratio, Slip, length of the belts and Power transmitted in a belt drive- Advantages and limitations of belt drive - Rope drive- Chain drive, types of chain drives – Advantages and limitations of chain drive over a belt drive – Gears and Gear trains- Nomenclature of gear-Types

of gears—spur, helical, bevel, and spiral - Simple and compound gear trains - Velocity ratio in gear drive and problems - Advantages and limitations of gear drive over a belt drive.

5.0 Fundamentals of Thermodynamics

Definitions of state and system - Types of thermodynamic systems – closed, open and isolated systems with examples - Properties of a system- Intrinsic and Extrinsic properties with examples- Definitions for properties: Pressure, Volume, Temperature, Enthalpy, Internal energy and their units - Definitions of quasi-static work, flow-work, Zeroth, first and second laws of thermodynamics - simple problems on conversion of Heat into Work and vice versa –Non flow energy equation - Steady flow energy equation (without proof) - simple problems of elementary type – Definition of Perfect Gas - Boyle's Law, Charle's Law, Avagadro's Law, Regnault's Law and Joule's Law - Characteristic gas equation & Universal Gas Equation – definition of Specific heat, Specific heat at constant pressure and Specific heat at constant volume and their relation.

6.0 Thermodynamic processes

Types of thermodynamic processes - Isochoric, Isobaric, Isothermal, Isentropic, Polytrophic and Adiabatic - Equations representing the processes - Derivation for work done and change in internal energy for the above processes - Calculation of heat supplied or rejected during the above processes - Simple problems on the above processes.

REFERENCEBOOKS:

1. Elements of Mechanical engineering by Roy&Choudary – Media Promoters Publishers Pvt. Ltd.
2. Engineering Thermodynamics P.K.Nag-TataMcGraw-Hill
3. Engineering Thermodynamics C.P.Arora -TataMcGraw-Hill
4. Theory of Machines by R. S Khurmi and J.K Gupta - S. Chand Publications.

Table specifying the scope of syllabus to be covered for Unit Tests

Unit Test	LearningOutcomestobecovered
UnitTest–I	From1.1to 2.8
UnitTest–II	From 3.1 to 4.11
Unit Test-III	From 5.1 to 6.5

ELEMENTS OF AUTOMOBILE ENGINEERING

Course Code	Course Title	No. of Periods/ week	Total Periods/ Semester	Marks for FA	Marks for SA
A-106	Elements of Automobile Engineering	03	90	20	80

TIME SCHEDULE

Chapter No	Title	Periods	Marks	Part – A Short Type (3 marks)	Part – B Essay Type (10 marks)	CO's Mapping
1	Introduction	6	3	1	0	CO1
2	Engines	16	23	1	2	CO1
3	Basics of Engine Systems	18	26	2	2	CO2
4	Basics of Transmission System	14	16	2	1	CO3
5	Basics of Electrical System	10	13	1	1	CO3
6	Basics of Chassis and Body	8	6	2	0	CO4
7	Basics of Steering, Braking and Suspension Systems	18	23	1	2	CO5
TOTAL		90	110	10	8	

Course Code: A - 106	Title: Elements of Automobile Engineering	Total No. of Periods:90
Course Objectives:	Upon completion of the course the student shall be able to	
	1. Understand the constructional details and basic systems of an IC Engine. 2. Apprehend the basics of Transmission, Electrical systems, Chassis and Body of vehicles. 3. Comprehend the basics of Steering, Braking and Suspension systems of automobile	

CO No.	Course Outcome
C0	A-106.1 Explain the history, recent trends in automobile and describe the constructional details of I.C engine.
C0	A- 106.2 Discuss the requirement of Inlet, Exhaust, Cooling and Lubrication systems of an I.C engine.
C0	A-106.3 Explain the requirement of Transmission and Electrical systems of a vehicle
C0	A-106.4 Discuss various types of chassis frames and vehicle bodies of an automotive
C0	A-106.5 Explain the functions of Steering, Braking & Suspension Systems of a Vehicle

LEARNING OUTCOMES:

Upon completion of the course the student shall be able to

1. Introduction

- 1.1 Define Automobile
 - 1.2 Know the history of Automobiles
 - 1.3 Know the developments of Automobile vehicles
 - 1.4 State the recent trends in Automobile vehicles
 - 2. Engine**
 - 2.1 Define an Engine, IC Engine and EC Engine
 - 2.2 State the difference between IC Engine and EC Engine.
 - 2.3 State the types of vehicles based on fuel used in an engine.
 - 2.4 Explain the functions of different IC Engine components.
 - 2.5 Explain the constructional details of IC Engine.
 - 2.6 State the classification of IC Engines.
 - 2.7 Explain the working Principle of 2 stroke and 4 stroke petrol engine with sketches.
 - 2.8 Explain the working Principle of 2 stroke and 4 stroke Diesel engine with sketches.
 - 2.9 State the differences between petrol engine and diesel engine.
 - 2.10 State the comparison between 2- stroke engine and 4-stroke engine.
 - 3. Basics of Engine Systems**
 - 3.1 List different systems of an Engine.
 - 3.2 State the necessity of Inlet and Exhaust Manifolds.
 - 3.3 State the functions of Air Cleaner and Mufflers.
 - 3.4 Draw the line diagram of a petrol engine fuel system and explain.
 - 3.5 List the elements of petrol engine fuel system and state their functions.
 - 3.6 Explain the working of simple carburettor with a sketch.
 - 3.7 Draw the line diagram of a diesel engine fuel system and explain.
 - 3.8 List the elements of diesel engine fuel system and state their functions.
 - 3.9 State the necessity of lubrication system and list types of lubricants used.
 - 3.10 List the types of lubrication systems and explain Petroil system of lubrication.
 - 3.11 State the necessity of cooling system and list types of cooling systems.
 - 3.12 Describe the air cooling system with a sketch.
 - 3.13 Explain the working of water cooling system of multi cylinder engine with a sketch.
 - 3.14 State the advantages of water cooling system over air cooling system.
 - 4. Basics of Transmission System**
 - 4.1 State the purpose of automobile transmission system.
 - 4.2 Draw the layout of automobile transmission system and explain.
 - 4.3 List the elements of transmission system of a vehicle.
 - 4.4 State the function of clutch.
 - 4.5 List different types of clutches.
 - 4.6 State the function of gear box.
 - 4.7 List different types of gear boxes.
 - 4.8 State the function of propeller shaft and universal joints.
 - 4.9 State the function of differential.
 - 4.10 State the function of front axles and rear axles.
 - 5. Basics of Electrical System**
 - 5.1 State the requirement of electrical system in automobiles.
 - 5.2 List different starting methods of engine.
 - 5.3 Explain the working of self-motor.
 - 5.4 Explain the working of alternator.
 - 5.5 Explain the Construction and working of a Lead acid battery.
 - 5.6 State the requirements of ignition system in engine.
-

5.7 List the elements of ignition system.

5.8 List the types of ignition systems.

5.9 State the functions of Spark Plug.

6. Basics of Chassis and Body:

6.1 State the functions of chassis frame.

6.2 Draw the layout of the chassis.

6.3 List the components of chassis frame.

6.4 List different types of chassis frames.

6.5 State various loads acting on the chassis frame.

6.6 State the requirements of automobile body.

6.7 List different types of car bodies.

7. Basics of Steering, Braking and Suspension Systems

7.1 State the requirement of steering system of a vehicle.

7.2 Draw the steering gear layout and indicate the components.

7.3 List different types of steering gear boxes.

7.4 State the functions and requirements of automobile brakes.

7.5 List different types of brakes.

7.6 Draw the layout of mechanical brakes and explain.

7.7 Draw the layout of hydraulic braking systems and explain.

7.8 State the advantages of hydraulic brakes over mechanical brakes.

7.9 State the functions of suspension system of a vehicle.

7.10 List different types of suspension system.

7.11 List different types of springs.

COs – POs Mapping

Course Code: A-106	Course Title: Elements of Automobile Engineering	No of Cos: 5			No. of periods: 90
POs	Mapped with CO No.	CO Periods Addressing PO in Col 1		Level (1,2,3)	Remarks
		No	%		
PO1	CO1, CO2, CO3, CO4, CO5	40	44	3	>40% Level 3 (Highly Addressed) 25% to 40% Level 2 (Moderately Addressed)
PO2	CO2, CO4, CO5	20	22	1	5% to 25% Level 1 (Low Addressed)
PO3	CO1, CO2, CO3, CO4,	15	17	1	<5% Not Addressed
PO4	CO2, CO3, CO4,	8	9	1	
PO5	CO1, CO2, CO3, CO4	7	8	1	
PO6					
PO7					

CO-PO/PSO MATRIX

CO No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
A-106.1	3		1		2			1	2	
A-106.2	3	2	1	2	2			1	2	
A-106.3	3		1	2	1			1	2	
A-106.4	3	2	1	1	1			1	2	2
A-106.5	3	2						1	3	
AVERAGE	3	1.2	0.8	1	1.2			1	2.2	0.4

COURSE CONTENTS

1.0 Introduction

Definition of Automobile - History of Automobiles - Development of Automobiles - Recent trends in Automobiles

2.0 Engine

Definition of Engine - Internal Combustion Engine (IC Engine), External Combustion Engine, (EC Engine) – types of engines based on fuel used - Engine Construction – functions of; Cylinder Block - Cylinder Head - Crank Case - Cylinder Bore and Stroke - Cylinder Liners - Oil Pan – Gaskets - Piston - Piston Pin - Piston Rings - Connecting Rod - Crankshaft - Cam shaft - Flywheel - Valves - Classification of I.C. Engines - working Principle of 2 stroke and 4stroke petrol and diesel engines - differences between petrol and diesel engines - comparison between 2-stroke and 4-stroke engines.

3.0 Basics of Engine Systems

Inlet and Exhaust System: Functions of Inlet and Exhaust Manifolds, Air Cleaner and Mufflers.

Fuel Supply System: Line diagram of petrol engine fuel systems - functions of fuel tank, fuel pump, filters, Working of Simple carburettor - Line diagram of diesel engine fuel system - Functions of fuel feed pump, fuel filter, fuel injection pump, fuel injector.

Lubrication System: Necessity of lubrication - Types of lubricants - Types of lubrication systems - Petroil system of lubrication.

Cooling System: Necessity of cooling system - Disadvantages of over cooling and under cooling – Working of air cooling - water cooling system.

4.0 Basics of Transmission System:

Purpose of automobile transmission system - layout of automobile transmission system - elements of transmission system – functions and types of clutch, gear box, propeller shaft, universal joints, differential, rear axles and front axles.

5.0 Basics of Electrical System:

Requirement of electrical system in automobiles - starting methods of engine - working of self-motor, alternator and Lead acid battery - Requirements of ignition system – elements of ignition system - types of ignition systems - functions of Spark Plug.

6.0 Basics of Chassis and Body:

Chassis frame - layout of the chassis and its main components - functions of the frame - types of chassis frames - various loads acting on the chassis frame - Requirements of automobile body – types of Car bodies.

7.0 Basics of Steering, Braking and Suspension Systems

Steering System: Requirements and functions of the vehicle steering system – Steering gear layout - Types of steering gears.

Braking System: Functions and requirements of automobile brakes - types of brakes - Mechanical and Hydraulic brakes - Working of mechanical and hydraulic braking systems - advantages of hydraulic brakes over mechanical brakes.

Suspension System: Functions of suspension system - types of suspension system - types of springs - Leaf spring & coil springs.

REFERENCEBOOKS

1. Heat Engines - by R.S. Kurmi—S. Chand & Company
2. Heat Engines - by Pandya and Shah—Charotar Publishing House
3. Automobile Engineering Vol-I&II- by Kirpal Singh
4. Automobile Engineering — by G.B.S. Narang—Khanna Publishers
5. Automobile Engineering — by R.B. Gupta —Standard Publishers
6. Automobile mechanics — by William Chrouse—McGraw Hill Education.
7. Internal Combustion Engines - by Mathur & Sharma—Dhanpat Rai Publications
8. Automobile Engineering Voll - Anil Chhikaria-Satya Prakashan Publishers

Table specifying the scope of syllabus to be covered for Unit Tests

Unit Test	Learning Outcomes to be covered
Unit Test-I	1.1 to 3.6
Unit Test-II	3.7 to 5.4
Unit Test-III	5.5 to 7.11

A-107 ENGINEERING DRAWING

Course Code	Course Title	No. of Periods/Week	Total Periods/ Semester	Marks for FA	Marks for SA
A-107	Engineering Drawing	06	180	40	60

TIME SCHEDULE

Chapter No	Title	No. of Drawing plates	Periods	Marks	Short Answer Questions	Essay type Questions	CO's Mapping
1	Importance of Engineering Drawing	--	01	-	-	-	CO1
2	Engineering Drawing Instruments	01	05	-	-	-	CO1
3	Free hand lettering & Numbering	01	06	5	1	-	CO2
4	Dimensioning Practice	01	09	5	1	-	CO2
5	Geometrical constructions	03	24	15	1	1	CO3
6	Projections of points, Lines, Planes & Auxiliary Views	03	21	5	1		CO4
7	Projections of Solids	01	12	10		1	CO4
8	Sections of Solids	01	21	10	-	1	CO4
9	Orthographic Projections	01	30	10	-	1	CO4
10	Isometric Views	01	30	10	-	1	CO4
11	Development of surfaces	01	21	10	-	1	CO5
Total		14	180	80	04	06	

Course Code: A-107	Course: Engineering Drawing	No. of Periods: 180
COURSE OBJECTIVES	Upon completion of the course the student shall able to understand the basic graphic skills and use them in preparation of engineering drawings, their reading and interpretation	

CO No.		Course Outcome
C01	A-107.1	Practice the use of engineering drawing instruments
C02	A-107.2	Familiarise with the conventions to be followed in engineering drawing as per BIS specifications.
C03	A-107.3	Construct the i) basic geometrical constructions ii) engineering curves
C04	A-107.4	Visualise and draw the orthographic projections of i) Points ii) Lines iii) Regular Planes iv) Regular Solids V) Sections of Regular Solids and the isometric views of machine components
C05	A-107.5	Draw the developments of surfaces of regular solids and thereby the components used in daily applications

LEARNING OUTCOMES

Upon completion of the course the student shall able to

1.0 Importance of Engineering Drawing

- 1.1 State the importance of drawing as an engineering communication medium
- 1.2 State the necessity of B.I.S. Code of practice for Engineering Drawing.
- 1.3 Explain the linkages between Engineering drawing and other subjects of Mechanical Engineering

2.0 Engineering Drawing Instruments

- 2.1 Select the correct instruments to draw the different lines / curves
- 2.2 Use correct grade of pencil to draw different types of lines and for different purposes
- 2.3 Select and use appropriate scales for a given application.
- 2.4 Identify different drawing sheet sizes as per I.S. and Standard Lay- outs.
- 2.5 Prepare Title block as per B.I.S. Specifications.
- 2.6 Identify the steps to be taken to keep the drawing clean and tidy.
Drawing Plate 1: (Having two exercises)

3.0 Free Hand Lettering and Numbering

- 3.1 Write titles using sloping lettering and numerals of 7mm, 10mm and 14mm height
- 3.2 Write titles using vertical lettering and numerals of 7mm, 10mm and 14mm height
- 3.3 Select suitable sizes of lettering for different layouts and applications
Drawing plate 2: (Having 5 to 6 exercises)

4.0 Understand Dimensioning Practice

- 4.1 Acquaint with the conventions, notations, rules and methods of dimensioning in engineering drawing as per the B.I.S.
- 4.2 Dimension a given drawing using standard notations and desired system of dimensioning.
Drawing Plate 3: (Having 08 to 10 exercises)

5.0 Apply Principles of Geometric Constructions

- 5.1 Practice the basic geometric constructions like i) dividing a line into equal parts
ii) Exterior and interior tangents to the given two circles iii) tangent arcs to two given lines and arcs
- 5.2 Draw any regular polygon using general method when i) side length is given
ii) Inscribing circle radius is given iii) describing circle radius is given
- 5.3 Draw the conics using general and special methods,
- 5.4 Draw the engineering curves like i) involute ii) cycloid iii) helix
- 5.5 Identify the applications of the above constructions in engineering practice.
Drawing Plate -4: Having problems up to construction of polygon
Drawing Plate -5: Having problems of construction of conics
Drawing Plate -6: Having problems of construction of involute, cycloid and helix

6.0 Apply Principles of Projection of points, lines, planes & auxiliary planes

- 6.1 Explain the basic principles of the orthographic projections
- 6.2 Visualise and draw the projection of a point with respect to reference planes (HP&VP)
- 6.3 Visualise and draw the projections of straight lines with respect to two reference Planes (up to lines parallel to one plane and inclined to other plane)

- 6.4 Visualise and draw the projections of planes (up to planes perpendicular to one plane and inclined to other plane)
- 6.5 Identify the need of Auxiliary views for a given engineering drawing.
- 6.6 Draw the auxiliary views of a given engineering component
 Drawing Plate -7: Having problems up to projection of points and Lines (15 exercises)
 Drawing Plate -8: Having problems of projection of planes (6 exercises)
 Drawing Plate -9: Having problems on auxiliary planes (Having 4 exercises)
- 7.0 Draw the Projections of Solids**
- 7.1 Visualise and draw the projections of solids (up to axis of solids parallel to one plane and inclined to other plane)
 Drawing plate No.10: Having problems of projection of solids (10 exercises)
- 8.0 Appreciate the need of Sectional Views**
- 8.1 Identify the need to draw sectional views.
- 8.2 Differentiate between true shape and apparent shape of section
- 8.3 Draw sectional views and true sections of regular solids by applying the principles of Hatching.
 Drawing Plate–11: Having problems of section of solids (6 exercises)
- 9.0 Apply principles of orthographic projection**
- 9.1 Draw the orthographic views of an object from its pictorial drawing.
- 9.2 Draw the minimum number of views needed to represent a given object fully.
 Drawing Plate 12 : (Having 10 to 12 exercises)
- 10.0 Prepare pictorial drawings**
- 10.1 identify the need of pictorial drawings.
- 10.2 Differentiate between isometric scale and true scale.
- 10.3 Prepare Isometric views for the given orthographic drawings.
 Drawing plate 13: (Having 10 to 12 exercises)
- 11.0 Interpret Development of surfaces of different solids**
- 11.1 State the need for preparing development drawing.
- 11.2 Draw the development of simple engineering objects (cubes, prisms, cylinders, cones, pyramid) using parallel line and radial line method.
- 11.3 Prepare development of surface of engineering components like i) funnel ii) 90° Elbow
 Drawing plate No. 14: (Having 05 exercises)

Competencies and Key competencies to be achieved by the student

S.No	Major topic	Key Competency
1.	Importance of Engineering Drawing	Explain the linkages between Engineering drawing and other subjects of study in Diploma course.
2.	Engineering Drawing Instruments	Select the correct instruments to draw various entities in different orientation
3.	Free hand lettering & Numbering	Write titles using sloping and vertical lettering and numerals as per B.I.S (Bureau of Indian standards)
4.	Dimensioning Practice	Dimension a given drawing using standard notations and desired system of dimensioning
5.	Geometrical construction	Construct ellipse, parabola, rectangular hyperbola, involute, cycloid and helix from the given data.
6.	Projection of points, Lines, Planes & Solids	Draw the projections of points, straight lines, planes & solids with respect to reference planes (HP& VP)
7.	Auxiliary views	- Draw the auxiliary views of a given Engineering component - Differentiate between Auxiliary view and apparent view
8.	Sections of Solids	- Differentiate between true shape and apparent shape of section - Apply principles of hatching. - Draw simple sections of regular solids
9.	Orthographic Projection	Draw the minimum number of views needed to represent a given object fully.
10.	Isometric Views	- Differentiate between isometric scale and true scale. - Draw the isometric views of given objects,.
11.	Development of surfaces	Prepare development of Surface of regular solids and other components like i) funnel ii) 90° elbow

COs – POs Mapping

Course Code : A-107	Course Title ENGINEERING DRAWING No. of CO's: 5				No. of Periods : 180
POs	Mapped with CO No.	CO Periods addressing PO in Column 1		Level (1,2,3)	Remarks
		No	%		
PO1	CO1, CO2, CO3, CO4, CO5	40	22	1	>40% Level 3 Highly addressed 25% to 40% Level 2 Moderately Addressed 5 to 25% Level 1 Low addressed <5% Not addressed
PO2	CO3, CO4, CO5	30	17	1	
PO3	CO1, CO2, CO3, CO4, CO5	50	28	2	
PO4	CO1, CO3, CO4, CO5	26	14	1	
PO5	CO1, CO2, CO3, CO4, CO5	34	19	1	
PO6					
PO7					

CO-PO/PSO MATRIX

CO No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
A-107.1	3		2	3	2			1	1	1
A-107.2	2		2		2			1	2	1
A-107.3	3	2	2	3	3			2	3	
A-107.4	3	2	3	3	3			2	3	2
A-107.5	3	2	3	3	2			3	2	
Average	2.8	1.2	2.4	2.4	2.8			1.8	2.2	0.8

Note: The gaps in CO and PO mapping will be achieved by one or more appropriate activities from the following:

- i) Seminars ii) Tutorials iii) Guest Lectures iv) Assignments v) Quiz competitions vi) Industrial visit vii) Tech Fest viii) Mini project ix) Group discussions x) Virtual classes xi) Library visit for e-books

COURSE CONTENT

NOTES:

- B.I.S Specification should invariably be followed in all the topics.**
- A-3 Size Drawing Sheets are to be used for all Drawing Practice Exercises.**

1.0 The importance of Engineering Drawing

Explanation of the scope and objectives of the subject of Engineering Drawing Its importance as a graphic communication -Need for preparing drawing as per standards – SP-46 –1988 – Mention B.I.S - Role of drawing in engineering education – Link between Engineering drawing and other subjects of study.

2.0 Engineering drawing Instruments

Classifications: Basic Tools, tools for drawing straight lines, tools for curved lines, tools for measuring distances and special tools like mini drafter & drafting machine – Mentioning of names under each classification and their brief description -Scales: Recommended scales reduced & enlarged -Lines: Types of lines, selection of line thickness - Selection of Pencils -Sheet Sizes: A0, A1, A2, A3, A4, A5, Layout of drawing

sheets in respect of A0, A1, A3 sizes, Sizes of the Title block and its contents - Care and maintenance of Drawing Sheet, Drawing plate:

Lay out of sheet – as per SP-46-1988 to a suitable scale.

Simple Exercises on the use of Drawing Instruments. Importance of Title Block.

3.0 Free hand lettering & numbering

Importance of lettering – Types of lettering -Guide Lines for Lettering

Practicing of letters & numbers of given sizes (7mm, 10mm and 14mm)

Advantages of single stroke or simple style of lettering - Use of lettering stencils

4.0 Dimensioning practice

Purpose of engineering Drawing, Need of B.I.S code in dimensioning -Shape description of an Engineering object -Definition of Dimensioning size description -Location of features, surface finish, fully dimensioned Drawing -Notations or tools of dimensioning, dimension line extension line, leader line, arrows, symbols, number and notes, rules to be observed in the use of above tools -Placing dimensions: Aligned system and unidirectional system (SP-46-1988)-Arrangement of dimensions Chain, parallel, combined progressive, and dimensioning by co-ordinate methods-The rules for dimensioning standard, features “Circles (holes) arcs, angles, tapers, chamfers, and dimension of narrow spaces.

5.0 Geometric Construction

Division of a line: to divide a straight line into given number of equal parts

Construction of tangent lines: to draw interior and exterior tangents to two circles of given radii and centre distance

Construction of tangent arcs:

i) To draw tangent arc of given radius to touch two lines inclined at given angle (acute, right and obtuse angles).

ii)Tangent arc of given radius touching a circle or an arc and a given line.

iii) Tangent arcs of radius R, touching two given circles internally and externally.

Construction of polygon: construction of any regular polygon by general method for given side length, inscribing circle radius and describing/super scribing circle radius

Conics: Explanation of Ellipse, Parabola, Hyperbola, as sections of a double cone and a loci of a moving point, Eccentricity of above curves – Their Engg. Applications viz. Projectiles, reflectors, P-V Diagram of a hyperbolic process,

Construction of any conic section of given eccentricity by general method

Construction of ellipse by concentric circles method, Oblong Method and Arcs of circles method,

Construction of parabola by rectangle method and Tangent method

Construction of rectangular hyperbola

General Curves: Involute, Cycloid and Helix, explanations as locus of a moving point, their engineering application, viz., Gear tooth profile, screw threads, springs etc. –their construction

6.0 Projection of points, lines and planes & Auxiliary views

Classification of projections, Observer, Object, Projectors, Projection, Reference Planes, Reference Line, Various angles of projections –Differences between first angle and third angle projections.

Projections of points in different quadrants

Projections of straight line -

(a) Parallel to both the planes.

(b) Perpendicular to one of the planes.

(c) Inclined to one plane and parallel to other planes

Projections of regular planes

(a) Plane parallel to one of the reference planes

(b) Plane perpendicular to HP and inclined to VP and vice versa.

Auxiliary views

Need for drawing auxiliary views -Explanation of the basic principles of drawing an auxiliary views
explanation of reference plane and auxiliary plane -Partial auxiliary view.

7.0 Projections of regular solids

- (a) Axis perpendicular to one of the planes
- (b) Axis parallel to VP and inclined to HP and vice versa.

8.0 Sections of Solids

Need for drawing sectional views – what is a sectional view - Hatching – Section of regular solids inclined to one plane and parallel to other plane

9.0 Orthographic Projections

Meaning of orthographic projection -Using a viewing box and a model – Number of views obtained on the six faces of the box, - Legible sketches of only 3 views for describing object -Concept of front view, top view, and side view sketching these views for a number of engg. objects - Explanation of first angle projection. – Positioning of three views in First angle projection -Projection of points as a means of locating the corners of the surfaces of an object – Use of mitre line in drawing a third view when other two views are given -Method of representing hidden lines -Selection of minimum number of views to describe an object fully.

10.0 Pictorial Drawings

Brief description of different types of pictorial drawing viz., Isometric, oblique, and perspective and their use - Isometric drawings: Iso axis, angle between them, meaning of visual distortion in dimensions - Need for an isometric scale, difference between Isometric scale, and ordinary scale difference between Isometric view and Isometric projection - Isometric and non-Isometric lines -Isometric drawing of common features like rectangles, circular - shapes, non-isometric lines - Use of box and offset methods

11.0 Development of Surfaces

Need for preparing development of surface with reference to sheet metal work-Concept of true length of a line with reference to its orthographic projection when the line is (i) parallel to the plane of projection (ii) inclined to one principal and parallel to the other -Development of simple solids like cubes, prisms, cylinders, cones, pyramid and truncation of these solids-Types of development: Parallel line and radial line development -Procedure of drawing development of funnels, 90° elbow pipes.

REFERENCE BOOKS

- Engineering Graphics by P I Varghese – (McGraw-hill)
- Engineering Drawing by BasantAgarwal& C.M Agarwal - (McGraw-hill)
- Engineering Drawing by N.D.Bhatt.
- T.S.M. & S.S.M on “ Technical Drawing” prepared by T.T.T.I., Madras
- SP-46-1998 – Bureau of Indian Standards.

WORKSHOP PRACTICE-I

Course Code	Course Title	No. of Periods/Week	Total Periods/Year	Marks for FA	Marks for SA
A-108	Workshop Practice-I	03	90	40	60

TIME SCHEDULE

Chapter No	Title	Periods	CO's Mapping
1	Workshop safety precautions	03	CO1
2.	Fitting shop	27	CO2
3.	Carpentry shop	24	CO3
4.	Sheet metal work	21	CO4
5.	Plumbing	15	CO5
	Total	90	

Course Code: A-108	Title: Workshop Practice-I	Total No. of Periods: 90
Course Objectives:	Upon completion of the course the student shall be able to 1.0 To Familiarize tools used in workshop processes 2.0 To handle the tools appropriately and safely 3.0 To learn theoretical concepts by practicing relevant exercises of workshop process	

CO No.	Course Outcome
CO1	A-108.1 Familiarize tools used in workshop processes and safety measures.
CO2	A-108.2 Practice the operations in Fitting Shop
CO3	A-108.3 Practice the operations in Carpentry Shop
CO4	A-108.4 Practice the operations in Sheet metal Shop
CO5	A-108.5 Practice the operations in Plumbing

Learning Outcomes:

Upon completion of the course the student shall able to

1. Workshop safety Precautions

1.1 Know the safety precautions and their importance in workshop.

2. Fitting shop

2.1 Identify the various marking tools, cutting tools, striking tools, holding devices and Miscellaneous tools used in fitting.

2.2 Handle appropriate chipping tools and measuring tools and mark the dimensions

2.3 Cutting with hack saw of MS flats of 6mm thick

2.3.1 Identify appropriate measuring tool

2.3.2 Choose the appropriate marking tool

2.3.3 Check the raw material for size

2.3.4 Fix the work piece in vice

2.3.5 Mark the work as per given dimensions

2.3.6 Perform dot punching

- 2.3.7 Load and unload hack saw blade from its frame
- 2.4 Drilling, chamfering on a MS flat of 12 mm thick
 - 2.4.1 Check the raw material for size
 - 2.4.2 Apply the chalk on the surface and on all sides of the flat
 - 2.4.3 Layout the dimensions and mark the lines using dot punch
 - 2.4.4 Chamfer the edges through filing
 - 2.4.5 Locate the whole centres using odd leg callipers and centre punching
 - 2.4.6 Identify appropriate drill bit
 - 2.4.7 Load and unload drill bit from the machine
- 2.5 Tapping and Dyeing on a MS flat of 12 mm thick
 - 2.5.1 Check the raw material for size.
 - 2.5.2 Identify appropriate tap and die
 - 2.5.3 Secure the tap in the wrench
 - 2.5.4 Perform Tapping
 - 2.5.5 Hold the bar in bench vice
 - 2.5.6 Fix the die in die stock
 - 2.5.7 Cut external threads using a Die
 - 2.5.8 Check the fit for accuracy
- 2.6 Assembling of two pieces, matching by filing
 - 2.6.1 Cut the pieces to size using hacksaw
 - 2.6.2 File surface of flat for trueness
 - 2.6.3 Mark the surfaces as per dimensions
 - 2.6.4 Perform cutting with hack saw as per marked lines
 - 2.6.5 Smooth the surfaces with file
 - 2.6.6 Assemble the two pieces
- 3. Carpentry shop**
 - 3.1 Identify the various tools used carpentry.
 - 3.2 Cutting of wood with handsaw
 - 3.2.1 Identify the orientation of grains
 - 3.2.2 Select appropriate saw for cutting in each of the directions viz. across and along the grains
 - 3.2.3 Select appropriate work holding device
 - 3.2.4 Handle appropriate measuring and marking tools(Steel rule, Try square, Marking gauge)
 - 3.2.5 Mark dimensions on work using Marking gauge
 - 3.2.6 Fix the work in the vice
 - 3.2.7 Perform cutting along the grains using Ripsaw
 - 3.2.8 Perform cutting perpendicular to the grains using cross cut saw
 - 3.3 Planning of wood
 - 3.3.1 Identify the direction for planning Woodstock
 - 3.3.2 Select appropriate jackplane
 - 3.3.3 Prepare the jack plane for planning
 - 3.3.4 Load and unload the blade of a jackplane
 - 3.3.5 Select appropriate work holding device
 - 3.3.6 Perform marking on work using appropriate tool
 - 3.3.7 Fix the work in the vice
 - 3.3.8 Plane the surfaces on all four sides using jackplane
 - 3.4 Chiselling of wood
 - 3.4.1 Select appropriate chisels and saw
 - 3.4.2 Select appropriate work holding device
 - 3.4.3 Select appropriate measuring and marking tools
 - 3.4.4 Fix the work in the vice
 - 3.4.5 Mark the position of grooves on work using marking gauge
 - 3.4.6 Cut sides of grooves by handsaw

- 3.4.7 Chip the material using firmer chisel by applying pressure with mallet
- 3.4.8 Finish the grooves with rasp file
- 3.5 Preparation of a Dove-tail joint
 - 3.5.1 Select the appropriate cutting tools and work holding devices
 - 3.5.2 Plane the wooden pieces on all sides
 - 3.5.3 Mark at an angle of 150 with bevel square
 - 3.5.4 Trim the dovetail by chisel to exact size
 - 3.5.5 Cut the dovetail groove on second piece
 - 3.5.6 Finish the groove
 - 3.5.7 Assemble the two pieces to prepare dovetail halving joint by using mallet
- 3.6 Preparation of Mortise and Tenon joint
 - 3.6.1 Select the appropriate cutting tools and work holding devices
 - 3.6.2 Plane the two pieces to the required size using jackplane
 - 3.6.3 Mark the dimensions to make Tenon using mortise gauge
 - 3.6.4 Cut tenon with tenon saw along the marked lines
 - 3.6.5 Use firmer chisel to remove the excess material to set finished tenon
 - 3.6.6 Mark the dimension to make mortise on the second piece with mortise gauge
 - 3.6.7 Use mortise chisel to provide recess in the second piece to accommodate tenon.
 - 3.6.8 Assemble the two pieces by fitting the tenon into mortise
- 3.7 Wood turning on lathe
 - 3.7.1 Select appropriate tools
 - 3.7.2 Plane the four corners of the work piece using jackplane
 - 3.7.3 Mark the centres of the work on either side
 - 3.7.4 Mount the work between head stock & tailstock centres
 - 3.7.5 Fix the tool in the tool post & Position it in appropriate height
 - 3.7.6 Start the lathe to make the work piece to revolve at desired speed
 - 3.7.7 Feed the bevel gauge against the rotating work to get the required size and shape
 - 3.7.8 Use outside callipers to check the diameter of the pin
 - 3.7.9 Use parting off tool to reduce the diameter on either ends of the pin
 - 3.7.10 Remove rolling pin between centres & cut excess material on either side
- 4. Sheet metal work**
 - 4.1 Identify the various tools used Sheet metal work.
 - 4.2 Know the various Sheetmetal operations: Cutting off, parting, blanking, punching, piercing, notching, slitting and trimming, bending, Deep drawing, coining and riveting.
 - 4.3 Practice on cutting of sheet
 - 4.3.1 Cut the required sheet from the stock using snip
 - 4.3.2 Mark the dimensions on the sheet using scribe & steel rule
 - 4.3.3 Draw the circular shapes using divider
 - 4.3.4 Perform rough cutting of the curved shapes using chisel and finish cutting using snips
 - 4.3.5 Cut the straight edges using straight snip
 - 4.4 Formation of joints like grooved joint, locked groove joint
 - 4.4.1 Cut the sheet into two halves
 - 4.4.2 Form the flange on the sheet by folding the sheet along scribed lines using mallet & stakes.
 - 4.4.3 Perform bending edges of sheets applying moderate pressure using mallet
 - 4.4.4 Interlock the bent edges and apply pressure with mallet to make required Joint.
 - 4.5 Preparation of a rectangular open tray
 - 4.5.1 Draw the development of the object to be made
 - 4.5.2 Place the pattern on the sheet
 - 4.5.3 Mark the dimensions using scribe
 - 4.5.4 Shear the required piece from the stock using straight snips
 - 4.5.5 Mark the lines on the sheet to form bends
 - 4.5.6 Strengthen the sides of sheet by single hem using hatchet stake
 - 4.5.7 Form the sheet into desired shape using stakes

- 4.5.8 Seam the corners by inserting laps of the adjacent sides with single hem.
- 4.6 Preparation of hollow cylinder
- 4.6.1 Draw the development of the object to be made
- 4.6.2 Place the pattern on the sheet
- 4.6.3 Mark the dimensions using scribe
- 4.6.4 Shear the required piece from the stock using straight snips
- 4.6.5 Mark the lines on the sheet to form bends
- 4.6.6 Strengthen the sides of sheet by single hem on top & bottom side using hatchet stake.
- 4.6.7 Form the flat sheet into cylindrical shape by cylindrical stake and apply pressure using mallet.
- 4.6.8 Prepare single hem on to longitudinal sides in opposite directions
- 4.6.9 Interlock the sides and apply pressure to make a strong joint
- 4.7 Preparation of funnel
- 4.7.1 Draw the development of upper and bottom conical parts
- 4.7.2 Place the pattern on the sheet and cut to required size
- 4.7.3 Form the sheet into conical shape using appropriate stake and mallet
- 4.7.4 Seam the top conical part and bottom conical part to obtain required funnel.
- 4.8 Preparation of utility articles such as dustpan
- 4.8.1 Draw the development of given dustpan
- 4.8.2 Scribe the lines on the sheet and cut to required size
- 4.8.3 Hem all the four sides to strengthen the edges
- 4.8.4 Form the sheet into designed shape using suitable stakes and mallet
- 4.8.5 Solder the corner lap joints to make the required dustpan
- 5. Plumbing**
- 5.1 Identify the various tools used plumbing work.
- 5.1.1 Preparation of pipe joint with pipe fittings
- 5.1.2 Select the plumbing tools: pipe wrench, pipe vice, Hack Saw,
- 5.1.3 Pipe Cutter, pipe threading dies
- 5.1.4 Select pipe fittings: coupling, union, nipple, elbow, tee, reducer
- 5.1.5 Perform pipe fitting operations on the pipe
- 5.2 Thread cutting on Pipe
- 5.2.1 Select the plumbing tools: pipe wrench, pipe vice, Hack Saw, Pipe Cutter, pipe threading Dies.
- 5.2.2 Perform thread cutting on pipe.

Course Code: A-108		Course Title: Workshop Practice - I		No. of COs:5	No. of Periods: 90
POs	Mapping with CO No	CO Periods Addressing PO in Col 1		Level (1,2,3)	Remarks
		No	%		
PO1	CO1,CO2, CO3,CO4,CO5	24	27	2	>40% Level 3 (Highly Addressed) 25% to 40% Level 2 (Moderately Addressed) 5% to 25% Level 1 (Low Addressed) <5% Not Addressed
PO2	CO1,CO2, CO3,CO4,CO5	09	10	1	
PO3	CO1,CO2, CO3,CO4,CO5	18	20	1	
PO4	CO1,CO2, CO3,CO4,CO5	24	27	2	
PO5	CO1,CO2, CO3,CO4,CO5	09	10	1	
PO6	CO2,CO5	06	06	1	
PO7					

CO-PO/PSO MATRIX

CO No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
A-108.1	3	1	1	3	3			3	2	3
A-108.2	3	1	2	3	3	3		3	1	3
A-108.3	3	2	2	3	3			2	1	3
A-108.4	3	2	3	2	3			3	2	3
A-108.5	3	1	1	3	3	1		3	1	3
AVERAGE	3	1.4	1.8	2.8	3	2		2.8	1.4	3

COURSE CONTENT

1. WORKSHOP SAFETY PRECAUTIONS

1. Wear tight clothes. Loose clothing or long hair will bring danger
2. Wear covered footwear. Never use sandals or chapels.
3. Never use files, screwdrivers without a handle.
4. Do not use a hammer without the wedge.
5. Never operate a machine unless otherwise you thoroughly know its mechanism.
6. Wear goggles while chipping, welding or grinding.
7. Always wear safety hand gloves, an apron or dungaree and leather shoes in the Workshop.
8. Never touch overhead lines unless you are sure that it is properly earthed and dead.
9. Don't touch /operate power tools without aid from instructors
10. Do not throw water on the equipment. It will be harmful since water is a good conductor of electricity.
11. Do not touch electrical circuits of welding machine.

2. FITTING SHOP

1. Identification of the various marking & measuring tools, cutting tools, striking tools, holding devices and miscellaneous tools used fitting.
2. Use the checking and measuring instruments: various types of callipers, Combination set, sine bar, Gauges- plug gauge, feeler gauge, Thread gauge, plate & wire gauge, ring gauge, snap gauge specifications & uses, Vernier callipers, Vernier height gauge, Vernier depth gauge, micrometre(outside& inside)
3. Marking and chipping on Mild – steel flat 12 mm thick.
4. Cutting with hack saw, M.S. Flats of 6 mm thick.
5. Marking, cutting, drilling, Chamfering and tapping on a M.S. Flat 12 mm thick.
6. Assembling of two pieces, Matching by filing (6 mm thick M.S.Plate).

3. CARPENTRY SHOP

1. Identification of the various marking & measuring tools, cutting tools, boring tools, Striking tools, holding devices and miscellaneous tools used fitting.
2. Cutting of wood with handsaw.
3. Planning of wood.
4. Planning and chiselling of wood.
5. Orientation of wood grain.
6. Preparation of dovetail joint.
7. Mortise and tenon joint.
8. Wood turning on a lathe.
9. Preparation of one household article.

4. SHEET METAL WORK

1. Metals used for sheet metal work
2. Sheet metal handtools:

Measuring tools-steel rule, circumference rule, thickness gauge, sheet metal gauge, straight edge, scribe, divider, trammel points, punches, chisels, hammers, snips or shears, straight snip, double cutting shear, squaring shear, circular shear, bench & block shears.

Stakes: Double seaming stake, beak horn stake, bevel edged square stake, hatches stake, needle stake, blow horn stake, hollow mandrel stake, groovers and rivet sets, soldering iron, specifications & uses.

3. Practice on cutting of sheet
4. Formation of joints like grooved joints, locked groove joint
5. Preparation of a rectangular open type tray
6. Preparation of hollow cylinder
7. Preparation of funnel
8. Preparation of utility articles such as dustpan.

5. PLUMBING PRACTICE

1. Familiarization of plumbing tools
2. Familiarization of pipe fitting
3. Familiarization of plumbing operations.

REFERENCE BOOKS

1. Manufacturing Technology (Vol-I) by P N Rao (McGrawHill)
2. Principles of Foundry Technology by P L Jain (McGrawHill)
3. Workshop Practice Vol- I by Hajra Choudhury Media Pro

ENGINEERING PHYSICS LAB

SUBJECT	SUBJECT CODE	TOTAL PERIODS	NUMBER OF PERIODS PER WEEK
PHYSICS LAB	A -109	45	03

Course objectives	(1) To provide strong practical knowledge of Physics to serve as a tool for various device applications in Engineering. (2) To enhance scientific skills of the students by incorporating new experiments so as to enrich the technical expertise of the students as required for industries.
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COURSE OUTCOMES	CO1	Improving accuracy in various measurements; understanding the nature of the forces keeping the body in equilibrium.
	CO2	Estimating the acceleration caused by the gravity of earth; Practical study of the concepts of refraction of light at curved/plane surface
	CO3	Understanding the pressure of the gas as function of its volume; study of the combined magnetic field of the earth and an artificial magnet to estimate its pole strength; Estimating the velocity of sound in air through resonance phenomenon.
	CO4	Applying Kirchoff's laws to evaluate the specific resistance of a wire; Study of exchange of heat from system to surrounding by graphical analysis; Conversion of light energy to micro currents as potential engineering application.

CO-PO MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	2	2	2	1	2
CO2	3		1	1	1	1	1
CO3	3	2			1		
CO4	3	2	2			1	2

CO-PO Mapping Strength

Course code A-109	Engineering Physics No of Course Objectives : 4				No of periods 45
POs	Mapped with CO No	CO periods addressing PO in Col 1 NO %		Level 1,2,3	remarks
PO1	CO1,CO2,CO3,CO4	15	33.3 %	2	>40% level 3 (highly addressed) 25% to 40% level2
PO2	CO1,CO3, CO4	8	17.8%	1	

PO3	CO1, CO2,CO4	6	13.3%	1	(moderately addressed) 5% to 25% level1 (Low addressed) < 5% (not addressed)
PO4	CO1, CO2	3	6.7%	1	
PO5	CO1,CO2, CO3	5	11.1%	1	
PO6	CO1, CO2,CO4	3	6.7%	1	
PO7	CO1, CO2, CO4	5	11.1%	1	

3 = strongly mapped, 2 = moderately mapped, 1 = slightly mapped

Note: The gaps in CO and PO mapping will be achieved by one or more appropriate activities from the following.

- (i) Seminars (ii) Viva-voce (iii) Assignments
 (iv) Quiz competitions (v) Industrial visits (vi) Tech fest (vii) Mini project
 (viii) Group discussions (ix) Virtual labs (x) Library visit for e-books

ENGINEERING PHYSICS LAB

SubjectTitle : **Engineering Physics Lab**
SubjectCode : **Common-109**
Periodsperweek : **03**
Totalperiodsperyear : **45**

TIMESCHEDULE

S.No	List of experiments	No.of Periods
1.	Verniercalipers	03
2.	Micrometer (Screw gauge)	03
3.	Verification of Lami's theorem using concurrent forces	03
4.	Determination of 'g' using simple pendulum	03
5.	Focal length and focal power of convex lens	03
6.	Refractive index of solid using travelling microscope	03
7.	Verification of Boyle's law using Quill tube	03
8	Determination of pole strength of the bar magnet through magnetic field lines	03
9	Resonance apparatus – Determination of velocity of sound in air	03
	Experiments for demonstration	
10	Meter bridge – Determination of resistance and specific resistance of a wire	03
11	Verification of Newton's law of cooling	03

12	Photo electric cell – Study of its characteristics	03
	Revision	06
	Test	03
	Total:	45

Learning Outcomes

Upon completion of the course the student shall be able to

- 1.0 Practice with Vernier calipers to determine the volumes of cylinder and sphere.
- 2.0 Practice with Screw gauge to determine thickness of a glass plate and cross sectional area of a wire.
- 3.0 Verify the Lami's theorem using concurrent forces.
- 4.0 Determine the value of acceleration due to gravity (g) using Simple Pendulum. To verify the result from $I-T^2$ graph.
- 5.0 Calculate the Focal length and focal power of convex lens using distant object method and U-V method. To verify the result from U-V graph and $1/U - 1/V$ graph methods.
- 6.0 Determine the refractive index of a solid using travelling microscope
- 7.0 Verify the Boyle's law using Quill tube. To draw a graph between P and $1/I$.
- 8.0 Determination of magnetic pole strength of a bar magnet by drawing magnetic lines of force and locating null points (either N - N or N - S method)
- 9.0 Determine the velocity of sound in air at room temperature and its value at zero degree Centigrade using resonance apparatus.
- 10.0 Determine the resistance and specific resistance of material of a wire using Meter Bridge
- 11.0 To verify the Newton's law of cooling.
- 12.0 To study the characteristics of photo electric cell.

Course Outcomes

S.No	List of experiments	No.of Periods	COs
1.	Vernier calipers	03	CO1
2.	Micrometer (Screw gauge)	03	
3.	Verification of Lami's theorem using concurrent forces	03	
4.	Determination of g using simple pendulum	03	CO2
5.	Focal length and power of convex lens	03	
6.	Refractive index of solid using travelling microscope	03	
7.	Verification of Boyle's law using Quill tube	03	CO3
8.	Determination of pole strength of the bar magnet through magnetic field lines	03	
9.	Resonance apparatus – Determination of velocity of sound in air	03	
10.	Meter bridge – Determination of resistance and specific resistance of a wire	03	CO4
11.	Verification of Newton's law of cooling	03	
12.	Photo electric cell – Study of its characteristics	03	

Competencies and Key competencies to be achieved by the student

Name of the Experiment (Nu)	Competencies	Key competencies
1. Practice on Vernier Calipers(03)	- Find the Least count - Fix the specimen in position - Read the scales - Calculate the physical quantities of given object	- Read the scales - Calculate the requisite physical quantities of given objects - Calculating volumes of the cylinder and sphere
2. Practice on Screw gauge(03)	- Find the Least count - Fix the specimen in position - Read the scales - Calculate thickness of glass plate and cross section of wire from radius	- Read the scales - Noting zero error - Calculate thickness of given glass plate - Calculate cross section of wire from radius
3. Verification of Lami's theorem forces(03)	- Making experimental set up - Fix suitable weights - Note the positions of threads on drawing sheet - Find the angles between the concurrent forces - Changing weights appropriately - Verify Lami's theorem	- Measuring angles between the forces - Marking the directions of forces on a paper - Verifying Lami's theorem from the weights and measured angles between the forces.
4. Simple pendulum(03)	- Fix the simple pendulum to the stand - Adjust the length of pendulum - Find the time for number of oscillations (say 20) - Find the time period - Calculate the acceleration due to gravity - Draw $I-T^2$ graph	- Find the time for number of oscillations - Find the time period - Calculate the acceleration due to gravity - Verify form $I-T^2$ graph
5. Focal length and Focal power of convex lens(03)	- Fix the object distance - Find the Image distance - Calculate the focal length and power of convex lens - Draw $u-v$ and $1/u - 1/v$ graphs	- Find focal length from distant object method. - Calculate the focal length and power of convex lens - Verify result from $u-v$ and $1/u - 1/v$ graphs

6 Refractive index of solid using traveling microscope(03)	- Find the least count of Vernier on microscope - Place the graph paper below microscope - Read the scales	- Reading the scales on Microscope. - Finding real and apparent thickness of the slab - Calculate the refractive index
7 . Boyle's law verification (03)	- Note the atmospheric pressure - Fix the Quill tube to retort stand - Find the length of air column - Find the pressure of enclosed air - Find and compare the calculated values of $P \times l$	- Fixing Quill tube in various positions on retort stand. - Find the length of air column - Find the pressure of enclosed air - Find the values of $P \times l$ - Verify Boyle's law.
8. Mapping of magnet lines of force(03)	- Draw magnetic meridian - Place the bar magnet in N-N or N-S directions - Draw magnetic lines of force - Locate the neutral points	- Draw the pattern of magnetic lines of force - Locate the neutral points - Calculating pole strength of the bar magnet
9. Velocity of sound in air –Resonance method (03)	- Arrange the resonance apparatus - Adjust the reservoir level for booming sound - Find the first and second resonating lengths - Calculate velocity of sound.	- Adjust the reservoir level - Find the first and second resonating lengths - Calculate velocity of sound at room temperature and at 0°C
10.Meter bridge(03)	- Make the circuit connections - Find the balancing length - Calculate unknown resistance - Find the radius of wire - Calculate the specific resistance	- Making connections as per circuit diagram. - Find the balancing length - Calculate unknown resistance - Calculate the specific resistance of the given wire
11. Verification of Newton's law of Cooling (03)	- Heating liquid in a beaker using a heating element - Inserting thermometer in liquid in calorimeter - Stirring liquid - Measuring temperatures as a function of time using thermometer - Plotting a cooling curve	- Measuring temperature of a liquid as function of time. - Plotting a cooling curve. - Verifying Newton's law of cooling.

12. Photo electric cell – Study of its Characteristics (03)	• Experimental set up and making connections • Verifying intensity of light by varying distances between light source and photocell. • Measuring Voltage and current values.	• Making connections for experimental set up. • Varying distances appropriately • Measuring Voltage and current values. • Study of V- I Characteristics from graph.
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Scheme of Valuation for End Practical Examination:

Activity	Marks
For writing, Apparatus, formulae, least count (if applicable)	5
Procedure & precautions	5
Drawing Tables	3
Readings, calculations, graph (if applicable), reporting the findings	12
Viva-voce	5
Total marks	30

CHEMISTRY LABORATORY
(C-23 curriculum common to all Branches)

Subject Title : Chemistry Laboratory
Subject Code : A-110
Periods per week : 03
Total periods per year : 45

CO1	Operate and practice volumetric apparatus and preparation of standard solution.
CO2	Evaluate and judge the neutralization point in acid base titration.
CO3	Evaluate the end point of reduction and oxidation reaction.
CO4	Judge the stable end point of complex formation, stable precipitation.
CO5	Judge operate and demonstrate and perform precise operations with instrument for investigation of water pollution parameters.

PO- CO mapping

Course code Common- 110	Chemistry Laboratory No. of CO's:5				No. of periods : 45
POs	Mapped with CO No.	CO periods addressing PO in Col. No. 1	%	Level 1,2,3	Remarks
PO1	CO1,CO2,CO3, CO4,CO5	12	26.66	2	>40% Level 3 (highly addressed)
PO2	CO1,CO2,CO3, CO4,CO5	9	20	1	
PO3					25% to 40% Level2 (moderately addressed)
PO4	CO1,CO2,CO3, CO4,CO5	12	26.66	2	
PO5	CO2,CO3, CO4,CO5	12	26.66	2	5% to 25% Level1 (Low addressed)
PO6	-	-	-	-	
PO7	-	-	-	-	< 5%(not addressed)

COs-POs mapping strength (as per given table)

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
CO1	2	1	-	2	-	-	-	-	-	-
CO2	2	1	-	2	2	-	-	-	-	-
CO3	2	1	-	2	2	-	-	-	-	-
CO4	2	1	-	2	2	-	-	-	-	-
CO5	2	1	-	2	2	-	-	-	-	-

3=strongly mapped 2= moderately mapped 1= slightly mapped

Note:The gaps in CO and PO mapping will be achieved by one or more appropriate activities from the following: i) Seminars ii) Tutorials iii) Guest Lectures iv) Assignments v) Quiz competitions vi) Industrial visit vii) Tech Fest viii) Mini project ix) Group discussions x) Virtual classes xi) Library visit for e-books

TIMESCHEDULE

S.No	Name of the Experiment	o.ofPeriods	Mapped with COs
1.	a) Recognition of chemical substances and solutions used in the laboratory by senses. b) Familiarization of methods for Volumetric analysis.	03	CO1
2.	Preparation of Std. Na_2CO_3 solution and making solutions of different dilutions	03	CO1
3.	Estimation of HCl solution using Std. Na_2CO_3 solution.	03	CO2
4.	Estimation of NaOH using Std. HCl solution.	03	CO2
5.	Determination of acidity of water sample.	03	CO2
6.	Determination of alkalinity of water sample.	03	CO2
7.	Estimation of Mohr's Salt using Std. KMnO_4 solution.	03	CO3
8.	Estimation of Ferrous ion by using Std. $\text{K}_2\text{Cr}_2\text{O}_7$ solution.	03	CO3
9.	Determination of total hardness of water sample using Std. EDTA solution.	03	CO4
10.	Estimation of Chlorides present in water sample by using Std. AgNO_3 solution.	03	CO4
11.	Estimation of Dissolved Oxygen (D.O) in water sample by using Std. hypo solution.	03	CO5
12.	Determination of pH using pH meter..	03	CO5
13.	Determination of conductivity of water and adjusting ionic strength required level.	03	CO5
14.	Determination of turbidity of water.	03	CO5
15.	Estimation of total solids present in water sample.	03	CO5
	Total:	45	

Objectives:

Upon completion of the course the student shall be able to

- 1.0 To identify the chemical compounds and solutions by senses.
Practice volumetric measurements (using pipettes, measuring jars, volumetric flask, burettes) and gravimetric measurements (using different types of balances), making dilutions, etc.
- 2.0 Practice making standard solutions with pre weighed salts and to make solutions of desired dilutions using appropriate techniques.
- 3.0 Conduct titrations adopting standard procedures and using Std. Na_2CO_3 solution for estimation of HCl .
- 4.0 Conduct titrations adopting standard procedures and using Std. HCl solution for estimation of NaOH .
- 5.0 Conduct titrations adopting standard procedures to determine the acidity of given samples of water (One ground water and one surface / tap water, and rain water if available).
- 6.0 Conduct titrations adopting standard procedures to determine the alkalinity of given samples of water (One ground water and one surface / tap water).
- 7.0 Conduct titrations adopting standard procedures and using Std. KMnO_4 solution for estimation of Mohr's Salt.
- 8.0 Conduct titrations adopting standard procedures and using Std. $\text{K}_2\text{Cr}_2\text{O}_7$ solution for estimation of Ferrous ion.
- 9.0 Conduct titrations adopting standard procedures to determine the total hardness of given samples of water (One ground water and one surface / tap water) using Std. EDTA solution.
10. Conduct titrations adopting standard procedures to determine the chlorides present in the given samples of water (One ground water and one surface / tap water) using Std. AgNO_3 solution.
11. Conduct the test using titrimetric / electrometric method to determine. Dissolved Oxygen (D.O) in the given water samples (One sample from closed container and one from open container / tap water) by Std. Hypo solution.
12. Conduct the test on given samples of water / solutions (like soft drinks, sewage, etc.) to determine their pH using standard pH meter.
13. Conduct the test on given samples of water / solutions.

- a) to determine conductivity.
b) to adjust the ionic strength of the sample to the desired value.
14. Conduct the test on given samples of solutions (coloured and non-coloured) to determine their turbidity in NTU.
15. Determine the total solids present in given samples of water (One ground water and one surface / tap water).

Competencies and Key competencies to be achieved by the student

Name of the Experiment (No of Periods)	Competencies	Key competencies
Recognition of chemical substances and solutions. Familiarization of methods for Volumetric analysis. (03)	-	--
Preparation of Std. Na_2CO_3 solution and making solutions of different dilutions. (03)	▪ Weighing the salt to the accuracy of .01 mg. ▪ Measuring the water with volumetric flask, measuring jar, volumetric pipette and graduated pipette. ▪ Making appropriate	▪ Weighing the salt to the accuracy of 0.01 mg. ▪ Measuring the water with volumetric flask, measuring jar, volumetric pipette
Estimation of HCl solution using Std. Na_2CO_3 solution. (03)	▪ Cleaning the glassware and rinsing with appropriate solutions. ▪ Making standard solutions. ▪ Measuring accurately the standard solutions and titrants. ▪ Filling the burette with titrant. ▪ Fixing the burette to the stand. ▪ Effectively Controlling the flow of the titrant. ▪ Identifying the end point. ▪ Making accurate observations. ▪ Calculating the results.	▪ Making standard solutions. ▪ Measuring accurately the standard solutions and titrants. ▪ Effectively Controlling the flow of the titrant. ▪ Identifying the end point. ▪ Making accurate observations.
Estimation of NaOH using Std. HCl solution. (03)		
Determination of acidity of water sample. (03)		
Determination of alkalinity of water sample. (03)		
Estimation of Mohr's Salt using Std. KMnO_4 solution. (03)		
Estimation of Ferrous ion by using Std. $\text{K}_2\text{Cr}_2\text{O}_7$ solution (03)		
Determination of total hardness of water using Std. EDTA solution. (03)		
Estimation of Chlorides present in water sample using Std. AgNO_3 solution (03)		
Estimation of Dissolved Oxygen (D.O) in water sample (By titration method) (03)		
Determination of pH using pH meter. (03)	▪ Familiarize with instrument.	▪ Prepare standard

Determination of conductivity of water and adjusting ionic strength to required level. (03)	- Choose appropriate 'Mode' / 'Unit'. - Prepare standard solutions / buffers, etc. - Standardize the instrument with appropriate standard solutions. - Plot the standard	solutions / buffers, etc. - Standardize the instrument with appropriate standard solutions. - Plot the standard curve.
Determination of turbidity of water. (03)	- Measuring the accurate volume and weight of sample. - Filtering and air drying without losing any filtrate. - Accurately weighing	Measuring the accurate volume and weight of sample. Filtering and air drying without losing any

SCHEME OF VALUATION

A) Writing Chemicals, apparatus, principle and procedure.	5M
B) Demonstrated competencies.	20M
Making standard solutions.	
Measuring accurately the standard solutions and titrants.	
Effectively controlling the flow of the titrant.	
Identifying the end point.	
Making accurate observations.	
C) Viva-voce.	5M
Total	30M

COMPUTER FUNDAMENTALS LAB

Course code	Course Title	No. of Periods/ Weeks	Total No. of periods	Marks for FA	Marks for SA
A-111 (common to all branches)	Computer Fundamentals Lab	3	90	40	60

Time schedule:

S.No.	Chapter/Unit Title	No. of sessions each of 3 periods duration	No.of Periods
1.	Computer hardware Basics	2	6
2.	Windows Operating System	2	6
3.	MS Word	8	24
4.	MS Excel	7	21
5.	MS PowerPoint	5	15
6.	Adobe Photoshop	6	18
Total periods		30	90

S.No.	Chapter/Unit Title	No.of Periods	CO's Mapped
1.	Computer hardware Basics	6	CO1
2.	Windows Operating System	6	CO1
3.	MS Word	24	CO2
4.	MS Excel	21	CO3
5.	MS PowerPoint	15	CO4
6.	Adobe Photoshop	18	CO5
Total periods		90	

Course Objectives	i)To know Hardware Basics ii)To familiarize operating systems iii)To use MS Office effectively to enable to students use these skills in future courses iv) To use Adobe Photoshop in image editing.
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Course Outcomes	At the end of the course students will be able to		
	CO1	A-111.1	Identify hardware and software components
	CO2	A -111.2	Prepare documents with given specifications using word processing software
	CO3	A -111.3	Use Spread sheet software to make calculation and to draw various graphs / charts.
	CO4	A -111.4	Use Power point software to develop effective presentation for a given theme or topic.
	CO5	A -111.5	Edit digital or scanned images using Photoshop

CO-PO/PSO MATRIX

CO NO.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
A -111.1	3	3	3	3	3	3	3	3	2	3
A -111.2	3	3	3	3	3	3	3	3	2	3
A -111.3	3	3	3	3	3	3	3	3	2	3
A -111.4	3	3	3	3	3	3	3	3	2	3
A -111.5	3	3	3	3	3	3	3	3	2	3
Average	3	3	3	3	3	3	3	3	2	3

3=Strongly mapped , 2=moderately mapped, 1=slightly mapped

Learning Outcomes:

I. Computer Hardware Basics

1. a).To Familiarize with Computer system and hardware connections
b).To Start and Shut down Computer correctly
c).To check the software details of the computer
2. To check the hardware present in your computer

II. Windows's operating system

3. To Explore Windows Desktop
4. Working with Files and Folders
5. Windows Accessories: Calculator – Notepad – WordPad – MS Paint

III. Practice with MS-WORD

6. To familiarize with Ribbon layout of MS Word
Home – Insert- Page layout – References – Review- View.
7. To practice Word Processing Basics
8. To practice Formatting techniques
9. To insert a table of required number of rows and columns
10. To insert Objects, Clipart and Hyperlinks
11. To use Mail Merge feature of MS Word
12. To use Equations and symbols features

IV. Practice with MS-EXCEL

13. To familiarize with MS-EXCEL layout
14. To access and enter data in the cells
15. To edit a spread sheet- Copy, Cut, Paste, and selecting Cells
16. To use built in functions and Formatting Data
17. To create Excel Functions, Filling Cells
18. To enter a Formula for automatic calculations
19. To sort and filter data in table.
20. To present data using Excel Graphs and Charts.
21. To develop lab reports of respective discipline.
22. To format a Worksheet in Excel, Page Setup and Print

V. Practice with MS-POWERPOINT

23. To familiarize with Ribbon layout features of PowerPoint 2007.
24. To create a simple PowerPoint Presentation
25. To set up a Master Slide in PowerPoint

26. To insert Text and Objects
27. To insert a Flow Charts
28. To insert a Table
29. To insert a Charts/Graphs
30. To insert video and audio
31. To practice Animating text and objects
32. To Review presentation

VI. Practice with Adobe Photoshop

33. To familiarize with standard toolbox
34. To edit a photograph.
35. To insert Borders around photograph.
36. To change Background of a Photograph.
37. To change colours of Photograph.
38. To prepare a cover page for the book in your subject area.
39. To adjust the brightness and contrast of the picture so that it gives an elegant look.
40. To type a word and apply the shadow emboss effects.

Key competencies:

Expt No	Name of Experiment	Competencies	Key competencies
1 (a).	To familiarize with Computer system and hardware connections	a. Identify the parts of a Computer system: i). CPU ii). Mother Board iii) Monitor iv) CD/DVD Drive v) Power Switch vi) Start Button vii) Reset Button b. Identify and connect various peripherals c. Identify and connect the cables used with computer system d. Identify various ports on CPU and connect Keyboard & Mouse	Connect cables to external hardware and operate the computer
1 (b).	To Start and Shut down Computer correctly	a. Log in using the password b. Start and shut down the computer c. Use Mouse and Key Board	a. Login and logout as per the standard procedure b. Operate mouse & Key Board
1 (c).	To Explore Windows Desktop	a. Familiarize with Start Menu, Taskbar, Icons and Shortcuts b. Access application programs using Start menu, Task manager c. Use Help support	a. Access application programs using Start menu b. Use taskbar and Task manager
2.	To check the software details of the computer	a. Find the details of Operating System being used	Access the properties of computer and find the details

		b. Find the details of Service Pack installed	
3.	To check the hardware present in your computer	a. Find the CPU name and clock speed b. Find the details of RAM and Hard disk present c. Access Device manager using Control Panel and check the status of devices like mouse and key board d. Use My Computer to check the details of Hard drives and partitions e. Use the Taskbar	a. Access device manager and find the details b. Type /Navigate the correct path and Select icon related to the details required
4.	Working with Files and Folders	a. Create folders and organizing files in different folders b. Use copy / paste move commands to organize files and folders	a. Create files and folders Rename , arrange and search for the required folder/file
	Working with Files and Folders Continued....	c. Arrange icons – name wise, size, type, Modified d. Search a file or folder and find its path e. Create shortcut to files and folders (in other folders) on Desktop f. Familiarize with the use of My Documents g. Familiarize with the use of Recycle Bin	b. Restore deleted files from Recycle bin
5.	To use Windows Accessories: Calculator – Notepad – WordPad – MS Paint	a. Familiarize with the use of Calculator b. Access Calculator using Run command c. Create Text Files using Notepad and WordPad and observe the difference in file size d. Use MS paint and create .jpeg, .bmp files using MS Paint	a. Use windows accessories and select correct text editor based on the situation. b. Use MS pain to create /Edit pictures and save in the required format.
6.	To familiarize with Ribbon layout of MS word. – Home – Insert- page layout- References- Review-View	a. Create/Open a document b. Use Save and Save as features c. Work on two Word documents	a. Create a Document and name appropriately and save b. Set paper size

		d. simultaneously Choose correct Paper size and Printing options	and print options
7.	To practice Word Processing Basics	a. Typing text b. Keyboard usage c. Use mouse (Left click / Right click / Scroll) d. Use Keyboard shortcuts e. Use Find and Replace features in MS- word f. Use Undo and Redo Features g. Use spell check to correct Spellings and Grammar	a. Use key board and mouse to enter/edit text in the document. b. Use shortcuts c. Use spell check/ Grammar features for auto corrections.
8.	To practice Formatting techniques	a. Formatting Text b. Formatting Paragraphs c. Setting Tabs d. Formatting Pages e. The Styles of Word f. Insert bullets and numbers g. Themes and Templates h. Insert page numbers, header and footer	a. Format Text and paragraphs and use various text styles. b. Use bullets and numbers to create lists c. Use Templates /Themes d. Insert page numbers date, headers and footers
9.	To insert a table of required number of rows and columns	a. Edit the table by adding the fields – Deleting rows and columns – inserting sub table – marking borders. Merging and splitting of cells in a Table b. Changing the background colour of the table c. Use table design tools d. Use auto fit – fixed row/ column height/length – Even distribution of rows / columns features e. Convert Text to table and Table to Text f. Use Sort feature of the Table to arrange data in ascending/descending order	a. Insert table in the word document and edit b. Use sort option for arranging data.
10.	To Insert objects, clipart and Hyperlinks	a. Create a 2-page document. &Insert hyperlinks and t Bookmarks. b. Create an organization	a. Insert hyperlinks &Bookmarks b. Create organization charts/flow

		c. chart Practice examples like preparing an Examination schedule notice with a hyperlink to Exam schedule table.	charts
11.	To Use Mail merge feature of MS Word	a. Use mail merge to prepare individually addressed letters b. Use mail merge to print envelopes.	Use Mail merge feature
12.	To use Equations and symbols features.	a. Explore various symbols available in MS Word b. Insert a symbol in the text c. Insert mathematical equations in the document	Enter Mathematical symbols and Equations in the word document
13.	To Practice with MS-EXCEL	a. Open /create an MS Excel spreadsheet and familiarize with MS Excel 2007 layout like MS office Button- b. Use Quick Access Toolbar- Title Bar- Ribbon-Worksheets- Formula Bar-Status Bar	a. Familiarize with excel layout and use b. Use various features available in toolbar
14.	To access and Enter data in the cells	a. Move Around a Worksheets-Quick access -Select Cells b. Enter Data-Edit a Cell- Wrap Text-Delete a Cell Entry-Save a File-Close Excel	a. Access and select the required cells by various addressing methods b. Enter data and edit
15.	To edit spread sheet Copy, Cut, Paste, and selecting cells	a. Insert and Delete Columns and Rows- Create Borders-Merge and Centre b. Add Background Colour- Change the Font, Font Size, and Font Colour c. Format text with Bold, Italicize, and Underline- Work with Long Text- Change a Column's Width	Format the excel sheet
16.	To use built in functions and Formatting Data	a. Perform Mathematical Calculations verify - AutoSum b. Perform Automatic Calculations-Align Cell Entries	Use built in functions in Excel
17.	To enter a Formula for automatic calculations	a. Enter formula b. Use Cell References in	Enter formula for

		c. Formulae Use Automatic updating function of Excel Formulae d. Use Mathematical Operators in Formulae e. Use Excel Error Message and Help	automatic calculations
18.	To Create Excel Functions, Filling Cells	a. Use Reference Operators b. Work with sum, Sum if , Count and Count If Functions c. Fill Cells Automatically	a. Create Excel sheets involving cross references and equations b. Use the advanced functions for conditional calculations
19.	To sort and filter data in table	a. Sort data in multiple columns b. Sort data in a row c. Sort data using Custom order d. Filter data in work sheet	a. Refine the data in a worksheet and keep it organized b. Narrow a worksheet by selecting specific choice
20.	To Practice Excel Graphs and Charts	a. Produce an Excel Pie Chart b. Produce c. Excel Column Chart	a. Use data in Excel sheet to Create technical charts and graphs Produce Excel Line Graph b. Produce a Pictograph in Excel
21.	To develop lab reports of respective discipline	Create Lab reports using MS Word and Excel	a. Insert Practical subject name in Header and page numbers in Footer
22.	To format a Worksheet in Excel, page setup and print	a. Shade alternate rows of data b. Add currency and percentage symbols c. Change height of a row and width of a column d. Change data alignment e. Insert Headers and Footers f. Set Print Options and Print	a. Format Excel sheet b. Insert headers & footers and print
23.	To familiarize with Ribbon layout & features of PowerPoint 2007.	Use various options in PowerPoint a. Home b. Insert c. Design d. Animation	Access required options in the tool bar

		e. Slideshow f. View g. Review	
24.	To create a simple PowerPoint Presentation	a. Insert a New Slide into PowerPoint b. Change the Title of a PowerPoint Slide c. PowerPoint Bullets d. Add an Image to a PowerPoint Slide e. Add a Textbox to a PowerPoint slide	a. Create simple PowerPoint presentation with photographs/ClipArt and text boxes b. Use bullets option
25.	To Set up a Master Slide in PowerPoint and add notes	a. Create a PowerPoint Design Template b. Modify themes c. Switch between Slide master view and Normal view d. Format a Design Template Master Slide e. Add a Title Slide to a Design Template f. The Slide Show Footer in PowerPoint g. Add Notes to a PowerPoint Presentation	a. Setup Master slide and format b. Add notes
26.	To Insert Text and Objects	a. Insert Text and objects b. Set Indents and line spacing c. Insert pictures/ clipart d. Format pictures e. Insert shapes and word art f. Use 3d features g. Arrange objects	Insert Text and Objects Use 3d features
27.	To insert a Flow Chart / Organizational Charts	a. Create a Flow Chart in PowerPoint b. Group and Ungroup Shapes c. Use smart art	Create organizational charts and flow charts using smart art
28.	To insert a Table	a. PowerPoint Tables b. Format the Table Data c. Change Table Background d. Format Series Legend	Insert tables and format
29.	To insert a Charts/Graphs	a. Create 3D Bar Graphs in PowerPoint b. Work with the PowerPoint Datasheet c. Format a PowerPoint Chart Axis	Create charts and Bar graphs, Pie Charts and format.

		d. Format the Bars of a Chart e. Create PowerPoint Pie Charts f. Use Pie Chart Segments g. Create 2D Bar Charts in PowerPoint h. Format the 2D Chart e. Format a Chart Background	
30.	To Insert audio & video, Hyperlinks in a slide Add narration to the slide	a. Insert sounds in the slide and hide the audio symbol b. Adjust the volume in the settings c. Insert video file in the format supported by PowerPoint in a slide d. Use automatic and on click options e. Add narration to the slide f. Insert Hyperlinks	a. Insert Sounds and Video in appropriate format. b. Add narration to the slide c. Use hyperlinks to switch to different slides and files
31.	To Practice Animation effects	a. Apply transitions to slides b. To explore and practice special animation effects like Entrance, Emphasis, Motion Paths & Exit	Add animation effects
32.	Reviewing presentation	a. Checking spelling and grammar b. Previewing presentation c. Set up slide show d. Set up resolution e. Exercise with Rehearse Timings feature in PowerPoint f. Use PowerPoint Pen Tool during slide show g. Saving h. Printing presentation (a) Slides (b) Hand-out	a. Use Spell check and Grammar feature b. Setup slide show c. Add timing to the slides d. Setup automatic slide show
33	To familiarize with standard toolbox	a. Open Adobe Photoshop b. Use various tools such as i. The Layer Tool ii. The Color & Swatches Tool iii. Custom Fonts & The Text Tool iv. Brush Tool v. The Select Tool	Open a photograph and save it in Photoshop

		vi. The Move Tool vii. The Zoom Tool viii. The Eraser ix. The Crop Tool x. The Fill Tool	
34	To edit a photograph	a. Use the Crop tool b. Trim edges c. Change the shape and size of a photo d. Remove the part of photograph including graphics and text	a. Able to edit image by using corresponding tools.
35	To insert Borders around photograph	a. Start with a single background layer b. Bring the background forward c. Enlarge the canvas d. Create a border color e. Send the border color to the back f. Experiment with different colors	Able to create a border or frame around an image to add visual interest to a photo
36	To change Background of a Photograph	a. open the foreground and background image b. Use different selection tools to paint over the image c. Copy background image and paste it on the foreground. d. Resize and/or drag the background image to reposition. e. In the Layers panel, drag the background layer below the foreground image layer.	Able to swap background elements using the Select and Mask tool and layers.
37	To change colors of Photograph	a. Change colours using: i) Colour Replacement tool ii) Hue/Saturation adjustment layer tool	Able to control colour saturation
38	To prepare a cover page for the book in subject area	a. open a file with height 500 and width 400 for the cover page. b. apply two different colours to work area by dividing it into two parts using Rectangle tool. c. Copy any picture and place it on work area→ resize it using free transform tool. d. Type text and apply colour and style	Able to prepare cover page for the book

		e. Apply effects using blended options	
39	To adjust the brightness and contrast of picture to give an elegant look	a. open a file. b. Go to image→ adjustments→ Brightness/Contrast. c. adjust the brightness and contrast. d. Save the image.	Able to control brightness/contrast.
40	To type a word and apply the shadow emboss effects	a. open a file b. Select the text tool and type text. c. Select the typed text go to layer→ layer style→ blended option→ drop shadow, inner shadow, bevel and emboss→ contour→ satin→ gradient overlay d. Save the image.	Able to apply shadow emboss effects

Table specifying the scope of syllabus to be covered for unit tests

Unit Test	Learning outcomes to be covered
Unit test-1	From 1 to 8
Unit test-2	From 9 to 22
Unit test-3	From 23 to 40

AUTOMOBILE LABORATORY – I

Course Code	Course Title	Periods per week	Total Periods/ Semester	Marks for FA	Marks for SA
A - 112	Automobile Laboratory- I	3	90	40	60

TIME SCHEDULE

Chapter No	Title	Periods	CO's Mapping
1	Demonstration on Safety practices in automobile workshop	6	CO1
2	Identification of tools, equipment and machinery used in automobile workshops	12	CO1
3	Demonstrate the function of IC engine components and engine systems	24	CO2
4	Demonstration on working of two stroke and four stroke petrol and diesel engines	12	CO3
5	Identification of transmission, electrical, chassis and body steering, braking and suspension system major components of a vehicle	18	CO4
6	To Measure the length/thickness/diameter of automobile components, bore diameter and depth using Vernier calliper, outside micrometer, inside micrometer and depth micrometer	18	CO5
Total:		90	

Course Code: A – 112 Course Title: AUTOMOBILE LABORATORY – I No. of Periods: 90	
COURSE OBJECTIVES	Upon completion of the course the student shall be able to 1. Understand the Safety practices in automobile workshop and identify the tools and equipment used in automobile laboratory. 2. Understand the functions of IC engine components and working of 2 & 4 stroke petrol and diesel engines. 3. Identify the transmission, electrical, chassis and body, steering, braking and suspension system major components of a vehicle. 4. Understand the Measuring of length/thickness/diameter of automobile components; bore diameter and depth using Vernier caliper, outside micrometer, inside micrometer and depth micrometer.

CO No.		Course Outcome
C01	A - 112.1	Safety practices in automobile workshop and Identify the tools and equipment used in automobile laboratory
C02	A - 112.2	Demonstrate the functions of IC engine components and engine systems
C03	A - 112.3	Demonstrate the working of two stroke and four stroke petrol and diesel engines

C04	A - 112.4	Identify the transmission, electrical, chassis and body, steering, braking and suspension system major components of a vehicle
C05	A - 112.5	Measure the length/thickness/diameter of automobile components, bore diameter and depth using Vernier calliper, outside micro meter, inside micro meter and depth micro meter

LEARNING OUTCOMES: Upon completion of the course the student shall able to
LIST OF EXPERIMENTS

1. Demonstration on Safety practices in automobile workshop
2. Identification of tools, equipment and machinery used in automobile workshops
3. Identification of IC engine components
4. Demonstration on working of two stroke petrol engine
5. Demonstration on working of two stroke diesel engine
6. Demonstration on working of four stroke petrol engine
7. Demonstration on working of four stroke diesel engine
8. Identification of transmission components of a vehicle
9. Identification of electrical, chassis and body major components of a vehicle
10. Identification of steering and braking system major components of a vehicle
11. Identification of suspension system major components of a vehicle
12. To measure the thickness/ diameter of automobile component using micro meter
13. To measure the diameter and thickness of automobile component using Vernier calliper
14. To measure the diameter of cylinder bore using inside micro meter/dial bore gauge.
15. To measure the depth of hole/slot of any automobile component using depth micro meter.

COs – POs Mapping

Course Code: A-112		Course Title: Automobile Lab-I			No of Cos:5	No. Of periods:90
POs	Mapped with CO No	CO Periods Addressing PO in Col 1		Level)	Remarks	
		No	%			
PO1	CO1,CO2,CO3,CO4	15	33	2	25% to 40% Level 2 (Moderately Addressed)	
PO2	CO2,CO3,CO4	15	33	2	5% to 25% Level 1 (Low Addressed)	
PO3					<5% Not Addressed	
PO4	CO1, CO2,CO3, CO4,CO5	5	12	1		
PO5	CO1, CO2,CO3, CO4,CO5	5	11	1		
PO6	CO2,CO3,CO4	5	11	1		
PO7						

CO-PO/PSO MATRIX

CO No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
A-112.1	3			3	1			3		3
A-112.2	3	2		3	3	3		3		3
A-112.3	3	2		3	3	3		3		2
A-112.4	3	2		2	2	3		2		3
A-112.5				3	3			3		3
Average	2.6	1.2		2.8	2.4	1.8		2.8		2.8

Note: The gaps in CO and PO mapping will be achieved by one or more appropriate activities from the following:

i) Seminars ii) Tutorials iii) Guest Lectures iv) Assignments v) Quiz competitions vi) Industrial visit vii) Tech Fest viii) Mini project ix) Group discussions x) Virtual classes xi) Library visit for e-books

The competencies and key competencies to be achieved by the student

S no	Title of the experiment	Key competencies	Competencies
1	Demonstration on Safety practices in automobile workshop	1) Understand the safety precautions to be followed in automobile workshops/garages	1) State the general safety precautions in workshops. 2) State the safety precautions to be followed in industries. 3) State the safety precautions while doing serving and repair works on vehicles
2	Identification of tools, equipment and machinery used in automobile workshops	1) Identify the tools required for a Particular task 2) Identify the equipment, machinery and gauges required for a particular task and also able to identify the them	1) State different types of tools and their applications 2) State different types of equipment and gauges (Like Magnetic stand, feeler gauge, micrometer, depth gauge etc). 3) State different types of machinery and their use
3	Identification of IC engine components	Identify the engine parts in cut model of petrol and diesel engine	1) State the functions of IC engine components 2) State the materials used for the components
4	Demonstration on working of two stroke petrol engine	Identify the two stroke petrol engine parts in cut model	Appreciate the working principle of two stroke petrol engines

5	Demonstration on working of two stroke diesel engine	Identify the two stroke diesel engine parts in cut model	Appreciate the working principle of two stroke diesel engines
6	Demonstration on working of four stroke petrol engine	Identify the four stroke petrol engine parts in cut model	Appreciate the working principle of four stroke petrol engines
7	Demonstration on working of four stroke diesel engine	Identify the four stroke diesel engine parts in cut model	Appreciate the working principle of four stroke diesel engines
8	Identification of transmission components of a vehicle	Identify the major transmission components of a given vehicle	1) Identify the major transmission components of a given vehicle 2) State the functions of those components
9	Identification of electrical, chassis and body major components of a vehicle	Identify the major electrical, chassis and body components of a given vehicle	1) Identify the major electrical, chassis and body components of a given vehicle 2) State the functions of those components
10	Identification of steering and braking system major components of a vehicle	Identify the major steering and braking system components of a given vehicle	1) Identify the major steering and braking system components of a given vehicle 2) State the functions of those components
11	Identification of suspension system major components of a vehicle	Identify the major suspension system components of a given vehicle	1) Identify the major suspension system components of a given vehicle 2) State the functions of those components
12	To measure the thickness/ diameter of automobile component using micro meter	1) Identify the parts of a micro meter 2) Measure the dimensions of a given component	1) State the principle of micro meter and its sensitivity/least count 2) Measure the dimensions of a given component
13	To measure the thickness/diameter of automobile component using Vernier calliper	1) Identify the parts of Vernier calliper 2) Measure the dimensions of a given component	1) State the principle of Vernier calliper and its sensitivity/least count. 2) Measure the dimensions of a given component
14	To measure the diameter of cylinder bore of component using inside micro meter and dial bore gauge.	1) Identify the parts of inside micro meter 2) Measure the dimensions of a given component	1) State the principle of inside micro meter and its sensitivity/least count. 2) Measure the dimensions of a given component
15	To measure the depth of hole/slot of automobile component using depth micro meter.	1) Identify the parts of depth micro meter 2) Measure the dimensions of a given component	1) State the principle of depth micro meter and its sensitivity/least count. 2) Measure the dimensions of a given component

III SEMESTER

DIPLOMA IN AUTOMOBILE ENGINEERING
SCHEME OF INSTRUCTIONS AND EXAMINATION
III SEMESTER

Subject Code	Name of the Subject	Instruction period/week		Total Periods/Semester	Scheme of Examination			
		Theory	Practical /Tutorial		Duration (hour s)	Sessional Marks	End Exam Marks	Total Marks
THEORY								
A-301	Engineering Mathematics–II	4	-	60	3	20	80	100
A-302	Automobile Power Plant	5	-	75	3	20	80	100
A-303	Automobile Transmission System	4	-	60	3	20	80	100
A-304	Autotronics	4	-	60	3	20	80	100
A-305	Engineering Materials & Manufacturing Processes	4	-	60	3	20	80	100
PRACTICAL								
A-306	Machine Drawing	-	6	90	3	40	60	100
A-307	CAD Lab –I	-	3	45	3	40	60	100
A-308	Automobile Engineering Lab –II	-	3	45	3	40	60	100
A-309	Workshop Practice – II	-	3	45	3	40	60	100
A-310	Autotronics Lab	-	3	45	3	40	60	100
Student Centric Activity (Library, Mini project, Physical Education, Preparation for competitive exams / placements, Seminars, Reporting etc)			3	45				
TOTAL		21	21	315+315=630		300	700	1000

A-301 is common to Group-I branches A/AA/CER/M/MET/MNG/TT

ENGINEERING MATHEMATICS-II
(common to A/AA/CER/C/EE/M/MET/MNG/TT)

Course Code	Course Title	No. of Periods/week	Total No. of periods	Marks for FA	Marks for SA
A-301	Engineering Mathematics-II	4	60	20	80

S.No.	Unit Title	No. of periods	COs mapped
1	Indefinite Integration	20	CO1
2	Definite Integration	10	CO2
3	Applications of Definite Integrals	10	CO3
4	Differential Equations	20	CO4
	Total Periods	60	

Course Objectives	(i) To understand the concepts of indefinite integrals and definite integrals with applications to engineering problems. (ii) To understand the formation of differential equations and learn various methods of solving first order differential equations. (iii) To learn the principles of solving homogeneous and non-homogeneous differential equations of second order.
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Course Outcomes	CO1	Integrate various functions using different methods.
	CO2	Evaluate definite integrals.
	CO3	Solve engineering problems by applying definite integrals.
	CO4	Obtain differential equations and solve differential equations of first order and first degree, and solve homogeneous and non-homogeneous differential equations of second order.

C-23A-301

ENGINEERING MATHEMATICS – II
(common to A/AA/CER/C/EE/M/MET/MNG/TT)

Learning Outcomes

Unit-I

C.O. 1 Integrate various functions using different methods.

L.O.1.1. Explain the concept of Indefinite integral as an anti-derivative.

1.2. State the indefinite integral of standard functions and properties of $\int (u + v) dx$ and $\int k u dx$ where u, v are functions of x and k is constant.

1.3. Solve problems involving standard functions using these properties.

1.4. Evaluate integrals involving simple functions of the following type by the method of substitution.

i) $\int f(ax + b) dx$, where $f(x)$ is in standard form.

ii) $\int (f(x))^n f'(x) dx$, $n \neq -1$

iii) $\int \frac{f'(x)}{f(x)} dx$

iv) $\int [f(g(x))]g'(x) dx$

1.5. Find the integrals of $\tan x$, $\cot x$, $\sec x$ and $\operatorname{cosec} x$ w.r.t x

1.6. Evaluate the Standard integrals of the functions of the type

$$\begin{aligned} \text{i)} & \frac{1}{a^2 + x^2}, \frac{1}{a^2 - x^2}, \frac{1}{x^2 - a^2} \\ \text{ii)} & \frac{1}{\sqrt{a^2 + x^2}}, \frac{1}{\sqrt{a^2 - x^2}}, \frac{1}{\sqrt{x^2 - a^2}} \\ \text{iii)} & \sqrt{a^2 + x^2}, \sqrt{a^2 - x^2}, \sqrt{x^2 - a^2} \end{aligned}$$

- 1.7. Evaluate integrals using decomposition method.
- 1.8. Solve problems using integration by parts.
- 1.9 Use Bernoulli's rule for evaluating the integrals of the form $\int u.v dx$.
- 1.10. Evaluate the integrals of the form $\int e^x [f(x) + f'(x)] dx$

Unit-II

C.O.2 Evaluate definite integrals.

- L.O.2.1. State the fundamental theorem of integral calculus
- 2.2. Explain the concept of definite integral.
- 2.3. Solve simple problems on definite integrals.
- 2.4. State various properties of definite integrals.
- 2.5. Evaluate simple problems on definite integrals using these properties.

Syllabus for Unit test-I completed

Unit -III

C.O.3 Solve engineering problems by applying definite integrals.

- L.O. 3.1. Find the area bounded by a curve and axes.
- 3.2. Obtain the mean and R.M.S values of the simple functions in given intervals.
- 3.3. Solve simple problems using Trapezoidal rule and Simpson's 1/3 rule for the approximation of definite integrals.

Unit -IV

C.O. 4 Form differential equations and solve differential equations of first order and first degree and Solve homogeneous and non-homogeneous differential equations of second order

- L.O.4.1. Define a Differential equation, its order and degree
- 4.2 Find order and degree of a given differential equation.
- 4.3 Form a differential equation by eliminating arbitrary constants.
- 4.4 Solve the first order and first degree differential equations by variables separable method.
- 4.5 Solve linear differential equation of first order of the form $\frac{dy}{dx} + Py = Q$, where P and Q are functions of x only or constants.
- 4.6 Solve homogeneous second order linear differential equations of the type $(aD^2 + bD + c) y = 0$ where $a(\neq 0)$, b, c are real numbers.
- 4.7 Define complementary function, particular integral and general solution of a non-homogeneous linear differential equation of second order with constant coefficients.
- 4.8 Describe the methods of solving $f(D) = X$, where $f(D)$ is a polynomial of second order and X is a function of the forms $k, e^{ax}, \sin ax, \cos ax$ and x and their linear combinations.

Syllabus for Unit test-II completed

C-23A-301

Engineering Mathematics – II (common to A/AA/CER/C/EE/M/MET/MNG/TT) CO/PO – Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
CO1	3	2	2	1				3	2	2
CO2	3	2	2	2				3	2	2
CO3	3	3	3	3				3	3	3

CO4	3	3	3	3				3	3	3
Avg.	3	2.5	2.5	2.25				3	2.5	2.5

3 =Strongly mapped (High), **2** = Moderately mapped (Medium), **1** = Slightly mapped (Low)

Note: The gaps in CO/PO mapping can be met with appropriate activities as follows:

For PO5: Appropriate quiz programmes may be conducted at intervals and duration as decided by concerned faculty.

For PO6: Seminars on applications of mathematics in various engineering disciplines are to be planned and conducted.

For PO7: Plan activities in such a way that students can visit the Library to refer standard books on Mathematics and access the latest updates in reputed national and international journals. Additionally, encourage them to attend seminars and learn mathematical software tools.

C-23A-301
Engineering Mathematics – II
(common to A/AA/CER/C/EE/M/MET/MNG/TT)
PO- CO – Mapping strength

PO no	Mapped with CO no	CO periods addressing PO in column I		Level (1,2 or 3)	Remarks
		Number	%		
1	CO1, CO2, CO3,CO4	60 (20+10+10+20)	100%	3	>40% Level 3 Highly addressed
2	CO1, CO2, CO3,CO4	37 (6+6+10+15)	61.6%	3	
3	CO1, CO2, CO3,CO4	37 (6+6+10+15)	61.6%	3	
4	CO1, CO2, CO3,CO4	35 (4+6+10+15)	58.3%	3	
5					5% to 25% Level 1 Low addressed
6					
7					
PSO 1	CO1, CO2, CO3,CO4	60 (20+10+10+20)	100%	3	<5% Not addressed
PSO 2	CO1, CO2, CO3,CO4	37 (6+6+10+15)	61.6%	3	
PSO 3	CO1, CO2, CO3,CO4	37 (6+6+10+15)	61.6%	3	

C-23A-301
ENGINEERING MATHEMATICS – II
(common to A/AA/CER/C/EE/M/MET/MNG/TT)
COURSE CONTENTS

Unit-I

Indefinite Integration

1. Integration regarded as anti-derivative, indefinite integrals of standard functions - Properties of indefinite integrals - Integration by substitution or change of variable - Integrals of $\tan x$, $\cot x$, $\sec x$, $\operatorname{cosec} x$.

Evaluation of integrals which are of the following forms:

$$i) \frac{1}{a^2 + x^2}, \frac{1}{a^2 - x^2}, \frac{1}{x^2 - a^2}$$

$$ii) \frac{1}{\sqrt{a^2 + x^2}}, \frac{1}{\sqrt{a^2 - x^2}}, \frac{1}{\sqrt{x^2 - a^2}}$$

$$iii) \sqrt{a^2 + x^2}, \sqrt{a^2 - x^2}, \sqrt{x^2 - a^2}$$

Integration by decomposition of the integrand into simple rational algebraic functions.

Integration by parts, Bernoulli's rule and integrals of the form $\int e^x [f(x) + f'(x)] dx$.

Unit-II

Definite Integration

2. Definite integral, fundamental theorem of integral calculus, properties of definite integrals, evaluation of simple definite integrals.

Unit-III

Applications of Definite Integrals

3. Area bounded by a curve and axes - Mean and RMS values of a function in given intervals - Trapezoidal rule, Simpson's 1/3 rule to evaluate an approximate value of a definite integral.

Unit -IV

Differential Equations

4. Definition of a differential equation - Order and degree of a differential equation- Formation of differential equations - Solutions of differential equations of first order and first degree using variables separable method and linear differential equation of the type $\frac{dy}{dx} + Py = Q$ - Solutions of homogenous and non-homogeneous linear differential equations of second order with constant coefficients.

Textbook:

Engineering Mathematics-II, a textbook for second year third semester diploma courses, prepared & prescribed by SBTET, AP.

Reference Books:

1. B. S. Grewal, Higher Engineering Mathematics, Khanna Publishers.
2. Schaum's Outlines Differential Equations, Richard Bronson & Gabriel B. Costa
3. M.Vygodsky, Mathematical Handbook: Higher Mathematics, Mir Publishers, Moscow.

C- 23 Engineering Mathematics – II

Subject Title : Engineering Mathematics – II
 Subject Code : A-301 (common to A/AA/CER/C/EE/M/MET/MNG/TT)
 Periods/Week : 04
 Periods/Semester : 60

TIME SCHEDULE

S.No.	Chapter/Unit title	No. of Periods	Marks Allotted	Short Type	Essay Type	COs mapped
Unit – I: Indefinite integration						
1	Indefinite integration	20	34	3	2 1/2	CO1
Unit – II: Definite Integration						
2	Definite Integrals	10	16	2	1	CO2
Unit-III: Applications of Definite Integrals						
3	Area of curves	3	3	1	0	CO3
4	Mean and RMS values	3	8	1	1/2	CO3
5	Numerical Integration	4	10	0	1	CO3
Unit – IV: Differential Equations						
6	Introduction to Differential Equations	5	6	2	0	CO4
7	Solutions of first order differential equations	4	13	1	1	CO4
8	Solutions of second order homogeneous differential equations	4	10	0	1	CO4
9	Solutions of second order non-homogeneous differential equations	7	10	0	1	CO4
Total		60	110	10	8	
Marks				30	80	

Unit Test Syllabus

Unit Test	Syllabus
Unit Test-I	From L.O 1.1 to L.O 2.5
Unit Test-II	From L.O 3.1 to L.O 4.8

AUTOMOBILE POWER PLANT

Course Code	Course Title	No. of Periods/Week	Total Periods/Year	Marks for FA	Marks for SA
A-302	Automobile Power Plant	05	75	20	80

TIME SCHEDULE

Chapter No.	Title	Periods	Marks	Part – A Short Type (3 marks)	Part – B Essay Type (10 marks)	Co's Mapping
1	Engine Construction	15	16	2	1	CO1
2	Inlet and Exhaust System	08	13	1	1	CO2
3	Petrol Engine Fuel System	10	16	2	1	CO3
4	Diesel Engine Fuel System	10	16	2	1	CO3
5	Lubrication and Cooling System	09	13	1	1	CO2
6	Fuels & Combustion	08	13	1	1	CO5
7	Air Standard Cycles and Performance of Engine	15	23	1	2	CO5
TOTAL		75	110	10	8	

Course Code: A - 302	Title: Automobile Power Plant	Total No. of Periods: 75
Course Objectives:	Upon completion of the course the student shall be able to	
	- Understand the constructional details of I.C. Engines - Understand the working of inlet, exhaust, cooling and lubrication system, fuel supply system in diesel and petrol engines, fuel and combustion and combustion phenomenon in petrol and diesel engine system of I.C engine - Understand the air standard cycles and performance of internal combustion engines.	

CO No.		Course Outcome
CO1	A-302.1	Explain the constructional details of I.C engine
CO2	A-302.2	Describe the working of inlet, exhaust, cooling and lubrication system of I.C engines.
CO3	A-302.3	Explain the fuel systems in diesel and petrol engines.
CO4	A-302.4	Describe the combustion phenomenon in diesel and petrol engines and explain the terms: diesel knock, detonation, pre ignition, LCV, HCV, Octane and Cetane rating of fuels
CO5	A-302.5	Solve simple problems in various air standard cycles and Evaluate performance of Internal combustion Engines.

LEARNING OUTCOMES:

Upon completion of the course the student shall be able to

1. Engine Construction

- 1.1 State the components of I.C engine and their materials
- 1.2 Explain the constructional details of an IC engine
- 1.3 Distinguish between wet and dry liners
- 1.4 List out different types of piston rings
- 1.5 Summarise the piston cooling jet system
- 1.6 State the requirements of firing order and firing order of 4, 6 and 8 cylinder inline and V-type engines
- 1.7 Summarise the necessity of multiple valves in engines and List different types of Valves
- 1.8 Explain valve time diagrams of 4-stroke petrol and diesel engines
- 1.9 State the function of components of valve actuating mechanism
- 1.10 Explain Straight poppet, overhead valve and electronic controlled Valve actuating mechanism (i-VTEC (intelligent variable valve timing and lift electronic control) mechanism with line diagrams
- 1.11 State the importance of tappet or valve clearance
- 1.12 Distinguish different types of SI and CI Engine Combustion chambers and state advantages and disadvantages of different Combustion chambers
- 1.13 List types of gaskets used in an engine and their material
- 1.14 State the functions of flywheel and describe the use of vibration dampers

2.0 Inlet and Exhaust System

- 2.1 Explain the constructional details of different types of Inlet and exhaust manifolds
- 2.2 State the requirements for good manifold design
- 2.3 Explain the constructional details of different types of air cleaners
- 2.4 Explain the methods of Super Charging
- 2.5 State the principle of Turbo Charging
- 2.6 State the necessity of inter cooler in turbo charger
- 2.7 Explain the working principle of Turbo charger with inter cooling system
- 2.8 Explain the working principle of Turbo Stratified Injection system
- 2.9 Explain the working Principle of fixed Geometry Turbo Charger (FGT), waste gate Turbo Charger and Variable Geometry Turbo Charger (VGT)
- 2.10 State the necessity of Mufflers
- 2.11 Describe the construction and working of different types of Mufflers with simple sketches

3.0 Fuel Supply System in Petrol (S.I) Engines

- 3.1 List the components of petrol engine fuel supply system
- 3.2 Describe each component of fuel supply system viz., fuel tank, fuel filter, fuel feed pump, etc.,
- 3.3 List the types of fuel pumps
- 3.4 Explain the construction and working of mechanical and electrical fuel pumps

- 3.5 Define stoichiometric air fuel ratio
- 3.6 Distinguish between rich mixture and lean mixture
- 3.7 State the air fuel ratio for different operating conditions of petrol engine
- 3.8 Explain the working Principle of EFI system
- 3.9 Explain the working Principle of MPFI system
- 3.10 Explain the working Principle of PDI or GDI system
- 3.11 State the merits and limitations of EFI, MPFI and GDI
- 4.0 Diesel Engines Fuel Systems**
- 4.1 Identify the Parts of fuel supply system in diesel engine
- 4.2 Explain Construction and Working of mechanical and electronic diesel fuel feed pump
- 4.3 State the function of fuel filter, fuel feed pump, fuel injection pump and fuel injector
- 4.4 List the types of fuel filters, fuel Injectors and injector nozzles
- 4.5 State the requirement of fuel injection pump
- 4.6 List different types of fuel injection system in C I Engines
- 4.7 Explain Unit injection system, individual pump, distributor pump system and Common rail diesel injection system (CRDI) with line diagrams
- 4.8 State the function of governor and classify them
- 4.9 Explain the working of mechanical, pneumatic and ECU Controlled governors.
- 4.10 Explain the layout of ECU controlled common rail diesel injection system (ECU CRDI)
- 4.11 Explain the layout of Electronic Unit Injector (EUI)
- 5.0 Lubrication and Cooling System**
- 5.1 State the petroleum and synthetic Lubricants and their properties of lubricants used in Automobiles
- 5.2 Define SAE number of lubricant
- 5.3 State the grades of Lubricants used in Engine, Gear-Box and Differential
- 5.4 State the Oil replacement intervals
- 5.5 Describe construction & working of Wet Sump, Dry Sump and Piston cooling jet Lubrication system with sketches
- 5.6 List the different types of oil filters used in IC Engines
- 5.7 Differentiate between Bypass filter and Full flow filter .
- 5.8 List the various additives of lubricants
- 5.9 Describe construction & working of Liquid cooling system in I.C engines with the help of line diagram
- 5.10 Explain construction and working of different types of water pump, Radiator, Pressure cap and thermostat
- 5.11 Describe construction & working of Thermo viscous fan
- 5.12 State the necessity of Anti-freeze and Anti-rusting liquids
- 5.13 List the Anti-freeze and Anti-rusting liquids
- 6.0 Fuels and Combustion**
- 6.1 Define the term fuel
- 6.2 State the classification of fuels
- 6.3 State different types of fuels used in I.C. Engine
- 6.4 List the properties of fuels
- 6.5 Define higher and lower calorific value of fuel
- 6.6 Know the importance of Octane, Cetane rating and HUCR and fuel additives
- 6.7 Describe the stages of combustion in S.I. engines
- 6.8 Explain the Phenomenon of pre-ignition & detonation and state their effects on the performance of S.I. engine
- 6.9 Explain the stages of combustion in C.I. engines
- 6.10 Define diesel knock and State the factors influencing diesel knock

7.0 Study of air standard cycles

- 7.1 Define the term Air Standard cycle, Reversible Cycle and Irreversible cycle.
- 7.2 State the assumptions made in Air Standard cycles.
- 7.3 Explain Carnot cycle, Otto Cycle and Diesel cycle with the help of a p-v and T-S diagrams
- 7.4 Derive the formula for the air standard efficiency of Carnot, Otto and Diesel cycles.
- 7.5 Solve simple problems on Carnot, Otto and Diesel cycles
- 7.6 State the need of Engine performance testing.
- 7.7 Define the terms: Mean Effective Pressure, Indicated Power, Brake Power, Frictional Power, Mechanical Efficiency, Thermal Efficiency, and Specific Fuel Consumption and Volumetric efficiency
- 7.8 Solve simple problems on Indicated Power, Brake Power, Frictional Power, Mechanical efficiency, Specific Fuel consumption, Indicated Brake Thermal Efficiency and Volumetric Efficiency
- 7.9 State the importance of Morse Test and Compute IP of the IC Engine by using Morse Test
- 7.10 Explain Heat Balance sheet for a given test result of an Internal Combustion engine.

CO – PO Mapping

Course Code: A-302	Course Title: Automobile Power Plant	No of Cos: 5		No. of periods: 75	
POs	Mapped with CO No.	CO Periods Addressing PO in Col 1		Level (1,2,3)	Remarks
		No	%		
PO1	CO1, CO2, CO3, CO4, CO5	30	40	3	>40% Level 3 (Highly Addressed)
PO2	CO2, CO3, CO4, CO5	15	20	1	25% to 40% Level 2 (Moderately Addressed)
PO3	CO1, CO2, CO3, CO4, CO5	12	16	1	5% to 25% Level 1 (Low Addressed)
PO4	CO1, CO3, CO4, CO5	10	13	1	<5% Not Addressed
PO5	CO1, CO2, CO3, CO4	8	11	1	
PO6					
PO7					

CO-PO/PSO MATRIX

CO No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
A-302.1	3	1	1	2	1			1	2	
A-302.2	3	2	1	1	2			1	2	
A-302.3	2	2	1	2	1			1	2	
A-302.4	3	2	1	2	1			1	2	2
A-302.5	3	3	2	3	2			1	3	
AVERAGE	3	2	1.2	2.2	1.4			1	2.2	0.4

COURSE CONTENT

1) Engine Construction

Engine Components – Constructional features, Manufacturing process and materials used of Cylinder Block, Cylinder Head-Cylinder head with engine brakes - Cylinder Liners — Crankcase – Oil Pan– Gaskets–Piston - Piston Pin - Piston Rings-Piston cooling jet– Connecting Rod– Crankshaft – Connecting Rod bearings– Flywheel– Vibration dampers– Valve –Valve types-Multiple valve engines–Timing gear and case-Arrangement of Cylinders -Firing Order- valve timing diagrams 4-stroke petrol and diesel engine – Valve advancement –

necessity - Valve actuating mechanisms – mechanical and electronically controlled valve actuating mechanisms - components and function of valve actuating mechanism: cam shaft, valve lifter, valve guide, valve spring, push rod, rocker arm, rocker arm shaft, rocker arm spring - Combustion chamber – Requirements of S.I. and C.I. Engine combustion chambers - Types of SI Engine combustion chambers T-head type, L-head type, I-head type -Bath Tub Type - Wedge Type and F-head type and C.I. engine combustion chambers and functions – Direct injection type, pre-combustion chambers, Turbulence chamber type and Air cell combustion chamber - Relative advantages and disadvantages.

2) Inlet and Exhaust Systems

Inlet and Exhaust Manifolds - Consideration for good manifold design - Types of Manifolds – constructional details – Air Cleaners – constructional details of different types of air cleaners- –Super Charging method - Turbo Charger - Fixed Geometry Turbo (FGT) – Waste gate Turbocharger - Variable Geometry Turbocharger (VGT)- Turbo Stratified Injection (TSI) – TC with inter cooling systems- turbo charged engine modifications - Mufflers–types of Mufflers–Constructional details and working principle of different types of Mufflers

3) Petrol Engine Fuel Systems

Components of Petrol engine fuel systems - Details of fuel tank and fuel filters – Working principle with constructional details of Mechanical and Electrical fuel feed pumps - Air fuel ratio's for different operating conditions - Electronic Fuel injection system (EFI) – Electronically controlled Multi point fuel Injection System (MPFI) – Petrol or Gasoline Direct Injection System (PDI or GDI) - Merits and limitations.

4) Diesel Engine Fuel Systems

Diesel engine fuel system – diesel fuel feed pumps – mechanical& Electronic - Types of Diesel filter – Requirements of fuel injection system – Types of fuel injection systems – Constructional details and working of fuel injection pump and injectors - types of injectors – Types of Nozzles – Governing system – Types of governors - Mechanical, Pneumatic, ECU Controlled Governors – Electronically controlled Common Rail Diesel Injection System (CRDI) - Electronic unit Injector (EUI)

5) Lubrication and Cooling Systems

Types of lubricants – Petroleum and Synthetic - Properties of a lubricant –S.A.E Number — Grades of Lubricants used in Engine, Gear-Box, Differential —Oil replacement intervals - Wet and Dry Sump Lubrication –Oil filters – Types of filters –Full flow and By-pass filter arrangements –Piston cooling jet lubrication system - Additives of Lubricants – Liquid cooling system - Types of Radiators – construction – Construction and Working principle of water pump - Thermostat – Wax pellet, Bellows type - Radiator pressure cap – Thermo viscous Fans - Anti freezing additives – Anti rusting additives

6) Fuels and combustion

Definition of fuel – classification of fuels – Properties of fuels – Calorific value of fuels–H.C.V & L.C.V–Fuels used in I.C. Engines–Rating of I.C engine fuels - Octane rating, Cetane rating, and HUCR – Fuel dopes or additives – Combustion in I.C. engines – Stages of combustion in S.I. engines –detonation in S.I. engines – pre-ignition in S.I. engines – stages of combustion in C.I. engines–Diesel knock, Factors influencing diesel knock.

7) Air standard cycles and Performance of I.C. Engines:

Air standard cycles - Reversible Cycle and Irreversible cycle - Assumptions made - Brief description of Carnot cycle, Otto Cycle, and Diesel cycle with P-V and T-S diagrams – Efficiency – Problems - Need for Engine performance testing - Mean effective Pressure - Indicated, Brake and Frictional Power - Mechanical Efficiency and Thermal Efficiency – Specific Fuel Consumption - Volumetric efficiency - Heat Balance Sheet – Problems pertaining to performance of IC Engines, Morse Test and heat balance.

REFERENCE BOOKS

1. Elements of Heat Engines-by R.C. Patel and C.J. Karamchandani.
2. Heat Engines - by R.S. Kurmi
3. Heat Engines – by Pandya and Shah
4. Automobile Engineering Vol-I&II – by Kirpal Singh
5. Automotive Mechanics - by Heitner
6. Automobile Engineering-by G.B.S. Narang
7. Automobile Engineering – by R.B. Gupta
8. Automobile Engineering - by Banga and Nathun Singh
9. Automobile mechanics-by William Chrouse
10. High speed combustion Engines-by P.M.Heldt
11. Heat power engineering –by P.L. Ballany
12. Internal Combustion Engines – by Mathur&Sharma
13. Automobile Engineering Voll-Anil Chhikaria (SatyaPrakashanPublisher)

Table specifying the scope of syllabus to be covered for Unit Test-I & Unit Test-II

Unit Test	Learning Outcomes to be covered
Unit Test – I	1.1 to 4.6
Unit Test – II	4.7 to 7.10

AUTOMOBILE TRANSMISSION SYSTEMS

Course Code	Course Title	No. of Periods/Week	Total Periods/Year	Marks for FA	Marks for SA
A-303	Automobile Transmission Systems	04	60	20	80

TIME SCHEDULE

Chapter No	Title	Periods	Marks	Part – A Short Type (3 marks)	Part – B Essay Type (10 Marks)	CO's Mapping
1	Clutch	10	16	2	1	CO1
2	Transmission Systems	15	26	2	2	CO2 & CO4
3	Propeller shaft and Universal joint	06	13	1	1	CO2
4	Front axle	06	13	1	1	CO3
5	Rear axle and Differential	13	26	2	2	CO2 & CO3
6	Wheels and Tyres	10	16	2	1	CO5
Total		60	110	10	8	

Course Code : A-303	Course Name : Automobile Transmission Systems No. of Periods : 60
Course Objectives	Upon completion of the course the student shall be able to - Understand the need and working of an Automobile Transmission systems. - Understand the function and working of clutch, gear box, gear shifting mechanism, over drive, fluid coupling, torque converter, universal joint, propeller shaft, differential, front axle and rear axle, transaxle, automatic transmission system, intelligent Manual Transmission (iMT), Continuously Variable Transmission System (CVT), Dual Shift Gear Box system (DSG) - Appreciate the term wheels and tyres

CO No.	Course Outcome
C01	A-303.1 Describe the components of Transmission System in an automobile
C02	A-303.2 Explain construction and working principles of automatic transmission, drive train components, final drive and differential systems.
C03	A-303.3 Comprehend the components of axles and their working
C04	A-303.4 Explain recent developments in transmission systems viz., iMT, CVT, DCT etc.,
C05	A-303.5 Distinguish between different tyres and wheels.

Learning Outcomes:

1.0 Clutch

- 1.1 State the general requirements of an automobile clutch
- 1.2 List any six types of clutches.
- 1.3 Explain the principles of operation of the clutch.
- 1.4 Draw the line diagram and explain the working of Single plate, Multi plate, Centrifugal, Semi Centrifugal and Diaphragm Clutches.
- 1.5 Explain the working principles of vacuum, electromagnetic, hydraulically operated clutches.
- 1.6 Describe the types of friction materials used in clutches.

2.0 Transmission Systems

- 2.1 State the objectives of the gear box in the transmission system and List four types of gear boxes.
- 2.2 Explain the principles and operation of sliding mesh, Constant mesh, Synchromesh and Epicyclic gear box and synchromesh gearbox with number of speeds
- 2.3 Describe floor shifting & steering column mounted shifting mechanism.
- 2.4 Explain about Inter locking device.
- 2.5 Explain free wheel unit and over drive mechanism.
- 2.6 List any four advantages of using fluid coupling.
- 2.7 Explain the working principle of the torque converter and fluid coupling.
- 2.8 List any six advantages of automatic transmission system
- 2.9 Explain the principle of automatic transmission system
- 2.10 Describe intelligent Manual Transmission (iMT) with line diagram
- 2.11 Describe Continuously Variable Transmission System (CVT) with line diagram
- 2.12 Describe Dual Shift Gear Box (DSG) with line diagram

3.0 Propeller shaft and Universal joints

- 3.1 Describe the components of propeller shaft and its components
- 3.2 Draw the line diagram of propeller shaft with slip joint and universal joints.
- 3.3 List different types of universal joints
- 3.4 Explain different types of universal joints – Cross type or Spider and two yoke types, Ball and trunnion type.
- 3.5 List the three types of Constant velocity type universal joints.
- 3.6 Explain the Rzeppa, Plunge and Tripod type Constant velocity joints.

4.0 Front axle

- 4.1 Differentiate between live and dead axles.
- 4.2 List various components of front axle assembly.
- 4.3 Explain front axle assembly.
- 4.4 List various types of stub-axles.
- 4.5 Explain Elliot, Reverse Elliot, Lamoine and Reverse Lamoine type of Stub axles
- 4.6 Draw the layout of lift axle used in heavy duty vehicles.
- 4.7 State the purpose of lift axle
- 4.8 Explain the lift axle mechanism with a sketch.

5.0 Rear Axle and Differential:

- 5.1 State loads acting on the rear axles.
- 5.2 List different types of rear axles and Explain semi floating, three quarter floating and fully floating rear axles
- 5.3 List types of rear axle casing and Explain Split, Banzo and Salisbury type of axle casings
- 5.4 Explain Hotchkiss drive and Torque-tube drive.
- 5.5 Distinguish the types of rear axle drives.

- 5.6 State the purpose of the final drive
- 5.7 Explain bevel, Worm & worm wheel and Hypoid type final drives.
- 5.8 Explain the working principle of Single and Double reduction final drives.
- 5.9 State the necessity of differential unit and explain the working principle of differential
- 5.10 Explain the differential system for tandem axle
- 5.11 State the function of differential locking system
- 5.12 Explain manual and automatic differential locking system
- 5.13 Explain the working principle of Limited slip differential.
- 5.14 State the purpose of the four-wheel drive and explain the working principle of four-wheel drive.
- 5.15 Explain the working of Transaxle.

6.0 Wheels and Tyres

- 6.1 State any four requirements of automobile wheel.
- 6.2 List three types of automobile wheels and wheel rims.
- 6.3 Explain the wheel assembly
- 6.4 State the functions and desirable properties of tyre.
- 6.5 State the classification of tyres based on type of tyre and tyre construction.
- 6.6 Explain Tube tyre and tubeless tyre
- 6.7 Describe tyre designation and construction of tyres
- 6.8 Explain carcass types – Cross ply, radial ply and belted bias ply tyres
- 6.9 Explain tyre tread pattern.
- 6.10 State tyre wear and its causes.
- 6.11 State the tyre designation in terms of speed rating, width, aspect ratio, load index, ply Rating
- 6.12 Describe tread wear indication
- 6.13 Describe the tyre pressure monitor system.
- 6.14 Explain the terms tyre rotation, re-trading and vulcanizing
- 6.15 Describe cold and hot re-tread procedure.

COs – POs Mapping

Course Code: A-303		Course Title: Automobile Transmission Systems		No. of COs: 5	No. of Periods:60
POs	Mapped with CO No.	CO Periods Addressing PO in Col. 1		Level (1,2,3)	Remarks
		Number	Percentage		
PO1	CO1,CO2,CO3,CO4,CO5	12	20	1	> 40% Level 3 (Highly Addressed) 25% to 40% level 2 (Moderately Addressed) 5% to 25% level 1 (Low Addressed) <5% Not Addressed
PO2	CO1,CO2,CO3,CO4,CO5	16	26	2	
PO3	CO2,CO3,CO4,CO5	8	14	1	
PO4	CO1,CO2,CO3,CO4,CO5	12	20	1	
PO5	CO1,CO2,CO3,CO4,CO5	12	20	1	
PO6					
PO7					

CO-PO/PSO MATRIX

CO No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
A-303.1	2	1		1	2			2	3	1
A-303.2	3	3	2	2	2			2	3	2
A-303.3	2	3	2	1	2			2	3	
A-303.4	3	2	2	2	1			2	3	1

A-303.5	3	2	1	3	2			2	2	1
AVERAGE	2.6	2.2	1.4	1.8	1.8			2	2.8	1

COURSE CONTENT

1.0 Clutch:

General requirements of an automobile clutch - Principle of Operation - constructional details of different types of clutches and diagram of clutches – Single plate – Multi plate – Centrifugal – Semi centrifugal – Diaphragm clutches – Working principles of vacuum, electromagnetic and Hydraulically operated clutches – friction materials of clutch.

2.0 Transmission Systems:

Need of the gear box in the transmission system – types of gear boxes –Principles and operation of sliding mesh, constant mesh and synchromesh gearbox with number of speeds - Epicyclic gear box - types of gear shifting mechanism – floor shifting and steering column shifting-Inter lock devices - principle and working of freewheeling – overdrive mechanism – transfer case - Fluid coupling - Advantages and disadvantages – details of fluid coupling - Torque converter and its principle –principle of automatic transmission system and advantages –intelligent Manual Transmission (iMT) - Continuously Variable Transmission System (CVT) - Dual Shift Gear Box (DSG) or Dual Clutch Transmission System (DCT).

3.0 Propeller shaft and Universal joint:

Propeller shaft components — Slip joint or sliding joint – Universal joint -. Types of universal joints – Cross type or spider and two yoke type – Ball and trunnion type – constant velocity type – Different types of Constant velocity joints – Rzeppa and Tri-pod type.

4.0 Front Axle

Live and Dead axles – Components of Front axle assembly –Stub axle – Types of stub axles –Elliot – Reverse Elliot – Lamoine – Reverse Lamoine. - Lift axle-purpose and layout.

5.0 Rear Axle and Differential:

Loads on the rear axle – Types of rear axles - semi floating – Three quarter floating– fully floating axles, Types of drives – Hotchkiss drive, Torque tube drive– Axle casings and types –Split, Banzo and Salisbury types. Final drive – Purpose of final drive, types of final drive – Bevel, worm and worm wheel, Hypoid gear -single and double reduction final drives –Differential gear-differential - differential lock –Manual and Automatic differential locks – Differential for Tandem axle – Four-wheel drive – Transaxle.

6.0 Wheels and Tyres:

Wheel assembly – functions of the wheels – Types of wheels, types of rims– Types of tyres – Construction of tyres-constituents of tyre- Cross ply tyre – Radial tyre –Belted bias ply - merits of radial tyre over cross ply tyre-tube tyre - tubeless tyre and its repair–tyre tread patterns – tyre pressure & wear –Desirable tyre properties – causes of tyre wear – Tyre designation- speed rating, width, aspect ratio, load index, ply rating, tread wear indication, tyre pressure monitor system -Tyre rotation –Vulcanizing – Re-treading.

REFERENCE BOOKS:

1. The motor vehicle - Newton steeds. & Garret
2. Automotive Chassis - P.M. Heldt.
3. Mechanism of the car - A.W. Judge

4. Automotive mechanism - Joseph Heitner.
5. The Automobile – HarbansinghReyat
6. Automotive Engineering - G.B.S. Narang
7. An introduction to Automobile Engineering - N.R. Khatawate
8. Problems in Automobile Mechanics - Dr. N K Giri

Table specifying the scope of syllabus to be covered for Unit Test-I & Unit Test-II

Unit Test	Learning Outcomes to be covered
Unit Test – I	From 1.1 to 3.6
Unit Test – II	From 4.1 to 6.15

AUTOTRONICS

Course Code	Course Title	No. of Periods/Week	Total Periods/Year	Marks for FA	Marks for SA
A-304	Autotronics	04	60	20	80

TIME SCHEDULE

Chapter No.	Title	Periods	Marks	Part – A Short Type (3 marks)	Part – B Essay Type (10 Marks)	CO's Mapping
1.	Electrical and Electronic Components	8	16	2	1	CO1
2.	Automotive Electrical systems	12	26	2	2	CO2
3.	Lighting circuits and Accessories	8	16	2	1	CO3
4.	Body Computer and instrumentation	14	26	2	2	CO4
5.	Vehicle communication Networks and Diagnostics	10	13	1	1	CO5
6	Modern Vehicle Electronics	8	13	1	1	CO5
Total		60	110	10	8	

Course Code: A-304	Title: Autotronics	Total No. Of Periods: 60
Course Objectives:	Upon completion of the course the student shall be able to	
	1.0	Apprehend the electrical and electronics systems of an automobile
	2.0	Understand the Instrumentation, communication, electronic safety, diagnosis systems used in vehicles.

CO No.		Course Outcome
CO1	A-304.1	Explain basic electrical and electronic components used in a vehicle
CO2	A-304.2	Explain battery, starting, generating and ignition systems
CO3	A-304.3	Describe the lighting circuits.
CO4	A-304.4	Explain the automotive electronics and instrumentation of automobiles.
CO5	A-304.5	Describe vehicle communication, electronics and diagnosis

Learner Outcomes:

Upon completion of the course the student shall be able to

1.0 Electrical and Electronic Components

- 1.1 Explain working principle of switch, relay, and solenoid with a simple electric circuit.
- 1.2 Explain working principle of relay with a simple diagram.
- 1.3 Define resistor and capacitors.
- 1.4 List different types of resistors and fuses.

- 1.5 Define P-N junction diode and mention its applications
- 1.6 Define P-N-P & N-P-N transistors and mention their applications
- 1.7 Define Integrated circuits, fuses, circuit breaker, ground straps, terminals and connectors.
- 1.8 Explain working principle of stepper motor with a neat diagram.
- 1.6 State automotive wiring, solid and stranded wire.
- 1.7 Define major factors that determine the proper size of wire to be used.
- 1.8 List different terminals and connectors used on the automobile.
- 1.9 Define Wiring harness, wiring protection devices, circuit numbering and electric symbols.
- 1.10 Explain wiring diagram with an example
- 1.11 Draw any six electrical and electronic symbols used in wiring diagrams.
- 1.12 Explain colour coding of wires used in automobile with suitable diagrams.
- 2.0 Automotive Electrical systems.**
- 2.1 Explain construction and working of Lead acid, alkaline, sodium sulphur battery with sketch.
- 2.2 Define maintenance free battery and super capacitors
- 2.3 State Battery ratings, capacity, cables and terminals.
- 2.4 List starter motor circuit components and Starter motors
- 2.5 Explain construction and working of pre-engaged starter, permanent magnet starter, integrated starters, bendix drive and over running clutch with a sketch
- 2.6 State Belt driven starter generator, water cooled alternator, regulator, AC to DC rectifier and charge indicators.
- 2.7 Explain Capacitor discharge, distributor less, coil on plug ignition systems and electronic spark advance system with a diagram.
- 2.8 Explain construction and working of spark plug with a neat sketch.
- 2.9 List the types of spark plugs.
- 2.10 State the necessity of electrode gap and materials of spark plugs
- 3.0 Lighting circuits and Accessories**
- 3.1 List the types of lamps used in automobile.
- 3.2 State the functions of Gas Discharge Lamps, Xenon Lighting, Ultra violet Lights, LED lights, Infrared Lights, Head light lenses and head light reflectors.
- 3.3 Explain working of head light levelling with a neat diagram.
- 3.4 State different types of exterior and interior lights used in automobiles
- 3.5 List the types of lighting circuits.
- 3.6 Explain the working principles of Head light, Dim-Dip, Stop light or reverse Light and interior lights with circuit diagrams
- 3.7 Explain the working principles of power door lock, power windows, Horn and Windshield wiper with circuit diagrams
- 3.8 State the functions of intelligent wipers, electronic defoggers and automotive rear-view mirror
- 4.0 Body Computer and instrumentation**
- 4.1 State the functions of vehicle computer, analog, digital, microprocessor, microcontroller, adaptive strategy, memory and programming.
- 4.2 List the types of Computer memory.
- 4.3 Explain microcomputer or ECU used in vehicles with a diagram.
- 4.4 Describe computer memory and its types.
- 4.5 Explain logic gates with simple diagrams.
- 4.6 State the classification of sensors and actuators.
- 4.7 List different sensors and actuators used in modern vehicles.
- 4.8 Explain different types of sensors and actuators with diagrams.
- 4.9 State the applications of different sensors types used in automobiles.
- 4.10 State the functions of thermal, air cored, moving iron gauges, and head up display, GPS, Quantization and holography.
- 4.11 Explain vehicle condition monitoring.

- 4.12 Explain digital instrumentation system and trip computer with a block diagram.
- 4.13 List Advanced Instrumentation Technology Types
- 4.14 Explain Telematics and Multiplexed displays.
- 5.0 Vehicle communication Networks and Diagnostics**
- 5.1 List the types of Multiplex Communication protocols.
- 5.2 Explain Multiplexing and Multiplex Communication protocols.
- 5.3 Explain Controller Area Network (CAN) and Local Interconnect Network (LIN).
- 5.4 Describe Media Oriented System Transport (MOST).
- 5.5 State the importance of Connected Cars, Smart Cars & Traffic systems, Wi-Fi Cars and Bluetooth.
- 5.6 Explain Oscilloscope and Diagnostic Equipment.
- 5.7 State the functions of OBD I, OBD II, OBD scanner tool and trouble codes.
- 6.0 Modern Vehicle Electronics**
- 6.1 State the functions of Memory seats, Remote Keyless entry, Keyless start, Electric mirrors and Electric sun roof
- 6.2 Explain the antitheft system and Obstacle avoidance radar with a diagram.
- 6.3 State the importance of 360-degree camera, Vehicle Navigation, Automotive IoT, Benefits of IoT and Artificial Intelligence in vehicles
- 6.4 Explain Tyre pressure monitoring system with a diagram.
- 6.5 State the importance of automotive lift gate, Biometric seat technology, safe exist assist technology, teen Driver Technology, Wireless smart phone connectivity and charging, video rear view mirror and connected car.

COs – POs Mapping

Course Code: A-304		Course Title: AUTOTRONICS		No. of COs:5	No. of Periods:60
POs	Mapped with CO No.	CO Periods Addressing PO in Col. 1		Level (1,2,3)	Remarks
		Number	Percentage		
PO1	CO1,CO2, CO3, CO4, CO5	25	42	3	> 40% Level 3 (Highly Addressed) 25% to 40% level 2 (Moderately Addressed) 5% to 25% level 1 (Low Addressed) <5% Not Addressed
PO2					
PO3	CO4, CO5	5	9	1	
PO4	CO1, CO2, CO3, CO4, CO5	20	33	2	
PO5	CO4, CO5	5	8	1	
PO6					
PO7	CO4, CO5	5	8	1	

CO-PO/PSO MATRIX

CO No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
A-304.1	3			2				2	3	1
A-304.2	3			2				2	3	2
A-304.3	3			3				2	3	2
A-304.4	3		2	3	2		2	2	3	3
A-304.5	3		2	3	2		2	2	3	3
AVERAGE	3		0.8	2.6	0.8		0.8	2	3	2.2

COURSE CONTENT

1.0 Electrical and Electronic Components

Switches - Relays - Solenoids - Resistors –Types - Capacitors-Electronic Components –Conductors-insulators-semiconductors-Diodes- Transistors - Applications - Integrated Circuits - Stepper Motor - Fuses - Circuit Breaker - Automotive wiring - factors that determine the proper size of wire to be used - ground straps -

Terminal and connectors, Printed circuits - wiring harness - wiring protection devices - wiring diagram - electrical symbols - color code –Automotive circuit numbering

2.0 Automotive electrical systems:

Lead Acid Battery– Construction and working - Maintenance free battery - Sodium Sulphur battery - Alkaline Batteries- lithium ion batteries - Super capacitors - Battery terminals - Battery rating and capacity- Battery cables - Starting system - starter motor circuit components - Types of starter motors -Pre-engaged starters - Permanent magnet starters - Integrated starters - Belt driven starter generator –bendix drive- Over running clutch drive - Charging System- Principle of operation - Alternators–water cooled alternators- regulators - Rectification AC to DC- charge indicators- Capacitor Discharge Ignition system (CDI)- computerized ignition system – Distributor less ignition - Coil on plug (COP) ignition - Spark plugs- Types - Construction - Electrode gap and materials.

3.0 Lighting circuits and Accessories

Lamps - Types - Gas Discharge Lamps - Xenon Lighting -Ultra violet Lights - LED lights- Infrared Lights - Head Lights- Head light reflectors - Head Light Lenses – adaptive head lights- Head light Levelling - Exterior Lights - Side - Rear - Brake- Reversing - Day time running light (DRL)- Rear and front fog - Interior Lights - Courtesy lights – reading lights- instrument cluster lights - panel lights- Types of Lighting circuits – Head light - Dim-Dip - Stop light or reverse Light -Circuit diagrams- Power door lock - power windows–Electric Horn – Electromagnetic Windshield wiper - intelligent wipers - Electronic defoggers - automotive rear view mirror.

4.0 Body Computer and instrumentation

Computer functions-Analog&Digital - Microprocessor - Microcontroller – Microcomputer- Programming- Computer Memory - Types - Adaptive Strategy and memory - Logic gates - Sensors - Classification of sensors – Voltage generating sensors – resistive sensors – switch sensors– Sensors used in Automobile - Thermistors - Thermocouples -Inductive Sensors -Hall effect -Strain gauges -Variable capacitance -variable resistance - Accelerometer - Linear variable differential transformer -Hot wire air flow sensor -thin film air flow sensor - vortex flow sensor -Pitot tube - Turbine fluid flow sensor -optical sensor -oxygen sensor -light sensor -thin-film air temperature sensor -Methanol sensor -rain sensor -Oil sensor -dynamic vehicle position sensor- Application of different sensors in automobiles - Actuators - Solenoid - EGR Valve -Motorized -stepper motor-synchronous motor-thermal actuator -Gauges -Thermal - Air Cored - Moving iron - Digital Instrumentation- modern instrument cluster - Head up display - Vehicle condition Monitoring - Trip computer - Global Positioning system (GPS) - Advanced Instrumentation Technology Types -Telematics-Multiplexed displays-Holography – Quantization.

5.0 Vehicle communication Networks and Diagnostics

Multiplexing- Multiplex Communication protocols - Types - Controller Area Network (CAN) - Local Interconnect Network(LIN) - Media Oriented System Transport (MOST) - Connected Cars - Smart Cars & Traffic systems - Wi-Fi Cars – Bluetooth-Oscilloscope- On board diagnostics - OBD I and OBD II -OBD Scanner tool- Diagnostic Equipment-OBD codes

6.0 Modern Vehicle Electronics

Memory seats - Remote key less entry- Keyless start –immobiliser - Antitheft system- Electric mirrors - Electric Sunroof - Obstacle avoidance radar - 360 degree camera - Vehicle Navigation - Tyre pressure monitoring system - Automotive IoT(Internet of Things) - Benefits of IoT - Artificial Intelligence - Automotive lift gate- Biometric Seat technology - safe exist assist technology - teen driver technology- Wireless smart phone connectivity and charging - Emergency Services- video rear view mirror.

REFERENCES

1. Automotive Electrical systems equipment: by N.R. Khatawate - (S.Chand& Co.)
2. Basic Electrical Engineering. : by M.L. Gupta
3. Electrical Technology: by B.L.Theraja
4. Electrical Equipment of Automobiles: by Parker Smith
5. Basic Automobile Electricity: by C.P. Nakra
6. Automobile Electrical Engineering: by Arora&Dass
7. Automobile Engineering: by K.K. Ramalingam
8. Automotive Electricity and Electronics by Barry Hollembeak
9. Automobile Electrical and Electronics systems by Tom Denton

Table specifying the scope of syllabus to be covered for Unit Test-I & Unit Test-II

Unit Test	Learning Outcomes to be covered
Unit Test – I	From 1.1 to 3.8
Unit Test – II	From 4.1 to 6.5

ENGINEERING MATERIALS AND MANUFACTURING PROCESSES

Course Code	Course Title	No. of Periods/Week	Total Periods/Year	Marks for FA	Marks for SA
A-305	Engineering Materials and Manufacturing Process	04	60	20	80

TIME SCHEDULE

Chapter No.	Title	Periods	Marks	Part – A Short Type (3 marks)	Part – B Essay Type (10 Marks)	CO's Mapping
1	Mechanical Properties and Testing of materials	12	26	2	2	CO1
2	Manufacture of Iron & steel and Heat treatment	12	26	2	2	CO2
3	Engineering Materials and their applications	10	16	2	1	CO3
4	Conventional manufacturing processes	16	26	2	2	CO4
5	Mechanical working of metals	10	16	2	1	CO5
Total:		60	110	10	8	

Course Code: A-305	Course Title: Engineering Materials and Manufacturing Processes	No. of Periods: 60
COURSE OBJECTIVES	Upon completion of the course the student shall be able to - Understand the mechanical properties, applications and testing procedure of engineering materials - Apprehend the manufacturing processes of Iron and steel, Iron and Carbon equilibrium diagram and heat treatment processes - Understand the various types of conventional manufacturing processes.	

CO No.	Course Outcome
C01	A-305.1 Explain mechanical properties of materials and their testing procedures
C02	A-305.2 Describe manufacturing of Iron and steel and various heat treatment processes
C03	A-305.3 Describe properties and applications of engineering materials and their alloys
C04	A-305.4 Explain various conventional manufacturing processes and their engineering applications.
C05	A-305.5 Describe the hot working and cold working processes used in manufacturing industries.

LEARNING OUTCOMES:

Upon completion of the course the student shall be able to

- 1. Mechanical properties and testing of materials:**
 - 1.1 State the importance of materials for engineering applications.
 - 1.2 List different properties of engineering materials.
 - 1.3 Define the terms ductility, malleability, hardness, brittleness, toughness Impact strength, Tensile and

compressive strength, stiffness, elasticity, Plasticity, fatigue and creep resistance.

- 1.4 Describe the tensile and compressive tests on mild steel specimen.
- 1.5 Describe the tests to measure impact strength of steel.
- 1.6 Explain the non-destructive testing methods like Dye penetrant, Ultrasonic, Magnetic particle, Radiography (X-ray and Gamma ray).
- 1.7 Explain the hardness test on a given specimen.
- 1.8 State the concept of tribology, wear and corrosion.
- 1.9 State the ASTM standards for mechanical testing.
- 2. Manufacturing methods of iron & steel and Heat treatment**
- 2.1 Describe the manufacturing of cast-iron in cupola furnace.
- 2.2 Explain the manufacture of steel by Bessemer converter, L.D. process, electric arc furnace and electric induction furnace.
- 2.3 State the meaning of space lattice and unit cell.
- 2.4 Explain the three main types of space lattice.
- 2.5 Illustrate the iron–carbon equilibrium diagram.
- 2.6 State the need of heat treatment process.
- 2.7 Explain heat treatment process of annealing, normalizing, hardening, tempering, case-hardening, Nitriding and carburizing
- 2.8 State the appropriate heat treatment methods for a given application.
- 3. Engineering Materials and their applications**
- 3.1 State the composition, properties and application of different types of cast iron and plain carbon steel.
- 3.2 State the advantages of adding alloying elements to the steel and list various alloying elements.
- 3.3 Describe the composition, properties and industrial applications of Stainless steel, HS Sand alloy steels.
- 3.4 Describe the composition, properties and industrial applications of the non-Ferrous alloys like Brass, Bronze, and Gun metal.
- 3.5 State the requirement of lightweight materials in automobiles.
- 3.6 List the lightweight materials used in engineering applications.
- 4. Conventional manufacturing processes**
- 4.1 List types of lathes and lathe operations.
- 4.2 List types of work holding & tool holding devices used in lathe.
- 4.3 Explain the working of turret lathe and capstan lathes.
- 4.4 Explain turret indexing mechanism.
- 4.5 List various operations on drilling machine
- 4.6 Describe the construction and working of radial drilling machines.
- 4.7 State the properties of good moulds.
- 4.8 List different types of moulding sands, patterns moulding processes.
- 4.9 Describe the casting process and defects of casting.
- 4.10 State the various welding processes and their applications.
- 4.11 Explain the working principle of arc welding process with sketch.
- 4.12 Explain the procedure of gas welding with sketch.
- 4.13 Explain the working principle of resistance welding process with sketch.
- 4.14 Differentiate between soldering and brazing
- 5. Mechanical working of metals**
- 5.1 Define hot working and cold working processes.
- 5.2 State the difference between hot and cold working.
- 5.3 List types of cold working and hot working processes.
- 5.4 Explain the working principle of hot rolling, piercing, spinning, Extrusion and drawing process with sketches.
- 5.5 Describe the various types of hot extrusion processes.
- 5.6 State advantages and limitations of hot working.
- 5.7 Explain the various cold working processes such as rolling, bending and squeezing with Sketches.
- 5.8 State advantages and limitations of cold working.

COs – POs Mapping

Course Code: A-305		Course Title: Engineering materials and manufacturing processes		No of Cos:5	No. Of periods:60
POs	Mapped with CO No.	CO Periods Addressing PO in Col 1		Level (1,2,3)	Remarks
		No	%		
PO1	CO1,CO2,CO3,CO4,CO5	20	33	2	>40% Level 3 (Highly Addressed)
PO2	CO1,CO4	09	15	1	25% to 40% Level 2 (Moderately Addressed)
PO3					5% to 25% Level 1 (Low Addressed)
PO4	CO1,CO2	08	13	1	<5% Not Addressed
PO5	CO1,CO2,CO4,CO5	16	27	2	
PO6	CO2,CO4	07	12	1	
PO7					

CO-PO/PSO MATRIX

CO No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
A-305.1	3	2		2	2	2		1	3	
A-305.2	3			1	2	2		2	3	
A-305.3	2							2	3	3
A-305.4	2	2			3	2		2	3	2
A-305.5	3				2			2	3	
AVERAGE	2.6	0.8		0.6	1.8	1.2		1.8	3	1

COURSE CONTENT

1. MECHANICAL PROPERTIES AND TESTING OF MATERIALS

Introduction–Importance of materials in modern Technology-Mechanical properties: Tensile strength, compressive strength, stiffness, elasticity, brittleness, ductility, malleability, hardness, toughness, fatigue resistance, creep, impact strength.

Destructive testing methods: Tensile strength, compressive strength (using UTM), Impact strength (Charpy and Izod tests) and Hardness tests (Brinell, Rockwell and Vicker's).

Non-destructive testing: Dye penetrant, Ultrasonic, Magnetic particle, Radiography (X-ray and Gamma ray) tests.

Tribology in automotive industry: Tribology - wear and corrosion - American Society for Testing Materials (ASTM) standards for mechanical testing.

2. MANUFACTURE OF IRON & STEEL AND HEAT TREATMENT

Manufacture of cast iron – cupola furnace - manufacture of steel – Bessemer converter – L.D. process – electric arc furnace- electric induction furnace.

Space lattice-unit cell- types of metallic space lattices like BCC, FCC and HCP–Iron–carbon equilibrium diagram .

Heat treatment - purpose of heat treatment – types of heat treatment processes: annealing, hardening, tempering, normalizing, case hardening, nitriding and carburizing–applications of these heat treatment processes.

3. ENGINEERING MATERIALS AND THEIR APPLICATIONS

Metallic materials:

Ferrous metals and alloys: Types of cast iron: white, grey, modular, malleable and alloy cast iron – properties and uses of white and grey cast iron-types of plain carbon steel: low carbon, medium and high carbon steel –

alloying elements in steel– composition, properties and uses of alloy steels: stainless steel and high speed steel.

Non-ferrous metals and alloys: Non-Ferrous metals and their alloys in engineering applications– composition-properties and uses of Tin, Zinc, Lead, Nickel, Copper and their alloys: Brass, Bronze, Gun metal– Industrial applications.

Lightweight materials for automotive: Lightweight materials and their metallurgy: aluminum and aluminum alloys, magnesium and magnesium alloys and titanium alloys.

4. **CONVENTIONAL MANUFACTURING PROCESSES**

Lathe machines: Principle of working- specifications -types of lathes - lathe construction and working –work holding devices–Tool holding devices –Working principle of capstan and turret lathes - Principle of turret indexing mechanism.

Drilling: Principle of working- specifications -types of drilling machines - construction and working of Radial drilling machines - drilling operations (Brief explanation): drilling, reaming, boring, counter boring, counter sinking, tapping, spot facing and trepanning.

Casting: Introduction to casting - types of castings, patterns, allowances, moulding sands - gating system - casting of metals and plastics- casting defects and remedies.

Welding: Introduction–types of welding: pressure and non-pressure welding-principle of arc welding, atomic hydrogen, TIG, MIG and submerged arc welding processes –types of welding rods - principle of resistance, spot, seam and butt welding processes – principle of oxy-acetylene gas welding -types of oxy-acetylene gas flames and their uses- soldering and brazing.

5. **Mechanical working of metals**

Definition of hot working and cold working

Hot working processes: rolling - types of rolling, two high mill, three high mills, four high mills, piercing or seamless tubing, drawing or cupping, spinning, extrusion-direct or forward extrusion, indirect or backward extrusion, tube extrusion, Impact extrusion - hot working operations, advantages & limitations.

Cold working processes: Rolling, drawing - wire drawing, tube drawing, bending, stretch forming, angle bending, spinning, extrusion, squeezing, cold heading, thread rolling, peening - advantages & limitations of cold working.

REFERENCE BOOKS

1. Engineering Material and Manufacturing Processes – by V.N.Kumar
2. Engineering Materials-by Ray & Chowdhury
3. Engineering Materials- by R.S.Deshpande
4. Elements of workshop Technology Volume I and II -by Hazra Chowdhary
5. A course in workshop Technology Vol. I & Vol. II- by Raghuvamsi

Table specifying the scope of syllabus to be covered for Unit Test-I & Unit Test-II

Unit Test	Learning Outcomes to be covered
Unit Test – I	From 1.0 to 3.4
Unit Test –II	From 3.5 to 5.8

MACHINE DRAWING

Course Code	Course Title	No. of Periods/Week	Total Periods/Year	Marks for FA	Marks for SA
A-306	MACHINE DRAWING	06	90	40	60

TIME SCHEDULE

Chapter No.	Title	Periods	Marks	Part A	Part B	CO's Mapping
1	Nuts and Bolts	18	15	5	10	CO1
2	Rivets and Riveted joints	21	15	5	10	CO2
3	Keys	12	10	10	-	CO3
4	Assembly Drawing*	39	40	-	40	CO4 & CO5
Total:		90	80	20	60	

Course Code: A-306	Title: MACHINE DRAWING	Total No. Of Periods: 90
Course Objectives:	Upon completion of the course the student shall be able to draw to scale the various views of Machine elements like	
	1.0	Various types of nuts and bolts.
	2.0	Various types of riveted joints & keys and
	3.0	Various types of joints,
	4.0	Couplings&bearings.

CO No.		Course Outcome
C01	A-306.1	Draw to scale various views of different types of nuts and bolts, from the given data or drawing.
C02	A-306.2	Draw to scale various views of different types of rivets & riveted joints from the given data or drawing.
C03	A-306.3	Draw to scale various views of different types of keys and joints- knuckle & cotter joints from the given data or drawing.
C04	A-306.4	Draw to scale various views of different types of couplings from the given data or drawing.
C05	A-306.5	Draw to scale various views of different types of bearings from the given data or drawing.

Learning Objectives:

Upon completion of the course the student shall be able to draw to scale of different views of

1.0 NUTS AND BOLTS

- 1.1 Screw nomenclature – front view.
- 1.2 External and internal threads – front view.
- 1.3 Left hand and right hand threads – front view.
- 1.4 Multiple threads – front view.
- 1.5 Forms of V-threads - B.S.W. threads, B.A. threads, multiple threads, Sellers threads, and metric threads –front view.
- 1.6 Able to draw the forms of square threads, knuckle threads Acme threads, Buttress threads

International metric thread form –front view.

- 1.7 Draw the hexagonal nut, square nut, cap nut, capstan nut, and flanged nut –front view, top view.
- 1.8 Draw the hexagonal headed bolt –front view, side view.
- 1.9 Square headed bolt –front view, side view, Cheese head bolt –front view, side view and
- 1.10 Cup headed bolt –front view, side view.
- 1.11 Hook bolt –front view, side view and Eye bolt –front view, side view.
- 1.12 Foundation bolts-Lewis foundation bolt, eye foundation bolt, Rag foundation bolt – front views.

2.0 Rivets and riveted joints

- 2.1 Draw the various types of rivets: Snap, cup, conical. Round conical, countersunk –frontviews.
- 2.2 Lap joint: single riveted and double riveted Lap joints (Chain and Zig-Zag) - front view, top views.
- 2.3 Butt joint: Single cover and double cover of single riveted and double riveted (Chain and Zig - Zag) Butt joints - front view, top view.

3.0 Keys

- 3.1 Draw the views of different types of Keys. Sunk Taper key –front view, side view.
- 3.2 Saddle keys – flat, hollow keys –front view, side view.
- 3.3 Feather keys and types – peg, single head, and double headed feather keys – front view.
- 3.4 Gib headed key–front view, side view.
- 3.5 Wood ruff key –front view, side view.

4.0 Assembly drawing

- 4.1. Joints: Knuckle joint –sectional front view, top view. Cotter joint –sectional front view, top view, side view.
- 4.2. Couplings: Muff coupling –sectional front view, side view.Split Muff coupling- sectional front view, side view.Oldham’s coupling - sectional front view, side view.Flange coupling-protected type flange coupling- sectional front view, side view.Universal coupling - sectional front view, top view.
- 4.3 Bearings: Solid bearing -front view, top view.Bushed bearings - sectional front view, top view. Pedestal bearings- sectional front view, top view.Footstep bearings - sectional front view, top view.
- 4.4 Simple machines:Screw Jack – Front View and Top View.

CO – PO MAPPING

Course Code: A-306	Course Title: MACHINE DRAWING	No of COs:05			No. Of periods: 90
Pos	Mapped with CO No	CO Periods Addressing PO in Col 1		Level (1,2,3)	Remarks
		No	%		>40% Level 3 (Highly Addressed)
PO1	CO1,CO2,CO3,CO4,CO5	15	17	1	25% to 40% Level 2 (Moderately Addressed)
PO2	CO1,CO2,CO3,CO4,CO5	15	17	1	5% to 25% Level 1 (Low Addressed)
PO3	CO1,CO2,CO3,CO4,CO5	10	11	1	<5% Not Addressed
PO4	CO1,CO2,CO3,CO4,CO5	40	44	3	
PO5	CO1,CO2,CO3,CO4,CO5	10	11	1	
PO6					
PO7					

CO-PO/PSO MATRIX

CO No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
A-306.1	3	2	1	3	2			1	3	
A-306.2	3	1	1	3	2			1	3	
A-306.3	3	2	1	3	3			1	3	
A-306.4	3	1	1	3	3			1	3	
A-306.5	3	1	1	3	3			1	3	
AVERAGE	3	1.4	1	3	2.2			1	3	

*Assembly drawing is compulsory question and its weightage is 20 marks, Short answer questions weightage is each 5 marks, remaining essay questions weightage is each 10 marks. All questions are compulsory.

COURSE CONTENT

1. Nuts and Bolts

Screw nomenclature, external and internal threads, left hand and right hand threads, multiple threads, forms of V-threads, B.S.W. threads, B.A threads, multiple threads, Sellers threads, metric threads and able to draw. Draw the forms of square threads, knuckle threads, Acme threads, Buttress threads International Metric thread form.

Draw the hexagonal nut square nut, cap nut, capstan nut, and flanged nut.

Draw the hexagonal headed bolt square headed bolt, cheese head bolt, cup headed bolt, hook bolt, eye bolt, and foundation bolts-Lewis foundation bolt, eye foundation bolt, Rag foundation bolt.

Draw the collar stud, square neck stud, round neck stud.

2. Rivets and Riveted joints:

Draw the various types of rivets, lap joint: single riveted and double riveted Lap joints (Chain and Zig- Zag), butt joint: Single cover and double cover of single riveted and double riveted (Chain and Zig -Zag) Butt joints.

3. Keys

Draw the views different types of Keys.

4. Assembly drawing

Joints: Knuckle joint, Cotter joint,

Couplings: Muff coupling, split Muff coupling, Oldham's coupling, flange coupling-protected type flange coupling- universal coupling

Bearings: Solid bearing, bushed bearings, pedestal bearings-Foot step bearings.

Simple machines: Screw Jack.

REFERENCE BOOKS

- Machine Drawing by N. D. Bhatt
- Machine Drawing by N. Siddeswar, Kannaiah, Sastri
- Machine Drawing by G.R. Nagpal
- Machine Drawing by R.B. Gupta

A-307 CADLAB – I

Course Code	Course Title	No. of Periods/Week	Total Periods/Year	Marks for FA	Marks for SA
A-307	CAD LAB -I	03	45	40	60

TIME SCHEDULE

Chapter No.	Title	Periods	CO's Mapping
1.	Introduction to Computer Aided Drafting	6	CO1
2.	Computer Aided Drafting Interface	9	CO2
3.	Computer Aided Drafting–2D	15	CO3
4.	Practice 2D drawings, Isometric View drawings and Plotting	15	CO4
TOTAL		45	

CourseCode:A-307	Course Name: CAD Lab-I	No.OfPeriods:45
Course Objectives	Upon completion of the course the student shall be able to 1. Understand basics of computer aided drafting. 2. Understand working in CAD environment. 3. Prepare drawings of standard components using CAD software in 2D.	

CO No.		Course Outcome
C01	A-307.1	Explain Computer Aided Drafting and its importance.
C02	A-307.2	Practice usage of Computer Aided Drafting Interface and its various tools
C03	A-307.3	Practice1.Creatingobjectscommands,2.Modifyoreeditingcommands, 3. Coordinatesystem,4.Textanddimensioning.
C04	A-307.4	Create2D drawings using CAD software

Learning Outcome**1.0 IntroductiontoComputerAidedDrafting**

- 1.1. Define Computer Aided Drafting
- 1.2. List the Advantages of CAD
- 1.3. List the CAD software's used for drafting.
- 1.4. Explain the importance of CAD software
- 1.5. Explain the features of Graphic Work station
- 1.6. State the system requirements of CAD software.

2.0 Computer Aided Drafting Interface

- 2.1 Explain CAD Environment 1. Screen, 2. Various tool bars, 3. Command line, 4. Drop down menus, 5. Ribbon tabs and 6. Icons
- 2.2 Practicerepeatingacommands,undoingandredoingaction
- 2.3 Practice mouse functions

- 2.4 Practice the Creating drawing, Opening, existing Files and saving the drawing
- 2.5 Practice the setting limits and drawing unit, Standard workspace settings
- 2.6 Practice selection methods – crossing and window, drawing accuracy methods – Ortho, Osnap, Snap and grid and use of grips
- 2.7 PracticetheuseofZoom,Scrollbar,pancommand,androtatingviewtomove aroundwithin drawing
- 2.8 Practicethedisplayingofmultipleviews,changingline-weight.
- 3.0 Computer Aided Drafting – 2D**
- 3.1** Drawthelines,circles,arcs,ellipses,ellipticalarcs,raysandinfinitelinesandshapes like rectangles, polygons, polylines, Splines, adding of hatch pattern
- 3.2** Practice how the coordinate system work, Two dimensional coordinates such as Absolute Cartesian,Relative CartesianandPolarcoordinates, useofright-handrule and Practice the how to enter into x, y, z – coordinates, defining user coordinate system
- 3.3** Calculatetheareas,distanceandangleanddisplaytheinformationaboutthe entities and drawing status
- 3.4** Practicetheentityselectionandde-selectionmethods,Deletion,Copyingofentities within a drawing and between drawings and making of parallel copies, Mirroring entitiesandarrayingentities.
- 3.5 Practice the Rearranging of entities by Moving, Rotating and Reordering and Resizing of entities by Stretching, Scaling, Extending, and Trimming, Editing the length, Braking and joining of entities.
- 3.6 Practicethecreating,modifyingthegroupsandungroupingofEntitiesandExploding of entities, Chamfering and Filleting of entities
- 3.7 Practicethesettingacurrentlayer,layerscolour,linetype, lineweight,print Style, locking and unlocking of layers and setting of current line type
- 3.8 Explain the purpose of a block, creating, inserting, redefining and exploding a block
- 3.9 Practice the creating of linear, Angular, Diameter, Radial, Ordinate Dimensions and creating leaders and annotations, dimensions oblique, Edit the dimension text, Controlling of dimension arrows and format.
- 3.10 Practice the creating, naming and modifying the text fonts, Creation of line text, paragraph text, Setting of line text style and its alignment and Practice the Setting of Paragraph text style and its alignment
- 4.0 Practice2Ddrawings,IsometricandPlotting**
- 4.1 Create2Ddrawingsofstandardmechanicalcomponents
- 4.2 CreateIsometricviewsofsimpleobjects
- 4.3 Understandthelayouts,viewingofdrawingsinpaperandmodel space
- 4.4 Displaythemodelandlayouttabs,newlayout,layoutviewports
- 4.5 Setthepapersizeandorientation
- 4.6 Selectaprinteroraplotter

Course Code: A-307	Course Title: CAD Lab – I	No. of Cos: 5			No. Of periods: 45
POs	Mapped with CO No	CO Periods Addressing PO in Col 1		Level (1,2,3)	Remarks
		No	%		
PO1	CO1,CO2,CO3,CO4	05	12	1	>40%Level3 (Highly Addressed) 25%to40%Level2 (ModeratelyAddressed) 5%to25%Level1 (LowAddressed) <5%Not Addressed
PO2					
PO3	CO3,CO4	10	22	1	
PO4	CO3,CO4	15	33	2	
PO5	CO2,CO3,CO4	15	33	2	
PO6					
PO7					

CO-PO/PSOMATRIX

CO No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
A-307.1	1									
A-307.2	1				2					
A-307.3	1		1	2	2			1		
A-307.4	1		1	2	2			2		
Average	1		0.5	1	1.5			0.75		

Note: The gaps in CO and PO mapping will be achieved by one or more appropriate activities from the following:

- i) Seminars ii) Tutorials iii) Guest Lectures iv) Assignments v) Quiz competitions
- vi) Industrial visit vii) Tech Fest viii) Mini project ix) Group discussions
- x) Virtual classes xi) Library visit for e-books

COURSE CONTENT

S.No	Experiment Title	Key Competency
1.	Introduction to Computer Aided Drafting	• Computer Aided Drafting, Advantages of CAD, CAD software's • Importance of CAD, system requirements of CAD software.
2.	Computer Aided Drafting Interface	• CAD Environment: Screen, Various toolbars, command line, drop down menus, ribbon tabs and icons, repeating a command, undoing and redoing action, mouse functions • Creating the drawing, Opening existing Files, saving the drawing- setting limits and drawing unit, Standard workspace settings • Selection methods – crossing and window, • Using Drawing accuracy methods – Ortho, Osnap, Snap and grid and use of grips Practice viewing commands
3	Computer Aided Drafting – 2D	• Practice Draw commands • Practice how the coordinate system work, Two dimensional coordinates such as Absolute Cartesian, Relative Cartesian and Polar coordinates, use of right-hand rule, how to enter into x, y, z – coordinates, defining user coordinate system • Practice Inquiry commands • Practice Modify/Edit commands • Practice layers and block commands. • Practice Dimensioning commands. • Practice text and Mtext • Draw simple geometrical shapes like circles, rectangles etc.,
4	Practice 2D drawings, Isometric drawings and Plotting	• Create 2D drawing of standard mechanical components • Create isometric view of simple objects • Practice layouts, paper size and orientation and plotting

REFERENCE BOOKS

1. Mastering AutoCAD 2017 by George Omura, Brain C. Benton
2. Up and Running with AutoCAD 2017 2D and 3D by Elliot J. Gindis.
3. AutoCAD 2020 Tutorial First Level 2D Fundamentals by Luke Jumper, Randy H. Shih

AUTOMOBILE LABORATORY-II

Course Code	Course Title	No. of Periods/Week	Total Periods/Year	Marks for FA	Marks for SA
A-308	AUTOMOBILE LABORATORY - II	03	45	40	60

TIME SCHEDULE

Chapter No.	Title	Periods	CO's Mapping
1.	Identification of tools, equipment & gauges and Dismantling cleaning and assembling of engine Fuel filters, oil filters, AC filters	15	CO1
2.	Dismantling and assembling of mechanical fuel pump, water feed pump, single and multi-plate clutch, constant and synchromesh gear box and differential, transaxle	18	CO2
3.	Demonstration of working procedure of fuel supply, cooling, lubrication and intake and exhaust system of an engine with the aid of simulator	6	CO3
4.	Demonstrate the working procedure of MPFI and CRDI with the aid of circuitry system	6	CO4
5.	Demonstration of driving skills(<i>One batch will go for driving practice and another batch will do the</i>		CO5
Total:		45	

Course Code: A-308	Course Title: AUTOMOBILE LABORATORY -II	No. of Periods: 45
COURSE OBJECTIVES	Upon completion of the course the student shall able to - Understand the dismantling of engine, Fuel filters, oil filters, AC filters, identify the components and assemble the engine components - Know the working procedure of fuel supply, cooling, lubrication and intake and exhaust system of an engine with the aid of simulator - Understand the dismantling and assembling of Mechanical and Electrical fuel pumps, clutch, gear box and differential also know the driving skills	

CO No.		Course Outcome
CO1	A-308.1	Perform the dismantling and cleaning of an Engine, Fuel filters, oil filters, AC filters and identify the components and assemble the engine and filters.
CO2	A-308.2	Demonstrate the working procedure of fuel supply, cooling, lubrication and intake and exhaust system of an engine with the aid of simulator
CO3	A-308.3	Demonstrate the working procedure of MPFI and CRDI with the aid of circuitry system
CO4	A-308.4	Perform the dismantling and assembling of mechanical & electrical fuel pump, single plate clutch, multi plate clutch (wet), constant mesh, synchromesh gear box differential and transaxle
CO5	A-308.5	Demonstrate driving skills on four wheeler vehicle

LEARNING OUTCOMES:

Upon completion of the course the student shall able to

LIST OF EXPERIMENTS

1. Identification of Tools, equipment's and gauges like Torque wrench, angular torque wrench, air pressure gauge, compression gauge and oil pressure gauges, air compressor, power tools etc. used in automobile labs
2. Dismantle a given I.C. Engine and identify the various basic components such as Cylinder, Cylinder block, Cylinder head, Piston, Piston rings, Connecting rod, Crank shaft, Valves, Valve actuating mechanism, Timing gears, Crankcase, Spark plug, Fuel pump, Fuel Injector, Fuel Injection pump, water pump
3. Dismantle and clean Fuel filters, oil filters, AC filters
4. Demonstrate the fuel system of a multi cylinder petrol and diesel engines with the aid of simulator
5. Demonstrate the Inlet and Exhaust system of a multi cylinder engine with the aid of a simulator
6. Demonstrate the lubrication system of a multi cylinder engine with the aid of a simulator
7. Demonstrate the water cooling system of a multi cylinder engine with the aid of a simulator
8. Demonstrate the Multi Point Fuel Injection System (MPFI) on circuitry system
9. Demonstrate the Common Rail Direct Injection System (CRDI) on circuitry system
10. Dismantling and Assembling of A.C. Mechanical and electrical fuel pump.
11. Dismantling and Assembling of a single plate clutch, multi plate clutch (wet)
12. Dismantling and Assembling of a gear box (Constant mesh & synchromesh)
13. Dismantling and Assembling of a Differential
14. Demonstrate the driving skills on four wheeler vehicle/driving simulator

COs – POs Mapping

Course Code: A-308		Course Title: Automobile Laboratory			No of Cos:5	No. Of periods:45
POs	Mapped with CO No	CO Periods Addressing PO in Col 1		Level (1,2,3)	Remarks	
		No	%			
PO1	CO1,CO2,CO3,CO4	10	22	1	>40% Level 3 (Highly Addressed)	
PO2	CO2,CO3,CO4	10	22	1	25% to 40% Level 2 (Moderately Addressed)	
PO3					5% to 25% Level 1 (Low Addressed)	
PO4	CO1, CO2,CO3, CO4,CO5	5	11	1	<5% Not Addressed	
PO5	CO1, CO2,CO3, CO4,CO5	15	34	2		
PO6	CO2,CO3,CO4	5	11	1		
PO7						

CO-PO/PSO MATRIX

CO No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
A-308.1	3			3	1			3		3
A-308.2	3	2		3	3	3		3		3
A-308.3	3	2		3	3	3		3		2
A-308.4	3	2		2	2	3		2		3
A-308.5				3	3			3		3
AVERAGE	2.4	1.2		2.8	2.4	1.8		2.8		2.8

The competencies and key competencies to be achieved by the student

Sl.no	Title of the experiment	Key competencies	Competencies
1	Identification of Tools, equipment's and gauges	Torque wrench, angular torque wrench, air pressure gauge, compression gauge and oil pressure gauges, air compressor, power tools etc. used in automobile labs	1. Identify proper tools, equipment's and gauges used in automobile lab 2. Identify proper tool for the given job
2	Engine Dismantling and assembling	1) Dismantle the Engine from the Vehicle. 2) Disassemble the engine 3) Report on the condition 4) Assembling the Engine 5) Check the Running Condition of the engine.	1) Identify the tools required for Dismantling. 2) Use of appropriate tools dismantle the engine parts and keep the parts without damage in order. 3) Clean all the parts with appropriate tool / medium. 4) Assemble all the parts in order. 5) Check the running condition of the engine.

3	Fuel System of a Multi Cylinder Engine.	1) Identify the components of a Fuel system. 2) Know the function of each component in the system	1) Draw a line diagram of a fuel System of a given Multi Cylinder Engine
4	Inlet and Exhaust System of a Multi Cylinder Engine.	1) Understand the need of inlet and exhaust system. 2) Know the function of each component in the system	1) Draw a line diagram of Inlet and Exhaust System of a given Multi Cylinder Engine
5	Lubrication System of a Multi Cylinder Engine	1) Identify the components of a Lubricating system. 2) Know the function of each component in the system	1) Draw a line diagram of a Lubrication System of a given Multi Cylinder Engine
6	Cooling System of a Multi Cylinder Engine	1) Identify the components of a Cooling system. 2) Know the function of each component in the system	1) Draw a line diagram of a Cooling System of a given Multi Cylinder Engine
7	Multi Point Fuel Injection System	1) Identify the components of a Cooling system. 2) Know the function of each component in the system	1) Draw a line diagram of Multi Point Fuel Injection System
8	Common Rail Direct Injection System	1) Identify the components of a Cooling system. 2) Know the function of each component in the system	1) Draw a line diagram of Common Rail Direct Injection System

9	AC Mechanical and Electrical fuel Pumps	1) Dismantle the component from the Vehicle. 2) Disassemble the component 3) Report on the condition 4) Assembling the component 5) Check the Running Condition of the component	1) Identify the tools required for Dismantling the component. 2) Use of appropriate tools to dismantle the component 3) Keep the dismantled parts without any damage in order 4) Clean all the parts with appropriate tool / medium 5) Assemble the component 6) Check and bring to the operating condition
10	Fuel filters, Lub-oil filters, AC filters		
11	Single plate and Multi plate clutch		
12	Constant mesh and synchromesh Gear box		
13	Differential		
14	Driving Practice	1) Know the foot controls of the vehicle 2) Know the hand controls of the vehicle 3) Familiarity of all Driving Controls 4) Operate all the above Controls simultaneously whenever needed. 5) To Know the various Road signs and signals used while driving a vehicle. 6) Legal requirements to	1) Identify the Foot Controls of a four Wheeler i.e A,B,C 2) State the Purpose of Foot Controls. 3) Operate the foot controls efficiently 4) Operate the steering controls efficiently 5) Operate to change of gear by gear Shift lever 6) Identify the various accessories Like Horn, Lights and Indicators. 7) Operate all the above Controls Simultaneously.

WORKSHOP PRACTICE – II

Course Code	Course Title	No. of Periods/Week	Total Periods/Year	Marks for FA	Marks for SA
A-309	WORKSHOP PRACTICE-II	03	45	40	60

TIME SCHEDULE

Chapter No.	Title	Periods	CO's Mapping
	Safety precautions(Industrial safety)	03	CO1
1.	Foundry	06	CO1
2.	Forging shop	06	CO2
3.	Machine Shop (Turning)	12	CO3
4.	Welding Practice	09	CO4
5.	Special Machine Shop	09	CO5
Total		45	

Course Code: A-309	Title: Workshop Practice- II	Total No. Of Periods: 45
Course Objectives:	Upon completion of the course the student shall be able to To perform foundry /casting processes To perform forging operations Perform various operations on lathe Practice the arc welding Practice on special machines	

CO No.		Course Outcome
CO1	A-309.1	Prepare a casting of required material
CO2	A-309.2	Producing forgings of different shapes
CO3	A-309.3	Reproduce an object as per model by performing on or more of the following Operations viz., plain turning, step turning, taper turning, knurling and thread cutting.
CO4	A-309.4	Produce a lap joint, a butt joint and a T – joint by fastening the plates with arc Welding.
CO5	A-309.5	Prepare a component by using shaping machine and milling machine Machine.

Upon the completion of the course the student shall be able to

1. Perform Foundry operation:
 - 1.1 Identify the hand tools and equipment used in foundry.

- 1.2 Prepare pattern.
- 1.3 Prepare the moulds using single piece or split patterns.
- 1.4 Prepare simple castings.
2. Perform Forging operations:
 - 2.1 Identify the various tools and equipment used in forging.
 - 2.2 Conversion of Round to Square
 - a) Identify the holding and striking tools
 - b) Heat the specimen to the appropriate temperature
 - c) Remove the specimen and hold it on the anvil
 - d) Hammer the specimen to the required shape
 - 2.3. Preparation of a chisel from round rod
 - a) Identify the holding and striking tools
 - b) Heat the specimen to the appropriate temperature
 - c) Remove the specimen and hold it on the anvil
 - d) Hammer the specimen to the required shape
 - 2.4 Preparation of ring and hook from M.S. round
 - a) Identify the holding and striking tools
 - b) Heat the specimen to the appropriate temperature
 - c) Remove the specimen and hold it on the anvil
 - d) Hammer the specimen to the required shape
3. Perform various operations on the Lathe
 - 3.1 Perform a plain turning operation on a lathe machine.
 - a) Select proper tool to perform the job.
 - b) Centring the job by dial gauge
 - c) Select the suitable speed for different operations
 - d) Make use of various measuring instruments for taking dimensions.
 - 3.2 Perform step turning operation on lathe.
 - a) Calculate the taper angle.
 - b) Practice the different taper turning methods on lathe
 - c) Turn the required tapers by swivelling the compound rest.
 - d) Produce articles of industrial application such as ring gauges, plug gauges, handle etc.
4. Practice the Arc Welding
 - a) Prepare the edges for welding
 - b) Select the suitable electrode, voltage and current
 - c) Handle the Electrode Holder for laying welding beads.
 - d) Understand the operation of welding transformer and generator.
 - e) Perform various weld joint operations.
5. Hands on practice on Special Machines
 - 5.1 Perform the operation of groove cutting by using Shaper.
 - 5.2 Perform the operation of gear cutting by using Milling Machine.

COURSE CONTENT

1. Foundry:

1. Foundry hand tools and equipment.
2. Preparation of moulds using two moulding boxes with patterns.
3. Preparation one casting (metal or wax).
1. Forging:
 1. Identification of forging tools:
 - Hand forging tools: Anvil, swage block, hand hammers - types; sledgehammer, specifications

and uses, tongs - types, specifications & uses, chisel -hot & cold chisels specifications & uses. Swages - types and sizes, fullers, flatters, punch and drift - sizes and uses.i

- Equipment: Forge-parts-Openandclosed hearth, heating furnaces,openand stock fire, fuels-charcoal, coal, oil and gaseous fuels-characteristicsof fuel.

1. Conversion of round to square.
2. Preparation of chisel from round rod.
3. Preparation of ring and hook from M.S.round.

3. Machine Shop (Turning)

1. Facing
2. Plain Turning
3. Step Turning
4. Taper Turning
5. Knurling

4. Welding

1. Practice of Handling of Electrode
2. Butt joints.
3. Lap joints.
4. T-Joint.

5. Special Machines

1. Shaping (Preparation of Groove)
2. Milling (Gear Cutting Operation)

COs – POs Mapping

Course Code: A-309		Course Title: Workshop Practice - II		No. of COs: 5	No. of Periods: 45
POs	Mapped with CO No.	CO Periods Addressing PO in Col. 1		Level (1,2,3)	Remarks
		Number	Percentage		
PO1	CO1,CO2	5	13	1	> 40% Level 3(Highly Addressed) 25% to 40% level 2 (Moderately Addressed) 5% to 25% level 1 (Low Addressed) <5% Not Addressed
PO2	CO1,CO2	5	11	1	
PO3	CO1,CO2	7	16	1	
PO4	CO1,CO2,CO3,CO4,CO5	6	13	1	
PO5	CO1,CO2,CO3,CO4,CO5	6	13	1	
PO6	CO1,CO2	8	18	1	
PO7	CO3,CO4,CO5	7	16	1	

CO-PO/PSO MATRIX

CO No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
A-309.1	3	1	2	3	3	3		3	1	3
A-309.2	3	1	2	3	3	3		3	1	3
A-309.3				3	3		3	3		3
A-309.4				3	3		3	3		3
A-309.5				3	3		3	3		3
AVERAGE	3	1	2	3	3	3	3	3	1	3

The competencies and key competencies to be achieved by the student

Title of the experiment	Key competency
Pattern Making	• Selecting pattern material • Marking dimensions on pattern material considering allowances • Cutting off extra material to get required pattern
Mould preparation	• Placing of pattern in drag box • Filling drag box with sand properly • Placing cope box on drag box • Placing Runner and Riser appropriately and filling cope box with sand • Removing Pattern, Runner and Riser
Casing Preparation	• Preparing mould of required shape and size • Gate cutting • Filling mould with molten material • Allowing it to cool to room temperature
Plain turning	• Centering of the work piece using dial gauge • Fixing the cutting tool at proper inclination • Selecting the suitable speed, feed and depth of cut for rough and finishing operations • Checking the dimensions
Step turning	• Centering of the work piece using dial gauge • Fixing the cutting tool at proper inclination • Selecting the suitable speed, feed and depth of cut for rough and finishing operations • Checking the dimensions
Taper turning	• Centering of the work piece using dial gauge • Fixing the cutting tool at proper inclination • Selecting the suitable speed, feed and depth of cut for rough and finishing operations • Checking the dimensions • Rotating the compound rest to the suitable angle
Knurling	• Centering of the work piece using dial gauge • Fixing the cutting tool at proper inclination • Selecting the suitable speed, feed and depth of cut for rough and finishing operations • Checking the dimensions • Fixing the knurling tool and selecting the suitable speed and feed
Welding	• Edge preparation • Holding the electrode at suitable angle and distance with respect to the work piece to maintain the arc • Checking the bead
Lap joint	• Edge preparation • Holding the electrode at suitable angle and distance with respect to the work piece to maintain the arc • Checking the bead

Butt joint	• Edge preparation • Holding the electrode at suitable angle and distance with respect to the work piece to maintain the arc • Checking the bead
T-Joint	• Edge preparation • Holding the electrode at suitable angle and distance with respect to the work piece to maintain the arc • Checking the bead
Shaping	• Prepare the work surface • Holding the component on a T slotted table fixed to the vice and check the distance with respect to the work piece to maintain tool clearance • Shape the component on given sides
Milling	• Prepare the work surface • Position the given milling cutter • Hold the work piece at suitable position and maintain the distance with respect to the milling cutter to attain milling operation • Clean the machine before and after use

REFERENCEBOOKS

1. ManufacturingTechnology(VolI) by P NRao (McGrawHill)
2. PrinciplesofFoundryTechnology by P LJain(McGrawHill)
3. WorkshopPractice Vol-IbyHajraChoudhuryMediaPromotersandPublishersPvtLtd.

AUTOTRONICS LABORATORY

Course Code	Course Title	No. of Periods/Week	Total Periods/Year	Marks for FA	Marks for SA
A-310	AUTOTRONICS LABORATORY	03	45	40	60

TIME SCHEDULE

Chapter No.	Title	Periods	CO's Mapping
1.	Identification of various components of electrical and electronic systems used in automobile	3	CO1
2.	Measure Voltage, Current, Resistance, check continuity in wiring harness by using Multimeter	3	CO2 & CO3
3.	Preparation of Electrolyte for Lead Acid battery	3	CO2
4.	Testing of Lead acid battery	3	CO2
5.	Diagnosis of Ignition System	6	CO3
6.	Testing, dismantling and assembling of starter motor and alternator used in automobile	9	CO3
7.	Diagnose the wiring of Head light, wiper motor Horn and wiring harness	6	CO3
8.	Testing of Vehicle Sensors	6	CO3
9.	Diagnosing the two wheeler using OBD Scanner	3	CO5
10.	Study of various features of vehicle electronic systems	3	CO4
Total		45	

Course Code: A-310	Title: Autotronics Laboratory	Total No. Of Periods: 45
Course Objectives:	Upon completion of the course the student shall be able to	
	1.0	Familiarize with Automobile Electrical and Electronic systems
	2.0	Handle the tools and equipment's appropriately and safely
	3.0	To reinforce theoretical concepts by practising relevant exercises

CO No.		Course Outcome
CO1	A-310.1	Identify various components used in automobiles
CO2	A-310.2	Service of lead acid batteries and understand the construction of other types of batteries.
CO3	A-310.3	Diagnose Ignition System, starting system, Charging System, Electrical and electronic accessories
CO4	A-310.4	Study of various features of vehicle electronics
CO5	A-310.5	Use diagnostic equipment for finding the root cause of the problem in a vehicle and rectifying the problem

Learning Objectives:

1. Identification of various components of electrical and electronic systems used in automobile.
2. Measure Voltage, Current, Resistance, check continuity in wiring harness by using Multimeter
3. Preparation of Electrolyte for Lead Acid battery
4. Testing of Lead acid battery
5. Diagnosis of Ignition System
6. Testing, dismantling and assembling of starter motor and alternator used in automobile.
7. Diagnose the wiring of Head light, wiper motor and Horn
8. Testing of Vehicle Sensors
9. Diagnosing the two wheeler using OBD Scanner
10. Study of various features of vehicle electronics

COs – POs Mapping

Course Code: A-310		Course Title: Autotronics laboratory		No. of COs: 4	No. of Periods: 45
POs	Mapped with CO No.	CO Periods Addressing PO in Col. 1		Level (1,2,3)	Remarks
		Number	Percentage		
PO1	CO1, CO2,CO3,CO4,CO5	20	44	3	> 40% Level 3 (Highly Addressed) 25% to 40% level 2 (Moderately Addressed) 5% to 25% level 1 (Low Addressed) <5% Not Addressed
PO2					
PO3					
PO4	CO2, CO3,CO4	15	34	2	
PO5	CO4,CO5	5	11	1	
PO6					
PO7	CO4,CO5	5	11	1	

CO-PO/PSO MATRIX

CO No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
A-310.1	3							1	1	1
A-310.2	3			3				2	2	
A-310.3	3			3				3	3	1
A-310.4	3			2	3		2	1	1	3
A-310.5	3				3		3	1	3	3
AVERAGE	3	0	0	1.6	1.2	0	1	1.6	2	1.6

Note: The gaps in CO and PO mapping will be achieved by one or more appropriate activities from the following:

- | | | | |
|-------------------------------|----------------------|-----------------------|-----------------|
| i) Seminars | ii) Tutorials | iii) Guest Lectures | iv) Assignments |
| v) Quiz competitions | vi) Industrial visit | | |
| vii) Tech Fest | viii) Mini project | ix) Group discussions | x) |
| Virtual classes | | | |
| xi) Library visit for e-books | | | |

Key competencies to be achieved by the students:

Exp No.	Name of the Experiment	Key Competencies
1.	Identification of various components of electrical and electronic systems used in automobile.	Comprehend the commonly used electrical and electronic components of an automobile 1. Starter Motor 2. Ignition System 3. Alternator 4. Relays 5. Switches 6. Fuses 7. Wiper Motor 8. Instrument Panel 9. Wiring Harness 10. Sensors 11. Battery 12. Combination Switch
2	Measure Voltage, Current, Resistance, check continuity in wiring harness by using Multimeter	1. Practice measuring Voltage, Current, Resistance by using multimeter 2. Identify different wiring (including colours) Connection of main wiring harness of an automobile and test its continuity using multimeter.
3	Preparation of Electrolyte for Lead – Acid battery	a. Prepare electrolyte for Lead – Acid battery by mixing proper ratio of acid and distilled water. b. Check the specific gravity of electrolyte.
4	Testing of Lead acid battery	a. Inspect battery condition b. Conduct open circuit voltage test c. Hydrometer test to get specific gravity of the electrolyte d. Battery load test
5.	Diagnosis of Ignition System	a. Check primary Circuit b. Secondary Circuit c. Spark Plug testing
6.	Testing, dismantling and assembling of starter motor used in automobile.	a. Familiarize with the parts of starter motor. b. Test the starting motor for continuity, faults, if any c. Conduct no load test on starter motor d. Step by step dismantling of starting motor e. Chronological order to be followed for assembling the starter motor.
	Testing, dismantling and assembling of alternator used in automobile.	a. Step by step dismantling of alternator. b. Chronological order to be followed for assembling the alternator. c. Check Alternator performance characteristics. d. Test Alternator output e. Check Cut – in and Cut – out voltage of alternator
7	Diagnose the wiring of Head light, wiper motor, Horn and wiring harness	a. Check the fuse in fuse box and replace b. Check the connection to the battery c. Check the brightness of the head lamps and horn d. Check the alignment of head lamps
8	Testing of Vehicle sensors	a. Coolant Temperature Sensor b. O₂ Sensor

		c. Throttle position sensor d. Manifold Absolute Pressure sensor e. Crankshaft Sensor f. Camshaft Sensor
9	Diagnosing two wheeler using OBD Scanner	1. Connecting of OBD Scanner 2. Checking different parameters received from the vehicle ECU 3. Checking Diagnostic Trouble Codes (DTC's) 4. Finding the fault and rectifying the problem 5. Deleting the DTC's
10	Study of various features of vehicle electronic systems	1. Memory Seats 2. Remote Keyless Entry 3. Electric Sunroof 4. Advanced Driver Assistance Systems 5. 360° Camera 6. Artificial Intelligence 7. Internet of Things 8. Tyre Pressure Monitoring System 9. Automotive Lift gate 10. Wireless Smart Phone connectivity

IV SEMESTER

DIPLOMA IN AUTOMOBILE ENGINEERING
SCHEME OF INSTRUCTIONS AND EXAMINATIONS
IV SEMESTER

Subject Code	Name of the Subject	Instruction period/week		Total Periods /Semester	Scheme of Examination			
		Theor y	Practical /Tutorial		Durat ion (hour s)	Sessional Marks	End Exam Mark s	Total Mark s
THEORY								
A-401	Two and Three Wheeler Technology	4		60	3	20	80	100
A-402	Strength of Materials And Fluid Mechanics	5		75	3	20	80	100
A-403	Automobile Chassis and Body Engineering	4		60	3	20	80	100
A-404	Vehicle Service and Maintenance Technology	4		60	3	20	80	100
A-405	Farm Tractors and Earth Movers	4		60	3	20	80	100
PRACTICAL								
A-406	Automobile Engineering Drawing	-	6	90	3	40	60	100
A-407	Vehicle Service and Maintenance Technology Lab	-	3	45	3	40	60	100
A-408	Communication Skills	-	3	45	3	40	60	100
A-409	Hydraulic and Pneumatic Controls Lab	-	3	45	3	40	60	100
A-410	CAD Lab-II	-	3	45	3	40	60	100
Student Centric Activity (Library, Mini project, Physical Education, Preparation for competitive exams/placements, Seminars, Reporting etc)			3	45				
TOTAL		21	21	315+315	-	300	700	1000

A-408 Common with all branches

TWO AND THREE-WHEELER TECHNOLOGY

Course Code	Course Title	No. of Periods/Week	Total Periods/Year	Marks for FA	Marks for SA
A-401	Two and Three Wheeler Technology	04	60	20	80

TIME SCHEDULE

Chapter No.	Title	Periods	Marks	Part – A Short Type (3 marks)	Part – B Essay Type (10 Marks)	CO's Mapping
1	Introduction and Power Unit	08	16	2	1	CO1
2	Fuel, Lubrication and Electrical Systems	14	26	2	2	CO2
3	Two-Wheeler Chassis and Sub-Systems	20	36	2	3	CO3
4	Three-Wheeler Chassis and Sub-Systems	10	16	2	1	CO4
5	Electric Two and Three-wheeler	08	16	2	1	CO5
Total		60	110	10	8	

Course Code : A-401	Course Name : Two and Three Wheeler Technology No. of Periods : 60
Course Objectives	Upon completion of the course the student shall be able to - Understand the need and working of Power Unit, Fuel, Lubrication and Electrical Systems of two and three wheelers - Understand the function and working of Two and Three-wheeler chassis and their sub systems. - Appreciate Electric Two and Three-wheeler technology

CO No.		Course Outcome
CO1	A-401.1	Explain the history and Power unit of Two and Three-wheeler.
CO2	A-401.2	Explain construction and working principles of Fuel, Lubrication and Electrical Systems of Two and Three-wheeler
CO3	A-401.3	Explain Two-wheeler chassis and their sub systems.
CO4	A-401.4	Explain Three-wheeler chassis and their sub systems.
CO5	A-401.5	Describe Electric Two and Three-wheeler vehicles.

Learning Outcomes:

1.0 Introduction and Power Unit

- 1.1 State the classification of Two and Three wheelers
- 1.2 List the basic systems in Two-wheeler vehicles
- 1.3 List out the arrangement of cylinders used in two wheelers.
- 1.4 Explain the layout of Motor Cycle, Scooter, Cargo Auto Rickshaw and Passenger Auto Rickshaw with a sketch.
- 1.5 Explain the engine selection criteria for two-wheeler and three-wheeler
- 1.6 List out the types of cooling systems used in two wheelers.

- 1.7 Explain the Water-cooled engine used in two wheelers.
- 1.8 Explain the Oil cooled engine used in two wheelers.
- 1.9 List two-wheeler exhaust system pipe and header types.

2.0 Fuel, Lubrication and Electrical Systems

- 2.1 Explain the layout of fuel injection system of two-wheeler with a sketch.
- 2.2 List different sensors used in two wheelers.
- 2.3 State the purpose of ECU used in two wheelers.
- 2.4 List the types of lubrication systems used in two wheelers.
- 2.5 Explain electronic ignition system of two wheeler with a sketch.
- 2.6 List the types of starting system of two wheelers
- 2.7 Explain Kick start mechanism with a sketch.
- 2.8 Explain Auto start mechanism with a sketch.
- 2.9 State the battery ratings of two and three wheelers.
- 2.10 Explain handle bar controls and instrument panel of Two wheelers
- 2.11 Explain Side stand / Ignition Interlock systems of two wheelers
- 2.12 Explain idle-start- Stop system(i3S technology of Hero two wheelers) used in two Wheelers.
- 2.13 Explain multiple sensor type (XSENS technology used in Hero two wheelers) engine management system.
- 2.14 Explain Digital twin spark ignition system (DTSi technology used in Bajaj two wheelers).

3.0 Two Wheeler Chassis and Sub-Systems

- 3.1 Explain single cradle, double cradle, tubular, twin spar, monocoque frames with sketch.
- 3.2 Explain Layout of transmission system in two wheelers with a sketch.
- 3.3 List types of primary drive and clutches in two wheelers.
- 3.4 Explain the working principle of Assist slipper clutch with a sketch.
- 3.5 List the types of gear box used in two wheelers.
- 3.6 Explain Ball Lock type gear locking mechanism and Turn-on type gear locking mechanism of Constant Mesh Gear box with sketch.
- 3.7 Explain sequential gear box with a neat sketch.
- 3.8 Explain Hand operated and Foot operated gear shifting mechanism with sketch.
- 3.9 Explain CVT of two wheelers.
- 3.10 Describe belt drive, chain drive, shaft drive and cush drive of final drive.
- 3.11 Explain steering column and Handle bar construction with a sketch.
- 3.12 List types of front and rear suspension.
- 3.13 Explain telescopic, single link and double link suspension with a sketch
- 3.14 Explain Hard tail type and Swing arm type rear suspension with a sketch.
- 3.15 Explain shock absorber with a sketch.
- 3.16 List the types of brakes used in two wheelers
- 3.17 Explain Hand operated & Foot operated Mechanical and Hydraulic brakes.
- 3.18 Explain antilock braking system with a sketch.

4.0 Three Wheeler Chassis and Sub-Systems

- 4.1 Explain conventional, frame construction and integral construction of three-wheeler chassis with a sketch.
- 4.2 List the types of 'Loading auto rickshaws'
- 4.3 Explain drive train layout for passenger and loading auto rickshaws with a sketch.

- 4.4 Explain rear suspension system of Passenger and Loading auto rickshaw with a sketch.
- 4.5 Explain Braking system and Hand brakes of three wheelers with a sketch.

5.0 Electric Two and Three wheeler

- 5.1 Explain drive train layout of electric two and three wheelers with a sketch.
- 5.2 List the components of Electric two-wheeler.
- 5.3 Explain the different types of electric motors used in two and three wheelers.
- 5.4 Explain about motor controller.
- 5.5 Explain the battery management system.
- 5.6 Explain charger and charging unit used for electric two and three wheelers.
- 5.7 State the function of battery balancer.
- 5.8 Explain the merits and demerits of Electric two wheelers.

COs – POs Mapping

Course Code:A-401		Course Title:Two and Three-Wheeler Technology		No. of COs: 5	No. of Periods:60
POs	Mapped with CO No.	CO Periods Addressing PO in Col. 1		Level (1,2,3)	Remarks
		Number	Percentage		
PO1	CO1,CO2,CO3,CO4,CO5	25	42	3	> 40% Level 3 (Highly Addressed) 25% to 40% level 2 (Moderately Addressed) 5% to 25% level 1 (Low Addressed) <5% Not Addressed
PO2	CO1,CO2	5	08	1	
PO3	CO1,CO2,CO3,CO4,CO5	10	17	1	
PO4					
PO5	CO1,CO2,CO3,CO4,CO5	15	25	2	
PO6					
PO7	CO1,CO2,CO3,CO5	5	08	1	

CO-PO/PSO MATRIX

CO No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
A-401.1	3	2	2		2			3	2	1
A-401.2	3	2	2		2			3	2	1
A-401.3	3		2		2			3	2	1
A-401.4	3		2		2			3	2	1
A-401.5	3		1		3			3	3	1
AVERAGE	3	0.8	1.8		2.2			3	2.2	1

COURSE CONTENT

1.0 Introduction and Power Unit

History of Two wheelers - classification of two and three wheelers - Layout of Motor cycle, Scooter Cargo Auto Rickshaw and Passenger Auto Rickshaw - basic systems in two wheelers – Arrangement of cylinders used in two wheelers-Engine selection criteria for two-wheeler and three-wheeler- Air cooled, water cooled and liquid cooled engines – Two wheeler exhaust system pipe and header types.

2.0 Fuel, Lubrication and Electrical Systems

Fuel system –Layout of fuel injection system - ECU- Sensors - Lubrication system -Types- Electronic ignition System, Starting system - Kick starter Mechanism – Auto-starter Mechanism, Battery - Ratings in Two and Three wheelers–Handle Bar controls – Side stand / Ignition Interlock systems - Recent technologies - i3S - XSENS Technology - DTSi

3.0 Two-Wheeler Chassis and Sub-Systems

Main frame for two wheelers- Single cradle frame – double cradle frame - Tubular frame - twin-spar frame-monocoque frame –Layout of transmission system in two wheelers – Primary Drive – Types Clutch- Types of

Clutches-Assist Slipper Clutch - Gear box- Types of Gear box –Constant Mesh Gear box – Ball Lock type gear locking mechanism – Trunnion type gear locking mechanism - Sequential Gear Box- Gear Shift Mechanism - Foot operated shifting Mechanism - CVT -Continuously variableTransmission-Final Drive - Belt Drive, Chain Drive, shaft drive and Cush drive - Front and rear suspensionsystems- Shock absorber - Steering column construction - Handle bar – Types and construction - Types -Brakes - Types - Control System- Hand operated Brake and Foot operated brake - Antilock Braking System

4.0 Three-Wheeler Chassis and Sub-Systems

Chassis for three wheelers- Conventionalchassis –Frame Construction - integral construction - Drive train layout and rear suspension system for passenger and loading auto rickshaws –Types of Loading auto rickshaws - Braking System - Hand brake

5.0 Electric Two and Three-wheeler

Drive train layout of Electric Two and Three wheelers- Batteries - Electric Motors - Motor Controller - Charger and charging - Battery Balancer - battery management system – Merits and demerits of Electric two wheelers

REFERENCE BOOKS:

1. Dhruv U. Panchal, "Two and Three-Wheeler Technology", McGraw Hill Books Co. Ltd New Delhi
2. BalajiChoudhury, "Introduction to Two and Three-wheeler", Amazon Publisher
3. Tony Foale, "Motor Cycle Handling and Chassis Design", TonyFoale Designs
4. Service Manuals of Manufacturers of Indian Two and Three wheelers
5. John Lowry, "Electric Vehilce Technology Explained", Wiley
6. G.B.S. Narang, "Automotive Engineering",Khanna Publishers Delhi
7. Siegfried Herrmann, "The Motor Vehicle", Asia Publishing House, Bombay

Table specifying the scope of syllabus to be covered for Unit Test-I & Unit Test-II

Unit Test	Learning Outcomes to be covered
Unit Test – I	From 1.1 to 3.7
Unit Test – II	From 3.8 to 5.8

STRENGTH OF MATERIALS AND FLUID MECHANICS

Course Code	Course Title	No. of Periods/Week	Total Periods/Year	Marks for FA	Marks for SA
A-402	Strength of Materials and Fluid Mechanics	05	75	20	80

TIME SCHEDULE

Chapter No.	Title	Periods	Marks	Part – A Short Type (3 marks)	Part – B Essay Type (10 Marks)	CO's Mapping
STRENGTH OF MATERIALS						
1.	Simple stresses and strains & Torsion	12	23	1	2	CO1
2.	Geometrical Properties of sections	12	16	2	1	CO2
3.	S.F and B.M diagrams	12	16	2	1	CO3
FLUID MECHANICS						
4.	Properties of Fluids	12	16	2	1	CO4
5.	Flow of fluids	12	16	2	1	CO4
6	Oil Power hydraulics & Pneumatics	10	13	1	1	CO5
7	Hydraulic Machinery	05	10	0	1	CO5
Total:		75	110	10	08	

Course Code: A-402	Title: Strength of materials and Fluid mechanics		Total No. Of Periods: 75
Course Objectives:	Upon completion of the course the student shall be able to		
	1.0 Understand the behaviour of materials under simple stresses and strains and Torsion.		
	2.0 Understand the centre of gravity, centroid, Moment of inertia & Radius of Gyration for a given area of cross section.		
	3.0 Understand types of Beams, properties of fluids and concept of flow of fluids		
	4.0 Understand the concept of oil power Hydraulic control unit, pneumatics and Hydraulic machinery.		
Course Out Comes:	Upon completion of the course the student shall be able to		
	CO1	A402.1	Explain the behaviour of materials under simple stresses and strains and Torsion
	CO2	A402.2	Determine centroid, Moment of Inertias and Radius of gyration.
	CO3	A402.3	Evaluate shear force and bending moments in various types of beams.
	CO4	A402.4	Explain the properties of fluids and Analyse the concept of fluid flow.
	CO5	A402.5	Explain oil power-hydraulic control unit, pneumatics and Hydraulic Machinery

Learning Objectives: Upon completion of the course the student shall be able to

- 1.0 **Simple stresses, strain and Torsion.**
- 1.1 Define the terms: Stress, strain, Modulus of elasticity, longitudinal strain, lateral strain, Poisson's ratio, Modulus of rigidity, Bulk modulus, working stress, factor of safety, Ultimate stress, Design stress.
- 1.2 Distinguish between different kinds of stresses and strains.
- 1.3 Explain the salient points in stress-strain curve for ductile materials (mild steel)

- 1.4 State Hooke's law and limits of Proportionality.
- 1.5 Solve problems on relationship between simple stress and simple strain under axial loading on uniform bars and stepped bars.
- 1.6 State the relationship between the elastic constants.
- 1.7 Solve Problems on relationship between elastic constants.
- 1.8 Explain the concept of torque equation.
- 1.9 State the assumptions made in pure torsion.
- 2.0 Geometrical properties of sections**
- 2.1 Define centroid, centre of gravity, centre of mass, Moment of inertia, and radius of gyration and polar moment of inertia.
- 2.2 Calculate positions of centroids for simple plane figures from first principles.
- 2.3 Explain the method of determining the centroid by 'method of moments'.
- 2.4 Determine the position of centroids of standard sections-T, I, C and L.
- 2.5 State the necessity of finding moment of inertia for various engineering Applications.
- 2.6 Determine MI of standard sections by applying parallel and perpendicular axes theorems (T, I, C and L)
- 3.0 Shear Force and Bending Moment Diagrams**
- 3.1 Define different types of beams Viz., a) Cantilever beam, b) simply supported beam. c) Overhanging beam
- 3.2 Define different types of loads viz., a) Point load, b) Uniformly distributed load
- 3.3 Explain the terms a) Shear force, b) Bending moment
- 3.4 Determine shear force and bending moments on cantilever and simply supported beam for Simple cases of loading (Point load, uniformly distributed load) analytically.
- 3.5 Describe the procedures for sketching the shear force diagrams (SFD) and Bending Moment diagrams (BMD)
- 3.6 Sketch Shear Force Diagrams (SFD) and Bending Moment Diagrams (BMD) for cantilever and Simply Supported Beams.
- 3.7 Define Point of contra flexure
- 4.0 Properties of fluids**
- 4.1 Define fluid, differentiate between compressible and Incompressible fluids.
- 4.2 Discuss various properties of fluids.
- 4.3 Analyse Pascal law and its applications
- 4.4 Evaluate atmospheric pressure, absolute Pressure and pressure head of fluid
- 4.5 Discuss the working principle of pressure Gauges piezometer, U-tube, inverted U-tube and differential manometers
- 4.6 Problems on manometers.
- 5.0 Flow of fluids**
- 5.1 Explain the fluid flow of steady, unsteady, laminar and turbulent flows.
- 5.2 Write the Equation of continuity of flow and work done in overcoming pressure.
- 5.3 Solve the total energy equation of fluid in motion. Bernoulli's theorem (proof not necessary) And its practical applications in venturi meter. Simple problems.
- 5.4 Explain the laws of fluid friction, Reynolds number, change of stream lined Flow into Turbulent flow and critical velocity.
- 5.5 Explain the various losses in a fluid flow. Loss of friction – Darcy's expressions (without proof) simple problems.
- 5.6 Explain hydraulic gradient line and total energy line
- 5.7 Calculate the power transmitted and maximum power transmitted
- 6.0 Oil power hydraulics and pneumatics**
- 6.1 Explain the function of the hydraulic power source
- 6.2 List the parts of the power source.
- 6.3 Explain the function of each part of the power source.
- 6.4 Classify the oil pumps like gear pump, vane pump (other pumps not necessary).
- 6.5 Explain the constructional details of gear pump (external and internal gear type), vane pump, and the working principle.
- 6.6 List the different types of actuators.
- 6.7 Explain the concept of pneumatic circuit

- 6.8 Write the basic symbols used in pneumatic circuit.
- 6.9 List Advantages and limitations of Pneumatics over hydraulic control.
- 7.0 Explain the Constructional details and working principle of**
 - 7.1 Hydraulic press
 - 7.2 Hydraulic jack
 - 7.3 Hydraulic lift
 - 7.4 Hydraulic crane
 - 7.5 Hydraulic ram

COURSE CONTENT:

STRENGTH OF MATERIALS

1.0 Simple stresses and strains & Torsion

Simple stresses and strains – strain diagram, Hooke's law, Factor of Safety- Ultimate stress and Design stress - Young's modulus, Modulus of Rigidity - Bulk Modulus – Poisson's ratio - Relation among elastic constant, Torsion equation and assumptions in it.

2.0 Geometrical properties of sections

Centroid, centre of gravity, centre of mass, Moment of inertia, and radius of Gyration and polar moment of inertia. Centre of gravity geometric figures - Square, Rectangle, Triangle and Circle - neutral axis- Moment of Inertia- Radius of gyration- Parallel and perpendicular axis theorems- Polar moment of inertia – Calculation of moment of inertia of I, T, C and L – sections

3.0 Shear Force and Bending Moment Diagrams

Types of beams – Cantilever, simply supported and overhanging beams -subjected to point or concentrated and U.D. loads - S.F. and B.M. calculations and diagrams for the above cases

FLUID MECHANICS

4.0 Properties of fluids

Definition of fluid – compressible and incompressible fluids – Properties - Density, Specific weight, viscosity, surface Tension – Pressure-capillarity- Pascal's law – Intensity of pressure – Gauge and absolute pressure – Measurement of pressure by Piezo meter, U- tube and differential manometers.

5.0 Flow of Fluids

Types of flow- concept of Reynolds's number – Equation of continuity of flow – Bernoulli's equation and practical applications – Problems on Bernoulli's equation – Loss of head in pipes due to friction – Darcy's (without proof) – Hydraulic Gradient – Total energy line – Power transmitted – Condition for maximum power transmission (without Derivation)- Simple problems.

6.0 Oil power hydraulics & Pneumatics

Basic components of an oil power hydraulic control circuit like power source, control element and actuator, Function of each basic component- Major parts of the power source – types of oil pumps and their constructional details-Definition of Pneumatics- Pneumatic control circuit –Basics symbols of Pneumatic circuit-Advantages and limitations of Pneumatics over hydraulic control.

7.0 Hydraulic Machinery

Constructional details and working principle of Hydraulic press, Hydraulic jack, Hydraulic lift, Hydraulic crane and Hydraulic ram.

REFERENCE BOOKS

- 1. Strength of Materials -by R. S. Khurmi
- 2. Solid Mechanics -by Pakirappa
- 3. Strength of Materials -by Ramamrutham
- 4. Hydraulics -by R.S. Khurmi
- 5. Hydraulics &Hydraulics Machines, Vol I &II -by JagadishLal

COs – POs Mapping
A-402, STRENGTH OF MATERIALS & FLUID MECHANICS

Course Code:A-402		Course Title: STRENGTH OF MATERIALS & FLUID MECHANICS		No. of COs: 5	No. of Periods: 75
POs	Mapped with CO No.	CO Periods Addressing PO in Col. 1		Level (1,2,3)	Remarks
		Number	Percentage		
PO1	CO1,CO2,CO3,CO4,CO5	20	26	2	> 40% Level 3 (Highly Addressed) 25% to 40% level 2 (Moderately Addressed) 5% to 25% level 1 (Low Addressed) <5% Not Addressed
PO2	CO1,CO2,CO3,CO4,CO5	12	16	1	
PO3	CO4,CO5	11	15	1	
PO4	CO3,CO4,CO5	20	27	2	
PO5	CO1,CO2,CO3,CO4,CO5	12	16	1	
PO6	CO1,CO2,CO3,CO4,CO5				
PO7	CO1,CO2,CO3,CO4,CO5				

Table specifying the scope of syllabus to be covered for Unit Test-I & Unit Test –II

Unit Test	Learning Outcomes to be covered
Unit Test-I	From 1.1 to 3.7
Unit Test-II	From 4.1 to 7.5

AUTOMOBILE CHASSIS AND BODY ENGINEERING

Course Code	Course Title	No. of Periods/Week	Total Periods/Year	Marks for FA	Marks for SA
A-403	Automobile Chassis and Body Engineering	04	60	20	80

TIME SCHEDULE

Chapter No.	Title	Periods	Marks	Part – A Short Type (3 marks)	Part – B Essay Type (10 Marks)	CO's Mapping
1	Steering System	12	26	02	02	CO1
2	Braking System	12	26	02	02	CO2
3	Suspension System	12	21	02	1 1/2	CO3
4	Body Design, Construction & safety	12	21	02	1 1/2	CO4
5	Automobile Air Conditioning System	12	16	02	01	CO5
Total		60	110	10	08	

Course Code : A-403	Course Title: Automobile Chassis and Body Engineering No. Of Periods : 60
Course Objectives	Upon completion of the course the student shall be able to 1. Understand steering, braking and suspension system. 2. Understand the various aspects of vehicle body, its structures and safety features of automobiles. 3. Understand the refrigeration and AC systems of automobiles

CO No.	Course Outcome
C01	A-403.1 Discuss the working of different steering systems and its components used in automobiles.
C02	A-403.2 Discuss the working of braking systems and its components used in automobiles.
C03	A-403.3 Analyse the working of suspension systems and its components used in automobiles.
C04	A-403.4 Explain the various vehicle body design, body parts, its construction and safety features provided and testing procedures of automobiles
C05	A-403.5 Explain working of air conditioning system of an automobile.

Learning Outcomes:

1.0 Steering System

- 1.1 Explain Steering linkage for independent and rigid axle system.
- 1.2 Explain Ackermann steering gear mechanism and Davis steering gear mechanism.
- 1.3 Explain the reversible and irreversible steering, Under-steering, over-steering and Turning radius
- 1.4 Explain Integral power steering system and linkage type power steering system.
- 1.5 Explain various Steering geometry – camber, King-pin inclination, Caster,

- Toe-in, Toe-out, Combined inclination and Scrub radius.
- 1.6 Explain Wheel alignment and its adjustment.
 - 1.7 Explain Collapsible steering column, Cornering force, Self-aligning torque and Cornering force
 - 1.8 Explain 1.ECU Controlled Power Assisted Steering 2.Electronic Power Assisted Steering-
 - 1.9 Explain Four Wheel Steering.
 - 2.0 Braking System:**
 - 2.1 Calculate the stopping time and distance – braking efficiency.
 - 2.2 Draw and explain the constructional details of master cylinder, wheel cylinder and Tandem master cylinder.
 - 2.3 State the types and importance of brake fluid for safe driving.
 - 2.4 Explain the construction of internal shoes brakes.
 - 2.5 Explain the fixed calliper and swinging calliper type of disc brakes.
 - 2.6 Compare advantages and disadvantages of disc brakes over drum brakes.
 - 2.7 Describe self-energizing brakes.
 - 2.8 Explain 1. Vacuum servo assisted brakes 2. Air brakes 3. Hand brakes 4. Parking brakes. 5. Wear Indicators and 6. Automatic Brake adjustment
 - 2.9 Explain the working of 1. ABS 2. EBD 3. Traction control 4. ESC
 - 2.10 Explain multi collision braking
 - 3.0 Suspension system:**
 - 3.1 Explain rigid axle suspension system.
 - 3.2 List and describe various types of leaf springs.
 - 3.3 Explain types of independent suspension system like McPherson, double wish bone type, Trailing arm type and multi-link type independent suspension system.
 - 3.4 Explain pneumatic brakes and its components
 - 3.5 Explain 1. Torsion bars, 2. Air springs 2. Telescopic type hydraulic shock absorber 3. Stabilizer bar.
 - 3.6 Explain the working Hydrolastic suspension system, Hydro gas suspension system
 - 3.7 Explain Hydro Pneumatic suspension system.
 - 3.8 Explain adaptive suspension system
 - 4.0 Body design and Construction**
 - 4.1 Explain framed and unitized construction of vehicles
 - 4.2 Describe various loads acting on body and its effects.
 - 4.3 Describe streamlining and aerodynamics of vehicle body.
 - 4.4 Identify major external dimensions of vehicle body
 - 4.5 List the materials used for construction of body
 - 4.6 Describe various body components.
 - 4.7 Describe interior finish of automobile body.
 - 4.8 Describe power windows and Central locking system.
 - 4.9 Describe body exterior finishing techniques and company painting method.
 - 4.10 State the importance of automobile safety
 - 4.11. Describe active and passive safety
 - 4.13 Explain supplementary restraint system and its components
 - 4.14 State the function of ADAS, its types, working. Merits and limitations of it.
 - 4.15 Explain the testing procedure of 1) Chassis dynamometer 2) Crash test 3) roll over test
 - 4.16 State about the bio-fidelity dummies
 - 5.0 Air Conditioning System**
 - 5.1 Define Refrigeration and air conditioning.

- 5.2 Express unit of refrigeration and define coefficient of performance.
- 5.3 List methods of refrigeration and understand the working of vapour compression Refrigeration system.
- 5.4 Identify commonly used refrigerants.
- 5.5 Explain the necessity of automobile as applicable to human comfort.
- 5.6 Define the terms 1. Humidity, 2. Relative humidity, 3. Dew point, 4. DBT, 5. WBT, 6. Absolute humidity ratio, 7. Sensible heat and 8. Latent heat.
- 5.7 Explain the functions of various components of auto air condition such as 1. Fans 2. Supply ducts/ vents 3. Outlets 4. Return ducts 5. Filters and 6. Dust collectors.
- 5.8 Locate the different components of Air conditioner in a car and understand its working.
- 5.9 Explain the working of 1. various types of auto air conditioning system 2. Automatic Temperature control and 3. Compressor clutch.
- 5.10 State the necessity of automotive heaters

COs – POs Mapping

Course Code: A-403		Course Title: Automobile Chassis and Body Engineering		No. of COs: 5	No. of Periods: 60
POs	Mapped with CO No.	CO Periods Addressing PO in Col. 1		Level (1,2,3)	Remarks
		Number	Percentage		
PO1	CO1,CO2,CO3,CO4,CO5	20	33	2	> 40% Level 3 (Highly Addressed) 25% to 40% level 2 (Moderately Addressed) 5% to 25% level 1 (Low Addressed) <5% Not Addressed
PO2	CO1,CO2,CO3,CO4,CO5	15	25	2	
PO3	CO1,CO2,CO3,CO4,CO5	15	25	2	
PO4	CO1,CO2,CO3,CO5	5	9	1	
PO5	CO2,CO5	5	8	1	
PO6					
PO7					

CO-PO/PSO MATRIX

CO No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
A-403.1	3	2	1	2				1	3	
A-403.2	3	3	1	1	2			1	3	
A-403.3	3	3	2	1				1	3	
A-403.4	3	2	2					1	3	
A-403.5	3	1	2	2	2			1	3	
Average	3	2.2	1.6	1.2	0.8			1	3	

Note: The gaps in CO and PO mapping will be achieved by one or more appropriate activities from the following:

- i) Seminars ii) Tutorials iii) Guest Lectures iv) Assignments v) Quiz competitions vi) Industrial visit vii) Tech Fest viii) Mini project ix) Group discussions x) Virtual classes xi) Library visit for e-books

COURSE CONTENTS:

1.0 Steering System:

Ackermann & Davis steering gear -Reversible and irreversible steering, slip angle- under steering and over steering-steering gear ratio-Turning radius- Steering geometry – camber, king-pin inclination, caster, Toe-in, Toe-out, combined inclination and scrub radius-Collapsible steering column – cornering force - self aligning torque, Power steering -Integral power steering and linkage type power steering-ECU Controlled Power Assisted Steering.-Electronic Power Assisted Steering-twin steer axles-Four Wheel Steering.

2.0 Braking System:

Stopping time – stopping distance –braking efficiency-(Simple problems)– Mechanical and Hydraulic brakes – master cylinder, wheel cylinder and Tandem master cylinder— brake fluids-constructional details of internal shoe brakes (drum brakes) - self energizing brakes - Disc brakes – Types of disc brakes - advantages of disc brakes over drum brakes – power assisted brakes – vacuum servo assisted brakes – airbrakes – Hand brakes / parking brakes-Wear Indicators and Automatic Brake adjustment-Antilock Braking System (ABS), Electronic brake force Distribution System (EBD),Traction Control System (TC),Electronic Stability Control System (ESC) and Multi-collision Braking System- Emergency brake assist-Electronic parking brake.

3.0 Suspension System:

Leaf spring types - Torsion bar – Independent suspension types- McPherson strut, and double wishbone type, trailing arm type and Multi link suspension system- – construction and working of telescopic type shock absorber-Pneumatic suspension system- construction and working of its components – stabilizer bar- Air springs -Hydro (lastic) Suspension- Hydro gas suspension – Hydro Pneumatic suspension – Adaptive Suspension System

4.0 Body design, Construction and Safety:

Framed construction, unitized construction-merits and demerits of unitized-Construction - Various loads acting on body – materials used for body-Streamlining of vehicles-.Major external dimensions of body – overall width, height, length, wheel track wheelbase and overhangs. major internal dimensions – like head room, leg room etc. body components – doors– types - windows-Power windows-window regulating mechanism - seats – types of seats-seat adjustment mechanism -Other body components like, hood, boot lid, roof , firewall, spoilers, pillars etc. - fabrication of body components- Interior finish – trimming-body trim items, upholstery -Exterior – body painting –types –company painting method - central locking system.

Automobile Safety- Active Safety, Passive Safety -Supplementary restraint systems (SRS) – Air bags- seat belt- Pre-Tensioners-. ADAS- driver assistance systems — Levels of driving automation - Autonomous Vehicles Over view, types, components & working – Advantages and limitations-blind spot collision warning-lane following assist. Testing- Chassis dynamometer- Crash test-roll over test-bio fidelity dummies.

5.0 Air Conditioning System

Introduction – Definition of refrigeration – unit of refrigeration – coefficient of performance – methods of refrigeration -Vapour compression refrigeration system-Refrigerants- Comfort Air conditioning – definition of terms human comfort, humidity, relative humidity, humidity ratio, DBT, WBT, sensible heat, latent heat. Fans - supply ducts/vents – outlets – return ducts – filters and dust collectors – cooling coils - Automotive heaters- Necessity – location of air conditioning components in a car —air conditioning system of automobile - Expansion valve type- fixed orifice tube type- Automatic climate control -Compressor clutch.

Reference Books:

1. The motor vehicle - Newton steeds. &Garret – KENNATH NEWTON
2. Automobile engineering - G.B.S.Narang – Khanna Publishers
3. Automobile engineering - Sethi.- McGraw Hill Education
4. Automotive Mechanics - William crouse. – McGraw Hill Education.

Unit Test	Learning Outcomes to be covered
Unit Test – I	From 1.1 to 3.8
Unit Test – II	From 4.1 to 5.10

VEHICLE SERVICE AND MAINTENANCE TECHNOLOGY

Course Code	Course Title	No. of Periods/Week	Total Periods/Year	Marks for FA	Marks for SA
A-404	Vehicle Service and Maintenance Technology	04	60	20	80

TIME SCHEDULE

Chapter No.	Title	Periods	Marks	Part – A Short Type (3 marks)	Part – B Essay Type (10 Marks)	CO's Mapping
1.	Servicing Equipment and Maintenance	14	26	2	2	CO1
2.	Engine Maintenance	8	16	2	1	CO2
3.	Cooling and Lubrication system Maintenance	8	16	2	1	CO3
4.	Transmission system Maintenance	8	13	1	1	CO4
5.	Braking, Steering and Suspension Maintenance	14	26	2	2	CO4
6.	Auto Electrical and Body maintenance	8	13	1	1	CO5
	Total	60	110	10	8	

Course Code:A-404 COURSE TITLE:VEHICLE SERVICE AND MAINTENANCE TECHNOLOGY

COURSE OBJECTIVES	Upon completion of the course the student shall be able to 1. Appreciate the importance of Vehicle Service and Maintenance and tools & equipment used for maintenance 2. Understand the procedure for maintenance of various systems of vehicle 3. Know the automobile air conditioning maintenance.
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CO No.	Course Outcome
C01	A-404.1 Describe the importance of Vehicle Service and Maintenance and tools & equipment's used for maintenance
C02	A-404.2 Describe the Engine and Fuel system maintenance
C03	A-404.3 Explain Cooling and Lubrication system Maintenance
C04	A-404.4 Explain Clutch, Transmission, Drive shaft, final drive, Braking, Steering, Suspension, wheels and tyres Maintenance
C05	A-404.5 Explain Auto Electrical, Air Conditioning and Chassis & body maintenance

LEARNING OUTCOMES:

Upon completion of the course the student shall be able to

1.0 Servicing Equipment and Maintenance

- 1.1 Explain about safety of personnel in automotive maintenance shops.
- 1.2 Explain about the Personal Protective Equipment (PPE) used in the workshop.
- 1.3 State safety requirements and First Aid.
- 1.4 Explain Fire safety and First aid.

- 1.5 Explain working of car washing equipment, vehicle hoist and Air Compressor
- 1.6 List different types of lubrication equipment and their function
- 1.7 State the need of Servicing and Maintenance
- 1.8 Explain different types of maintenance.
- 1.9 Explain about timing light and Nozzle Testing machine
- 1.10 Explain fuel pump testing and calibration machine.
- 1.11 List types of forks and lifts
- 1.12 Explain OBD Scanner and OBD Diagnostic tool.
- 1.13 State the function of growler and wiring harness tester
- 1.14 Explain working of High-pressure lubrication equipment
- 2.0 Engine and Fuel System Maintenance**
- 2.1 Explain diagnosis of Noise, vibration, harshness (NVH) issues.
- 2.2 Explain different Engine smoke and diagnose the problem.
- 2.3 Explain dismantling and assembly of engine components
- 2.4 State the cleaning methods of the engine components.
- 2.5 State the visual and dimensional inspections
- 2.6 State about cylinder liner fitting
- 2.7 Explain Cylinder head resurfacing.
- 2.8 Explain Engine balance and testing methods
- 2.9 Explain Valve timing setting and Tappet Setting.
- 2.10 State about replacement of Air filter
- 2.11 Explain blow by test and cylinder compression test
- 2.12 Explain checking and testing of fuel injectors and pump
- 2.13 Explain Fuel pump timing.
- 3.0 Cooling and Lubrication system Maintenance**
- 3.1 State the need for maintenance of cooling and lubrication systems.
- 3.2 Explain the causes of overheating of the engine.
- 3.3 State the general cleaning of cooling system
- 3.4 Enumerate trouble shooting of cooling and lubrication system
- 3.5 State coolant replacement, bleeding and flushing.
- 3.6 Explain the maintenance of Thermostat, water pump and radiator.
- 3.7 State about checking of Level of engine oil, draining of engine oil, refilling and replacing of engine oil and filter.
- 3.8 State about engine oil leak, low oil pressure, excessive oil consumption.
- 3.9 State about oil pump, pressure relief valve maintenance.
- 3.10 Explain oil coolers test.
- 4.0 Transmission system Maintenance**
- 4.1 Explain the trouble shooting of clutch.
- 4.2 Explain clutch assembly and disassembly.
- 4.3 State about clutch troubles -Slipping, grabbing, drag, chatter, noise, hard pedal effort, vibration, free play adjustment.
- 4.4 Explain troubleshooting and maintenance of transmission, drive shaft and final drive
- 4.5 Explain dismantling and assembly of transmission, drive shaft and final drive
- 4.6 State about common faults, service and lubrication of the gear box, drive shafts and final drive.
- 5.0 Braking, Steering, Suspension system maintenance**
- 5.1 Explain the maintenance of Hydraulic brake, Drum brake, Disc brake.
- 5.2 Explain bleeding of brakes
- 5.3 Explain maintenance of Air brakes and Anti-lock braking system.

- 5.4 Explain the maintenance and troubleshooting of steering linkage, steering column, steering gear box, Rack and pinion, Recirculating ball and Worm type and Power steering system and Twin steering system.
- 5.5 Explain lubrication of steering system.
- 5.6 Explain the maintenance of strut, coil spring, leaf spring, shock absorbers, stabilizer bar, torsion bars and ball joints.
- 5.7 State about greasing of the steering and suspension parts.
- 5.8 Explain the maintenance of wheels and tyres.
- 5.9 Explain the procedure of Wheel alignment and wheel balance.
- 5.10 State about static and dynamic balance of wheels.
- 5.11 Explain procedure of removing and refitting of tyres using tyre changer equipment.
- 5.12 State about the tyre wear, tyre rotation and tubeless tyres
- 5.13 State tyre pressure monitor system maintenance
- 5.14 Explain Tyre re-treading process.
- 6.0 Auto Electrical and Body maintenance**
- 6.1 Explain AC performance test.
- 6.2 Explain Air conditioning system servicing equipment.
- 6.3 State about leak detection and repair of HVAC system.
- 6.4 State about defects of vehicle chassis and body.
- 6.5 Explain Denting process.
- 6.5 Explain Periodical cleaning, washing, waxing of vehicle body.
- 6.6 Explain Interior body cleaning.
- 6.7 State about body coatings.
- 6.8 Explain the maintenance and servicing of Auto Electrical and Electronics components.

COs – POs Mapping Format

Course Code: A-404		Course Title: Vehicle Service and Maintenance Technology		No of Cos: 5	No. Of periods: 60
POs	Mapped with CO No	CO Periods Addressing PO in Col 1		Level (1,2,3)	Remarks
		No	%		
PO1	CO1, CO2, CO3,CO4,CO5	22	37	2	>40% Level 3 (Highly Addressed)
PO2					25% to 40% Level 2 (Moderately Addressed)
PO3					5% to 25% Level 1 (Low Addressed)
PO4	CO1, CO2, CO3,CO4,CO5	22	36	2	<5% Not Addressed
PO5	CO1,CO2,CO3 ,CO4,CO5	16	27	1	
PO6					
PO7					

CO-PO/PSO MATRIX

CO No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
A-404.1	3			2	2				2	3

A-404.2	3			2	2				2	3
A-404.3	3			2	2				2	3
A-404.4	3			2	2				2	3
A-404.5	3			2	2				2	3
AVERAGE	3			2	2				2	3

COURSE CONTENT

1.0 Vehicle Service and Maintenance Technology , Servicing and Equipment:

Vehicle Service and Maintenance Technology and Servicing- Safety – PPE - Personnel, machines and equipment, vehicles, fire safety - First aid - Need for Maintenance – importance, Types of maintenance – Periodic/Preventive – breakdown - Total Productive Maintenance - Servicing and its necessity.

Vehicle Service and Maintenance Technology Equipment-, engine tune-up instruments- Timing light (Neon light), , specification and working -car washer, air compressor, fuel injection pump testing and calibration machine, nozzle testing machine, grease guns, Types of lifts, battery charger, growler, wiring harness tester, Diagnostic tool.- OBD Scanner

2.0 Engine and Fuel System Maintenance

Engine Maintenance- Engine smoke and NVH diagnosis-Dismantling of engine components and cleaning-cleaning methods - visual and dimensional inspections - Cylinder head resurfacing -Cylinder liner fitting - Engine balance and testing methods - engine assembly - Valve timing- Tappet setting - Air filters - blow by test - cylinder compression test -

Fuel system Maintenance - checking and testing of fuel injectors –fuel injection pump - fuel pump timing. Trouble shooting of engine and fuel system..

3.0 Cooling and Lubrication system Maintenance

Cooling system - Need for cooling system maintenance - over heating causes - General cleaning - - coolant - replacement, bleeding and flushing - Thermostat, Water pump and radiator maintenance.

Lubrication System - Need for Lubrication system maintenance - Level of Engine oil, Draining of Engine oil, refilling and replacing of oil, filter - Engine oil leak - Low oil pressure - Excessive oil consumption -oil pump - pressure relief valve - oil coolers test. –trouble shooting of cooling and lubrication system

4.0 Clutch, Transmission, Drive shaft and final drive system Maintenance

Clutch - Clutch troubles - Slipping, grabbing, drag, chatter, noise, hard pedal effort, vibration, free play adjustment - clutch assembly and disassembly.

Transmission, drive shaft and final drive -Common faults, service and lubrication - gear box, drive shafts and final drive - Dismantling, checking and reassembling.

5.0 Braking, Steering and Suspension maintenance

Braking - Inspection, Maintenance and Service - Hydraulic brake, Drum brake, Disc brake – brake shoes - Bleeding of brakes - Parking brake- air brakes –ABS – troubles and rectification.

Steering - Inspection, Maintenance and Service of steering linkage, steering column, steering gear box service- Rack and pinion, Recirculating ball and Worm type and Power **steering system** - Twin steering system –Power assisted steering -Lubrication of steering system-

Suspension - Inspection, Maintenance and Service of suspension components - strut, coil spring, leaf spring, shock absorbers, stabilizer bar, torsion bars and ball joints - Greasing of all parts

Wheels and Tyres - Inspection and maintenance of wheel and tyre - Dismantling and assembly procedures of Tyres- Tyre Changer -Wheel alignment, wheel balance- Static and dynamic balance- removing and fitting of tyres, tyre wear and tyre rotation – Re-treading - Tyre pressure monitoring system- Tubeless tyres.

6.0 Auto Electrical and Body maintenance:

Auto Electrical - Battery failure - Battery tests - Types- Maintenance Schedule of starter motor, Alternator, ignition system, wiper motor, electrical fuel pump, alternator, horn, flasher unit, heaters and defoggers- charging systems - DC generator and alternator - lighting systems - maintenance of modern electronic controls, checking and servicing of dash board instruments.

Air Conditioning - Possible faults of HVAC system - AC performance test - leak detection and repair - Air conditioning system servicing equipment.

Chassis & body - Defects of vehicle chassis and body - chassis alignment equipment - dent repairs- Periodical cleaning of vehicle body, washing, waxing - Interior body cleaning by vacuum cleaning, Cleaning carpet, dashboard, seats- Body coatings.

REFERENCE BOOKS

1. Automobile Engineering : G.B.S. Narang.
2. Automobile Engineering : Harbanth Singh
3. Manual on Depot Maintenance : A.P.S.R.T.C Manuals
4. Transmission Overhaul : Station Abbey.
5. The Modern motor Engineer Vol I, II : Arthen Judge
6. Practical Auto Engineering : Odamis.

➤ Table specifying the scope of syllabus to be covered for Unit Test-I & Unit Test-II

Unit Test	Learning Outcomes to be covered
Unit Test – I	From 1.1 to 3.10
Unit Test – II	From 4.1 to 6.8

FARM TRACTORS AND EARTH MOVERS

Course Code	Course Title	No. of Periods/ Week	Total Periods/ Year	Marks for FA	Marks for SA
A-405	Farm Tractors and Earth Movers	04	60	20	80

TIME SCHEDULE

Chapter No.	Title	Periods	Marks	Part – A Short Type (3marks)	Part –B Essay Type (10 marks)	CO's Mapping
1	Introduction	06	6	2	0	CO1
2	Farm Tractors	12	26	2	2	CO2
3	Special Features & Recent Trends in Tractors	18	26	2	2	CO3
4	Special Purpose Vehicles for Industrial Applications	12	26	2	2	CO4
5	Earth Movers	12	26	2	2	CO5
TOTAL		60	110	10	8	

Course Code: A– 405	Title: Farm Tractors and Earth Movers	Total No. of Periods: 60
Course Objectives:	Upon completion of the course the student shall be able to	
	1.0 Know the types of special purpose vehicles and their applications.	
	2.0 Understand the types of farm tractors, special features in tractors and Recent trends in Tractors.	
	3.0 Understand the types of special purpose vehicles used for Industrial Applications.	
	4.0 Know the working of various earthmovers and their applications	

CO No.	Course Outcome
CO1	A-405.1 List types of special purpose vehicles and their applications
CO2	A-405.2 Discuss different types of tractors and special features in tractors like PTO shaft, draw bar, hydraulic system and lift mechanism.
CO3	A-405.3 Discuss different special purpose vehicles for industrial applications.
CO4	A-405.4 Explain working of earth movers like Bulldozers, Shovels, Excavators etc.,
CO5	A-405.5 Propose a suitable earth moving vehicle for a given job.

LEARNING OUTCOMES:

Upon completion of the course the student shall be able to

1. **Introduction**
 - 1.1 State the meaning of Special Purpose Vehicle.
 - 1.2 List different types of Special Purpose Vehicles.
 - 1.3 State the applications of Special Purpose Vehicles.
 - 1.4 State the design features of Special Purpose Vehicles.

2. Farm Tractors

- 2.1 State the classification of tractors
- 2.2 Describe wheel type and track type tractor
- 2.3 List out different types of tractors based on their purpose.
- 2.4 Describe the purposes of Utility tractor, Row crop tractor, Orchard tractor, Industrial tractor, Garden tractor, Rotary tiller, Implement carrier and Earth moving tractors.
- 2.5 List tractor manufacturing companies in India.
- 2.6 State the elements in transmission system of a tractor.
- 2.7 Explain the transmission system of a tractor.
- 2.8 Describe the differential assembly of a tractor.
- 2.9 State the function of differential lock.
- 2.10 List out different types of differential locks and describe its working.
- 2.11 Explain steering system of a tractor with a sketch.
- 2.12 Explain braking system of a tractor with a sketch.

3. Special Features and Recent trends in Tractors

- 3.1 Define P.T.O shaft.
- 3.2 State the types of drives used for P.T.O. shafts.
- 3.3 State safety precaution to be taken for various P.T.O shaft drives.
- 3.4 Explain belt pulley drive and precaution to be taken while using belt pulley.
- 3.5 Draw the general layout of Hydraulic system of a tractor and explain.
- 3.6 Describe the hydraulic lift mechanism in a tractor.
- 3.7 Explain the troubleshooting of Hydraulic system of a tractor.
- 3.8 Explain different types of Drawbars.
- 3.9 Explain different types of hitches.
- 3.10 Discuss the draft and depth control mechanism in a tractor.
- 3.11 Explain working of electric tractors.
- 3.12 Explain automations in tractors.
- 3.13 Explain the concept of IoT in Tractors.
- 3.14 Explain the servicing and maintenance of tractors.

4. Special Purpose Vehicles for Industrial Applications

- 4.1 State the specifications of SPV used for industrial applications.
- 4.2 Explain the construction and working of scissor lift trucks.
- 4.3 Explain the construction and working of fork lift trucks.
- 4.4 Explain the construction and working of multi axle trucks.
- 4.5 Explain the construction and working of jib cranes.
- 4.6 Explain the construction and working of vibratory compactors.
- 4.7 Explain the construction and working of articulated vehicles.
- 4.8 Describe fifth wheel coupling of articulated vehicles.
- 4.9 Explain the construction and working of tillers.

5. Earth Moving Vehicles

- 5.1 List different types of earth movers.
- 5.2 Explain the functions and specifications of different types of earth moving vehicles.
- 5.3 Discuss the working and applications of Bulldozers.
- 5.4 Describe the working and applications of Dumpers.
- 5.5 Explain different types of Dumpers.
- 5.6 Describe the working and applications of Tippers.

- 5.7 DescribetheworkingandapplicationsofShovels.
- 5.8 DiscusstheworkingandapplicationsofExcavators.
- 5.9 DescribetheworkingandapplicationsofGraders.
- 5.10 DiscusstheworkingandapplicationsofScrapers.
- 5.11 Describe the working and applications of Drills.
- 5.12 Explain the servicing and maintenance of earth movers.

Cos – Pos Mapping Format

Course Code: A-405	Course Title: Farm Tractors and Earth Movers	No of Cos: 5		No. Of periods: 60	
POs	Mapped with CO No	CO Periods Addressing PO in Col 1		Level (1,2,3)	Remarks
		No	%		>40% Level 3 (Highly Addressed)
PO1	CO1, CO2, CO3,CO4,CO5	10	17	1	25% to 40% Level 2 (Moderately Addressed)
PO2	CO5	14	23	1	5% to 25% Level 1 (Low Addressed)
PO3					<5% Not Addressed
PO4	CO2, CO3,CO4,CO5	16	27	2	
PO5	CO2, CO3,CO4,CO5	20	33	2	
PO6					
PO7					

CO-PO/PSO MATRIX

CO No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
A-405.1	1									
A-405.2	3			1	2			2	3	1
A-405.3	3			1	2			2	3	1
A-405.4	3			2	1			1	2	1
A-405.5	3	3		2	2			1	3	3
AVERAGE	2.6	0.6		1.2	1.4			1.2	2.2	1.2

COURSE CONTENT

1. Introduction

Special purpose vehicles – types–Farm tractors – Earthmovers – Design features - advantages – applications

2. Farm Tractors

Classification of tractors–Wheel type and Track type–Types of drives – types of tractors based on their purpose – Tractor manufacturing companies in India – Transmission system –Differential– Differential lock– types– Steering and braking system of tractors

3. Special Features & Recent Trends in Tractors

P.T.O shaft–Types of drive for P.T.O. shafts–Belt pulley drive– Precautions to be taken while using belt pulley and P.T.O. shaft – General layout of hydraulic system of a tractor – Hydraulic Lift mechanism in a tractor – Trouble shooting of hydraulic system – Draw bar – Types (2 & 3 point linkage) – Types of hitches - Draft and Depth control mechanism – Electric Tractors - Automation in tractors - IoT in Tractors – servicing and maintenance of tractors.

4. Special Purpose Vehicles for Industrial Applications

Specifications – applications – construction and working of scissor lift trucks – fork lift trucks – multi axle trucks – jib cranes – vibratory compactors – concrete mixtures – articulated vehicles – fifth wheel coupling of articulated vehicles – tillers.

5. Earth Moving Vehicles

Types of Earth movers –Dumpers –Tippers – Bulldozers – Shovels – Excavators – Graders – Scrapers – Drills - their working and applications – servicing and maintenance of earth movers.

REFERENCEBOOKS

1. FarmMachines&Equipments by C.P.Nakra–DhanpatRaiPublishingCompany
2. AutomobileEngineering by K.K.Ramalingam-Scitech
3. TruckCranes by A.Astakhov-(MIR)
4. MotorGraders by E.G.Ronioson-(MIR)
5. Constructionequipment by Y.Pokras&Tushnyakov–(MIR)

Tables specifying the scope of syllabus to be covered for Unit Test-I & Unit Test-II

Unit Test	Learning Outcome to be covered
Unit Test-I	1.1 to 3.14
Unit Test-II	4.1 to 5.12

AUTOMOBILE ENGINEERING DRAWING

Course Code	Course Title	No. of Periods/ Week	Total Periods/ Year	Marks for FA	Marks for SA
A-406	AUTOMOBILE ENGINEERING DRAWING	06	90	40	60

TIME SCHEDULE

Chapter No.	Title	Periods	Marks	Part - A	Part - B	Total Marks	CO's Mapping
1	Automobile Engine Components	24	15	2	1	10	CO1
2	Chassis and Transmission	24	15	2		10	
3	Cams and Cam profiles	12	10	0	1	10	CO3
4	Automobile Parts assembly drawings	30	40	0	2	40	CO4
	Total	90	80	4	4	80	

Course Code : A-406	Course Title : Automobile Engineering Drawing No. of Periods : 90
Course Objectives	Upon completion of the course the student shall be able to visualise and draw various views of the given automobile components.

CO No.	Course Outcome
CO1	A-406.1 Apply the drawing methods to various automobile engine components
CO2	A-406.2 Able to Visualise and draw various views of a chassis and transmission system components.
CO3	A-406.3 Able to draw cam profile for a given follower motion.
CO4	A-406.4 Acquire the knowledge and draw assembly of different Automobile components.

Learning Outcomes:

- 1.0 Draw to scale various views like front view, side view, top view and Sectional Views of Automobile Engine Components**
 - 1.1 Pistons for diesel and petrol.
 - 1.2 Connecting rods of I.C.Engines.
 - 1.3 Cylinder Head for 2 stroke petrol engine.
 - 1.4 Crank shaft.
 - 1.5 Cam shaft.

- 1.6 Valves.
- 1.7 Rocker Arms.
- 1.8 Spark plugs.
- 2.0 **Draw to scale various views like front view, side view, top view and Sectional Views of Chassis and transmission components**
- 2.1 Linkage Bracket
- 2.2 Chassis Bracket
- 2.3 Gear box selector fork
- 2.4 Engine Mount
- 2.5 Bell crank lever
- 3.0 **Cams and followers**
- 3.1 Draw types of cams.
- 3.2 Draw the cam profile for the following motion of the follower
 - (i) Uniform velocity
 - (ii) Uniform Acceleration and retardation
 - (iii) Simple harmonic motion.
- 3.3 Draw cam profiles in above three cases for a knife edge follower, roller follower and flat follower. (Inline cams)
- 4.0 **Draw the automobile parts of the assembly drawing.**
- 4.1 Connecting Rod
- 4.2 Master Cylinder
- 4.3 Wheel cylinder

CO-PO MAPPING

Course Code: A-406	Course Title: Automobile Engineering Drawing	No of Cos: 4			No. Of periods:90
POs	Mapped with CO No	CO Periods Addressing PO in Col 1		Level (1,2,3)	Remarks
		No	%		
PO1	CO1,CO2,CO3,CO4	15	17	1	>40% Level 3 (Highly Addressed) 25% to 40% Level 2 (Moderately Addressed)
PO2	CO1,CO2,CO3,CO4	15	17	1	5% to 25% Level 1 (Low Addressed)
PO3	CO4	10	11	1	<5% Not Addressed
PO4	CO1,CO2,CO3,CO4	30	33	2	
PO5	CO1,CO2,CO3,CO4	20	22	1	
PO6					
PO7					

CO-PO/PSO MATRIX

CO No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
A-406.1	3	2		3	3				3	1
A-406.2	3	2		3	3				3	1
A-406.3	3	3		3	3				3	1
A-406.4	3	3	2	3	3				3	
AVERAGE	3	2.5	0.5	3	3				3	0.75

COURSE CONTENT

1.0 Draw to scale various views like front view, side view, top view and Sectional Views of Automobile Engine Components

- 1.1 Pistons for diesel and petrol.
- 1.2 Connecting rods of I.C. Engines.
- 1.3 Cylinder Head for 2 stroke petrol engine.
- 1.4 Crank shaft.
- 1.5 Cam shaft.
- 1.6 Valves.
- 1.7 Rocker Arms.
- 1.8 Spark plugs.

2.0 Draw to scale various views like front view, side view, top view and Sectional Views of Chassis and transmission components

- 2.1 Linkage Bracket
- 2.2 Chassis Bracket
- 2.3 Gear box selector fork
- 2.4 Engine Mount
- 2.5 Bell crank lever

3.0 Cams and followers

- 3.1 Draw types of cams.
- 3.2 Draw the cam profile for the following motion of the follower
 - (i) Uniform velocity
 - (ii) Uniform Acceleration and retardation
 - (iii) Simple harmonic motion.
- 3.3 Draw cam profiles in above three cases for a knife edge follower, roller follower and flat Follower. (Inline cams)

4.0 Draw the automobile parts of the assembly drawing.

- 4.1 Connecting Rod
- 4.2 Master Cylinder
- 4.3 Wheel cylinder

REFERENCE BOOKS:

- 1. Automobile Engineering Drawing by R.B Gupta
- 2. Machine Drawing by N.D Bhat

VEHICLE SERVICE AND MAINTENANCE TECHNOLOGY LAB

Course Code	Course Title	No. of Periods/ Week	Total Periods/ Year	Marks for FA	Marks for SA
A-407	VEHICLE SERVICE AND MAINTENANCE TECHNOLOGY LAB	03	45	40	60

TIME SCHEDULE

Chapter No.	MAJOR TOPIC	PERIODS	CO's Mapping
1	Servicing and Maintenance of various systems of an Automobile	30	CO1, CO2, CO3 & CO5
2	Engine maintenance and OBD scanner	15	CO4
3	Demonstration of driving skills (<i>One batch will go for driving practice and another batch will do the experiment</i>)		CO5
	TOTAL	45	

Course Code: A – 407

COURSE: VEHICLE SERVICE AND MAINTENANCE TECHNOLOGY LAB

COURSE OBJECTIVES	Upon completion of the course the student shall able to 1. Understand servicing and maintenance of various systems of an automobile 2. Know the procedure of OBD scanner 3. Know the procedure of washing of a vehicle
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CO No.	Course Outcome
C01	A-407.1 Perform the maintenance of a given engine
C02	A-407.2 Perform the Fuel system maintenance of given MPFI or CRDI Engine
C03	A-407.3 Perform the maintenance, Troubleshoot and rectify the problems in braking system, suspension system, steering system, cooling and lubrication system of an engine.
C04	A-407.4 Diagonise automobile using OBD Diagnostic Equipment
C05	A-407.5 Perform water washing of a given vehicle and Demonstrate driving skills on four wheeler vehicle

LIST OF EXPERIMENTS

Servicing and Maintenance of various systems of an Automobile:

1. Study of different types of tools and instruments required for repair and maintenance
2. Engine Maintenance
3. Fuel system of MPFI and CRDI engine maintenance
4. Braking system maintenance
5. Steering system maintenance
6. Suspension system maintenance
7. Cooling system maintenance
8. Fault diagnosis using OBD Diagnostic Equipment
9. Water washing practice
10. Driving practice on the vehicle

COs – POs Mapping Format

Course Code:A-407		Course Title: Vehicle Service and Maintenance Technology Lab		No of Cos:5	No. of periods:90
POs	Mapped with CO No.	CO Periods Addressing PO in Col 1		Level (1,2,3)	Remarks
		No	%		
PO1	CO1,CO2,CO3,CO4,CO5	25	28	2	>40% Level 3 (Highly Addressed)
PO2	CO1,CO3,CO4	10	11	1	25% to 40% Level 2 (Moderately Addressed)
PO3	CO1,CO3,CO4	10	11	1	5% to 25% Level 1 (Low Addressed)
PO4	CO1,CO2,CO3,CO4	15	17	1	<5% Not Addressed
PO5	CO1,CO2,CO3,CO4,CO5	15	17	1	
PO6	CO1,CO2,CO3,CO4	15	16	1	
PO7					

CO-PO/PSO MATRIX

CO No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
A-407.1	3	3	2	3	3	2		3	1	3
A-407.2	2			1	3	1		3		3
A-407.3	3	3	3	2	2	2		3		3
A-407.4	3	3	3	2	2	2		3		3
A-407.5	2				3			3		3
AVERAGE	2.6	1.8	1.6	1.6	2.6	1.4		3	0.2	3

Competencies and key competencies to be achieved by the student

Sl. No	Title of the Experiment	Key competencies expected	Competencies expected
1	Tools and instruments required for repair and maintenance	Identify different tools and instruments for repairing of vehicle	Understand different tools and equipment's used for various vehicle repair and maintenance
2	Engine Maintenance	• Valve timing adjustment • Tappet setting • Air filter replacement • Oil change • Oil filter replacement • Blow by test • Cylinder compression test.	Understand value timing, tappet setting, blow by test and cylinder compression test
3	Fuel system of MPFI and CRDI engine maintenance	• Fuel injector checking and testing • Fuel Timing adjustment • Pump Timing adjustment • Checking (Diagnostic Trouble Codes)DTC's using OBD scanner • Rectification of faults in fuel system • Removal of DTC's from the vehicle	Understand fuel system of MPFI and CRDI engine maintenance
4	Braking systems Maintenance	• Identify the tools and gauges required • Dismantling and assembling • Fault finding and rectification of different troubles • Servicing & adjustments • Brake bleeding	• Understand the construction and working of braking system • Understand the braking systems used in different types of vehicles • Understand the grades of oils used in automobile braking systems

5	Steering systems Maintenance	- Identify the tools and gauges required - Dismantling & assembling - Fault finding and rectification of different troubles - Servicing & adjustments.	- Understand the construction and working - Understand the steering systems used in different types of vehicles - Understand the grades of oils used in automobile steering systems.
6	Suspension systems maintenance	- identify the tools and Gauges required - dismantling and assembling - Fault finding and rectification of different troubles - Servicing and adjustments	- Understand the construction and working - Understand the independent suspension systems - Used in different types of vehicles.
7	Cooling system maintenance	- Coolant Strength test using refractor meter - Descaling by using descaling agents and flushing - Coolant filter replacement - Radiator servicing and testing for leaks	Understand cooling system maintenance and coolant replacement

8	OBD Diagnostic Equipment	• Technical knowledge: A deep understanding of the vehicle's various systems • Diagnostic skills: The ability to read and interpret diagnostic trouble codes (DTCs) and other diagnostic information provided by the OBD Diagnostic Equipment • Problem-solving skills: This may include adjusting the settings on the scanner, replacing faulty components, or updating software. • Communication skills: Clear communication with other members of the engineering team or customers • Technical skills: Proficiency in using OBD Diagnostic Equipment, including the ability to navigate the interface, interpret data, and understand the various settings and options, Analytical skills: ability to interpret data, identify trends, and draw conclusions, which are crucial for identifying the root cause of issues. • Attention to detail: accurately recording data, tracking the progress of troubleshooting, and ensuring that all steps have been taken to resolve the issue.	• Used OBD Diagnostic Equipment • Understand the problems of vehicle
9	Water washing practice	• Operate hydraulic hoist for lifting the vehicle.	• Identify the components of washing equipment. • Set suitable water pressure. • Clean with suitable liquid. • Blow high pressure air.

10	Driving practice on the vehicle or driving simulator	• Foot controls • Driving controls • Operate all controls simultaneously.	• Identify foot controls of a four wheeler. • State the purpose of foot controls. • Operate foot controls efficiently. • Operate steering controls. • Operate gear shift lever. • Identify various accessories. • Operate all controls simultaneously.
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ENGLISH COMMUNICATION SKILLS

Course Title : English Communication Skills	Course code: C23-A- 408 (Common to all Branches)
Year/ Semester : IV Semester	Number of Periods : 45 (3 periods per week)
Type of Course : Practical	Max Marks : 100 (Internal 40 + External 60)

Course Objectives:	- to communicate effectively in diverse academic, professional and everyday situations
	- exhibit appropriate body language and etiquette at workplace
	- be employable through preparing appropriate job applications and attend interviews confidently with all necessary skills

CO No.	
CO1	Listen and comprehend the listening inputs related to different genres effectively
CO2	Communicate effectively in interpersonal interactions, interviews, group discussions and presentations
CO3	Acquire employability skills: job hunting, resume writing, attending interviews
CO4	Practise appropriate body language and professional etiquette

Course Delivery: Text book: “English Communication Skills”

by State Board of Technical Education and Training, AP

SI No	Unit	Teaching Hours
1	Listening Skills	6
2	Workplace Etiquette	3
3	Introducing Oneself	3
4	Short presentation (JAM)	6
5	Group Discussion	6
6	Resume Writing and Cover Letter	3
7	Interview Skills	9
8	Presentation Skills	9
<i>Total</i>		<i>45</i>

Course Content:

UNIT I: Listening Skills

6 periods

Pre – While- Post-listening activities- Listening to audio content (dialogues/ speech/ narrations) - answering the questions and fill in the blanks- vocabulary

UNIT 2: Workplace Etiquette

3 periods

Basics of Etiquette- politeness/ courtesy, good manners- features of work place etiquette- adaptability, positive attitude, body language.

UNIT 3: Introducing Oneself

3 periods

Speak about oneself - introduce oneself to a gathering/ formal & informal situations- Know about others- filling in the grid- introducing oneself in interviews

UNIT 4: Short Presentation

6 periods

Dos and Don'ts in short presentation- speak for a minute without repetition, deviation & hesitation - the techniques to speak fluently – defining and describing objects, people, phenomena, events.- speaking on randomly chosen topics.

UNIT 5: Group Discussion

6 periods

Fundamentals of Group Discussion- Dos and Don'ts- filling the Grid- possible list of topics- practice sessions- sample videos-Group activity

UNIT 6: Resume Writing and Cover Letter

3 periods

Pre activity: answer the questions- jotting down biographical information- sample resumes- tips, Dos and Don'ts- model resumes- practice exercises on Resume writing

UNIT 7: Interview Skills

9 periods

Pre –while-post activities: - things to do at three stages – respond to notifications- know the information about the organisation-practice FAQs - preparation of good/ suitable C V, Body language, tips for success in interviews, model / mock interviews.

UNIT 8: Presentation Skills

9 periods

Preparatory work: observe pictures and answer questions- different kinds of presentations- PPTs, Flash cards, Posters, Charts. - tips to prepare aids, slide show, model PPTs, - checklist on pre, while and post presentations.

Mapping Course Outcomes with Programme Outcomes:

PO	1	2	3	4	5	6	7
CO	POs 1 to 5 are applications of Engineering Principles, can't be directly mapped to English Communication Skills					1,2,3,4	1,2,3,4

Unit wise Mapping of CO –PO

CO	Course Outcome	COs / Unit Mapped	POs mapping	Cognitive levels as per Bloom's Taxonomy R/U/A/An (Remembering / Understanding / Applying/ Analysing)
CO 1	Listen and comprehend listening inputs related to different genres effectively	Unit 1	6,7	R/U/A
CO2	Communicate effectively in interpersonal interactions, interviews, group discussions and presentations	Units 3,4,5,7,8	6,7	R/U/A/An
CO3	Acquire employability skills: job hunting, resume writing, attending interviews	Units 6,7	6,7	R/U/A/An
CO4	Practise appropriate body language and professional etiquette	Units 2, 3, 4,5,7,8	6,7	R/U/A

HYDRAULIC AND PNEUMATIC CONTROLS LAB

Course Code	Course Title	No. of Periods/ Week	Total Periods/ Year	Marks for FA	Marks for SA
A-409	HYDRAULIC AND PNEUMATIC CONTROLS LAB	03	45	40	60

TIME SCHEDULE

Chapter No.	Title	Periods	CO's Mapping
1	Venturimeter	12	CO1
2	Orifice meter	12	CO1
3	Identification of Oil Power Hydraulic controls.	09	CO2,CO3, CO4 & CO5
4	Identification of Pneumatic Power controls.	12	CO2,CO3, CO4 & CO5
Total		45	

Course Title : HYDRAULIC AND PNEUMATIC CONTROLS LAB	
Course Objectives	1.0 Determination of coefficient of discharge by using 1. Orifice meter 2. Venturi meter 2.0 Familiarise with the experimental setup for hydraulic and pneumatic actuators and motors 3.0 Familiarise with the experimental setup for hydraulic and pneumatic flow control valve, pressure control valve and direction control valve.

CO No.	Course Outcome
C01	A-409.1 Determination of coefficient of discharge by using 1. Orifice metre 2. Venturi meter
C02	A-409.2 Operate the experimental setup for the hydraulic and pneumatic actuators and take required observations
C03	A-409.3 Operate the experimental setup for the hydraulic and pneumatic motors and take required observations
C04	A-409.4 Operate the experimental setup for the hydraulic and pneumatic direction control valve and take required observations
C05	A-409.5 Operate the experimental setup for the hydraulic and pneumatic flow control valve and pressure control valve by taking required observations

Learning outcomes

1.0 Practice the determination of C_d of Venturi meter

- 1.1 State the practical applications of venturi meter.
- 1.2 Record the manometric head readings from U-tube manometer
- 1.3 Record the time taken for collecting discharge by varying the discharge
- 1.4 Calculate the areas of the pipe and throat of the given venturi meter
- 1.4 Calculate coefficient of discharge of venturi meter.

2.0 Practice the determination of C_d of Orifice meter

- 2.1 State the practical applications of Orifice meter.
- 2.2 Record the manometric head readings from U-tube manometer
- 2.3 Record the time taken for collecting discharge by varying the discharge
- 2.4 Calculate the areas of the pipe and throat of the given Orifice meter
- 2.5 Calculate coefficient of discharge of Orifice meter.

3.0 Identification of Oil Power Hydraulic controls

- 3.1 Identify the principal components of oil power hydraulic circuits.
- 3.2 Understand the principle of working of hydraulic calculator.
- 3.3 Understand the principle of working of hydraulic motor.
- 3.4 Understand the principle of working of direction control valve.
- 3.5 Understand the principle of working of flow control valve.
- 3.6 Understand the principle of working of pressure control valve.

4.0 Identification of Pneumatic Power controls

- 4.1 Identify the principal components of pneumatic power circuits.
- 4.2 Understand the principle of working of pneumatic actuator.
- 4.3 Understand the principle of working of pneumatic motor.
- 4.4 Understand the principle of working of direction control valve.
- 4.5 Understand the principle of working of flow control valve.
- 4.6 Understand the principle of working of pressure control valve.

CO-PO Mapping

Course Code: A-409	Course Title : HYDRAULIC AND PNEUMATIC CONTROLS LAB	No of Cos:5			No. Of periods:45
POs	Mapped with CO No	CO Periods addressing PO in Col 1		Level)	Remarks
		No	%		>40% Level 3 (Highly Addressed)
PO1	CO1, CO2, CO3, CO4, CO5	15	33	2	25% to 40% Level 2 (Moderately Addressed)
PO2	CO1, CO2, CO3, CO4, CO5	10	22	1	5% to 25% Level 1 (Low Addressed)
PO3					<5% Not Addressed
PO4	CO1, CO2, CO3, CO4, CO5	10	22	1	
PO5	CO1, CO2, CO3, CO4, CO5	10	23	1	
PO6					
PO7					

CO-PO/PSO MATRIX

CO No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
A-409.1	3	1		3	3				3	
A-409.2	3	3		3	2				3	
A-409.3	3	3		3	2				3	
A-409.4	3	3		3	2				3	
A-409.5	3	3		3	2				3	
Average	3	2.6		3	2.2				3	

Note: The gaps in CO and PO mapping will be achieved by one or more appropriate activities from the following:

i) Seminars ii) Tutorials iii) Guest Lectures iv) Assignments v) Quiz competitions vi) Industrial visit vii) Tech Fest viii) Mini project ix) Group discussions x) Virtual classes xi) Library visit for e-books

COURSE CONTENT

1. Determination of Coefficient of discharge of Venturi meter.
2. Determination of Coefficient of discharge of Orifice meter
3. Hands on experience on oil power hydraulic trainer.
4. Hands on experience on pneumatic power trainer.

CADLAB – II

Course Code	Course Title	No. of Periods/ Week	Total Periods/ Year	Marks for FA	Marks for SA
A-410	CAD LAB-II	03	45	40	60

TIME SCHEDULE

Chapter No.	Title	Periods	CO's Mapping
1.	Introduction to Computer Aided Drafting Interface and its various tools –3D	3	CO1
2.	Practice usage of various Computer Aided Drafting tools – 3D	6	CO2
3.	Computer Aided Drafting–3 D Part Drawings	15	CO3
4.	Computer Aided Drafting–3 D Assembly Drawing	21	CO4
TOTAL		45	

CourseCode:A-410	Course Name: CADLAB - II	No.OfPeriods:45
Course Objectives	Upon completion of the course the student shall be able to 1. Understand working in 3D CAD environment. 2. Create 3D drawings of automotive engine/chassis/transmission components using CAD software. 3. Create 3D assembly drawings of standard automobile parts using CAD software.	

CO No.	Course Outcome
CO1	A-410.1 Explain 3D Computer Aided Drafting Interface and its various tools
CO2	A-410.2 Practice usage of 3D Computer Aided Drafting tools
CO3	A-410.3 Create 3D drawings using CAD software
CO4	A-410.4 Create 3D assembly drawings using CAD software

Learning Outcome

1.0 Introduction to Computer Aided Drafting Interface and its various tools – 3D

- 1.1 Explain CAD Environment 1. Screen, 2. Various tool bars, 3. Task pane, 4. Feature manager Design tree
- 1.2 Understand 3D Navigation Tools used to move around and view the 3D model from different angles and perspectives.
- 1.3 Understand Modeling Tools used to create 3D shapes and objects.
- 1.4 Understand Editing Tools used to modify existing 3D objects.
- 1.5 Understand Materials and Textures tools used to apply materials and textures to 3D objects, giving them a realistic look and feel.
- 1.6 Understand Lighting and Rendering tools used to set up lighting and create realistic 3D renderings of the model.

2.0 Practice usage of various Computer Aided Drafting tools – 3D

- 2.1 Create simple 3D parts with the lines/circles/arcs/ellipses/elliptical arcs/infinite lines and shapes like rectangles, polygons, polylines, Splines.
- 2.2 Practice moving around and viewing the 3D model from different angles and perspectives using 3D Navigation Tools.
- 2.3 Practice creating 3D shapes and objects using Modeling Tools.
- 2.4 Practice modifying existing 3D objects using Editing Tools.
- 2.5 Practice applying materials and textures to 3D objects to give them a realistic look and feel using Materials and Textures tools.
- 2.6 Practice setting up lighting and create realistic 3D renderings of the model using Lighting and Rendering tools.

3.0 Create 3D drawings

- 3.1 Create 3D drawings of Bolts, Nuts, Springs
- 3.2 Linkage Bracket
- 3.3 Chassis Bracket
- 3.4 Gear box selector fork
- 3.5 Engine Mount
- 3.6 Bell crank lever
- 3.7 Cylinder Block
- 3.8 Petrol engine piston
- 3.9 Inlet and Exhaust Valves
- 3.10 Crank shaft

4.0 Create 3D assembly drawings

- 4.1 Leaf spring
- 4.2 Knuckle joint
- 4.3 Universal joint
- 4.4 Screw jack
- 4.5 Piston (with rings and gudgeon pin)
- 4.6 Connecting rod

Course Code: A-410	Course Title: CAD Lab - 2	No of Cos: 4			No. Of periods: 45
POs	Mapped with COs	CO Periods Addressing PO in Col 1		Level (1,2,3)	Remarks
		No	%		
PO1	CO1,CO2,CO3,CO4	12	26	2	>40%Level3 (Highly Addressed) 25%to40%Level2 (Moderately Addressed) 5%to25%Level1 (Low Addressed) <5%Not Addressed
PO2					
PO3	CO3,CO4	27	60	3	
PO4	CO2,CO3,CO4	3	7	1	
PO5	CO2,CO3,CO4	3	7	1	
PO6					
PO7					

CO-PO/PSOMATRIX

CO No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
A-410.1	2									
A-410.2	2			1	1					
A-410.3	2		3	1	1			2		
A-410.4	2		3	1	1			2		
Average	2		1.5	0.75	0.75			1		

Note: The gaps in CO and PO mapping will be achieved by one or more appropriate activities from the following:

- | | |
|-----------------------|-------------------------------|
| ii) Seminars | ii) Guest Lectures |
| iv) Tutorials | iv) Assignments |
| v) Quiz competitions | vi) Industrial visit |
| ix) Group discussions | vii) Tech Fest |
| | viii) Mini project |
| | x) Virtual classes |
| | xi) Library visit for e-books |

COURSE CONTENT

S.No	Experiment Title	Key Competency
1.	Introduction to Computer Aided Drafting Interface and its various tools – 3D	- Explain the concept of 3D - Explain CAD Environment 1. Screen, 2. Various tool bars, 3. Task pane, 4. Feature manager Design tree - Move the 3D model around and view from different angles and perspectives using 3D Navigation Tools
2.	Practice usage of various Computer Aided Drafting tools – 3D	- Create 3D shapes and objects using Modelling Tools - Modify existing 3D objects using Editing Tools. - Apply materials and textures to 3D objects using Materials and Textures tools. - Set up lighting and create realistic 3D renderings of the model using Lighting and Rendering tools. - Create simple 3D parts with the lines / circles/arcs/ellipses/elliptical arcs/infinite lines and shapes like rectangles, polygons, polylines, Splines.
3.	Create 3D drawings	Create 3D drawings of - Bolts, Nuts, Springs - Linkage Bracket - Chassis Bracket - Gear box selector fork - Engine Mount - Bell crank lever - Cylinder Block - Petrol engine piston - Inlet and Exhaust Valves - Crank shaft
4.	Create 3D assembly drawings	- Create 3D parts of given component. - Import 3D parts of into assembly drawing module. - Mate the different parts of the component to make complete assembly. - Create exploded view of the assembled component

REFERENCEBOOKS

1. 3D CAD Principles and Applications by Hiroshi Toriya, Hiroaki Chiyokura
2. UpandRunningwithAutoCAD20172Dand3DbyElliotJ.Gindis.
3. Commands Guide Tutorial for SolidWorks A comprehensive reference guide with over 230 tutorials David C. Planchard and Marie P. Planchard
4. CATIA V5 Workbook Release 19 by Richard Cozzens
5. Creo Parametric 1.0 Tutorial and MultiMedia DVD by Roger Toogood, Jack Zecher P.E.

V SEMESTER

DIPLOMA IN AUTOMOBILE ENGINEERING
SCHEME OF INSTRUCTIONS AND EXAMINATION
V SEMESTER

Subject Code	Name of the Subject	Instruction period/week		Total Periods /Semester	Scheme of Examination			
		Theory	Practical/ Tutorial		Duration (hours)	Sessional Marks	End Exam Marks	Total Marks
THEORY								
A-501	Industrial and Transport Management	4	-	60	3	20	80	100
A-502	Design of Machine Elements	5	-	75	3	20	80	100
A-503	Advanced Manufacturing Processes	4	-	60	3	20	80	100
A-504	Alternative Fuels & Emission Controls	4	-	60	3	20	80	100
A-505	Electric and Hybrid Electric Vehicles	4	-	60	3	20	80	100
PRACTICAL								
A-506	IC Engines Testing & Vehicle Diagnostics Lab	-	6	90	3	40	60	100
A-507	Electric & Hybrid Electric Vehicles Lab	-	3	45	3	40	60	100
A-508	Life Skills	-	3	45	3	40	60	100
A-509	Project Work	-	6	90	3	40	60	100
Student Centric Activity (Library, Mini project, Physical Education, Preparation for competitive exams/placements Seminars, Reporting etc.)			3	45				
TOTAL		21	21	315+315	-	260	640	900

A-508 Common with all branches

INDUSTRIAL AND TRANSPORT MANAGEMENT

Course Code	Course Title	No. of Periods/ Week	Total Periods/ Year	Marks for FA	Marks for SA
A-501	INDUSTRIAL AND TRANSPORT MANAGEMENT	04	60	20	80

TIME SCHEDULE

Chapter No.	Title	Periods	Marks	Part – A Short Type (3 Marks)	Part –B Essay Type (10 marks)	CO's Mapping
1	Principles and functions of Industrial Management	8	13	1	1	CO1
2	Organization structure& Organizational behavior	10	16	2	1	CO2
3	Production Management & Industrial safety	10	13	1	1	CO3
4	Entrepreneurship Development	10	16	2	1	CO4
5	Transport Management	12	26	2	2	CO5
6	Legal Aspects of MV Act	10	26	2	2	CO5
Total:		60	110	10	8	

Course Code : A-501	Course Title : Industrial and Transport Management	No of Periods : 60
Course Objectives	Upon completion of the course the student shall be able to 1. Understand the principles and functions of industrial management, organisation structure and organisational behaviour. 2. Understand the production management, industrial safety and entrepreneurship development. 3. Understand the Transport Management and Legal Aspects of MV Act.	

Course Outcomes	Upon completion of the course the student shall be able to		
	CO1	A501.1	Explain principles and functions of industrial management.
	CO2	A501.2	Explain organisation structure and organisational behaviour.
	CO3	A501.3	Solve problems in CPM and PERT techniques in production management and explain industrial safety.
	CO4	A501.4	Discuss 1. Entrepreneurship 2. Demand and Market Survey 3. Feasibility Study
	CO5	A501.5	Discuss 1. Transport Management and 2. Legal Aspects of MV Act

Learning outcomes:

Upon completion of the course the student shall be able to

1.0 Principles and functions of Industrial Management

- 1.1 Define industry, commerce (Trade) and business.
- 1.2 State the need for management.
- 1.3 Explain functions of Management.
- 1.4 Explain the principles of scientific management (F.W.Taylor).
- 1.5 Differentiate between management and administration.
- 1.6 Differentiate between low, middle and top level management.
- 1.7 Explain the importance of managerial skills. (1. Technical, 2.Human & 3. Conceptual skills)

2.0 Organisation Structure & organisational behaviour

- 2.1 Explain the need of organisation structure of an industry.
- 2.2 Discuss the line, staff and Functional organisations.
- 2.3 Differentiate between Delegation and decentralization
- 2.4 Describe the communication process and motivational theories (Maslow's Hierarchy Theory)
- 2.5 Explain about leadership theories (Trait's theory and behavioural theory).
- 2.6 Explain the process of recruitment, selection, training and development.
- 2.7 State the importance and classify the Motor Transport Organisation.
- 2.8 Differentiate between passenger transport and goods transport, publictransport and private transport.
- 2.9 Explain types of business ownerships.
- 2.10 Explain the different types of ownership of Motor Transport Organisation.
- 2.11 Explain the functional wings of a transport system.

3.0 Production management

- 3.1 Define Production Management.
- 3.2 State the objectives and principles of plant Layout
- 3.3 State the factors of Plant Location
- 3.4 Explain the types of plant Layouts
- 3.5 State and explain different types of production.
- 3.6 Explain Break Even Analysis and Project Scheduling.
- 3.7 Draw CPM and PERT networks and identify the critical path.
- 3.8 Simple numerical problems on CPM and PERT.
- 3.9 Explain the importance of safety at Work place.
- 3.10 Explain about industrial hazards.
- 3.11 State the causes of accidents and safety provisions.

4.0 Entrepreneurship Development.

- 3.1 Define the word entrepreneur.
- 4.1 State the requirements of an entrepreneur.
- 4.2 Determine the role of entrepreneurs in promoting Small Scale Industries.
- 4.3 Describe the details of self-employment schemes.
- 4.4 List the financial assistance programmes.
- 4.5 List the organisations that help an entrepreneur
- 4.6 Explain the concept of Make in India, Zero Effect and Zero Defect
- 4.7 State the importance for start-ups
- 4.8 Explain the conduct of Demand Survey and Market Survey.
- 4.9 Explain about feasibility study.
- 4.10 Explain the function of material management in industry
- 4.11 Explain 1. ABC Analysis 2.Safety stock 3. Re order level.
- 5.0 Transport Management:**
- 5.1 State and explain the types of operations.
- 5.2 State and Explain the operating characteristics: utility, capacity, dependability, Safety, distance, speed, road condition, fuel economy, traffic interference and flexibility .
- 5.3 Explain about traffic data – published data
- 5.4 Explain the terms duty rosters, trip sheet, way bill and log book.
- 5.5 State and explain the factors to be considered in Bus scheduling.
- 5.6 State and explain the factors to be considered in Crew scheduling.
- 5.7 State and explain the types of fare methods.
- 5.8 Explain peak hour demand.
- 5.9 Define the term IoT and List the key features of IoT.
- 5.10 List the advantages, disadvantages and applications of IoT
- 5.11 Explain Smart Transportation.
- 5.12 Explain Value added services: TAT and Telematics.
- 6.0 Legal aspects of Motor Transport:**
- 6.1 Explain traffic signs and signals.
- 6.2 Explain registration procedure for the vehicle.
- 6.3 Explain the necessity of permit and meaning of fitness certificate.
- 6.4 Explain the procedure for obtaining driving license.
- 6.5 Explain insurance coverage and types of insurance coverage.
- 6.6 Explain the motor claim settlement procedure.

COURSE CONTENT:

1.0 Principles and Functions of Industrial Management

Introduction - Industry, Commerce and Business; Definition of management; Functions of management - Principles of scientific management by F.W.Taylor.-Administration and Management, levels of Management, Managerial skills.

2.0 Organisation Structure and Organisational Behaviour

Organization, Principles of Organization-Types of Organizations; Line, Staff and functional Organizations, Decentralization and Delegation, Communication, Motivational Theories, Leadership Theories; Human resources development; recruitment selection training and development, - Importance of motor transport in present day transport system – types of motor transport – Passenger transport and Goods transport – public transport and private transport - Types of business ownership-transport ownership- State, Public Under Taking, Private and

Municipal –functional wings of motor transport - administrative, traffic, secretarial and engineering function.

3.0 Production Management& Industrial safety

Definition of production management; objectives and principles of plant layout, Plant location and types of layout; Types of production -job, batch and mass production - Break even analysis; Project scheduling; Application of CPM and PERT techniques; simple numerical problems – importance of safety at work places – industrial hazards – causes of accidents- safety provisions-Material management-function – ABC Analysis-Safety stock –Reorder level.

4.0 Entrepreneurship Development

Definition of Entrepreneur; Requirements of entrepreneur, Role of Entrepreneur, , Details of self-employment schemes, financial assistant programmes, organizations that help entrepreneurs (SSI, MSME, DIC, Banks) - Concept of Make In India, ZERO defect, Zero Effect, Concept of Start-up Company, Demand survey and Market survey; Preparation of Feasibility study reports

5.0 Transport Management

Types of operations – city operation – mofussil operation –operating characteristics – utility – capacity , Dependability – safety, distance, speed – road condition- fuel economy, Traffic interference, flexibility – traffic data – published data – duty rosters –Trip sheet – way bill – log book – factors to be considered in bus scheduling – traffic demand, running time, Stand time, maintenance time- factors to be considered in crew scheduling–Hours of work–Daily and weekly intervals of rest –Spread over, spell of duty, overtime allowance – Handing over and taking over time – peak hour demand – Fare methods – Straight line method, taper scale method, flat rate, concessional fare, luggage fare. Overview of IoT - Define IoT - key features of IoT - advantages and disadvantages of IoT - IoT Applications - Smart Transportation - Value added services: TAT (Turnaround Time), Telematics

6.0 Legal aspects of Motor Transport

Traffic signs and signals–procedure for registering of a vehicle – necessity of permits – Fitness certificate – Procedure for issue of driving license – Motor insurance– basic principle – Scope and coverage of different policies –Claim settlement procedure.

REFERENCE BOOKS

S.no	Name of text book	Author	Publisher
1.	Industrial Engineering and Management	P KhannaDhanpatrai	-2019
2.	Production Management Buffa.	Abe books	
3.	Engineering Economics and Management Science -	Banga& Sharma.	Khanna
4.	Production and Operations Management –	S.N. Chary	Tata McGraw Hill
5.	Motor vehicle act of state and central Govt.		
6.	Traffic Engineering and Transport Planning -	L.R. Kadiyali	Khanna
7.	Motor vehicle workshop organisation and administration -	Bernard Chandler	Oxford
8.	Bus Operation and Bus Crew Scheduling -	CIRT. (Pune)	
9.	An introduction to Transport Engineering -	Wilton W Hey	
10.	Elements of Transport -	R.J. Eatan	CIRT. (Pune)

COs – POs Mapping**A-501, Industrial and Transport Management**

Course Code: A-501		Course Title: Industrial and Transport Management			No of Cos:5	No. Of periods:60
POs	Mapped with CO No	CO Periods Addressing PO in Col 1		Level (1,2,3)	Remarks	
		No	%		>40% Level 3 (Highly Addressed)	
PO1	CO1,CO2,CO3	16	27	2	25% to 40% Level 2 (Moderately Addressed)	
PO2	CO3	4	7	1	5% to 25% Level 1 (Low Addressed)	
PO3					<5% Not Addressed	
PO4						
PO5	CO4, CO5	24	40	3		
PO6	CO2,CO5	8	13	1		
PO7	CO2,CO5	8	13	1		

Table specifying the scope of syllabus to be covered for Unit Test-I &Unit Test –II

Unit Test	Learning Outcomes to be covered
Unit Test-I	From 1.1 to 3.11
Unit Test-II	From 4.1 to 6.6

Course Code	Course Title	No. of Periods/ Week	Total Periods/ Year	Marks for FA	Marks for SA
A-502	DESIGN OF MACHINE ELEMENTS	05	75	20	80

TIMESCHEDULE

Chapter No.	Title	Periods	Marks	Part–A ShortType (3 marks)	Part – B EssayType (10 marks)	CO's Mapping
1	Bolts and Nuts	8	11	2	1/2	CO1 & CO3
2	Shafts	15	26	2	2	CO1 & CO3
3	Keys and Couplings	12	23	1	2	CO1 & CO3
4	Balancing and Vibrations	8	11	2	1/2	CO2
5	Governors and Flywheel	14	16	2	1	CO4
6	Design of Auto mobile Components	18	23	1	2	CO5
TOTAL		75	110	10	8	

Course Code: A-502	Title: Design of Machine Elements	Total No. of Periods: 75
Course Objectives:	Upon completion of the course the student shall be able to	
	1. Know the functions of bolts, nuts, shafts, keys, couplings and automobile components	
	2. Understand the design of bolts, nuts, shafts, keys, couplings and automobile components	
	3. Understand the concept of balancing and vibrations	
	4. Solve problems on design of above components	

CO No.	Course Outcome
C01	A-502.1 Explain functions of Bolts and Nuts, Shafts and Couplings
C02	A-502.2 Describe the concept of balancing and vibrations
C03	A-502.3 Design Bolts and Nuts, Shafts, Keys and couplings with given parameters.
C04	A-502.4 Designing governors and flywheel with the given data.
C05	A-502.5 Design automobile components with given parameters.

LEARNING OUTCOMES:

Upon completion of the course the student shall be able to understand

1. Bolts and Nuts

1.1 List design factors considered while designing a machine component

- 1.2 Describe the terminology used in bolts and nuts—its proportions
- 1.3 State the effect of forces involved when the nut is tightened and the maximum load that asset of bolt and nut can take up before its failure
- 1.4 Calculate the dimensions of bolt and nut by using empirical formula
- 2. Shafts**
- 2.1 Explain the concept of torsion and torque equation
- 2.2 State the assumptions made in pure torsion
- 2.3 Describe the functions of a shaft and various materials used in manufacture of shaft with respect to its adaptability
- 2.4 Calculate the diameters of hollow and solid shafts
- 2.5 Calculate the diameters of hollow & solid shafts considering both strength & stiffness
- 3. Keys and Couplings**
- 3.1 State the function of keys and splines with the help of neat sketches and material used for them
- 3.2 Explain the types of keys
- 3.3 Design rectangular sunk key considering failure against shear & crushing using empirical proportions
- 3.4 Describe the proportions of a spline for various applications
- 3.5 State the function of a coupling
- 3.6 Calculate the proportions of muff coupling and flange coupling
- 3.7 Calculate the proportions of muff coupling, C.I. flange coupling considering the failure of the bolts against shear for a given torque by using the empirical formula
- 4. Balancing and Vibrations**
- 4.1 State the types of balancing
- 4.2 Explain the balancing of masses
- 4.3 Solve problems on balancing of masses
- 4.4 Describe vibrations
- 4.5 Describe types of vibrations
- 5. Governors and Flywheel**
- 5.1 State the types of Governors and their applications
- 5.2 Describe the working of Porter Governor
- 5.3 Solve problems on design of porter governor
- 5.4 State the purpose and applications of flywheel
- 5.5 Differentiate between the governor and flywheel.
- 5.6 Explain the turning moment diagram
- 5.7 Solve problems on design of flywheel
- 6. Design of Automobile components**
- 6.1 Design of Engine Cylinder
- 6.2 Design of Piston
- 6.3 Design of Connecting Rod
- 6.4 Design of Crankshaft
- 6.5 Design of Engine Valve
- 6.6 Design of Clutch
- 6.7 Design of Brake
- 6.8 Design of Universal coupling

Cos – Pos Mapping Format

Course Code: A-502	Course Title: Design of Machine Elements	No of Cos: 5		No. Of periods: 75	
POs	Mapped with CO No	CO Periods Addressing PO in Col 1		Level (1,2,3)	Remarks
		No	%		>40% Level 3 (Highly Addressed)
PO1	CO1, CO2,CO3,CO4,CO5	12	16	1	25% to 40% Level 2 (Moderately Addressed)
PO2	CO1, CO2,CO3,CO4,CO5	20	27	2	5% to 25% Level 1 (Low Addressed)
PO3	CO2	12	16	1	<5% Not Addressed
PO4	CO1, CO2,CO3,CO4,CO5	15	20	1	
PO5	CO1, CO2,CO3,CO4,CO5	16	21	1	
PO6					
PO7					

CO-PO/PSO MATRIX

CO No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
A-502.1	3	2		1	2				3	
A-502.2	3	1	1	1	2				3	
A-502.3	3	1		1	3				3	
A-502.4	1	3		1	3				3	
A-502.5	3	3		1	3				3	
AVERAGE	2.6	2	0.2	1	2.6				3	

COURSECONTENT

1. BoltsandNuts

Definition of terms: pitch, lead, lead angle, thread angle, nominal diameter, crest , root, depth of thread etc, related to bolts and nuts — classification ofbolts — loads acting on the bolts and the consequent failures — calculation ofvarious dimensions of a bolt and nut for a given diameter using empiricalproportions.

2. Shafts

Theory of pure torsion— Torque (torsion) equation (No Derivation)—assumptions made in pure torsion - Function of shafts— materials used —Power transmitted by solid and hollow circular shafts — Design of solid and hollow shafts subjected to (i) shear stress and angle of twist and (ii) shear stress and bending stress.

3. Keys and couplings

Functions of keys and splines - materials used - nature of failure of a key -Types of keys -saddle, flat round and sunk keys - Design of a rectangular sunk key— proportion of splines-Function of coupling—classification of couplings — problems on design of muff coupling and flange coupling.

4. Balancing and Vibrations

Necessity of balancing - types of balancing - balancing of single revolving mass— balancing of number of rotating masses —balancing of reciprocating masses – simple problems on balancing of masses – Vibrations – types of vibrations.

5. Governors and Flywheels

Function of Governor — types — applications - working of Porter Governor - simple problems on design of Porter Governor - Function of Flywheel - types – purpose and applications - Comparison between governors and flywheels – turning moment diagram-design of flywheel—problems on design of flywheel.

6. Design of Automobile Components

Design Engine Cylinder—Piston—Connecting Rod—Crankshaft—Engine Valve—Clutch— Brake—Universal coupling.

REFERENCE BOOKS

1. Auto Design R.B Gupta 2016 Satya Prakash publications
2. Machine Design R.S Khurmi 2005 S.Chand publications

Unit Test – I & Unit Test - II

Unit Test	Learning Outcome to be covered
Unit Test-I	1.1 to 3.7
Unit Test-II	4.1 to 6.8

ADVANCED MANUFACTURING PROCESSES

Course Code	Course Title	No. of Periods/ Week	Total Periods/ Year	Marks for FA	Marks for SA
A-503	ADVANCED MANUFACTURING PROCESSES	04	60	20	80

TIME SCHEDULE

Chapter No.	Title	Periods	Marks	Part – A Short Type (3 marks)	Part –B Essay Type (10 marks)	CO's Mapping
1	Introduction to modern manufacturing processes	8	16	2	1	CO1
2	Advanced casting, welding and surface finishing processes	16	26	2	2	CO2
3	Additive manufacturing (3D Printing) process	10	21	2	1 ½	CO3
4	Processing of Plastics	12	21	2	1 ½	CO4
5	Automation and robotics in manufacturing	14	26	2	2	CO5
Total:		60	110	10	8	

Course Code: A-503 Course Title: ADVANCED MANUFACTURING PROCESSES	
No. of Periods:60	
Course Objectives	Upon completion of the course the student shall be able to 1. Understand working principle of various modern manufacturing processes 2. Appreciate advanced casting, welding and surface finishing processes 3. Comprehend flexible manufacturing systems and Robotics

CO No.	Course Outcome
CO1	A-503.1 Understand working principle of various modern manufacturing processes
CO2	A-503.2 Explain advanced casting, welding and surface finishing processes
CO3	A-503.3 Describe the process of 3D printing
CO4	A-503.4 Describe processing of plastics
CO5	A-503.5 Explain flexible manufacturing systems and Robotics

LEARNING OUTCOMES:

Upon completion of the course the student shall be able to

1. Introduction to modern manufacturing processes

- 1.1 Explain need for modern manufacturing processes
- 1.2 Define modern manufacturing processes
- 1.3 Classify modern manufacturing processes
- 1.4 Differentiate traditional and modern manufacturing processes
- 1.5 List mechanical energy based processes
- 1.6 List electrical energy based processes
- 1.7 List chemical energy based processes
- 1.8 List thermal energy based processes
- 1.9 Explain working principle of Ultrasonic Machining and Abrasive Water Jet Machining with sketch
- 1.10 Explain working principle of Chemical Milling and Electrochemical Machining with sketch
- 1.11 Explain working principle of Laser Beam machining and Plasma Arc Machining with sketch

2. Advanced casting, welding and surface finish processes

- 2.1 Explain metal mould casting
- 2.2 Explain continuous casting and squeeze casting
- 2.3 Explain vacuum mould casting, evaporative pattern casting and ceramic shell casting
- 2.4 Explain electron beam welding with sketch
- 2.5 Explain laser beam welding with sketch
- 2.6 Explain ultrasonic welding with sketch
- 2.7 Explain friction stir welding with sketch
- 2.8 List different types of conventional surface finishing processes
- 2.9 List modern surface finishing processes
- 2.10 Differentiate in surface finish with conventional and modern processes
- 2.11 Explain principle of abrasive flow machining
- 2.12 Explain principle of magnetic abrasive finishing

3. Additive manufacturing (3D Printing) process

- 3.1 State development of additive manufacturing technologies
- 3.2 Explain advantages and limitations of 3D printing
- 3.3 State the applications of 3D printing
- 3.4 Explain general additive manufacturing processes
- 3.5 Explain Fused deposition modelling with sketch
- 3.6 Explain Selective laser sintering with sketch

4. Processing of Plastics

- 4.1 State the advantages and limitations of plastics as engineering materials
- 4.2 Differentiate between thermoplastics and thermosetting plastics
- 4.3 Describe the principle of making plastic products by compression moulding, Transfer moulding, blow moulding and injection moulding with sketches
- 4.4 Explain the principle of extruding with sketch
- 4.5 Explain the principle of calendaring with sketch

5. Automation and robotics in manufacturing

- 5.1 Define NC and CNC machines
- 5.2 Differentiate between NC and CNC machines

- 5.3 Explain the principle of open loop NC machine and closed loop NC Machine
- 5.4 State the advantages and limitations of NC machines over the Conventional machines
- 5.5 Explain the principle of computerized numerical control (CNC) machining
- 5.6 Explain direct numerical control (DNC) system
- 5.7 Define FMS
- 5.8 Describe the major elements and features of FMS
- 5.9 State the advantages and limitations of FMS
- 5.10 State the meaning of concept 'Robot'
- 5.11 Describe the basic elements common to all robots
- 5.11 State advantages and typical applications of Robots in production and Manufacturing
- 5.12 Assembly line for two wheelers and four wheelers- Trim assembly, chassis line and final Assembly
- 5.13 State the application of robots in assembly

COs – POs Mapping

Course Code: A-503		Course Title: Advance Manufacturing Processes			No. of Cos: 5	No. of periods: 60
POs	Mapped with CO No	CO Periods Addressing PO in Col 1		Level (1,2,3)	Remarks	
		No	%		>40% Level 3 (Highly Addressed)	
PO1	CO1,CO2,CO3,CO4,CO5	15	25	2	25% to 40% Level 2 (Moderately Addressed)	
PO2	CO3,CO5	5	07	1	5% to 25% Level 1 (Low Addressed)	
PO3	CO3,CO5	10	17	1	5% to 25% Level 1 (Low Addressed)	
PO4	CO1,CO2,CO3,CO4,CO5	10	17	1	5% to 25% Level 1 (Low Addressed)	
PO5	CO1,CO2,CO3,CO4,CO5	10	17	2	25% to 40% Level 2 (Moderately Addressed)	
PO6					<5% Not Addressed	
PO7	CO1,CO2,CO3,CO4,CO5	10	17	1	5% to 25% Level 1 (Low Addressed)	

CO-PO/PSOMATRIX

CO No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
A-503.1	3			2	2		2	3		3
A-503.2	3			1	2		3	3		3
A-503.3	3	1	1	2	2		3	3		3
A-503.4	2				1			3		3
A-503.5	3	2	3		2			3		3
AVERAGE	2.8	0.6	0.8	1	1.8		1.6	3		3

COURSECONTENT

1. Introduction to Modern Manufacturing Processes:

Introduction-Need of modern manufacturing processes-definition–classification -difference between traditional and modern manufacturing processes-

Mechanical energy based processes: Jet Machining, Water Jet Machining, and Abrasive water jet machining & Ultrasonic Machining-their working principle.

Electrical energy based processes: working principle of Electrical discharge machining.

Chemical energy based processes: working principle of Chemical Milling&Electrochemical Machining.

Thermal energy based processes: working principle of Laser Beam machining, Plasma Arc Machining&Electron Beam machining.

2. Advancedcasting, welding and Surface Finishing processes

Introduction-working of: metal mould casting-continuous casting-squeeze casting

Vacuum mould casting-evaporative pattern casting-ceramic shell casting.

Electron beam welding-Laser beam welding-Ultrasonic welding-Friction stir welding- types of conventional surface finishingprocesses: Lapping and Honing-modern surface finishing processes: abrasive flow machining (AFM)- magnetic abrasive finishing(MAF) and their principle - surface finish of conventional and modern processes.

3. Additive manufacturing (3D printing)process

Introduction-development of additive manufacturing technologies-advantages and limitations-general additive manufacturing processes: Fused deposition modelling,Selective laser sintering-Applications

4. Processing of Plastics

Advantages and limitations of plastics over other engineering Materials - differences between thermoplastics and thermosetting plastics - Processing of plastics: (a) Compression moulding (b) Transfer moulding (c) Extruding (d) Blow moulding (e) Injection moulding (f) calendaring

5. Automation and Robotics in manufacturing

Basicconcepts of NCmachines–comparison with operator controlled Machine - Types of NC system – open loop and closed loop - Classification of NC systems– Pointtopointpositioningcontrolsystem,straightcutpositioning system,continuous path or contouring system - Programming for NC machines – absolute programming, incremental

programming – Advantages and limitations of NC system over the conventional system - Basic concept of CNC and DNC systems.

FMS-definition-Features of FMS - advantages and limitations of FMS - Concept of Robot – Basic elements: Manipulator, controller, end – effectors, sensors, energy source and their functions - Basic construction of Robot and types of Robots- Assembly line for two wheelers and four wheelers- Trim assembly, chassis line and final Assembly-application of robots in assembly.

REFERENCE BOOKS

1. Advanced Machining Processes, V.K.Jain, Allied Publications
2. Manufacturing Process -B. M. L.Begeman.
3. Production Technology -H.M.T.
4. Automation production system - Michel P Groover (Prentice CIM Hall of India Ltd, New Delhi)
5. CAD / CAM – Michel P.Groover
6. Non-Conventional Machining, P.K.Mishra, The Institution of Engineers (India) Test book series, Narosa Publishing House – 2005
7. Introduction to Rapid Prototyping, A Ghosh, North West Publication

Tables specifying the scope of syllabus to be covered for Unit Test-I & Unit Test-II

Unit Test	Learning Outcome to be covered
Unit Test-I	From 1.1 to 2.12
Unit Test-II	From 3.1 to 5.13

ALTERNATIVE FUELS & EMISSION CONTROL

Course Code	Course Title	No. of Periods/ Week	Total Periods/ Year	Marks for FA	Marks for SA
A-504	ALTERNATIVE FUELS & EMISSION CONTROL	04	60	20	80

TIME SCHEDULE

Chapter No.	Title	Periods	Marks	Part–A Short Type (3marks)	Part –B Essay Type (10marks)	CO's Mapping
1	Introduction	06	06	2	0	CO1
2	Bio Diesel and Alcohol Fuels	14	26	2	2	CO2
3	Gaseous Fuels	08	13	1	1	CO3
4	Emissions from SI Engines and Their Control	10	23	1	2	CO4
5	Emissions from CI Engines and Their Control	12	26	2	2	CO4
6	Emission Measurement and Emission Standards	10	16	2	1	CO5
TOTAL		60	110	10	8	

Course Code : A-504	Course Name: Alternative Fuels & Emission Control	No. Of periods : 60
Course Objectives	Upon completion of the course student shall be able to 1. Explain the need, availability, properties of alternative fuels to use in IC engines. 2. Understand emission formation and control methods in SI and CI engines. 3. Understand working of emission measuring instruments and emission standards.	

CO No.	Course Outcome
CO1	A-504.1 Understand the need of alternative fuels.
CO2	A-504.2 Understand production and properties of biodiesel and alcohol fuels.
CO3	A-504.3 Understand the properties and suitability of gaseous fuels in IC engines.
CO4	A-504.4 Explain emissions formation and their control in SI and CI engines
CO5	A-504.5 Explain Bharat Stage (BS) & EURO emission standard.

LEARNING OUTCOMES:

Upon completion of the course the student shall be able to

1. INTRODUCTION

- 1.1. Define a fuel and an alternative fuel.
- 1.2. Understand the need of alternative fuels
- 1.3. Define a renewable fuel and non-renewable fuel.
- 1.4. Understand about their availability and properties.
- 1.5. List the types of alternate fuels that can be used for propulsion of automobile.
- 1.6. State the merit and demerits of these alternative fuels.

2. BIO DIESEL AND ALCOHOL FUELS

- 2.1. Define a Biodiesel
- 2.2. List the properties of Biodiesel
- 2.3. List the feed stocks used for manufacturing of Biodiesel
- 2.4. Compare the characteristics with diesel
- 2.5. List the materials compatible to Biodiesel
- 2.6. List the methods of producing Biodiesel
- 2.7. State the Chemical equation for Transesterification reaction
- 2.8. Explain Batch Production Process for manufacturing of Biodiesel with a block diagram
- 2.9. Effects of using B 100 and modifications required in the engine
- 2.10. Justify the use of B20
- 2.11. Explain the emission characteristics of Biodiesel blends in C.I engines.
- 2.12. Advantages and Limitations of Biodiesel blends as automobile fuels.
- 2.13. Understand the general properties of alcohols
- 2.14. List out the various types of alcohols that can be used as fuel in automobiles
- 2.15. Understand the effects if directly used as fuel in S.I engine.
- 2.16. Explain the effects of blending of methanol or ethanol with gasoline or diesel.
- 2.17. Summarize the emission characteristics of these blends.
- 2.18. Advantages and Limitations of Alcohol blends as automobile fuels.

3. GASEOUS FUELS

- 3.1. Define a Gaseous fuel
- 3.2. List the gaseous fuels that can be used in automobile engines.
- 3.3. State the properties of CNG, LPG, Hydrogen and LNG.
- 3.4. Explain the modifications required in the fuel system of the engine.
- 3.5. Describe the working of the fuel system of a gas engine
- 3.6. Advantages and Limitations of gaseous fuels as automobile fuels.
- 3.7. Explain handling, storage and safety of gaseous fuels.

4. EMISSIONS FROM SI ENGINES AND THEIR CONTROL

- 4.1. State the need for control of engine emissions
- 4.2. Explain emissions effects on health and environment
- 4.3. List the sources of emissions from SI engine
- 4.4. Understand emission formation in SI engine
- 4.5. Explain the effects of design variables on emission formation in SI engines
- 4.6. Explain the effects of operating variables on emission formation in SI engines
- 4.7. Explain charcoal canister control for evaporative emission control
- 4.8. Explain positive crank case ventilation system for UBHC emission reduction.
- 4.9. Explain controlling of pollutants in engines exhaust after treatment-2-way and 3-way catalytic converters

5. EMISSIONS FROM CI ENGINES AND THEIR CONTROL

- 5.1. Understand emission formation in CI engine
- 5.2. Explain the effects of design variables on emission formation in CI engines
- 5.3. Explain the effects of operating variables on emission formation in CI engines
- 5.4. Explain controlling of emissions using Exhaust Gas Recirculation (EGR) method.
- 5.5. Explain controlling of emissions using Diesel Oxidation Catalysts (DOC)
- 5.6. Explain controlling of emissions using Diesel Particulate Filters (DPF)
- 5.7. Explain controlling of emissions using Selective Catalytic Reduction (SCR)
- 5.8. Explain controlling of emissions using Lean NOx Trap (LNT)
- 5.9. Explain controlling of emissions using Diesel exhaust Fluid (DEF)
- 6. EMISSION MEASUREMENT AND EMISSION STANDARDS**
- 6.1. List the emission measuring instruments.
- 6.2. Explain the Techniques/Instruments used for Measurement of Hydrocarbons (HCs)
- 6.3. Explain the Techniques/Instruments used for measurement of Carbon monoxide (CO) and carbon dioxide (CO₂)
- 6.4. Explain the Techniques/Instruments used for Measurement of Nitrogen oxide (NOx) and Particulate Matter (PM)
- 6.5. Understand the pollution norms in India
- 6.6. Explain EURO norms – I, II, III, IV, V and VI
- 6.7. Explain Bharat Stage (BS) norms – II, III, IV and VI
- 6.8. Understand the concept of Real Driving Emission (RDE)

COs–POs Mapping

CourseCode:A-504		CourseTitle:ALTERNATIVE FUELS & EMISSION CONTROL		No.of COs:5	No.ofPeriods:60
POs	MappedwithCONo.	COPeriodsAddressingPOin Col.1		Level 2,3)	Remarks
		Number	Percentage		
PO1	CO1,CO2,CO3,CO4,CO5	5	8	1	>40%Level3(Highly Addressed) 25%to40%level2 (Moderately Addressed) 5%to25%level1 (Low Addressed) <5%Not Addressed
PO2	CO5	4	7	1	
PO3	CO2,CO3,CO4	12	20	1	
PO4	CO2,CO3,CO4,CO5	15	25	2	
PO5	CO2,CO3,CO4,CO5	24	40	3	
PO6					
PO7					

CO-PO/PSOMATRIX

CO No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
A-504.1	1				1			1		
A-504.2	1	1	1	1	3			1		
A-504.3	1	1	1	1	3			1		
A-504.4	1	1	2	3	3			2		
A-504.5	1	1		3	3					
Average	1	0.8	0.8	1.6	2.6			1		

Note: The gaps in CO and PO mapping will be achieved by one or more appropriate activities from the following:

i) Seminars	ii) Tutorials	iii) Guest Lectures	iv) Assignments
v) Quiz competitions	vi) Industrial visit	vii) Tech Fest	
viii) Mini project	ix) Group discussions	x) Virtual classes	
xi) Library visit for e-books			

Course Content:

1. Introduction

Definition of fuel, alternative fuel - necessity of alternative fuels - definition of renewable and nonrenewable fuels - existing alternative fuels like Alcohols, vegetable oils, Bio diesel, LPG, CNG, LNG, Hydrogen, Bio gas- availability of these fuels - properties - alternative fuels that can be used in automobiles – merits and demerits of these alternative fuels

2. Bio Diesel and Alcohols

Definition of Bio diesel - general properties of bio diesel, materials compatible to Biodiesel – feed stocks used for manufacturing of Bio diesel - comparison with petroleum diesel - methods of producing Bio diesel - chemical equation for Transesterification reaction - manufacturing Bio diesel using Batch production process with a block diagram - effects of using B100 - modifications required in the engine to use B100 - use of B 20 - emission characteristics of B20 - advantages and limitations of using Bio diesel blends as CI engine fuel.

General properties of alcohols - types of alcohols – alcohols that can be used in automobiles - effects of using alcohols as fuel in SI and CI engines - Blending of methanol or ethanol with gasoline and diesel - emission characteristics of these blends – advantages and limitations of alcohol blends as automobile fuels.

3. Gaseous Fuels

Definition of Gaseous fuels - types of gaseous fuels – short listing of gaseous fuels used in automobiles - properties of CNG, LPG, Hydrogen and LNG - modifications required in the fuel system of the engine - working of fuel system of a gas engine with a block diagram - advantages and limitations of using gaseous fuels as automobile fuels.

4. Emissions from SI engines and their control

Need for control the emissions - effects on health and environment, Emission formation in SI engines – Hydrocarbons (HCs), carbon monoxide (CO), oxides of nitrogen (NO_x), oxides of Sulphur (SO_x), effects of design and operating variables on emission formation in SI engines, charcoal canister control for evaporative emission control, positive crank case ventilation system for Unburnt Hydrocarbon (UBHC) emission reduction, controlling of pollutant formation in engines exhaust after treatment-2-way and 3-way catalytic converters,

5. Emissions from CI engines and their control

Emission formation in CI engines – Hydrocarbons (HCs), carbon monoxide (CO), oxides of nitrogen (NO_x), oxides of Sulphur (SO_x), Particulate Matter (PM), effect of design and operating variables on pollutant formation, controlling of emissions - Exhaust Gas Recirculation (EGR), exhaust after treatment – Diesel Oxidation Catalysts (DOC), Diesel Particulate Filters (DPF), Selective Catalytic Reduction (SCR) and Lean NO_x Trap (LNT), Diesel exhaust Fluid (DEF).

6. Emission measurement and emission standards

Emission measuring instruments for petrol and diesel engines. Techniques used for Measurement of Hydrocarbons (HCs), Carbon monoxide (CO) and carbon dioxide (CO₂), Nitrogen oxide (NO_x), pollution norms as per motor vehicle act 1989 –and EURO norms – I, II, III, IV, VI and Bharat Stage (BS) – II, III, IV, VI and Real Driving Emission (RDE)

Reference books

1. V. Ganesan, Internal combustion engines, 4/e, McGraw Hill, 2015.
2. J. Erjavec, A systems approach to automotive technology, 2/e, Cengage Learning, 2013.
3. J. B. Heywood, Internal Combustion Engines Fundamentals, McGraw Hill, 2017.
4. M.F. Hordeshi, Alternative Fuels: The Future of Hydrogen, The Fairmont Press, Inc., 2008.
5. R.K. Rajput, A textbook of Internal Combustion Engines, 2/e, Laxmi Publications (P) Ltd, 2007.
6. S.S. Thips, Alternative Fuels: Concepts, Technologies and Developments, Jaico Publishing House, 2010

Table specifying the scope of syllabus to be covered for Unit Test-I & Unit Test-II

Unit Test	Learning Outcomes to be covered
Unit Test-I	From 1.1 to 3.7
Unit Test-II	From 4.1 to 6.8

ELECTRIC ANDHYBRID ELECTRIC VEHICLES

Course Code	Course Title	No. of Periods/ Week	Total Periods/ Year	Marks for FA	Marks for SA
A-505	ELECTRIC AND HYBRID ELECTRIC VEHICLE	04	60	20	80

TIME SCHEDULE

Sl. No	Major topic	Periods Allocated	Weight age of marks	Part – A Short Type (3 marks)	Part – B Essay Type(10 Marks)	CO's Mapping
1	Introduction to Electric and Electric hybrid vehicles	10	16	2	1	CO1
2	Energy Storage	16	26	2	2	CO2
3	Design of Drive train	12	26	2	2	CO3
4	Electric Propulsion	12	26	2	2	CO4
5	Regenerative Braking	10	16	2	1	CO5
	Total	60	110	10	8	

Course Code : A-505	Course Name: Electric and HYBRID Electric Vehicles No. Of periods : 60
Course Objectives	Upon completion of the course student shall be able to 1. Study the concepts and drive train configurations of Electric and Electric hybrid vehicles. 2. Comprehend Electric propulsions systems and energy storage devices. 3. Appreciate regenerative braking systems.

CO No.	Course Outcome
CO1	A-505.1 Explain the concepts and configurations of Electric and Electric Hybrid Vehicles
CO2	A-505.2 Explain the concepts and principles of Energy storage systems used for Electric and Electric Hybrid Vehicles
CO3	A-505.3 Design hybrid electric drive trains used for Electric and Electric Hybrid Vehicles
CO4	A-505.4 Discuss the principle and working of electric propulsions used for Electric and Electric Hybrid Vehicles
CO5	A-505.5 Discuss the principle and working of regenerative braking systems used for Electric and Electric Hybrid Vehicles

Learning Outcomes:

Upon completion of the course student shall be able to

1.0 Introduction to Electric and Electric Hybrid Vehicles

- 1.1 Describe the history of electric and electric hybrid vehicles.
- 1.2 Describe the Social and environmental importance of Electric and Electric Hybrid vehicles.
- 1.3 Explain basic components of Electric and Electric Hybrid vehicles.
- 1.4 Explain fuel cell vehicles.
- 1.5 Explain the principle of solar power vehicles.
- 1.6 List the advantages of Electric and Electric Hybrid vehicles.

2.0 Energy Storage

- 2.1 List the requirements of energy storage.
- 2.2 Explain the working principle of Nickel and Lithium based batteries.
- 2.3 List the types of fuel cells.
- 2.4 State the characteristics of fuel cells.
- 2.5 State the applications of fuel cells.
- 2.6 Explain principle and operation of proton exchange membrane and alkaline fuel cells.
- 2.7 Describe ultra-capacitors.
- 2.8 Explain basic principle of ultra-capacitor.
- 2.9 Explain Hybridisation of Energy storage and battery management and safety.
- 2.10 Explain selection of batteries and battery charging methods.

3.0 Fundamentals of Drive train

- 3.1 List out four hybrid electric drive trains.
- 3.2 Explain Series hybrid, parallel hybrid, series parallel and complex drive trains.
- 3.3 Explain torque coupling parallel and speed coupling parallel hybrid electric drive.
- 3.4 List out advantages of parallel hybrid drive train.
- 3.5 Describe control strategy of parallel hybrid drive train.
- 3.6 Describe plug in hybrid electric drive.
- 3.7 Describe transmission system used in EV/EHV.

4.0 Electric Propulsion

- 4.1 Explain the principle and operation of permanent magnet brushless dc motor drive.
- 4.2 Explain the principle and operation of switch reluctance motor drive.
- 4.3 Describe sizing and selection of motors.
- 4.4 Explain methods of control of motors.
- 4.5 Explain configuration of drives.

5.0 Regenerative Braking

- 5.1 Describe regenerative braking.
- 5.2 Explain the braking power and energy on front and rear wheels.
- 5.3 Describe energy consumption in regenerative braking.
- 5.4 Explain components of braking system of EV and EHV.
- 5.5 Explain optimal feel series brake.
- 5.6 Explain optimal energy recovery series brake.
- 5.7 Explain parallel braking system used in EV/EHV.
- 5.8 Explain ABS system used in EV/EHV.

COs – POs Mapping

Course Code: A-505		Course Title: ELECTRIC AND HYBRID ELECTRIC VEHICLES		No. of 5	No. of Periods: 60
POs	Mapped with CO No.	CO Periods Addressing PO in Col. 1		Level)	Remarks
		Number	Percentage		
PO1	CO1, CO2, CO3,CO4,CO5	25	42	3	> 40% Level 3 (Highly Addressed) 25% to 40% level 2 (Moderately Addressed) 5% to 25% level 1 (Low Addressed) <5% Not Addressed
PO2	CO2, CO3,CO4,CO5	15	25	2	
PO3	CO3,CO4	10	17	1	
PO4	CO2	5	8	1	
PO5	CO2, CO3,CO4,CO5	5	8	1	
PO6					
PO7					

CO-PO/PSO MATRIX

CO No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
A-505.1	3							1	1	
A-505.2	3	1		1	1			1	3	
A-505.3	3	1	2		1				3	
A-505.4	3	1	3		1				3	
A-505.5	3	1			1				3	
Average	3	0.8	1	0.2	0.8			0.4	2.6	

Note: The gaps in CO and PO mapping will be achieved by one or more appropriate activities from the following:

- i) Seminars ii) Tutorials iii) Guest Lectures iv) Assignments v) Quiz competitions vi) Industrial visit vii) Tech Fest viii) Mini project ix) Group discussions x) Virtual classes xi) Library visit for e-books

Course Content:

1. Introduction to Electric and Electric hybrid vehicles :

History of Electric and Electric Hybrid vehicles – Social and environmental importance – basic components of Electric and Electric Hybrid vehicles - Fuel cell vehicles – solar power vehicles

2. Energy Storage:

Requirement of Energy storage – battery technologies – Nickel based (Nickel-iron, Nickel-cadmium, Nickel Metal hybrid) and lithium based (Lithium-polymer, Lithium-ion) batteries – types, characteristics and applications of fuel cells – principle and operation of fuel cells (proton exchange membrane and alkaline fuel cells)– ultra capacitors : basic principle of ultra-capacitors - Hybridisation of energy storage – battery management – battery safety – monitoring cells – thermal management – selection of batteries - battery charging methods.

3. Design of Drive train :

Hybrid Electric drive train – a) Series hybrid b) parallel hybrid c) series – parallel d) complex drive train – torque coupling parallel hybrid electric drive – speed coupling parallel hybrid electric drive - advantages of parallel hybrid drive train – control strategy of parallel hybrid drive train – plug in hybrid electric drive – transmission for EV/EHV.

4. Electric Propulsion:

DC motor drives – permanent magnet brushless dc motor drive (principle and operation) – switch reluctance motor drive (SRM) – sizing and selection of motors – control of motor and configuration of drives.

5. Regenerative Braking:

Energy consumption – braking power and energy on front and back wheels – braking system of EV and EHV – Optimal feel series brake – optimal energy recovery series brake – parallel brake – ABS for EV/EHV.

Reference books:

1. Vehicular Electric Power Systems – Mehrdad Ehsani, Yimin Gao CRC Press 2003
2. Electric and Hybrid Vehicles – I. Husain CRC Press 2003

Table specifying the scope of syllabus to be covered for Unit Test-I & Unit Test-II

Unit Test	Learning Outcomes to be covered
Unit Test – I	From 1.1 to 2.10
Unit Test – II	From 3.1 to 5.8

I.C. ENGINE TESTING AND VEHICLE DIAGNOISING LAB

Course Code	Course Title	No. of Periods/ Week	Total Periods/ Year	Marks for FA	Marks for SA
A-506	IC ENGINE TESTING AND VEHICLE DIAGNOISING LAB	06	90	40	60

TIME SCHEDULE

Chapter No.	Title	Periods	CO's Mapping
1.	IC Engine Testing	45	CO1&CO2
2.	Vehicle Diagnosing	45	CO3 & CO4
3	Demonstration of driving skills (One batch will go for driving practice and another batch will do the experiment)		CO5
TOTAL		90	

Course Code: A-506 COURSE: IC ENGINE TESTING AND VEHICLE DIAGNOISING LAB		Total No. of Periods: 90
COURSE OBJECTIVES	Upon completion of the course the student shall able to - Practice load test, Morse test on given petrol and diesel engines and draw the performance curves and heat balance sheet - Practice wheel balancing, wheel alignment, engine analysing, exhaust gas analysing and spark plug testing - Practice driving on four wheelers	

CO No.	Course Outcome	
CO1	A-506.1	Draw performance curves by conducting load test on a given petrol and diesel engine
CO2	A-506.2	Find mechanical efficiency by conducting Morse test on a given I.C engine and draw the heat balance sheet.
CO3	A-506.3	Perform 1. Wheel balancing, 2. Wheel alignment, 3. Exhaust gas analyser, 4. OBD scanner and 5. Spark plug cleaning and testing.
CO4	A-506.4	Troubleshoot the engine using engine analyser
CO5	A-506.5	Demonstrate driving skills on four wheelers

Learning Outcomes: Upon completion of the course student shall be able to

IC ENGINE TESTING

- Load tests and performance curves on the available diesel engine
- Load tests and performance curves on the available petrol engine
- Morse Test
- Drawing of heat balance sheet of an IC engine

VEHICLE DIAGNOISING

1. Wheel Balancing
2. Wheel Alignment
3. Engine Analyser
4. Exhaust gas Analyser for Petrol Engines
5. Exhaust Gas Analyser for Diesel Engine
6. Spark Plug Cleaning and Testing

POs - COs Mapping Format

Course Code: A-506		Course Title: IC ENGINE TESTING AND VEHICLE DIAGNOISING LAB			No of Cos:5	No. Of periods: 90
POs	Mapped with CO No	CO Periods Addressing PO in Col 1		Level	Remarks	
		No	%		>40% Level 3 (Highly Addressed)	
PO1	CO1, CO2, CO3, CO4, CO5	20	22	1	25% to 40% Level 2 (Moderately Addressed)	
PO2	CO1, CO2, CO3, CO4, CO5	20	22	1	5% to 25% Level 1 (Low Addressed)	
PO3	CO3, CO4, CO5	10	11	1	<5% Not Addressed	
PO4	CO1, CO2, CO3, CO4, CO5	30	34	1		
PO5	CO1, CO2, CO3, CO4, CO5	10	11	1		
PO6						
PO7						

CO-PO/PSO MATRIX

CO No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
A-506.1	1	1		2	1			2	1	
A-506.2	2	2		2	1			2	1	
A-506.3	2	2	1	2	1			2	1	1
A-506.4	2	2	1	2	1			2	1	1
A-506.5	1	1	1	2	1			2	1	1
Average	1.6	1.6	0.6	2	1			2	1	0.6

Note: The gaps in CO and PO mapping will be achieved by one or more appropriate activities from the following:

i) Seminars ii) Tutorials iii) Guest Lectures iv) Assignments v) Quiz competitions vi) Industrial visit vii) Tech Fest viii) Mini project ix) Group discussions x) Virtual classes xi) Library visit for e-books

The competencies and key competencies to be achieved by the student

Sl.	Name of the Experiment	Key Competencies	Competencies
1	Load tests and performance curves on the available diesel engine and Petrol engine	• Through knowledge about diesel engine and its various systems • Knowledge about test equipment • Safety awareness • Collection of data • Calibration of test rig. • Equipment maintenance • ☐ Data collection and analysis	• Planning and organization: The tester must organize the testing process, including selecting appropriate test procedures, setting up equipment • Attention to detail: ensure that all measurements are taken correctly and accurately recorded. • Technical knowledge: solid understanding of diesel engine under test and its other systems. • Testing skills: must familiar with various available dynamometer testing, emissions testing, • Data analysis skills: should be able to analyze data collected during testing to identify trends, anomalies, and potential issues. • Safety awareness: have a thorough understanding of safety procedures and protocols. • Problem-solving skills: able to troubleshoot issues may arise during testing, such as equipment malfunctions or unexpected results. Should able to calculate using appropriate formulas and plot required curves. Communication skills: should able to communicate the test results effectively.

3.	Morse test	• Cranking the engine and operating the decompression lever • Varying the load • Maintaining constant speed • Disconnecting engine cylinders one by one by operating the lever • Record the readings in spring balance without parallax error	• Define I.P., B.P. and F.P. • Write down the formulae for I.P., B.P. and F.P. • Cranking the engine and operating the decompression lever • Varying the load • Maintaining constant speed • Disconnecting engine cylinders one by one by operating the lever • Record the readings in spring balance without parallax error
4.	Heat Balance Sheet	• Identify the locations of cooling water inlet and outlet • Allow cooling water to enter the cylinder jacket. • Cranking the engine and operating the decompression lever • Adjust the load • Record the load and speed • Record the reading of cooling water temperature at inlet and outlet • Record time taken for 10c.c fuel consumption • Record exhaust gas temperature	• State various parameters to be measured • Identify the locations of cooling water inlet and outlet • Allow cooling water to enter the cylinder jacket • Cranking the engine and operating the decompression lever • Adjust the load • Record the load and speed • Record the reading of cooling water temperature at inlet and outlet • Record time taken for 10c.c fuel consumption • Record exhaust gas temperature

VEHICLE DIAGNOISING

The competencies and key competencies to be achieved by the student

Sl.N.	Name of the Experiment	Key Competencies	Competencies
1	Wheel Balancing	• Check Static Balancing • Check Dynamic Balancing	• Identify tools required for removing wheel • Determination of Wheel diameter and width • Entering all the parameters in computerized wheel balancer • Conduct of test • Add weights if necessary • Run the same procedure until the balancing is perfect
2	Wheel Alignment	• Check Toe-in • Check Toe-out • Check castor • Check camber	• Set up of the vehicle on a level surface • Installation of the wheel targets on each wheel

		• Check steering axis inclination	• Connection of the computerized wheel alignment tool to the targets • Preparation of tool for test. • Measurement of current alignment settings of the vehicle, • Adjustment of the alignment • Verification of the alignment settings are within the recommended range. • Document the alignment settings before and after adjustments • Test drive the vehicle to ensure that it handles and drives smoothly and without any pulling or vibration.
3	Engine Analyser	• Ignition timing • Compression pressure • Engine RPM • Exhaust emissions • Fuel injection timing • Fuel injection pressure • Battery voltage • Charging system output	• Identify various tests to be conducted. • Connect various probes • check various parameters • Record Various parameters • Compare with standards • Identify problems • Repair accordingly
4	Exhaust gas Analyzer for Petrol & Diesel Engines	• Note readings of pollutant levels • Compare with standards.	• State pollutants in exhaust emissions. • Connect the probe to the exhaust tail pipe • Note readings of pollutant levels • Compare with standards • Adjust or repair for compliance
5	Spark Plug Testing by Using Spark Plug Cleaning & testing Machine	• Check intensity of spark	• Check spark plug condition and electrode gap • Clean spark plug using machine • Check intensity of spark
6	Driving practice on light motor vehicle Or on driving simulator	• Foot controls • Driving controls • Operate all controls simultaneously.	• Identify foot controls of a four wheeler. • State the purpose of foot controls. • Operate foot controls efficiently. • Operate steering controls. • Operate gear shift lever. • Identify various accessories. • Operate all controls simultaneously.

NOTE:

1. Sessional Marks will be 20 in each laboratory
2. The End Examination will be conducted separately in two laboratories each (30 + 30)
3. Internal assessment marks will be awarded based on the performance of the candidate in each laboratory (20 + 20)
4. Both labs has to be appeared at a time. If a candidate did not appear for any one of the labs, it will be treated as Absent for the entire subject.
5. A candidate is declared passed, if he gets required percentage of marks to pass a practical subject by adding all the marks obtained in both the labs i.e., 50 % in the Practical end examination and combined minimum of 50 % of both sessional and practical end examination marks put together.
6. If a student fails in the subject, he has to appear for both the practicals in the subsequent examinations.

ELECTRIC AND HYBRID ELECTRIC VEHICLES LAB

Course Code	Course Title	No. of Periods/ Week	Total Periods/ Year	Marks for FA	Marks for SA
A-507	ELECTRIC AND HYBRID ELECTRIC VEHICLES LAB	03	45	40	60

TIME SCHEDULE

Chapter No.	Title	Perio	CO's Mapping
1.	Demonstration of the working of electric vehicles, solar powered vehicle and fuel cell vehicle	3	CO1
2.	Demonstration of working of hybrid electric vehicle, plug in hybrid electric vehicle	3	CO2
3.	Study the operation of dual fuel IC engine, performance and emission analysis	6	CO3
4.	Demonstration of the working, construction of batteries and its types	9	CO4
5.	Study of hybrid electric vehicle drive train configurations.	6	CO4
6.	Study of Auxiliary systems including charging, starter motor, on board power supply, lighting and environmental sensing and conducting repairs.	9	CO4
7.	Repair & Replacement of Electric/ Hybrid Vehicle body and drive train	6	CO5
8.	Understand Fault diagnosis & repair / replacement of Battery, DC & AC Electrical Machines of Electric Vehicles	3	CO5
Total:		45	

Course Code: A-508 Course Title: ELECTRIC AND HYBRID ELECTRIC VEHICLES LAB No. of Periods: 45		
COURSE OBJECTIVES	Upon complete on of the course the student shall able to	
	1. Understand the working of Electric vehicle, hybrid vehicle and plug in hybrid vehicles and their components 2. Understand the operation, performance and emission of dual fuel engine 3. Repair and replace of electric/hybrid vehicles body and drive train components	
CO No.	Course Outcome	
C01	A-507.1	Identify the components of electric vehicle, solar vehicle and fuel cell vehicle.
C02	A-507.2	Identify the components of hybrid vehicle and plug in hybrid electric vehicle
C03	A-507.3	Understand the operation, performance and emission of dual fuel engine
C04	A-507.4	Understand the hybrid electric vehicle drive train configurations, batteries and auxiliary systems of Electric vehicles
C05	A-507.5	Know the repair and replace of electric/hybrid vehicles body and drive train components

LEARNING OUTCOMES:

Upon completion of the course the students shall be able to

LIST OF EXPERIMENTS

1. Demonstration of electric vehicle, solar vehicle and fuel cell vehicle working and their components
2. Demonstration of hybrid electric vehicles and plug in hybrid electric vehicles working and their components
3. Study of dual fuel engine operation and its performance
4. Study of dual fuel engine emission characteristics
5. Demonstration of working, construction and properties of Lead Acid battery
6. Demonstration of working, construction and properties of Nickel batteries
7. Demonstration of working, construction and properties of Lithium batteries
8. Study of series and parallel hybrid drive systems.
9. Study of series parallel hybrid drive systems.
10. Study of Electric/hybrid electric/plug in hybrid electric vehicles charging system.
11. Demonstration of working of various motors used in Electric/hybrid electric/plug in hybrid electric vehicles
12. Repair of on-board power supply, lighting and environmental sensing and conducting systems.
13. Repair & Replacement of Electric/ Hybrid Vehicle body
14. Repair & Replacement of Electric Vehicle Drive Train
15. Fault diagnosis & repair / replacement of Battery, DC & AC Electrical Machines, Hybrid Electric Vehicles

POs - COs Mapping Format

Course Code: A- 507		Course Title: Electric and hybrid electric vehicles lab		No. of Cos:5	No. Of periods:45
POs	Mapped with CO No	CO Periods Addressing PO in Col 1		Level (1,2,3)	Remarks
		No	%		
PO1	CO1,CO2,CO3,CO4	5	12	1	>40%Level3(HighlyAddressed) 25%to40%Level2(ModeratelyAddressed) 5%to25%Level1 (LowAddressed) <5%Not Addressed
PO2	CO2,CO3,CO4	5	11	1	
PO3		12	26	2	
PO4	CO1,CO2,CO3,CO4,CO5	5	11	1	
PO5	CO1,CO2,CO3,CO4,CO5	18	40	3	
PO6					
PO7					

CO-PO/PSOMATRIX

CO No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
A-507.1	1		1		3			2		
A-507.2	1	1	1		3			2		
A-507.3	1	1	2	1	2			3		
A-507.4	1	1	2	1	2			2		
A-507.5			1	2	2			1		
Average	0.8	0.6	1.4	0.8	2.4			2.2		

Note: The gaps in CO and PO mapping will be achieved by one or more appropriate activities from the following:

- i) Seminars ii) Tutorials iii) Guest Lectures iv) Assignments
- v) Quiz competitions vi) Industrial visit vii) Tech Fest
- viii) Mini project ix) Group discussions x) Virtual classes
- xi) Library visit for e-books

The competencies and key competencies to be achieved by the student

S no	Title of the experiment	Key competencies	Competencies
1	Demonstration of electric vehicle, solar vehicle and fuel cell vehicle working and their components	1) Understand the working of electric vehicle, solar vehicle and fuel cell vehicle	1) State the functions of electric vehicle, solar vehicle and fuel cell vehicle key components 2) State the similarities and dissimilarities among electric vehicle, solar vehicle and fuel cell vehicle
2	Demonstration of hybrid electric vehicles and plug in hybrid electric vehicles working and their components	1) Understand the working of hybrid electric vehicles and plug in hybrid electric vehicles	1) State the functions of hybrid electric vehicles and plug in hybrid electric vehicles key components 2) State the similarities and dissimilarities between hybrid electric vehicles and plug in hybrid electric vehicles
3	Study of dual fuel engine operation and its performance	1) Understand the operation and performance of dual fuel engine.	1) State the constructional difference between SI/CI engine and a dual fuel engine. 2) Compare the performance of dual fuel engine with SI/CI engine.
4	Study of dual fuel engine emission characteristics	1) Understand the emission characteristics of dual fuel engine.	1) State the emission formation in dual fuel engine. 2) State the effect of operating variables on emission formation.
5	Demonstration of working, construction and properties of batteries	1) Understand the working, constructional features and properties of Lead acid battery.	1) State the materials used for various components of lead acid battery. 2) State energy/power density and life of lead acid battery. 3) State the life cycle and recycling of lead acid batteries.
6	Demonstration of working, construction and properties of Nickel batteries	1) Understand the construction and working of Nickel batteries.	1) State types of nickel batteries. 2) State the materials used for various components of nickel batteries. 3) State energy/power density and life of nickel batteries. 4) State the life cycle and recycling of nickel batteries.

7	Demonstration of working, construction and properties of Lithium batteries	1) Understand the construction and working of Lithium batteries.	1) State types of Lithium batteries. 2) State the materials used for various components of Lithium batteries. 3) State energy/power density and life of Lithium batteries. 4) State the life cycle and recycling of Lithium batteries.
S no	Title of the experiment	Key competencies	Competencies
8	Study of series and parallel hybrid drive systems.	1) Understand the power flow series and parallel hybrid drive system.	1) State the components of series and parallel hybrid drive system.
9	Study of series parallel hybrid drive systems.	1) Understand the power flow series parallel hybrid drive system.	1) State the components of series parallel hybrid drive system.
10	Study of Electric/hybrid electric/plug in hybrid electric vehicles charging system	1) Understand the charging system Electric/hybrid electric/plug in hybrid electric vehicles	1) State the components used for charging the Electric/hybrid electric/plug in hybrid electric vehicles. 2) State the different methods used for charging the Electric/hybrid electric/plug in hybrid electric vehicles.
11	Study the various motors used in Electric/hybrid electric/plug in hybrid electric vehicles	1) Understand the construction and working of permanent magnet brushed and brushless DC and Switched Reluctance Motors used Electric/hybrid electric/plug in hybrid electric vehicles.	1) State the components of permanent magnet brushed and brushless DC and Switched Reluctance Motors. 2) State the similarities and dissimilarities among these motors. 3) State the speed/ torque/ power control of these motors.
12	Repair of on-board power supply, lighting and environmental sensing and conducting systems.	1) Remove the systems from the vehicle. 2) Dismantle the components of the system. 3) Check the condition 4) If necessary repair/ replace the components 5) Assemble the component 6) Check the component Running Condition	1) Identify the tools required for 2) Dismantling the component. 3) Use of appropriate tools to dismantle the component 4) Keep the dismantled parts without any damage in order 5) Clean all the parts with appropriate tool / medium 6) Assemble the component 7) Check and bring to the operating condition
13	Repair & Replacement of Electric/ Hybrid Vehicle body		
14	Repair & Replacement of Electric Vehicle Drive Train		
15	Fault diagnosis & repair / replacement of Battery, DC & AC Electrical Machines, Hybrid Electric Vehicles		

LIFE SKILLS

Course Title : Life Skills	Course code : C23- A-508 (Common to all Branches)
Year/ Semester : V/ VI Semester	Total periods : 45
Type of Course : Lab Practice	Max Marks : 100 (Sessional 40 + External 60)

Course Objectives:	understand the relevance of life skills in both personal and professional lives
	practise life skills complementarily in life-management to lead a happy and successful life

	Course Outcomes:
CO1	exhibit right attitude and be adaptable in adverse and diverse situations
CO2	set appropriate goals and achieve them through proper planning, time management and self-motivation
CO3	solve diverse real-life and professional problems with critical thinking and creativity for a stress-free life
CO4	be an ideal team player and manifest as a leader

Course Delivery:

Text book: “Life Skills” – by State Board of Technical Education and Training, AP

Sl no	Unit	Teaching Hours
1	Attitude	4
2	Adaptability	4
3	Goal Setting	4
4	Motivation	4
5	Time Management	4
6	Critical Thinking	4
7	Creativity	4
8	Problem Solving	5
9	Team work	4
10	Leadership	4
11	Stress Management	4
	Total	45

Course Content:

UNIT 1: Attitude *matters!*

Preparatory activity-Role play; Generating word bank; Types of attitude. Read the passage and answer the related questions, read the story and discuss issues raised; Express opinions on the given topic and fill the grid with relevant words.

UNIT 2: Adaptability... *makes life easy!*

Pair work-Study the given pictures and understand adaptability -read the anecdote and discuss, read the story and answer the questions, role play

UNIT 3: Goal Setting... *life without a goal is a rudderless boat!*

Short term goals and long term goals-SMART features, observe the pictures and answer questions- matching- read the passage and answer questions-filling the grid.

UNIT 4: Motivation... *triggers success!*

Types of motivation-difference between motivation and inspiration- matching different personalities with traits - dialogue followed by questions - writing a paragraph based on the passage.

UNIT 5: Time Management ... *the need of the hour!*

Effective Time Management- Time quadrant - Group task on management of time- Time wasters-fill in the grid, read the story and answer the questions- prioritising tasks.

UNIT 6: Critical Thinking... *Logic is the key!*

Preparatory activity-read the passage and answer the questions- differentiate between facts and assumptions- components of critical thinking- complete the sets of analogies- choose the odd one out- true or false statements- decide which of the conclusions are logical.

UNIT 7: Creativity.... *The essential YOU!!*

Definition- Pre-activity-read the anecdote and answer the questions- matching celebrities with their fields of specialisation- think of creative uses of objects- think creatively in the given situations.

UNIT 8: Problem Solving... *there is always a way out!*

Preparatory activity-read the story and answer the questions- discuss the given problem and come out with three alternative solutions- group activity to select the best solution among available alternatives- discuss the problem and plan to analyse it.

UNIT 9: Team Work... *Together we are better!*

Advantages of team work- Characteristics of a team player- Activity-Observe the pictures and classify them into two groups- team game - read the story and answer the questions- fill in the grid.

UNIT 10 : Leadership... *the making of a leader!*

Characteristics of effective leadership- styles of leadership- Activity-read the dialogue and answer the questions- identify the people in the picture and describe them- discuss leadership qualities of the given leaders- filling the grid- read the quotes and write the name of the leader.

UNIT 11: Stress Management ... *live life to the full !!*

Types of stress- Strategies for Stress Management- Activity-read the passage and answer the questions, read the situation and write a paragraph about how to manage stress.

Mapping COs with POs

POs	1	2	3	4	5	6	7
COs	POs 1 to 5 are applications of Engineering Principles, can't directly be mapped with Life Skills					1,2,3,4	1,2,3,4

Unit wise Mapping of COs- POs

CO	Course Outcome	CO Unit Mapped	PO mapped	Cognitive levels as per Bloom's Taxonomy R/U/Ap/An/Ev/Cr (Remembering / Understanding/ Applying/Analysing/ Evaluating/ Creating)
CO 1	To exhibit right attitude and be adaptable to adverse and diverse situations	All Units (1 to 11)	6,7	U/Ap/ An
CO2	To set appropriate goals and achieve them through proper planning, time management and self-motivation	Units 3,4,5	6,7	U/Ap/An
CO3	To solve diverse real-life	Units	6,7	U/Ap/An/ Ev/ Cr.

	and professional problems with critical thinking and creativity for a stress-free life	6,7,8,11		
CO4	To be an ideal team player and manifest as a leader	Units 9,10	6,7	U/Ap/An/ Ev

PROJECT WORK

Course Code	Course Title	No. of Periods/ Week	Total Periods/ Year	Marks for FA	Marks for SA
A-509	PROJECT WORK	06	90	40	60

Course Code: A-509	COURSE: PROJECT WORK	Total No. of Periods: 90
COURSE OBJECTIVES	Upon completion of the course the student shall able to 1. Provide with the opportunity to synthesize knowledge from various areas of learning 2. Identify the problems in a given automobile and modify the design or structure of a given automobile to enhance the performance 3. Critically and creatively apply it to real life situations 4. Prepare proposal for a new design or modify and to prepare a feasibility report for the same	

CO No.	Course Outcome
C01	A-509.1 To build the strength, teamwork spirit and self-confidence
C02	A-509.2 To Improve independent learning
C03	A-509.3 To Apply theory to practical work situations
C04	A-509.4 To gain knowledge in writing report in technical projects

LEARNING OUTCOMES:

Upon Completion of the course the student shall be able to

1.0 Problem solving and Critical Thinking

- 1.1. Generate Ideas from Automobile courses.
- 1.2. Develop these Ideas.
- 1.3. Gather relevant Information.
- 1.4. Evaluate Ideas.
- 1.5. Apply these ideas to a specific task.
- 1.6. Execute appropriate Laboratory skills
- 1.7. Draw Appropriate Conclusions

2.0 Communication

- 2.1 Communicate effectively.
- 2.2 Present Ideas Clearly.
- 2.3 Present Ideas Coherently.
- 2.4 Report writing

3.0 Collaboration

- 3.1. Discuss the ideas.
- 3.2 Coordinate with team members
- 3.3. Team work in accomplishing the task.

4.0 Independent Learning

- 4.1. Involves in the group task.
- 4.2. Analyze the appropriate actions.
- 4.3. Compares merits and demerits
- 4.4. Analyse the activities for sustainability.
- 4.5. Analyze the activities to ensure ethics

5.0 Ethics

- 5.1 Give respect and value to all classmates, educators, colleagues, and others
- 5.2 Understand the health, safety, and environmental impacts of their work
- 5.3 Recognize the constraints of limited resources
- 5.4 Develop sustainable products and processes that protect the health, safety, and prosperity of future generations
- 5.5 Maintain integrity in all conduct and publications and give due credit to the contributions of others

COURSE CONTENT

- 1.0 Design/Fabrication/Analysis/ Case Study Projects in the areas of Automobile Engineering and other related areas

Weightage of marks for Assessment of Learning Outcomes of Project work

S.No	Item	Marks
1	Internal Marks Demonstration of Assigned task in the group to complete the project	40
	End Exam Marks: Demonstration of skill relevant to the project (30) Project Report(20) Viva Voce(10)	60
Total marks		100

End Examination assessment shall be done by both internal, external examiners and faculty members who guided the students during project work.

POs – Cos Mapping Format
A-509, Project Work

Course Code:A-509		Course Title: PROJECT WORK		No of Cos:3	No. Of periods:45
Pos	Mapped with CO No	CO Periods Addressing PO in Col 1		Level	Remarks
		No	%		>40% Level 3 (Highly Addressed)
PO1					25% to 40% Level 2 (Moderately Addressed)
PO2	CO2, CO3	5	12	1	
PO3	CO3	6	13	1	5% to 25% Level 1 (Low Addressed)
PO4	CO2, CO3,CO4	10	22	1	
PO5	CO2, CO3,CO4	10	22	1	<5% Not Addressed
PO6	CO1, CO4	8	17	1	
PO7	CO2	6	14	1	

CO-PO/PSO MATRIX

CO No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
A-509.1						3				
A-509.2		3		3	2		2			2
A-509.3		3	3	3	2					2
A-509.4				3	3	2				
Average		1.5	0.75	2.25	1.75	1.25	0.5			1

Note: The gaps in CO and PO mapping will be achieved by one or more appropriate activities from the following:

- | | | | |
|----------------------|-----------------------|---------------------|-------------------------------|
| i) Seminars | ii) Tutorials | iii) Guest Lectures | iv) Assignments |
| v) Quiz competitions | vi) Industrial visit | | vii) Tech Fest |
| viii) Mini project | ix) Group discussions | x) Virtual classes | xi) Library visit for e-books |

VI SEMESTER

INDUSTRIALTRAINING

Subject Title	Subject Code	Duration
Industrial Training	A-601	6 Months

Time schedule

S.NO	Code	TOPICS	Duration
1	A-601	1 Practical training in Industry 2 Training Report Preparation 3 Report Preparation: Title Page, Certificate, Acknowledgements, Abstract, Contents (introduction of Industry, Plant Layout, Organization Chart, List of Major Equipment's) List of Processes: Skills Acquired, Conclusions and References	Six Months

Scheme of evaluation

Sl.No.	Subject	Duration	Scheme of evaluation		
			Item	Nature	Max. Marks
1	Industrial Training	6 months	1.First Assessment at Industry(After12 Weeks)	Assessment of Learning outcomes by both the faculty and training Mentor of the industry	120
			2.Second Assessment at the Industry(After22 weeks)	Assessment of Learning outcomes by both the faculty and training Mentor of the industry	120
			Final Summative assessment at institution level	Training Report	20
				Demonstration of any one of the skills listed in learning outcomes	30
				Viva Voce	10
			TOTALMARKS		

INDUSTRIAL TRAINING

Course Title	Course Code	DURATION
INDUSTRIAL TRAINING	A-601	6 MONTHS

Time schedule

S.NO	Code	TOPICS	Duration
1	A-601	- Practical training in Industry - Training Report Preparation Report Preparation: Title Page, Certificate, Acknowledgements, Abstract, Contents(introduction of Industry, Plant Layout, Organization Chart, List of Major Equipment's List of Processes: Skills Acquired, Conclusions and References	Six Months

Course Objectives and Course Outcomes

Upon completion of the course the student shall be able to		
Course Objectives		1.Expose to real time working environment 2. Enhance knowledge and skill already learnt in the institution. 3. Acquire the required skills of manufacturing processes, assembling, servicing, and supervising in the engineering fields. 4. Install the good qualities of integrity, responsibility and self-confidence.
Course Out Comes	CO1	Apply theory to practical work situations
	CO2	Cultivate sense of responsibility and good work habits
	CO3	Exhibit the strength, teamwork spirit and self-confidence
	CO4	Write report in technical projects

PO-CO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3
CO1	2				2		1	3		2
CO2						3		3		2
CO3						3		3		2
CO4						3		3		2
	0.5				0.5	2.25	0.25	3		2

3: High, 2: Moderate,1: Low

Learning Outcomes

The student shall be able to display the following skill sets:

- 1) Technical Skills(Manufacturing/Service/Diagnosing the trouble /Maintenance)
- 2) Reading drawings and analysing Specifications
- 3) Recognize and Practice safety Measures
- 4) Handling Tools/Instruments/Materials/Machines
- 5) Assess and Control of quality parameters
- 6) Customer relationship and recording Skills
- 7) Ability to work collaboratively with others in a team-based setting, and communicate technical information effectively both orally and in writing.
- 8) Knowledge of safety standards and regulations related to automobiles, and apply this knowledge to ensure the safe and efficient operation of vehicles.
- 9) understanding of the latest trends and advancements in automobile engineering,
- 10) Appreciation for the impact of the automobile industry on society and the environment, and exploration of ways to improve the sustainability and efficiency of the automotive industry.

Scheme of evaluation

Sl.No.	Subject	Duration	Scheme of evaluation		
			Item	Nature	Max. Marks
1	Industrial Training	6 months	1.First Assessment at Industry (After 12 Weeks)	Assessment of Learning outcomes by both the faculty and training Mentor of the industry	120
			2.Second Assessment at the Industry (After 22 weeks)	Assessment of Learning outcomes by both the faculty and training Mentor of the industry	120
			Final Summative assessment at institution level	Training Report	20
				Demonstration of any one of the skills listed in learning outcomes	30
				Viva Voce	10
TOTAL MARKS					300

Weightage of marks for Assessment of Skill sets during first and second assessment.

Skill Set Sl.No	SKILL SET	Max Marks Allotted For each parameter
1	Technical Skills (Manufacturing/Service/Diagnosing the trouble /Maintenance etc)	25
2	Reading drawings and analysing Specifications	20
3	Recognize and Practice safety Measures	15
4	Handling Tools/Instruments/Materials/Machines	25
5	Assess and Control of quality parameters	15
6	Customer relationship and recording Skills	20
	<i>Total</i>	<i>120</i>

During assessment the performance of the students shall be assessed in those skills in which the student has been trained and be awarded the marks as per the weightage assigned as above. In case the student has undergone training in a few skill sets then the total marks obtained shall be raised to 120 marks for the given assessment i.e. either assessment 1 or 2. However the performance of the student shall be assessed at the most skill sets listed above but not less than three skill sets.

ILLUSTRATION

If the student has undergone training in only 4 skill sets (namely S.No. 1,3,4 and 5), and marks awarded during assessment is 50 out of 80, then the marks of 50 shall be enhanced to 120 proportionately as $50/80 \times 120 = 75$.

GUIDELINES FOR INDUSTRIAL TRAINING OF DIPLOMA IN AUTOMOBILE ENGINEERING PROGRAMME

- Duration of the training: 6 months (24 Weeks)
- Eligibility: The As per SBTET norms
- Training Area: Students may be trained in the fields
Fabrication/Manufacturing/Service/Diagnosing the trouble /Maintenance etc.
- The candidate shall put a minimum of 90% attendance, during industrial training.
- If the student fails to secure 90% attendance during industrial training, the student shall reappear for 6 months industrial training.
- Formative assessment at industry level shall be carried out by the Mentor from the industry, where the student is undergoing training and the in faculty incharge (Guide) from the concerned section in the institution.
- The Industrial training shall carry a weightage of 300 marks and pass marks is 50% in assessments at industry (first and second assessment) and final summative assessment at institution level put together i.e. 150 marks out of 300 marks.
- If the student fails to secure 50% marks in final summative assessment at institution level, the student should reappear for final summative assessment in the subsequent board examination.
- Final summative assessment at institution level is done by a committee including 1. Head of the section (**of concerned discipline ONLY**), 2.External examiner from an industry and 3. Faculty member who assessed the student during Industrial Training as members. Final Summative assessment at institution level is done by a committee including 1. Head of section, 2.External examiner from the industry and 3. Faculty members who assessed the students during Industrial Training, as members.

2. Shall create awareness regarding safety measures to be followed in the industry during the training period, and shall check it is followed scrupulously.
3. Shall check the logbook of the students during the time of their visit for the assessment.
4. Shall monitor progress at regular intervals and make appropriate suggestions for improvement.
5. Shall visit the industry and make first and second assessments as per stipulated schedule.
6. Shall assess the skill sets acquired by the students during their assessment.
7. Shall award the marks for each skill set as per the marks allotted for that skill set during final assessment at institution.
8. Shall voluntarily supplement students learning through appropriate materials like photographs, articles, videos etc.
9. Shall act as co-examiner along with external examiner.
10. Shall act as liaison between the student and mentor.
11. Shall maintain a dairy indicating his observation with respect to the progress of students learning in all three domains (Cognitive, Psychomotor and Affective)

Guidelines to the training mentor in the industry:

- Shall train the students in all the skill sets as far as possible.
- Shall assess and award the marks in both the assessments along with the faculty member.
- Shall check and approve the log books of the students.
- Shall approve the attendance of each student at the end of the training period.
- Shall report to the guide about student's progress, personality development or any misbehaviour as the case may be.
- **Every Teacher (including HoD if not holding any FAC) shall be assigned a batch of students of 10 to 15 for industrial training irrespective of student's placements for training.**

Rubrics for Industrial Training Assessment

Department of Technical Education

Name of the Institution/ College:

PIN:

Name of the student:

<i>Skill</i> <i>Sl.No</i>	<i>SKILL SET</i>	<i>Max Marks Allotted For each parameter</i>	<i>Precisely complete s the task</i>	<i>Completes the task, mistakes are absent, but not Precise</i>	<i>Completes the task, Mistakes are a few</i>	<i>Makes attempt, Mistakes are many</i>
1	Technical Skills (15 M) (i) Analytical Thinking (ii) Creativity. (iii) Problem-solving.	5 5 5	5 5 5	3 3 3	2 2 2	1 1 1
2	Reading Drawings and Analysing Specifications (20 M) ② (i) 2 D and 3D (Computer Aided Design) CAD/Drawings. (ii) Knowledge of specifying the machine member/Automobile Components	10 10	10 10	7 7	6 6	3 3
3	Handling Tools / Instruments / Safety Practices (15) (i) Identification of tools and instruments. (ii) Measuring the parameter (iii) Aware of safety Management	5 5 5	5 5 5	3 3 3	2 2 2	1 1 1
4	Trouble Shooting / Dismantling & Assembling and of the Unit (30) (i) Fault-Finding (ii) Dismantling the given Machine/ Equipment / Engine /Automobile components (iii) Repair or Replacement with spare parts (iv) Assembling of the given Machine/ Equipment / Engine /Automobile components after testing for working condition.	5 10 5 10	5 10 5 10	3 7 3 7	2 6 2 6	1 3 1 3
5	Quality Assessment and Control (15 M)					

	(i) Use of various Inspection and Quality Control Tools (OBD CODES)	5	5	4	3	2
	(ii) Identification of Proper Testing Method and Tool for the manufactured or Serviced or repaired Components	5	5	4	3	2
	(iii) Analyze Test Results	5	5	4	3	2
6	Soft skills and Recording skills(25)					
	(i) Communication Skills (oral/writing skills)	5	5	4	3	2
	(ii) Human relations.	5	5	4	3	2
	(iii) Supervisory abilities.	5	5	4	3	2
	(iv) Recording technical issues	5	5	4	3	2
	(v) Proper Maintenance of records in the industry.	5	5	4	3	2

*Mistakes are with reference to Technique, Procedure & Precautions, while precision refers to Technique, Procedure, Precautions, Time & Result

Total Marks: 120

Marks Awarded: _____

(Marks awarded in words: _____)

Signature of the Training In-charge (Mentor)
Name
Designation

Signature of the faculty in-charge (Guide)
Name
Designation
